

ORIGIN AND EVOLUTION OF AMPHIBIANS

About 360 million years ago, amphibians became the first vertebrates to live on land. The name amphibian comes from the Greek words meaning “double” and “life” and reflects the fact that many amphibians spend part of their life on land and part in water.

ADAPTATION TO LAND

The first amphibians to spend a significant part of their life on land most likely evolved from lobe-finned fishes. Lobe-finned fishes had several preadaptations that allowed them to transition to life on land. **Preadaptations** are adaptations in an ancestral group that allow a shift to new functions which are later favored by natural selection.

Lobe-finned fishes ancestral to amphibians had a bone structure within their fins that worked as legs that could walk on land. Ancient lobe-finned fishes also had modified pouches in their digestive tracts, which evolved into the lungs of lungfish and swim bladders in most modern fishes. Some groups of ancient lobe-finned fishes also had internal nostrils that functioned in air breathing. In lobe-finned fishes ancestral to amphibians, these structures worked as a means of breathing on land.

Early amphibians required more oxygen than their fish ancestors. Because gravity makes movement on land more difficult than movement in water, early amphibians were likely to have had a higher metabolism than their fish ancestors. As a result, efficient hearts were an important adaptation that allowed oxygen to be delivered to the body more efficiently.

Characteristics of Early Amphibians

Amphibians and lobe-finned fishes share many anatomical similarities, including features of the skull and vertebral column. Also, the bones in the fin of a lobe-finned fish are similar in shape and position to the bones in the limb of an amphibian. Figure 40-1 shows a sarcopterygian (sar-KOP-te-RIJ-ee-uhn), an extinct lobe-finned fish that is thought to be closely related to amphibians. This fish probably lived in shallow water and used its sturdy pelvic and pectoral fins to move along the bottom and to support its body while resting.

OBJECTIVES

- **Describe** three preadaptations involved in the transition from aquatic to terrestrial life.
- **Describe** two similarities between amphibians and lobe-finned fishes.
- **List** five characteristics of living amphibians.
- **Name** the three orders of living amphibians, and give an example of each.

VOCABULARY

preadaptation
tadpole

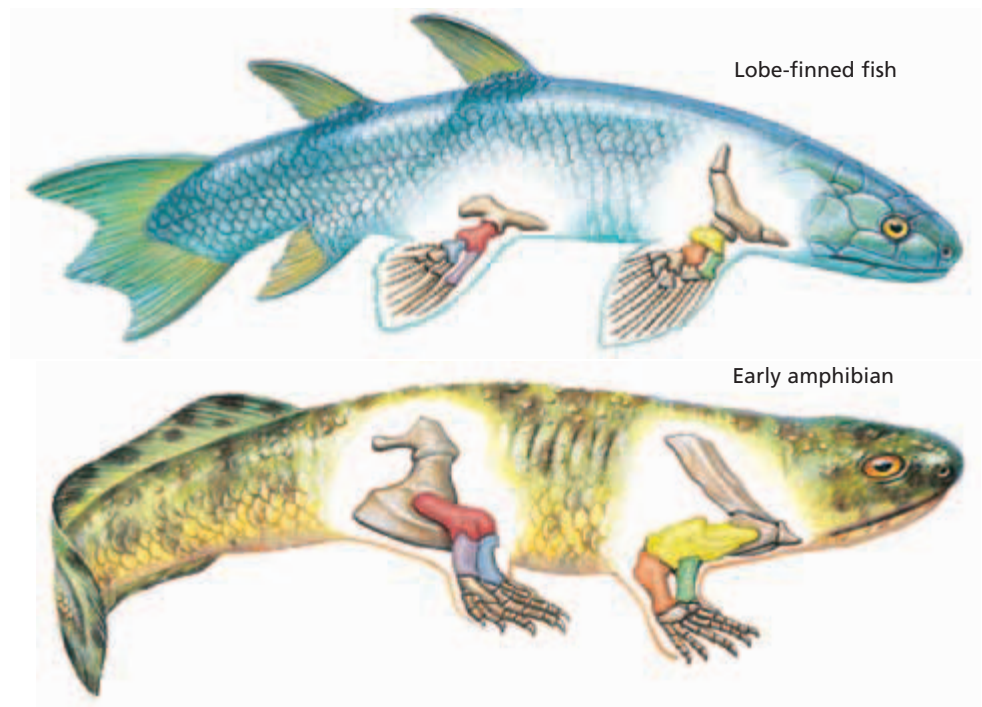
FIGURE 40-1

Early lobe-finned fishes, such as this sarcopterygian, are thought to be ancestors of the first land vertebrates.



FIGURE 40-2

Amphibians have similar bone structure to primitive lobe finned fishes. The colored bones in the figure are homologous structures between lobe-finned fish and amphibians. Homologous structures are anatomical structures that share a common ancestry.



The oldest known amphibian fossils date from about 360 million years ago. All of the early amphibians had four strong limbs, which developed from the fins of their fish ancestors, as shown in Figure 40-2. The forelimbs of amphibians (and all other terrestrial vertebrates) are homologous to the pectoral fins of fishes, and the hind limbs are homologous to the pelvic fins. The early amphibians also breathed air with lungs. Lungs arose early in the history of fishes and are found in the descendants of these early fishes—including terrestrial vertebrates.

Although the early amphibians showed several adaptations for life on land, such as sense organs for detecting airborne scents and sounds, they probably spent most of their time in the water. For example, some of the first amphibians had a large tail fin and lateral-line canals on their head. Their teeth were large and sharp, indicating a diet of fish, not insects. In addition, some of the early amphibians appear to have had gills like those of fishes.

Diversification of Amphibians

During the late Devonian period and the Carboniferous period (359 million to 299 million years ago), amphibians split into two main evolutionary lines. One line included the ancestors of modern amphibians, and the other line included the ancestors of reptiles. Amphibians have been a diverse, widespread, and abundant group since this early diversification.

Today there are about 4,500 species of amphibians, belonging to three orders. The largest order, with more than 3,900 species, is Anura, which includes the frogs and toads. The order Caudata contains about 400 species of salamanders. And the third order, Gymnophiona, consists of about 160 species of caecilians, which are legless tropical amphibians. Figure 40-3 on the following page shows hypotheses for the phylogenetic relationships between these three groups.



Quick Lab

Comparing Fish and Amphibian Skin

Materials disposable gloves, lab apron, safety goggles, paper, colored pencils, living or preserved specimens of a fish and a frog

Procedure



1. Put on your disposable gloves, lab apron, and safety goggles.
2. Handle living animals gently. Touch and examine the skin of the specimens provided by your teacher. Record your observations.
3. When you are finished with your observations, remove your disposable gloves, lab apron, and safety goggles. Wash your hands with soap and water.

Analysis Why can a frog use its skin as a respiratory membrane, while a fish cannot? What behaviors in amphibians enable them to maintain moist skin?

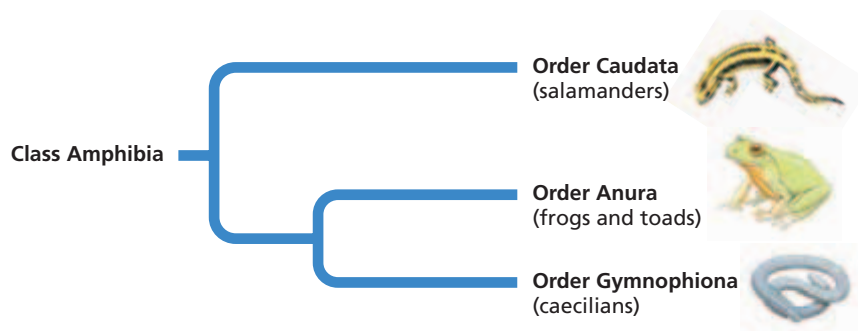


FIGURE 40-3

This phylogenetic diagram shows hypotheses of the evolutionary relationship among modern amphibian orders. For updates on phylogenetic information, visit go.hrw.com and enter the keyword **HM6 Phylo**.

MODERN AMPHIBIANS

Modern amphibians are a very diverse group, but they do share several key characteristics:

- Most change from an aquatic larval stage to a terrestrial adult form. This transformation is called *metamorphosis*.
- Most have moist, thin skin with no scales.
- Feet, if present, lack claws and often are webbed.
- Most use gills, lungs, and skin in respiration.
- Eggs lack multicellular membranes or shells. They are usually laid in water or in moist places and are usually fertilized externally.

Order Anura

Anurans (frogs and toads) are found worldwide except in polar climates and a few isolated oceanic islands. They live in a variety of habitats, from deserts and tundra to tropical rain forests. Many anurans spend at least part of their life in water, and some species are permanently aquatic. Many other species live and reproduce on land. Figure 40-4 shows two examples of anurans. The term *toad* is commonly used for any anuran that has rough, bumpy skin, as seen in Figure 40-4a. The term *frog* commonly refers to anurans having smooth, moist skin, such as in Figure 40-4b. These terms are general descriptions, however, and do not refer to any formal groups of anurans.



FIGURE 40-4

Anurans include toads and frogs such as the plains spadefoot toad (a), *Scaphiopus bombifrons*, which can be found throughout the United States, and the White's tree frog (b), *Litoria caerulea*, which is common in Australia.



(a)



(b)

Word Roots and Origins

caudata

from the Latin *cauda*,
meaning "tail"

Anurans are characterized by a body adapted for jumping. Long, muscular legs provide power for the jump. The anuran body is compact, with a short, rigid spine and strong forelimbs that help absorb the shock of landing. The word *anuran* means "tailless" and reflects the fact that no adult anuran has a tail.

Adult anurans are carnivores that feed on any animal they can capture. Some frogs have a sticky tongue that can be extended to catch prey. Many species of anurans return to water to reproduce. In nearly all species, eggs are fertilized externally. The fertilized eggs hatch into swimming, tailed larvae called **tadpoles**.

Order Caudata

Salamanders have elongated bodies, long tails, and moist skin. Except for a few aquatic species, they have four limbs. The smallest salamanders are only a few centimeters long, while the largest reach lengths of 1.5 m (4.5 ft). Like anurans, salamander species range from fully aquatic to permanently terrestrial. Terrestrial salamanders usually live in moist places, such as under logs and stones. Larval and adult salamanders are carnivores. They are active mainly at night. Figure 40-5 shows two representative salamanders.

Most salamander species live in North America and Central America. There are very few species in Africa and South America, several species are found in Asia and in Europe, and there are no species found in Australia. With more than 300 species, the lungless salamanders (family Plethodontidae) are the largest group of salamanders. As their name suggests, these salamanders lack lungs. They absorb oxygen and release carbon dioxide through their skin.

Like most anurans, many salamanders lay their eggs in water, and the eggs hatch into swimming larval forms. Other species can reproduce in moist land environments. Eggs laid on land usually hatch into miniature adult salamanders and do not pass through a free-living larval stage. Most salamander species have a type of internal fertilization by which females pick up sperm packets deposited by males. In some terrestrial species, the female stays with the eggs until they hatch, which can take up to several weeks.



FIGURE 40-5

The flatwoods salamander (a), *Ambystoma cingulatum*, and the spotted salamander (b), *Ambystoma maculatum*, are members of the order Caudata. *A. cingulatum*, which lives only in Florida, is an endangered species; *A. maculatum* can be found from eastern Canada to eastern Texas.



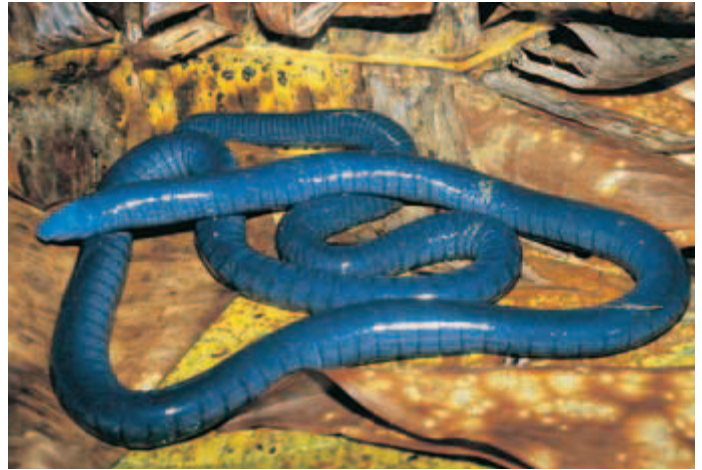
(a)



(b)



(a)



(b)

Order Gymnophiona

The common name used to refer to members of the order Gymnophiona is caecilian (see-SIL-yuhn). Caecilians are a highly specialized group of legless amphibians that resemble small snakes, as you can see in Figure 40-6. Caecilians live in tropical areas of Asia, Africa, and South America. Caecilians average about 30 cm (12 in.) in length, but some species reach lengths of 1.5 m (4.5 ft). Because they have very small eyes that are located beneath the skin or even under bone, caecilians often are blind.

Caecilians are rarely seen, and little is known about their ecology and behavior. Most species burrow in the soil, but some species are aquatic. All species have teeth in their jawbones that enable them to catch and consume prey. They eat worms and other invertebrates, which they detect by means of a chemosensory tentacle located on the side of their head. All species are thought to have internal fertilization. Some species lay eggs, which the female guards until they hatch. In a few species, the young are born alive. These caecilians provide nutrition to their developing embryos. The young use their jaws and teeth to scrape secretions, called “uterine milk,” from the walls of the female’s reproductive tract.

FIGURE 40-6

Caecilians, such as *Ichthyophis kohtaoensis* (a) and *Caecilia nigricans* (b), are primarily carnivores. They are burrowing amphibians that are usually blind, and a few species have scales embedded in their skin.

SECTION 1 REVIEW

1. Describe three adaptations that allowed early sarcopterygians to move onto land.
2. Identify two characteristics that amphibians share with modern lobe-finned fishes.
3. Name five key characteristics that are common to modern amphibians.
4. Differentiate each of the three living orders of amphibians.

CRITICAL THINKING

5. **Applying Information** Early amphibian fossils do not indicate the presence of lungs. Why do scientists think that early amphibians had lungs?
6. **Forming Reasoned Opinions** Is the lack of legs a primitive characteristic for amphibians or a later amphibian adaptation? Support your answer.
7. **Analyzing Information** Why do modern amphibians not have a lateral line?

SECTION 2

OBJECTIVES

- **Relate** the structure of amphibian skin to the types of habitats in which amphibians can survive.
- **Identify** three adaptations for life on land shown by the skeleton of a frog.
- **Sequence** the flow of blood through an amphibian's heart.
- **Describe** how a frog fills its lungs with air.
- **Describe** the digestive and excretory systems of amphibians.
- **Discuss** an amphibian's nervous system.

VOCABULARY

mucous gland
pulmonary circulation
systemic circulation
pulmonary respiration
cutaneous respiration
duodenum
ileum
mesentery
vent
nictitating membrane
tympanic membrane
columella

CHARACTERISTICS OF AMPHIBIANS

As you have already seen, terrestrial vertebrates face challenges that are far different from those faced by aquatic vertebrates. In this section, you will learn about some of the ways amphibians meet the challenges of living on land.

SKIN

The skin of an amphibian serves two important functions—respiration and protection. The skin is moist and permeable to gases and water, allowing rapid diffusion of oxygen, carbon dioxide, and water. Numerous **mucous glands** supply a lubricant that keeps the skin moist in air. This mucus is what makes a frog feel slimy. The skin also contains glands that secrete foul-tasting or poisonous substances that provide protection from predators.

However, the same features that allow efficient respiration also make amphibians vulnerable to dehydration, the loss of body water. Therefore, amphibians live mainly in wet or moist areas on land. Many species are active at night, when loss of water through evaporation is reduced. Although some species of frogs and toads survive in deserts, they spend most of their life in moist burrows deep in the soil. Only after heavy rains do these amphibians come to the surface to feed and reproduce.

Amphibians are affected by pollution. Chemicals present in water can be absorbed by amphibian skin. As a result, amphibians can serve as indicators of the health of an ecosystem.

SKELETON

While water supports the body of an aquatic vertebrate against the force of gravity, terrestrial vertebrates must rely on the support of their strong internal skeleton. The vertebrae of the spine interlock and form a rigid structure that can bear the weight of the body. Strong limbs support the body during walking or standing. The forelimbs attach to the pectoral girdle (the shoulder and supporting bones), while the hind limbs attach to the pelvic girdle (the “hips”). The pectoral and pelvic girdles transfer the body's weight to the limbs. The cervical vertebra at the anterior end of the spine allows neck movement.

The frog skeleton in Figure 40-7 shows several specializations for jumping and landing. In frogs, the bones of the lower forelimb are fused into a single bone, the radio-ulna. The bones of the lower hind limb are fused into the tibiofibula. Frogs have few vertebrae, and the vertebrae at the posterior end of the spine are fused into a single bone called the urostyle. The pectoral girdle has thick bones that are braced to absorb the impact of landing.

CIRCULATORY SYSTEM

The circulatory system of an amphibian is divided into two separate loops. The **pulmonary circulation** carries deoxygenated blood from the heart to the lungs and back to the heart. The **systemic circulation** carries oxygenated blood from the heart to the body and back to the heart.

The three-chambered heart of an amphibian reflects the division of the circulatory system into pulmonary and systemic circulation. Deoxygenated blood from the body first enters the right side of the heart, as shown in step 1 of Figure 40-8. Blood moves into the right atrium. In step 2, oxygenated blood from the lungs enters the left atrium. In step 3, contraction of the atria forces the deoxygenated and oxygenated blood into the single ventricle, the main pumping chamber of the heart. Although the ventricle is not divided, its spongy interior surface and the coordinated contractions of the atria keep the oxygenated and deoxygenated blood from mixing. In step 4, ventricular contraction expels both kinds of blood into the conus arteriosus, which directs deoxygenated blood to the lungs and oxygenated blood to the body.

FIGURE 40-7

The skeleton of the frog has adapted to absorb shocks when the frog jumps and lands.

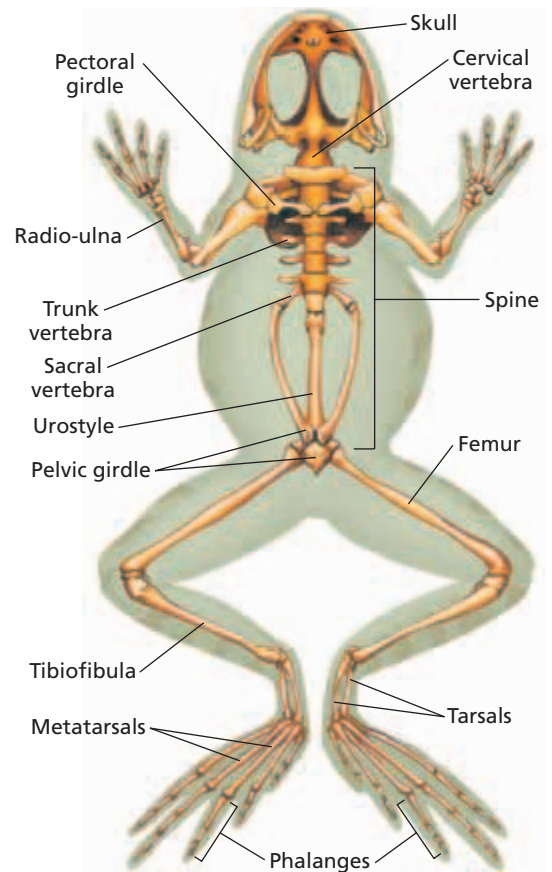
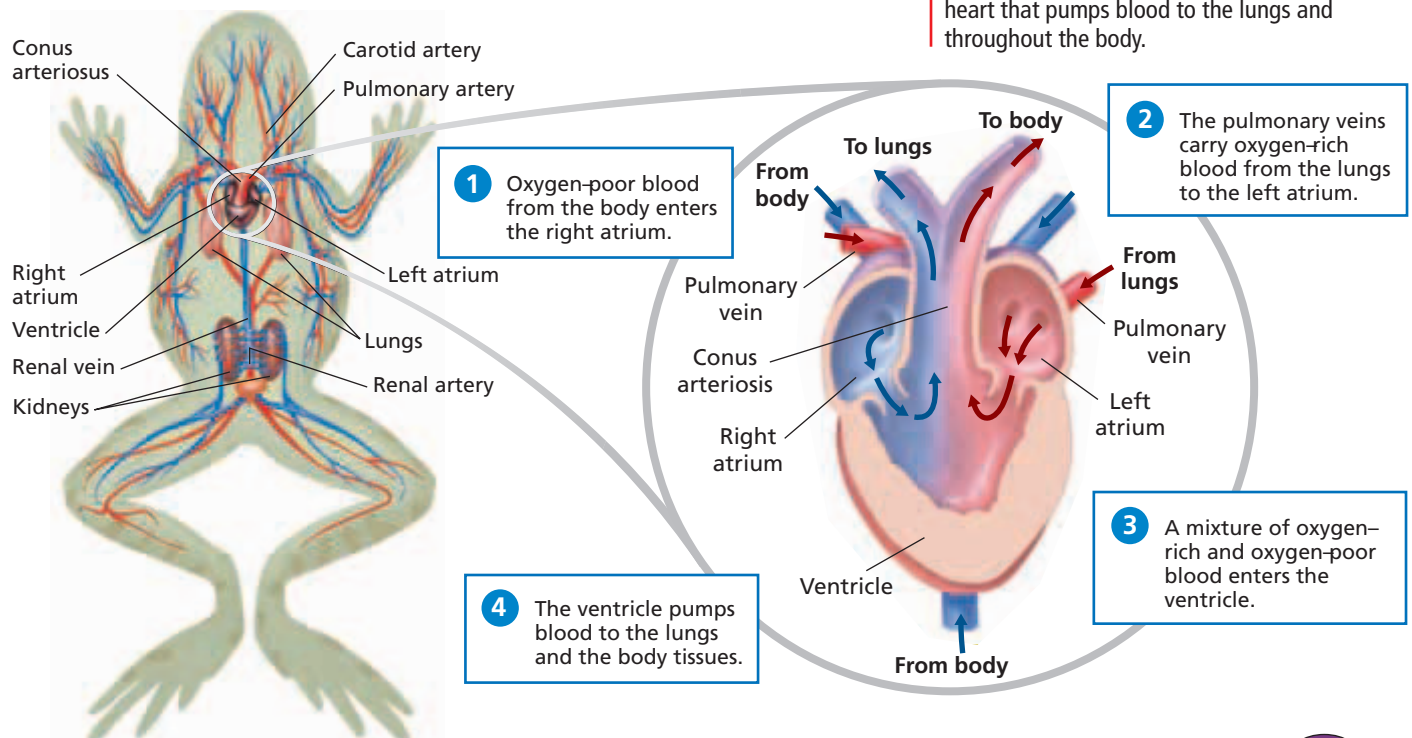


FIGURE 40-8

Amphibians have a three chambered heart that pumps blood to the lungs and throughout the body.



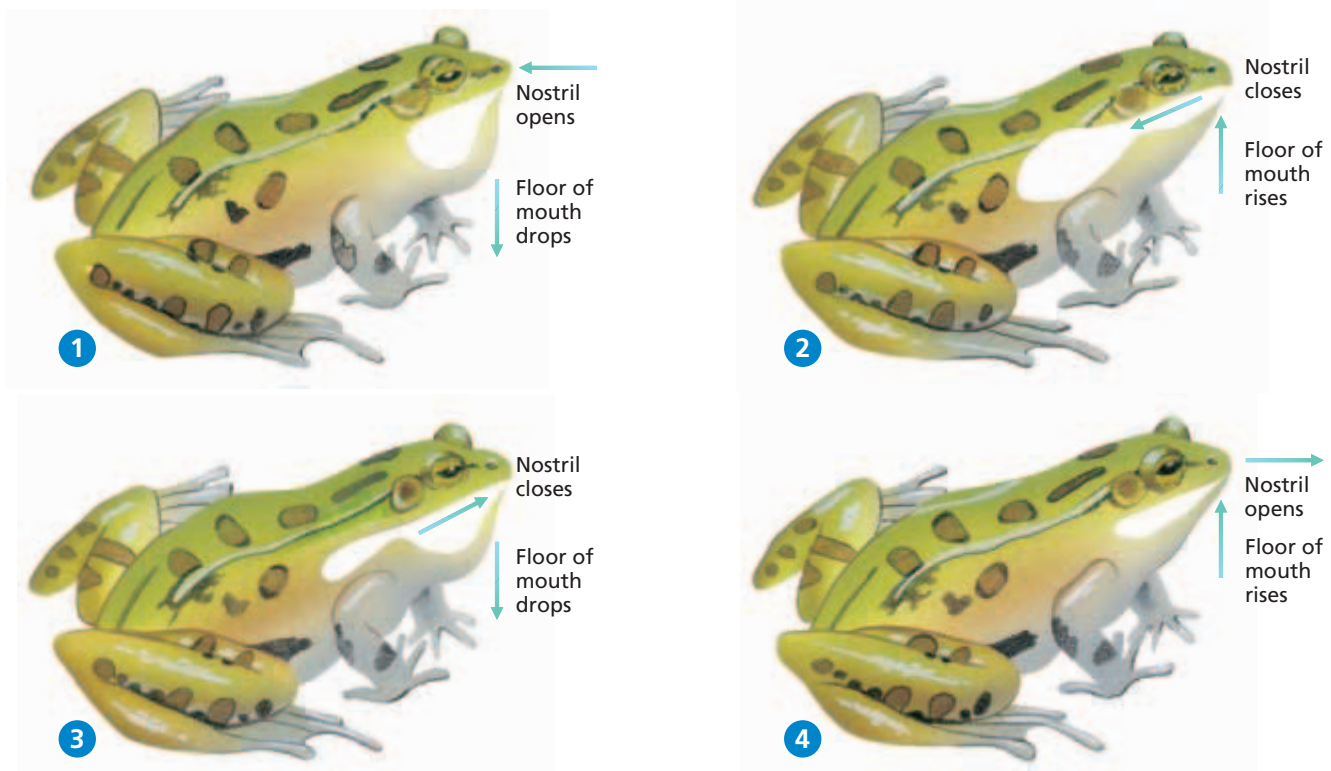
All other terrestrial vertebrates also have a “double-loop” circulatory pattern. This pattern of circulation provides a significant advantage over the “single-loop” circulation of a fish—faster blood flow to the body. In a fish, the blood loses some of its force as it passes through the narrow capillaries of the gills, and blood flow slows as a result. The lungs of an amphibian also contain narrow capillaries that slow blood flow. But after passing through the capillaries of the lung, blood returns to the heart to be pumped a second time before circulating to the body.

RESPIRATION

Larval amphibians respire, or exchange carbon dioxide and oxygen, through their gills and skin. Most adult amphibians lose their gills during metamorphosis, but they can respire in two ways: through the lungs and through the skin. Respiration through the lungs is called **pulmonary respiration**. Amphibians ventilate their lungs with a unique mechanism that pumps air into the lungs; this is called *positive-pressure breathing*. For example, a frog breathes by changing the volume and pressure of air in its mouth while either opening or closing its nostrils, as shown in Figure 40-9. Both inhalation and exhalation involve a two-step process during which the floor of the frog’s mouth is raised and lowered. The frog controls the direction of air flow by opening or closing its nostrils. Because amphibians have a small surface area in the lungs for gas exchange, respiration through the skin, or **cutaneous respiration**, is very important to most aquatic and terrestrial amphibians.

FIGURE 40-9

Frogs breathe by creating pressure that forces air into their lungs. ❶ When the floor of the frog’s mouth drops, air capacity increases in the frog’s mouth, and air rushes in. ❷ When the nostril is closed and the mouth floor rises, the air is forced into the lungs of the frog. ❸ The mouth floor lowers and air is forced out of the lungs. ❹ Then the nostril opens and the mouth floor rises again, forcing air out the nostril.



DIGESTIVE SYSTEM

All adult amphibians are carnivorous. Because most amphibians are small, insects and other arthropods are their most commonly consumed prey. Larger amphibians sometimes eat mice, snakes, fish, other amphibians, and even sometimes birds. Many amphibian larvae, such as those of frogs, are herbivorous, feeding on algae, bacteria, or plants. The larvae of some species, such as those of salamanders, are carnivorous, and some feed on the larvae of other species.

The amphibian digestive system includes the pharynx, esophagus, stomach, liver, gallbladder, small intestine, large intestine, and cloaca. Figure 40-10 shows a ventral view of the digestive system of a frog.

The elastic esophagus and stomach allow an amphibian to swallow large amounts of food. Once food reaches the stomach, tiny glands in the stomach walls secrete gastric juices that help break down, or digest, the food. A muscle called the pyloric sphincter at the lower end of the stomach relaxes, which allows digested food to move into the small intestine. The upper portion of the small intestine is called the **duodenum** (DOO-oh-DEE-nuhm). The coiled middle portion of the small intestine is the **ileum** (IL-ee-uhm). A membrane resembling plastic wrap, called the **mesentery**, holds the small intestine in place. Inside the small intestine, digestion is completed and the released nutrients pass through capillary walls into the bloodstream, which carries them to all parts of the body.

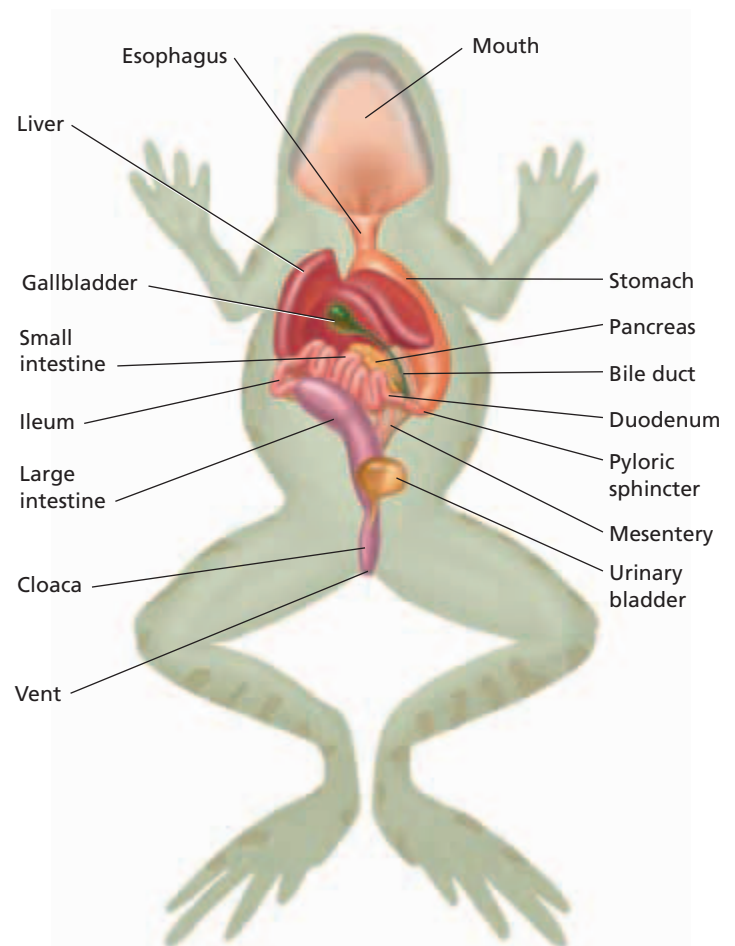
The lower end of the small intestine leads into the large intestine. Here indigestible wastes are collected and pushed by muscle action into a cavity called the *cloaca* (kloh-AY-kuh). Waste from the kidneys and urinary bladder, as well as either eggs or sperm from the gonads, also passes into the cloaca. Waste materials exit the body through the **vent**.

Accessory Glands

Other glands and organs aid in the digestion process. The liver produces bile, which is stored in the gallbladder. Bile helps break down fat into tiny globules that can be further digested and absorbed. A gland called the pancreas, located near the stomach, secretes enzymes that enter the small intestine and help break down food into products that can be absorbed by the blood.

FIGURE 40-10

The frog digestive system is shown in ventral view. Notice how the short small intestine is an adaptation for a carnivorous diet.



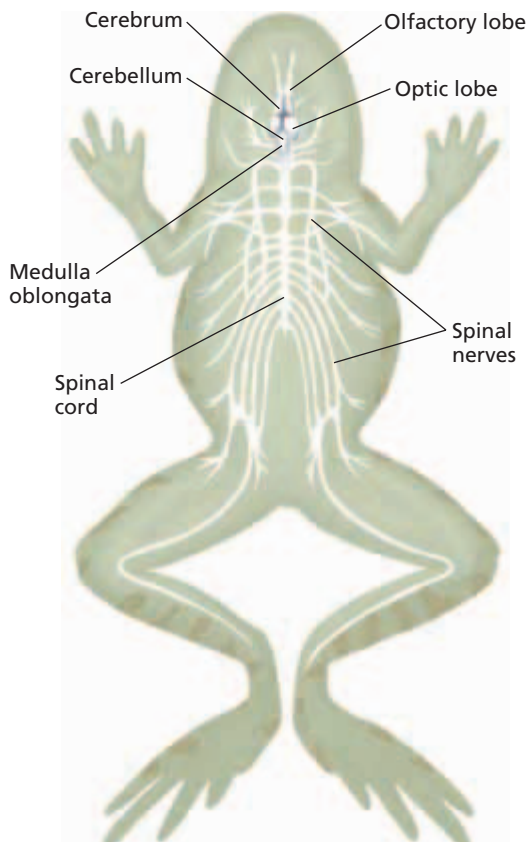
EXCRETORY SYSTEM

The kidneys are the primary excretory organs. One kidney lies on either side of the spine against the dorsal body wall. The kidneys filter nitrogenous wastes from the blood. These wastes, flushed from the body with water, are known as urine. Urine flows from the kidneys to the cloaca through tiny tubes called urinary ducts. From the cloaca, it flows into the urinary bladder, which branches from the ventral wall of the cloaca. For many terrestrial amphibians, the urinary bladder serves as a water-storage organ. During dry periods, water can be reabsorbed from the urine in the bladder.

Like the larvae of fishes, most amphibian larvae excrete the nitrogen-containing wastes as ammonia. Because ammonia is very toxic, it must be removed from the body quickly or diluted with large amounts of water in the urine. To conserve water, adult amphibians transform ammonia into urea, a less-toxic substance that can be excreted without using as much water. Although this transformation uses energy, it helps save water. During metamorphosis, larval amphibians change from excreting ammonia to excreting urea.

FIGURE 40-11

This diagram shows the frog's nervous system in ventral view. The brain of the frog is sufficiently developed to cope with both land and water environments.



NERVOUS SYSTEM

Use the diagram in Figure 40-11 to find the main components of the amphibian nervous system. An amphibian's brain is about the same size as that of a fish of similar size. The olfactory lobes, which are the center of the sense of smell, are larger in amphibians than in fish, and they lie at the anterior end of the brain. Behind the olfactory lobes are the long lobes of the cerebrum, the area of the brain that integrates behavior and is responsible for learning.

The optic lobes, which process information from the eyes, lie behind the cerebrum. The cerebellum, a small band of tissue that lies at a right angle to the long axis of the brain, is the center of muscular coordination and is not as well developed in amphibians as it is in other tetrapods. The medulla oblongata lies at the back of the brain and joins the spinal cord. It controls some organ functions, such as heart rate and respiration rate.

There is continuous communication among most areas of the brain. Ten pairs of cranial nerves extend directly from the brain. The spinal cord conducts signals from all parts of the body to the brain and from the brain back to the body. Encased in protective bony vertebrae, the spinal cord extends down the back. As in fishes, the spinal nerves branch from the spinal cord to various parts of the body.



FIGURE 40-12

Frogs have a well-developed sense of sight, which is necessary for locating prey and avoiding predators. Frogs also have good hearing. Note the tympanic membrane (the circle below and to the left of the eye), which transmits sound to the inner ear.

Sense Organs

Some sense organs work as well in air as in water, but others do not. For example, the lateral line system, used by fishes to detect disturbances in the water, works only in water. Thus, while larval amphibians have a lateral line, it is usually lost during metamorphosis. Only a few species of aquatic amphibians have a lateral line as adults.

The senses of sight, smell, and hearing are well developed in most amphibians. Visual information is often important in hunting and in avoiding predators. The eyes are covered by a transparent, movable membrane called a **nictitating** (nik-ti-tayt-eeng) **membrane**.

Sound is detected by the inner ear, which is embedded within the skull. Sounds are transmitted to this organ by the **tympanic** (tim-PAN-ik) **membrane**, or eardrum, and the **columella** (CAHL-yoo-mel-uh), a small bone that extends between the tympanic membrane and the inner ear. Sounds first strike the tympanic membrane, which is usually located on the side of the head, just behind the eye as shown in Figure 40-12. Vibrations of the tympanic membrane cause small movements in the columella that are transmitted to the fluid-filled inner ear. In the inner ear, the sound vibrations are converted to nervous impulses by sensitive hair cells. These impulses are transmitted to the brain through a nerve.

Word Roots and Origins

nictitating

from the Latin *nictare*,
meaning "to wink"

SECTION 2 REVIEW

1. Identify an advantage and a disadvantage to the permeability of amphibian skin.
2. Explain three ways in which a frog's skeleton is adapted for jumping.
3. Identify the relationship between the structure of a frog's heart and "double-loop" circulation.
4. Explain how a frog uses positive pressure to move air into and out of its lungs.
5. What feature do the digestive and excretory systems in frogs have in common?

CRITICAL THINKING

6. **Applying Information** Which are the two largest features of an amphibian's brain? Why do you think these lobes are the largest?
7. **Making Comparisons** Which sense organ of a terrestrial amphibian resembles the fish's lateral line in function? Explain your answer.
8. **Recognizing Relationships** Why are the skeletal bones in a frog's shoulders thicker than the bones in a frog's pelvic girdle?

SECTION 3

OBJECTIVES

- **Explain** how a male frog attracts a female of the same species.
- **Discuss** the reproductive system of a frog.
- **Describe** the life cycle of a frog.
- **Describe** the changes that occur during metamorphosis in frogs.
- **Identify** two examples of parental care in amphibians.

VOCABULARY

amplexus

REPRODUCTION IN AMPHIBIANS

One of the biggest differences between aquatic and terrestrial life-forms is their method of reproduction. Most amphibians depend on water for reproduction. They lay their eggs in water and spend the early part of their lives as aquatic larvae.

COURTSHIP AND FERTILIZATION

In the first warm days of spring in the temperate zones, frogs emerge from hibernation. They migrate in great numbers to ponds and slow-moving streams. Males call to attract females of their own species and to warn off other males, as shown in Figure 40-13. Each species has its own mating call. The frog's croak is produced by air that is driven back and forth between the mouth and the lungs, vibrating the vocal folds. Male frogs have vocal sacs that amplify their calls. The female responds only to the call from a male of the same species.

When a female approaches, the male frog climbs onto her back. He grasps her firmly in an embrace called **amplexus** (am-PLEKS-uhs). The male clings to the female until she lays her eggs. When the female finally releases her eggs into the water, the male frog discharges his sperm over them, and direct external fertilization takes place. The frogs then separate and resume their solitary lives. Courtship behavior and fertilization often differ between species.

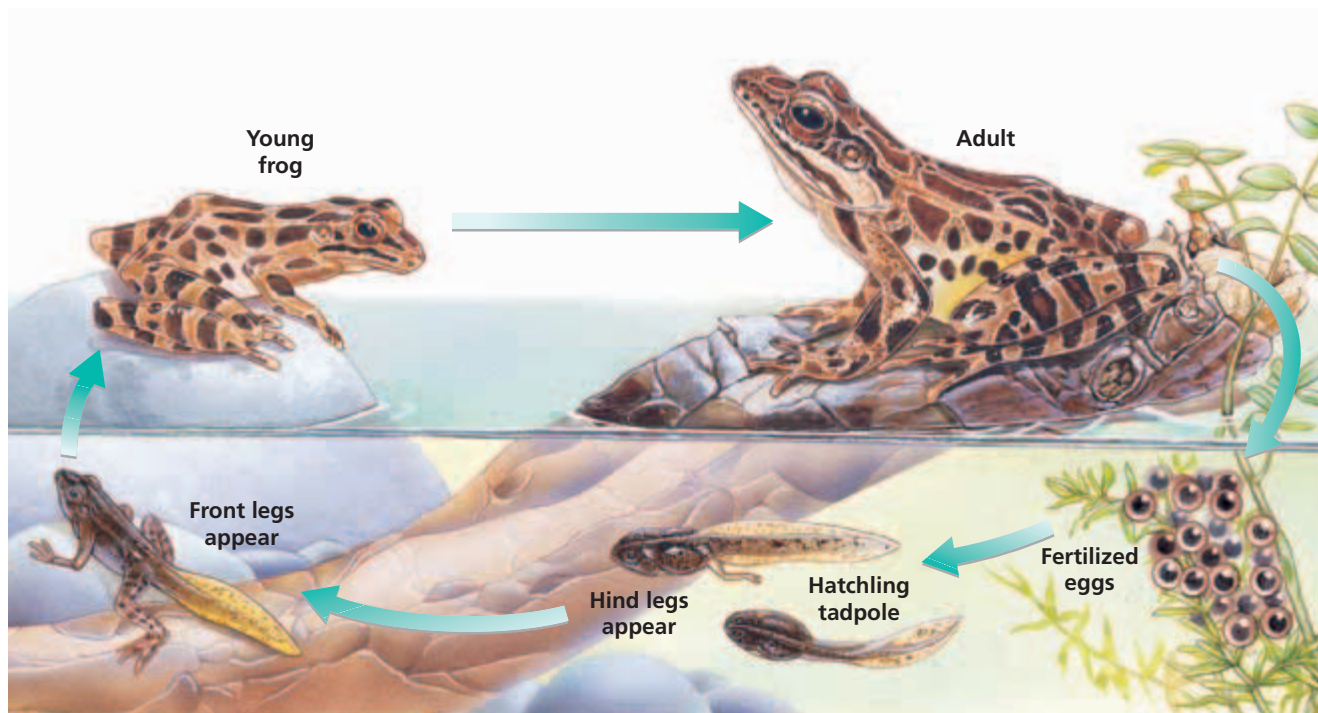
Reproductive System

The reproductive system of the male frog includes two bean-shaped testes located near the kidneys. During the breeding season, sperm cells develop in the testes and pass through tubes to the kidneys and urinary ducts. During mating, sperm leave the body through the cloacal opening. In female frogs, a pair of large, lobed ovaries containing thousands of tiny immature eggs lie near the kidneys. During the breeding season, the eggs enlarge, mature, and burst through the thin ovarian walls into the body cavity. Cilia move the eggs forward into the funnel-like openings of the oviducts. As the eggs pass down the oviducts, they are coated with a protective jellylike material. The eggs exit by the cloaca to the external environment, where they are fertilized.

FIGURE 40-13

Most frogs that sing are males. Their songs attract females and warn off males of the same species.





LIFE CYCLE

Within a few days of fertilization, the eggs hatch into tadpoles. A newly hatched tadpole lives off yolk stored in its body. It gradually grows larger and develops three pairs of gills. Eventually, the tadpole's mouth opens, allowing it to feed. The tadpole grows and slowly changes from an aquatic larva into an adult. This process of change is called *metamorphosis*. Legs grow from the body, and the tail and gills disappear. The mouth broadens, developing teeth and jaws, and the lungs become functional.

Biologists have long studied the process of metamorphosis and regeneration to learn what controls such dramatic physical changes. A hormone called *thyroxine* is produced by the thyroid gland and circulates throughout the bloodstream to stimulate metamorphosis.

The life cycles of many amphibians are similar to that of the frog shown in Figure 40-14. But there are a variety of alternative reproductive patterns among amphibians. For example, many amphibians do not lay their eggs in water. They select a moist place on land, such as under a rock, inside a rotting log, or in a tree. One or both parents may even construct a nest for the eggs. A number of frog species make a nest of mucus, whipping it into a froth by kicking their hind legs. And not all amphibians undergo metamorphosis. Some salamanders, such as the axolotl in Figure 40-15, remain in the larval stage for their entire life. The axolotl's thyroid does not produce thyroxine. Other amphibians bypass the free-living larval stage and hatch from the egg as a small version of the adult.

FIGURE 40-14

The life cycle of a frog begins with mating. When the eggs hatch, a tadpole is released. One of the first developments of metamorphosis is the growth of hind legs. When the tadpole completes metamorphosis, a small adult emerges from the water onto the land.

FIGURE 40-15

The axolotl does not produce thyroxine, and as a result, does not undergo metamorphosis. It retains its gills and lives a completely aquatic life.



FIGURE 40-16

Rhinoderma darwinii male frogs exhibit parental care by holding the maturing eggs and larvae in their vocal sacs. This frog has already released his offspring, whose tails are still visible. Not all frogs express parental care in this way, and some frogs express no parental behavior at all.



PARENTAL CARE

Parental care is common among amphibians. Eggs and larvae are vulnerable to predators, but parental care helps increase the likelihood that some offspring will survive. Most often, one parent (often the male) remains with the eggs, guarding them from predators and keeping them moist until they hatch. The male Darwin's frog (*Rhinoderma darwinii*) of Chile takes the eggs into his vocal sacs, where they hatch and eventually undergo metamorphosis. The young frogs climb out of the vocal sacs and emerge from the male's mouth, as shown in Figure 40-16.

Female gastric-brooding frogs of Australia swallow their eggs, which hatch and mature in the stomach. The eggs and tadpoles are not digested because the stomach stops producing acid and digestive enzymes until the young pass through metamorphosis and are released. Two species of gastric-brooding frogs are known, but both appear to have become extinct within the last two decades. Females of some species of frogs, such as *Eleutherodactylus*, sit on their eggs until they hatch, not to provide warmth but to prevent the eggs from desiccating. The female normally lays the eggs in the leaves of trees or bushes, where they may dry up.

SECTION 3 REVIEW

1. Identify two functions of the male frog's call.
2. Sequence the stages of a frog's life cycle.
3. Identify the features that are lost and the features that are gained during metamorphosis in frogs.
4. Name the hormone responsible for stimulating tadpoles to undergo metamorphosis.
5. Describe two strategies that have been found in frogs for protecting eggs and developing young.

CRITICAL THINKING

6. **Applying Information** Why is standing water not always necessary for frog reproduction?
7. **Recognizing Relationships** How do you think the number of eggs produced by amphibians relates to the amount of parental care invested?
8. **Justifying Conclusions** Would you expect an amphibian that bypasses the larval stage to produce more or less thyroxine? Explain.

CHAPTER HIGHLIGHTS

SECTION 1

Origin and Evolution of Amphibians

- Preadaptations are inherited traits used for new functions for which the traits are later selected.
- Fleshy fins with strong bone structure, nostrils, and lungs were preadaptations that allowed the transition from aquatic to terrestrial life.
- Early amphibians had lungs and four legs. Like lobe-finned fish, they were predominantly aquatic, they had a lateral line and a fishlike tail fin, similar skull and vertebral column, similar limb bones, and some had gills.
- Modern amphibians share several characteristics. Most change from aquatic larvae to terrestrial adult. Most have moist thin skin, with no scales. Feet, if present, lack claws and are often webbed. Most use gills, lungs, and skin in respiration. Most lay eggs that lack multicellular membranes or shells.
- Modern amphibians are divided into three orders: Anura (frogs and toads), Caudata (salamanders), and Gymnophiona (caecilians).

Vocabulary

preadaptation (p. 799)

tadpole (p. 802)

SECTION 2

Characteristics of Amphibians

- Adult amphibians respire through their lungs and skin. Mucous glands produce slimy mucus that helps retain moisture.
- The skeleton of an amphibian supports the body against the pull of gravity. In addition, the spine interlocks to provide rigid structure, strong limbs support the body, and the pectoral and pelvic girdle transfer body weight to the limbs.
- The amphibian pulmonary circuit carries blood between the heart and lungs. The systemic circuit carries blood to the body and returns blood to the heart.
- The heart of an amphibian consists of two atria and one ventricle. The right atrium receives blood from the body, pumps blood to the ventricle, and then pumps blood to the lungs. The left atrium receives blood from the lungs, pumps blood to the ventricle, and then to the body.
- Amphibians pump air into their lungs by raising and lowering the floor of their mouth cavity while closing and opening their nostrils.
- In an amphibian, food passes through the mouth, esophagus, stomach, small intestine, large intestine, and cloaca. The kidneys remove wastes from the blood. Adult amphibians eliminate nitrogenous wastes as urea.
- The brain of a frog has large optic lobes that process visual information, large olfactory lobes that control the sense of smell, a cerebrum that integrates behavior and controls learning, a cerebellum that coordinates movement, and a medulla oblongata that controls the heart and respiration rate. Fish have brains of similar size, but with smaller olfactory lobes.

Vocabulary

mucous gland (p. 804)

pulmonary
circulation (p. 805)

systemic circulation (p. 805)

pulmonary
respiration (p. 806)

cutaneous
respiration (p. 806)

duodenum (p. 807)

ileum (p. 807)

mesentery (p. 807)

vent (p. 807)

nictitating
membrane (p. 809)

tympanic membrane (p. 809)

columella (p. 809)

SECTION 3

Reproduction in Amphibians

- Male frogs call to attract females of the same species.
- The male frog grasps the female in an embrace called amplexus and fertilizes her eggs as they are released.
- A life cycle is the stages through which an organism passes from fertilization to reproductive maturity. Most amphibians lay their eggs in water, and have an aquatic larval stage and a terrestrial adult stage.
- The hormone thyroxine triggers metamorphosis. During metamorphosis, the tadpole loses its tail and gills and grows legs and lungs.
- Many amphibians show parental care by guarding their eggs and keeping their eggs moist. Some take their eggs into their body to develop.

Vocabulary

amplexus (p. 810)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - systemic circulation* and *pulmonary circulation*
 - amplexus* and *metamorphosis*
 - pulmonary respiration* and *cutaneous respiration*
- Use the following key terms in the same sentence: *duodenum*, *mesentery* and *ileum*.
- Explain the relationship between preadaptation and adaptation.
- Word Roots and Origins** The word *columella* comes from the Latin *columna*, which means “pillar.” Using this information, list two characteristics of this amphibian ear bone.

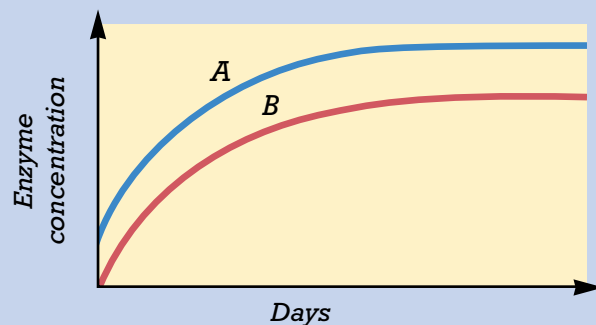
UNDERSTANDING KEY CONCEPTS

- Discuss** the preadaptations important in the evolution of amphibians.
 - Identify** two characteristics that indicate that amphibians are descendants of ancient lobe-finned fish.
 - Name** five major characteristics of amphibians.
 - Identify** the order to which each of the following belong: frogs, salamanders, caecilians.
 - Explain** how some species of frogs survive in desert environments even though they do not have watertight skin.
 - Relate** the adaptations in the amphibian skeleton to the change in gravity with life on land.
 - Outline** the route of blood flow through the body of a frog, beginning with the right atrium.
 - Determine** the next step for air within the oral cavity of a frog when it closes its nostrils and raises the mouth floor.
 - Trace** the digestive process as food passes through a frog’s digestive system.
 - Explain** how an amphibian’s nervous system is similar to that of a bony fish.
 - Explain** how male frogs attract the attention of female frogs during mating season.
 - Describe** the reproductive system of a frog.
 - Discuss** the life cycle of a frog.
 - Contrast** the physical characteristics of a larval frog to those of an adult frog.
 - Describe** two different examples of parental care found in frogs.
- CONCEPT MAPPING** Use the following terms to create a concept map that shows the methods of respiration in frogs: *frogs*, *skin*, *mouth floor*, *body wall muscles*, *nostrils*, *lungs*, *air pressure*, *oral cavity*, *environment*.

CRITICAL THINKING

- Inferring Relationships** Charles Darwin noticed that frogs and toads are often absent from oceanic islands, such as the Galápagos Islands, even though they may be found on the nearby mainland. Darwin conducted some experiments that showed that frogs’ eggs cannot tolerate exposure to salt water. What hypothesis do you think Darwin was trying to test?
- Recognizing Relationships** In the brains of amphibians, the largest parts are the olfactory lobes and the optic lobes, the centers of smell and sight. This is very important to amphibians in hunting prey and avoiding predators. Why else is the capacity for hearing important?
- Recognizing Relationships** The female gastric-brooding frogs of Australia did not produce stomach acid or digestive enzymes while brooding their young in their stomachs until the tadpoles completed metamorphosis and left. If the mother frog did not eat during this period, from where did she get her energy?
- Analyzing Graphics** When tadpoles undergo metamorphosis, their bodies begin to produce an enzyme that converts ammonia into urea. The time that a tadpole takes to produce this enzyme varies among species. In the graph below, the rate of enzyme production is shown for a species that inhabits a desert-like environment and a species that inhabits a forest environment. Which curve represents which frog? Explain.

Enzyme Production in Two Frog Species



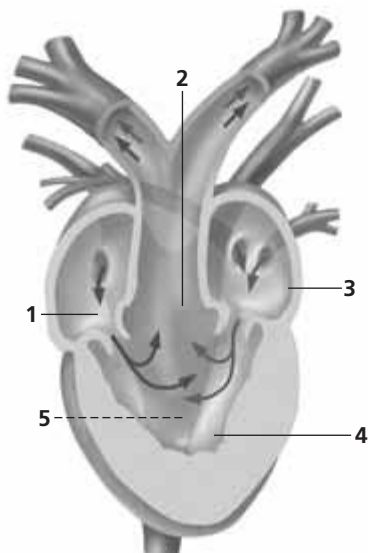


Standardized Test Preparation

DIRECTIONS: Choose the letter that best answers the question or completes the sentence.

- The forelimbs of vertebrates evolved from which structures in lobe-finned fishes?
 - anal fin
 - pelvic fins
 - pectoral fins
 - pectoral girdle
- Amphibians must lay eggs in water primarily for what reason?
 - The eggs are not laid in nests.
 - The eggs need oxygen from water.
 - The eggs need protection from predators.
 - The eggs do not have multicellular membranes and a shell.
- Metamorphosis must take place before amphibians are able to do what?
 - swim
 - live on land
 - respire with gills
 - feed themselves

INTERPRETING GRAPHICS: The figure below shows a longitudinal section, dorsal view, of a frog heart. Use the figure below to answer question 4.

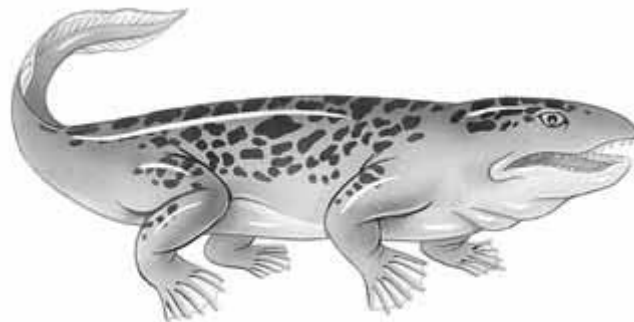


- Identify the source of blood flow in the section of the heart labeled 1.
 - the body
 - the aorta
 - the lungs
 - both lungs and body

DIRECTIONS: Complete the following analogy.

- Anura : frogs :: Gymnophiona :
 - toads
 - newts
 - caecilians
 - salamanders

INTERPRETING GRAPHICS: The figure below shows an artist's rendering of *Ichthyostega*. Use the figure to answer the question that follows.



- Ichthyostega* is an early amphibian. Which of the following characteristics is most likely to help it live on land?
 - fishlike tail
 - seven-toed feet
 - four strong limbs
 - lateral-line canals on the head

SHORT RESPONSE

Modern amphibians are a diverse group, but they do have some common characteristics.

Describe five key characteristics shared by modern amphibians.

EXTENDED RESPONSE

Frogs breathe by a positive pressure system.

Part A Describe how frogs move air into their lungs. Which part of inhaling is "positive pressure?"

Part B Describe how frogs move air out of their lungs and into the atmosphere.

Test TIP

Be sure to read all of the answer choices carefully. Do not assume that the first correct answer is the only one.

Observing Live Frogs

OBJECTIVES

- Observe the behavior of a frog.
- Explain how a frog is adapted to life on land and in water.

PROCESS SKILLS

- observing
- relating structure to function
- recognizing relationships

MATERIALS

- live frog in a terrarium
- aquarium half-filled with dechlorinated water
- live insects (crickets or mealworms)
- 600 mL beaker

Background

1. What does *amphibious* mean?
2. Describe how amphibians live part of their life on land and part in water.
3. What are some major characteristics of amphibians?



PART A Observing the Frog in a Terrarium

1. Observe a live frog in a terrarium. Closely examine the external features of the frog. Make a drawing of the frog in your lab report. Label the eyes, nostrils, tympanic membranes, front legs, and hind legs. The tympanic membrane, or eardrum, is a disc-like membrane behind each eye.
2. In your lab report, make a table similar to the one on the facing page to note all your observations of the frog in this investigation.
3. Watch the frog's movements as it breathes. Record your observations in your data table.
4. Look closely at the frog's eyes, and note their location. Examine the upper and lower eyelids as well as a third transparent eyelid called a *nictitating membrane*. The upper and lower eyelids do not move. The nictitating membrane moves upward over the eye. This eyelid protects the eye when the frog is underwater and keeps the eye moist when the frog is on land.
5. Study the frog's legs. Note in your data table the difference between the front and hind legs.
6. Place a live insect, such as a cricket or a mealworm, in the terrarium. Observe how the frog reacts.
7. Gently tap the side of the terrarium farthest from the frog, and observe the frog's response.

PART B Observing the Frog in an Aquarium

8.  **CAUTION** You will be working with a live animal. Handle it gently and follow instructions carefully. Frogs are slippery. Do not allow the frog to injure itself by jumping from the lab bench to the floor. Place a 600 mL beaker in the terrarium. Carefully pick up the frog and examine its skin. How does it feel? The skin of a frog acts as a respiratory organ, exchanging oxygen and carbon dioxide with the air or water. A frog also takes in and loses water through its skin.
9. Place the frog in the beaker. Cover the beaker with your hand, and carry it to a freshwater aquarium. Tilt the beaker and gently submerge it beneath the surface of the water until the frog swims out of the beaker.

OBSERVATIONS OF A LIVE FROG

Characteristic	Observation
Breathing	
Eyes	
Legs	
Response to food	
Response to noise	
Skin texture	
Swimming behavior	
Skin coloration	

10. Watch the frog float and swim in the aquarium. How does the frog use its legs to swim? Notice the position of the frog's head.
11. As the frog swims, bend down and look up into the aquarium so that you can see the underside of the frog. Then look down on the frog from above. Compare the color on the dorsal and ventral sides of the frog. When you are finished observing the frog, your teacher will remove the frog from the aquarium.
12. Record your observations of the frog's skin texture, swimming behavior, and skin coloration in your data table.
13.   Clean up your materials and wash your hands before leaving the lab.



Analysis and Conclusions

1. From the position of the frog's eyes, what can you infer about the frog's field of vision?
2. How does the position of the frog's eyes benefit the frog while it is swimming?
3. How does a frog hear?
4. How can a frog take in oxygen while it is swimming in water?
5. Why must a frog keep its skin moist while it is on land?
6. How are the hind legs of a frog adapted for life on land and in water?
7. What adaptive advantage do frogs have in showing different coloration on their dorsal and ventral sides?
8. What features provide evidence that an adult frog has an aquatic life and a terrestrial life?
9. What adaptations does the frog display in order to eat? What senses are involved in catching prey?
10. What movement does the frog make in order to breathe?

Further Inquiry

Observe other types of amphibians, or do research to find out how they are adapted to life on land and in water. How do the adaptations of other types of amphibians compare with those of the frog you observed in this investigation?

REPTILES

A marine iguana, *Amblyrhynchus cristatus*, of the Galápagos Islands warms itself by basking on a rock. These are the only marine lizards in the world, and they feed exclusively on seaweed.



SECTION 1 *Origin and Evolution of Reptiles*

SECTION 2 *Characteristics of Reptiles*

SECTION 3 *Modern Reptiles*

ORIGIN AND EVOLUTION OF REPTILES

The reptiles (class Reptilia) are one of the largest groups of terrestrial vertebrates. This chapter will discuss the diversity of reptiles and some of the characteristics that make the reptiles a successful group.

HISTORY OF REPTILES

From studies of fossils and comparative anatomy, biologists infer that reptiles arose from amphibians. The oldest known fossils of reptiles were found in deposits from the early Carboniferous period (359 million to 299 million years ago). The earliest reptiles were small, four-legged vertebrates that resembled lizards and had teeth adapted for eating insects. The abundance of insects at the time may have been one reason the early reptiles flourished. By the Permian period (299 million to 251 million years ago) reptiles had become the dominant land vertebrates.

By the end of the Permian nearly all of the continents had joined to form the supercontinent *Pangaea* (pan-JEE-uh), shown in Figure 41-1. The interior of Pangaea had a dryer climate than the coastal regions. Reptiles were suited to dry climates and were highly successful. The end of the Permian period (and of the Paleozoic era) is marked by a mass extinction during which a large number of species became extinct. The cause of this extinction is still debated. Reptiles that survived diversified to take over the ecological roles of the extinct species. The Mesozoic era (251 million to 66 million years ago) is often called the Age of Reptiles because nearly all of the large vertebrates during that time were reptiles.

EVOLUTION OF DINOSAURS

Beginning about 235 million years ago, **dinosaurs**, a group of extinct reptiles, dominated life on land for roughly 150 million years. They evolved from *thecodonts*, an extinct group of crocodile-like reptiles. Dinosaurs include a wide variety of reptiles that were adapted to very different environments. One factor that affected dinosaur evolution was the movement of the continents. Early in the Mesozoic era, Pangaea started to break apart. The climates of the separate landmasses changed. Some species of dinosaurs could not adapt and became extinct, while new kinds flourished.

OBJECTIVES

- **Summarize** the factors that led to the rise of reptiles as the dominant land vertebrates.
- **Identify** three factors that contributed to the success of dinosaurs.
- **Compare** two hypotheses to explain the extinction of the dinosaurs.
- **Identify** examples of the four modern orders of reptiles.
- **Describe** three characteristics of modern reptiles that make them well adapted to life on land.

VOCABULARY

dinosaur
asteroid impact hypothesis
amniotic egg
amnion
yolk sac
allantois
chorion
albumen
keratin



FIGURE 41-1

When dinosaurs first appeared, all of Earth's landmasses were joined in a single supercontinent called *Pangaea*, shown above.

Mass extinction	
Mesozoic era	Cretaceous period 146–66 mya 
	Jurassic period 200–146 mya 
	Mass extinction
Paleozoic era	Triassic period 251–200 mya 
	Mass extinction
	Permian period 299–251 mya
	Carboniferous period 359–299 mya 
	(mya = million years ago)

FIGURE 41-2

The evolution of reptiles has been marked by three mass extinction events. For more information about the geologic time scale, see the Appendix at the back of this textbook.

Triassic Dinosaurs

The oldest known dinosaur fossils are in rocks from the early Triassic period, about 235 million years ago. By the end of the Triassic, small, carnivorous dinosaurs, were very common and had largely replaced the thecodonts.

There are at least three reasons why dinosaurs were so successful. First, legs positioned directly under the body provided good support for the dinosaur's body weight, enabling dinosaurs to be faster and more agile runners than the thecodonts. Second, dinosaurs were well adapted to the dry conditions found in Pangaea during the late Triassic period. Finally, at the end of the Triassic period, another mass extinction wiped out about 80% of living species, including the thecodonts and the last of the large amphibians. Recent research suggests that asteroid impacts in what are now Canada, France, the Ukraine, and Minnesota occurred almost simultaneously and contributed to this mass extinction event. This event marked the end of the Triassic period, outlined in Figure 41-2. Reduction in competition for resources may have allowed dinosaurs to flourish.

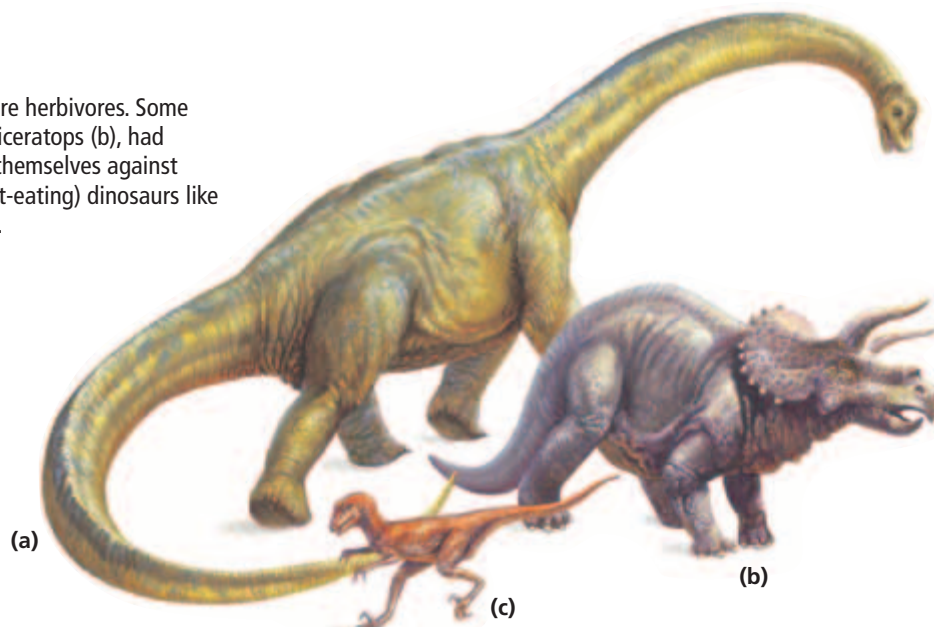
Jurassic and Cretaceous Dinosaurs

The Jurassic period, graphed in Figure 41-2, is considered the golden age of dinosaurs because of the variety and abundance of dinosaurs that lived during this time. They included the largest land animals of all time, the sauropods (SAWR-uh-PAHDZ), shown in Figure 41-3a.

By the late Jurassic period, a new type of dinosaur had evolved. The carnivorous theropods (THER-uh-PAHDZ), shown in Figure 41-3c stood on two powerful legs and had short arms. Their large heads were equipped with sharp teeth, and each foot had sickle-shaped claws used for ripping open prey. Theropods preyed on the large herbivorous dinosaurs and were the dominant terrestrial predators until the end of the Cretaceous (kruh-TAY-shus) period.

FIGURE 41-3

Sauropods (a) were herbivores. Some herbivores, like triceratops (b), had armor to defend themselves against carnivorous (meat-eating) dinosaurs like the theropods (c).



Dinosaur Diversity

Dinosaurs were not limited to terrestrial habitats. Some Mesozoic reptiles called *pterosaurs* (TER-uh-SAWRS), shown in Figure 41-4a, evolved the ability to fly. Several other groups of reptiles, including the ichthyosaurs (IK-thee-uh-SAWRS) and plesiosaurs, lived in the oceans. Ichthyosaurs, illustrated in Figure 41-4b, were sleek aquatic reptiles that resembled modern bottlenose dolphins. Plesiosaurs had long, flexible necks and compact bodies.

EXTINCTION OF DINOSAURS

Although the fossil record provides many clues about what dinosaurs were like, paleontologists who study dinosaurs still have many unanswered questions. For example, why did the dinosaurs and many other species become extinct 66 million years ago, at the end of the Cretaceous period?

Asteroid Impact Hypothesis

Until recently, most scientists thought that a single catastrophic event was responsible for the mass extinction of dinosaurs. The **asteroid impact hypothesis** suggests that a huge asteroid hit Earth and formed a crater on the Yucatán Peninsula in southern Mexico. The impact sent so much dust into the atmosphere that the amount of sunlight reaching the Earth's surface was greatly reduced. The reduced sunlight caused severe climatic changes that led to the mass extinction. The asteroid impact hypothesis is supported by the fact that sediments from the end of the Cretaceous period contain unusually high concentrations of iridium. Iridium is a metal that is very rare in the Earth's crust but more abundant in asteroids and other meteoroids. According to this hypothesis, the dinosaurs would have become extinct very quickly.

Multiple Impact Hypothesis

Another hypothesis, called the *multiple impact hypothesis*, proposes that asteroid impacts began before the mass extinction took place. Along with other unfavorable environmental conditions, multiple asteroid impacts, including the impact on the Yucatán peninsula, led to a decline in numbers of organisms among species. In 2004, paleontologist Gerta Keller and her colleagues provided evidence to support this hypothesis. Results of their research suggest that conditions hostile to dinosaur survival were spread over many years. It is likely that these reptiles were not wiped out by a single asteroid impact.

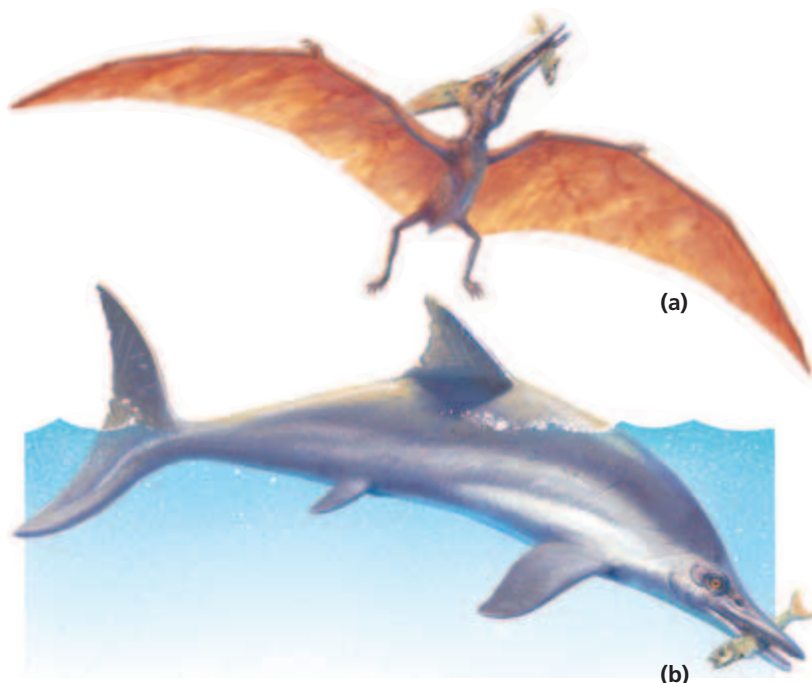


FIGURE 41-4

(a) Pterosaurs ranged in size from the smallest, which were only the size of sparrows, to the largest, which were about the size of a small airplane, with wingspans of 12 m (about 39 ft).
(b) Like dolphins, ichthyosaurs were probably fast swimmers and fed on fish.

Word Roots and Origins

crocodile

from the Greek *krokodilos*,
meaning "lizard of the Nile"

SUCCESS OF REPTILES

Representatives of the four modern orders of reptiles—Chelononia (turtles and tortoises), Squamata (lizards and snakes), Rhynchocephalia (tuataras), and Crocodilia (crocodiles, alligators, caimans)—survived the mass extinction of the Cretaceous period. These four orders of reptiles have diversified to more than 6,000 species. Reptiles successfully occupy a variety of terrestrial and aquatic habitats on all continents except Antarctica. Figure 41-5 is a phylogenetic diagram that shows hypotheses for the relationship among reptiles, birds, and mammals.

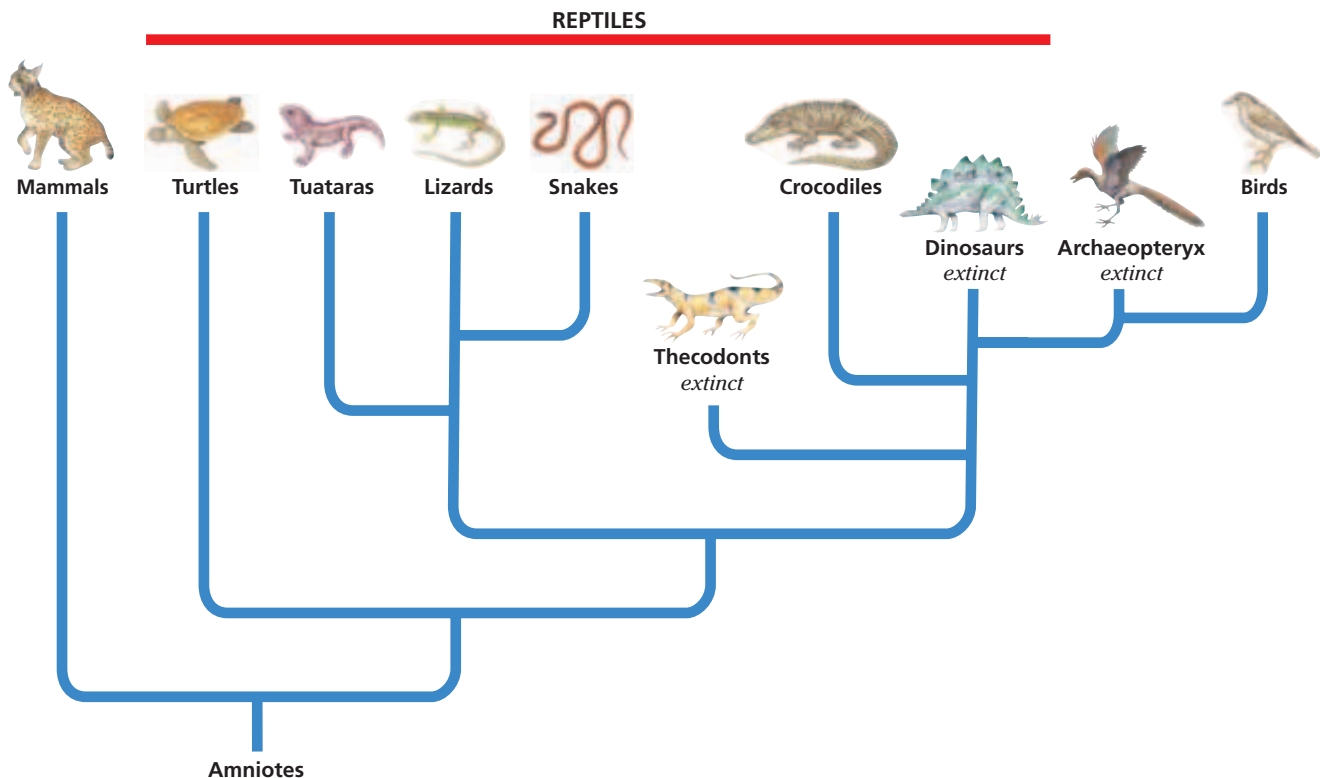
Modern Reptiles

The turtles have the most ancient origins and have changed very little in structure since before the time of the dinosaurs. Tuataras belong to a small group of lizard-like reptiles. The vast majority of living reptiles belong to the group snakes and lizards. The fourth line of living reptiles includes the crocodiles and their relatives. Crocodilians have changed very little in more than 200 million years. Like dinosaurs, crocodilians are descendants of the thecodonts.

In some ways, such as the structure of their heart, crocodilians resemble birds far more than they resemble other living reptiles. And crocodilians are the only living reptiles that care for their young. What does this mean in terms of their relationships to other vertebrate species? Today, many biologists think that birds are direct descendants of the dinosaurs.

FIGURE 41-5

This phylogenetic diagram represents hypotheses for the relationship among reptiles, birds, and mammals. For updates on phylogenetic information, visit go.hrw.com. Type in the keyword **HM6 Phylo**.



The Amniotic Egg

Although amphibians were the first vertebrates to successfully invade land, they did not make a full transition to terrestrial life. They still require water in their environment to reproduce. Reptiles are considered the first fully terrestrial vertebrates because they do not need to reproduce in water, as most amphibians do. Reptiles produce **amniotic eggs**, which encase the embryo in a secure, self-contained aquatic environment. Amniotic eggs provide more protection for the developing embryo than do the jellylike eggs of amphibians.

Figure 41-6 shows the internal structure of the amniotic egg, including its four specialized membranes: the amnion, yolk sac, allantois, and chorion. The egg is named for the **amnion** (AM-nee-uhn), the thin membrane enclosing the fluid in which the embryo floats. The **yolk sac** encloses the yolk, a fat-rich food supply for the developing embryo. The **allantois** (uh-LAN-toh-is) stores the nitrogenous wastes produced by the embryo. Its blood vessels, which lie near the porous shell, function in the exchange of oxygen and carbon dioxide gases. The **chorion** (KAWR-ee-AHN) surrounds all the other membranes and helps protect the developing embryo. Protein and water needed by the embryo are contained in the **albumen** (al-BYOO-muhn). You are familiar with albumen as the egg white in a chicken's egg. In most reptiles, the leathery outer shell provides protection from physical damage, limits the evaporation of water from the egg, and allows diffusion of oxygen and carbon dioxide.

The amniotic egg first evolved in reptiles, but it also occurs in mammals and birds. The presence of this feature is strong evidence that reptiles, birds, and mammals evolved from a common ancestor. The eggs of some reptiles and nearly all mammals lack shells, and the embryo develops within the mother's body.

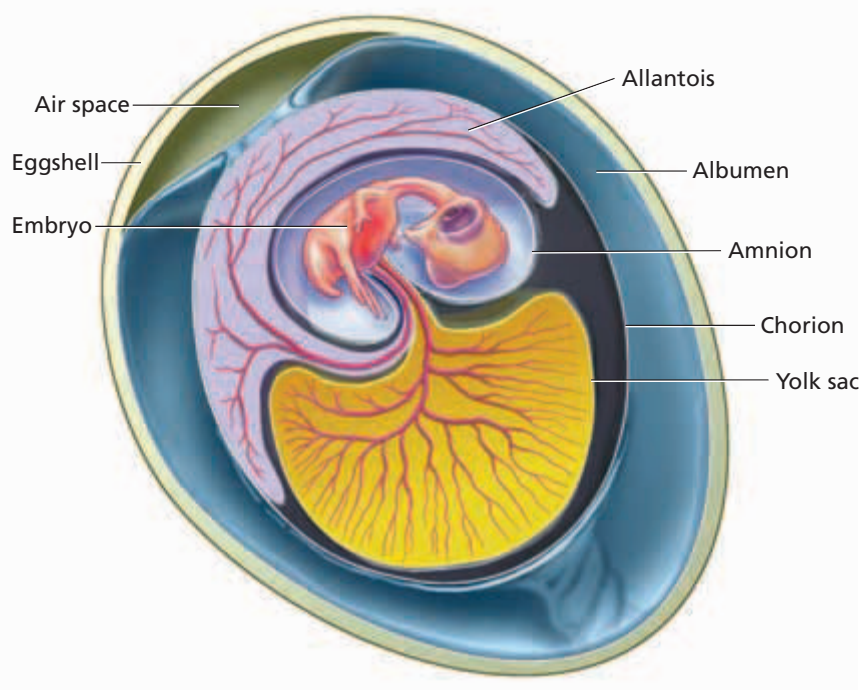
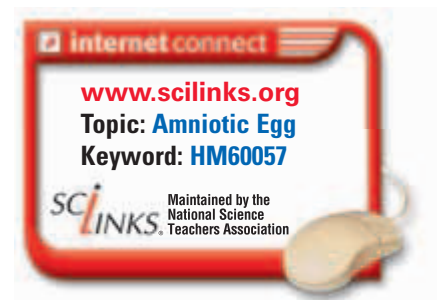


FIGURE 41-6

Amniotic eggs have four major membranes. The tough but porous shell provides protection while allowing the exchange of oxygen and carbon dioxide. The amniotic eggs of reptiles and birds (shown here) are very much alike internally.

FIGURE 41-7

The skin of this spiny lizard, *Sceloporus poinsetti*, protects it from the rugged terrestrial environment and from water loss.



Watertight Skin

Because amphibians exchange gases through their skin, the skin must be moist and thin enough to allow rapid diffusion. A drawback of this kind of skin is that amphibians face the loss of body water through evaporation. Reptiles, such as the lizard shown in Figure 41-7, are covered by a thick, dry, scaly skin that prevents water loss. This scaly covering develops as surface cells fill with **keratin**, the same protein that forms your fingernails and hair. Lipids and proteins in the skin help make the skin watertight. The tough skin of a reptile not only helps conserve body water but also protects the animal against infections and injuries.

Respiration and Excretion

Modern reptiles have developed efficient respiratory and excretory systems that help them conserve water. All reptiles have lungs for gas exchange. All of the tissues involved in gas exchange are located inside the body, where they can be kept moist in even the driest environments. The excretory system of reptiles also helps them conserve body water. Land-dwelling reptiles excrete nitrogenous wastes in the form of uric acid which requires little water for dilution. Reptiles lose small amounts of water in their urine.

SECTION 1 REVIEW

1. Explain the role of climate in the success of early reptiles.
2. Compare and contrast characteristics of Triassic and Jurassic dinosaurs.
3. Explain the importance of iridium found in sediments from the end of the Cretaceous period.
4. Summarize the two asteroid impact hypotheses.
5. Describe three characteristics that contribute to the success of reptiles on land.

CRITICAL THINKING

6. **Analyzing Information** Why might dinosaurs have survived the asteroid impact at the end of the Triassic period but not one at the end of the Cretaceous?
7. **Relating Concepts** Dinosaurs are now thought to have been warmblooded. How does this support the theory about their relationship to birds?
8. **Inferring Relationships** In what ways are the adaptations of reptiles to land similar to the adaptations of plants to land?

CHARACTERISTICS OF REPTILES

Reptiles live in many different habitats and show a great deal of diversity in size and shape. Think of the differences between a snake and a turtle or between a lizard and a crocodile. This section discusses some of the anatomical, physiological, and behavioral characteristics of reptiles.

CIRCULATORY SYSTEM

The circulatory system of a reptile, like those of all terrestrial vertebrates, is composed of two loops. The pulmonary loop carries deoxygenated blood from the heart to the lungs and returns oxygenated blood to the heart. The systemic loop transports oxygenated blood to the tissues of the body, where oxygen and nutrients are unloaded and where carbon dioxide and wastes are picked up, and returns deoxygenated blood to the heart.

Heart Structure and Function

In lizards, snakes, tuataras, and turtles, the heart has two atria and a single ventricle partially divided by a wall of tissue called a **septum**. In crocodiles, there are two atria and two separate ventricles. The sinus venosus and the conus arteriosus, which are major accessory structures to the heart of a fish, are much smaller in reptiles. The sinus venosus is absent in some species. When it is present, it collects blood from the body and channels it into the right atrium. The conus arteriosus forms the base of the three large arteries exiting from the reptilian heart.

Because the ventricle is not completely divided (except in crocodiles), it might seem that deoxygenated and oxygenated blood would mix. However, very little blood mixing occurs when a reptile is active. Deoxygenated and oxygenated blood are kept separate during contraction of the heart by the actions of the heart valves and the movement of the septum and ventricular walls.

Pumping blood through lungs requires energy. Under some conditions, it is advantageous for a reptile to divert blood away from the lungs to conserve energy. For example, an inactive reptile needs so little oxygen that it may go a long time without breathing. Similarly, aquatic reptiles do not breathe while they are underwater. Under these conditions, the heart pumps blood to the body while reducing circulation through the lungs.

OBJECTIVES

- **Identify** advantages associated with the structure of a reptile's heart.
- **Describe** the respiratory system of reptiles.
- **Describe** four methods reptiles use to sense their environment.
- **Explain** how reptiles regulate their body temperature.
- **Compare** oviparity, ovoviviparity, and viviparity as reproductive strategies.

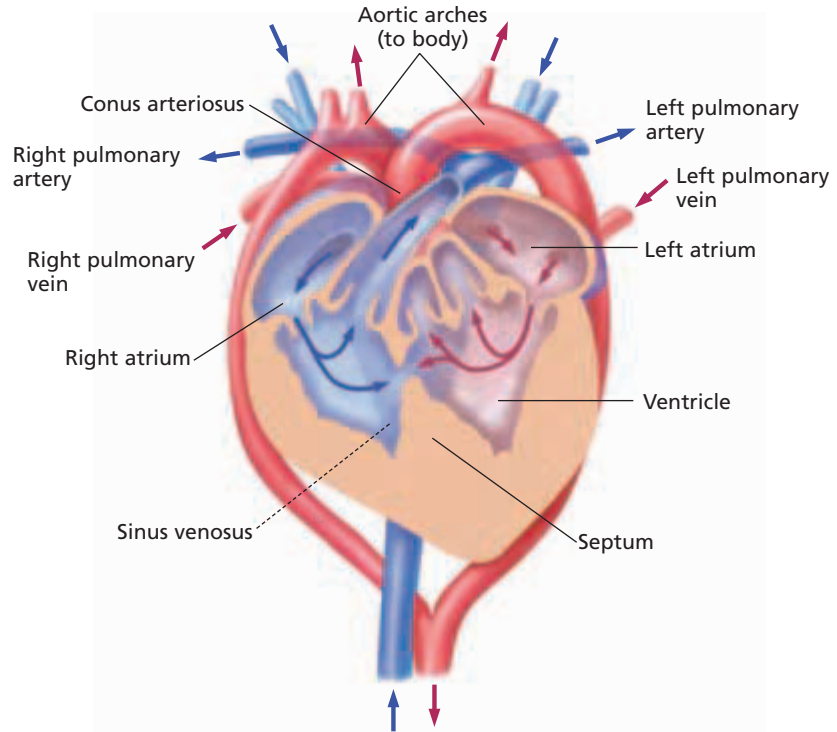
VOCABULARY

septum
alveolus
Jacobson's organ
thermoregulation
ectotherm
endotherm
oviparity
ovoviviparity
viviparity
placenta



FIGURE 41-8

The turtle's heart, shown in cross section, has a partially divided ventricle, unlike an amphibian's three-chambered heart or a crocodile's four-chambered heart. Because the flow of blood through a turtle's heart is asynchronous, deoxygenated blood and oxygenated blood pass through the upper part of the ventricle at different times and so mix very little.



By constricting the pulmonary arteries, a reptile's blood flow through the heart can be redirected to send some deoxygenated blood back to the body instead of to the lungs. Bypassing the lungs may help a reptile raise its body temperature quickly—warm blood from the skin can be directed to the organs deep within the body. During periods of activity, almost all deoxygenated blood is directed to the lungs to meet the muscles' demand for oxygen. The reptilian heart has a degree of circulatory flexibility that the hearts of birds and mammals do not. Instead of being a handicap, this flexibility is actually well suited to reptilian physiology and activity patterns. Figure 41-8 shows a schematic diagram of the heart of a turtle.

RESPIRATORY SYSTEM

The lungs of reptiles are large, and they are often divided internally into several chambers. The lining of the lungs may be folded into numerous small sacs called **alveoli**. Alveoli greatly increase the internal surface area of the lungs, thus increasing the amount of oxygen that can be absorbed. In most snakes, only the right lung actively functions. It is elongated and may be half as long as the body. The left lung is either reduced to a small nonfunctional sac or absent entirely.

A reptile fills its lungs by expanding its rib cage. This expansion reduces the pressure within the thorax and draws air into the lungs. When the ribs return to their resting position, pressure within the thorax increases, and air is forced out of the lungs. Similar movements help humans to breathe.

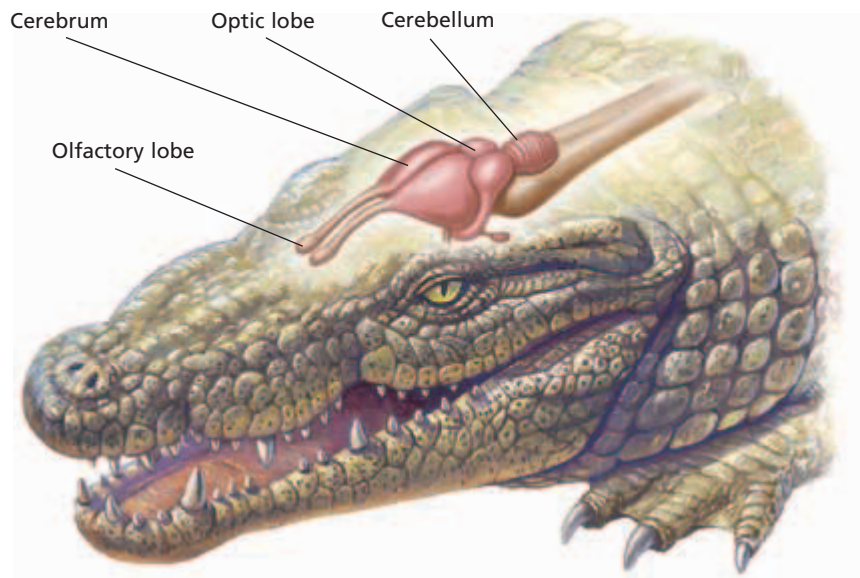


FIGURE 41-9

The crocodile's sense of smell is very important for its survival. The olfactory lobe of the reptile's brain, where the sense of smell is located, is highly developed.

NERVOUS SYSTEM

The brain of a reptile is about the same size as that of an amphibian of the same size. However, the reptilian cerebrum is much larger. This region of the brain is involved in controlling and integrating behavior. Because vision is an important sense for most reptiles, the optic lobes, which receive input from the eyes, are large. Figure 41-9 shows the structure of a crocodile's brain.

Most reptiles rely on their sense of sight to detect predators and prey. The eyes of reptiles are usually large, and many species have keen vision. Hearing is also an important sense. As in amphibians, sound waves first strike the tympanum, or eardrum, and are transmitted to the inner ear through the movements of a small bone called the *columella*. The inner ear contains the receptors for sound. Snakes lack a tympanum and are sensitive only to low-frequency sounds. They are able to detect ground vibrations, which are transmitted to the *columella* by the bones of the jaw.

Jacobson's organ is a specialized sense organ located in the roof of the mouth of reptiles. Jacobson's organ is sensitive to odors. Like the snake shown in Figure 41-10, reptiles use their tongue to collect small particles from the environment. These particles are transferred to the Jacobson's organ when the tongue is drawn back into the mouth. Jacobson's organ is found in all reptiles except crocodiles and most turtles, but it is highly developed in lizards and snakes.

Pit vipers, such as rattlesnakes, copperheads, and water moccasins, are able to detect the heat given off by warm-bodied prey, such as mammals and birds. These snakes have one heat-sensitive pit below each eye, as shown in Figure 41-10. Input from these pits allows a snake to determine the direction of and distance to a warm object.

FIGURE 41-10

Some snakes have reduced senses of sight and hearing. They compensate with a sensitive forked tongue that is an organ of touch and smell. As the tongue darts in and out of the mouth, it picks up particles that are taken into the Jacobson's organ inside the snake's mouth, where even extremely low concentrations of odors can be detected.



THERMOREGULATION

The control of body temperature is known as **thermoregulation**. Vertebrates regulate their body temperature in two different ways. An **ectotherm** warms its body by absorbing heat from its surroundings. Reptiles, fishes, and amphibians are ectotherms. **Endotherms**, such as mammals and birds, have a rapid metabolism, which generates heat needed to warm the body. Most endotherms have insulation, such as hair, feathers, or fat, to retain heat. The body temperatures of many aquatic ectotherms, such as fishes and amphibians, remain close to the temperature of their surroundings. Terrestrial ectotherms, such as lizards and snakes, usually keep their body temperatures about the same as the body temperatures of endotherms.

Most reptiles live in warm climates and regulate their body temperature by controlling how much heat they absorb. For example, when a lizard emerges from its nest after a cool night, its body temperature is low and must be raised before it can become active. The lizard warms itself by basking in the sun, as shown in Figure 41-11a. The lizard's warm blood is diverted from the skin to the interior of the body. As the graph in Figure 41-11b shows, a lizard can maintain its body temperature within a narrow range despite variations in air temperature. The lizard uses a variety of behaviors to accomplish this.

FIGURE 41-11

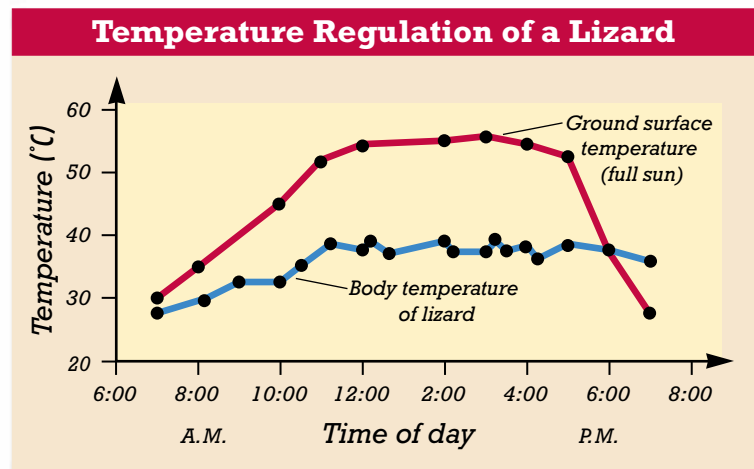
(a) A lizard regulates its body temperature throughout the day, basking in the sun to warm and seeking shade to prevent overheating. If its body temperature rises too high, the lizard may pant to accelerate heat loss (b). The graph shows an early-morning increase in the lizard's body temperature. The body temperature fluctuates only slightly during the remainder of the day, despite wide fluctuations in ground temperature.

Advantages and Limitations of Ectothermy

Because their metabolism is very slow, ectotherms require very little energy and need only about one-tenth as much food as an endotherm of the same size. Ectotherms cannot live in very cold climates, and they can survive temperate climates only by becoming dormant during the coldest months. Furthermore, ectotherms can run or swim at maximum speed only for short periods of time. Ectothermic metabolism cannot provide enough energy for sustained exertion.



(a)



(b)

REPRODUCTION AND PARENTAL CARE

There are three patterns of reproduction among reptiles. The differences between these three patterns lie in how long the eggs remain within the female and in how she provides them with nutrition.

In **oviparity**, the female's reproductive tract encloses each egg in a tough protective shell. The female then deposits the eggs in a favorable place in the environment. Oviparity is characteristic of most reptiles, all birds, and three species of mammals.

One way to protect delicate eggs is to retain the eggs within the female's body for a time. This strategy, seen in some reptiles including American pit vipers, is called **ovoviviparity**. The eggs may be laid shortly before hatching, or they may hatch within the female's body. The eggs absorb water and oxygen from the female, but they receive no nutrition other than the yolk.

In **viviparity**, a shell does not form around the egg, and the young are retained within the female's body until they are mature enough to be born. Nutrients and oxygen are transferred from mother to embryo through a structure called the **placenta**. The placenta forms from the membranes within the egg, and it brings blood vessels from the embryo near the vessels of the mother. Viviparity is the reproductive pattern shown by most mammals, but it is also found in a few species of lizards and snakes.

Many reptiles provide no care for their eggs or young. However, some species of lizards and snakes guard and warm the eggs until they hatch. Crocodiles and alligators provide the greatest amount of parental care. A female crocodilian, for example, builds a nest for her eggs. She remains nearby while the eggs incubate, guarding against nest-robbing predators. After the young hatch, she breaks open the nest and carries the hatchlings to the water in her mouth, as shown in Figure 41-12. The mother crocodile may protect her young for a year or more.

Word Roots and Origins

ovoviviparous

from the Latin *ovum*, meaning "egg," *vivus*, meaning "alive," and *parere*, meaning "to bring forth"

FIGURE 41-12

Reptiles generally do not provide care for their young, but the hatchlings are usually able to fend for themselves as soon as they emerge from the shell. Crocodiles and alligators, however, care for their young for up to two years. The female crocodile in the photograph is transporting her baby in her mouth.



SECTION 2 REVIEW

1. Describe how the heart of a turtle differs from the heart of a crocodile.
2. Describe how reptiles inhale and exhale.
3. Identify two sense organs reptiles and amphibians share and two sense organs that are unique to reptiles.
4. Compare thermoregulation in animals that are endothermic versus thermoregulation in animals that are ectothermic.
5. Contrast oviparity with viviparity.

CRITICAL THINKING

6. **Applying Information** Which of the patterns of reproduction in reptiles best serves to protect the eggs from predators? Explain.
7. **Making Comparisons** Both crocodiles and turtles are ectothermic. Why do you think crocodile hearts are different from turtle hearts?
8. **Justifying Conclusions** Would change in body temperature over the course of a day prove that an animal is an ectotherm? Why or why not?

SECTION 3

OBJECTIVES

- **Compare** the anatomy of turtles with that of other reptiles.
- **Describe** the structure that allows crocodilians to swallow prey under water.
- **Explain** three antipredator defenses of lizards.
- **Describe** two ways snakes subdue their prey.
- **Identify** two reasons that tuataras are rarely seen.

VOCABULARY

carapace
plastron
autotomy
constrictor
elapid
viper

FIGURE 41-13

(a) The Galápagos tortoise, *Geochelone gigantops*, is protected from predators by its high domed carapace. (b) The green sea turtle, *Chelonia mydas*, is streamlined for life in the sea.



(a)



(b)

MODERN REPTILES

Modern reptiles are classified into four orders: Chelonia, Crocodilia, Squamata, and Rhynchocephalia. As different as two species of reptile—such as a turtle and a snake—appear to be, all species of modern reptiles share the following characteristics: an amniotic egg; internal fertilization of eggs; dry, scaly skin; respiration through lungs; and ectothermic metabolism.

ORDER CHELONIA

The order Chelonia consists of about 250 species of turtles and tortoises. The term *tortoise* is generally reserved for the terrestrial members of the order, such as the Galápagos tortoise shown in Figure 41-13a. *Turtle* usually refers to chelonians that live in water, such as the green sea turtle shown in Figure 41-13b.

The earliest known turtle fossils, which are more than 200 million years old, show that ancient chelonians differed little from today's turtles and tortoises. This evolutionary stability may be the result of the continuous benefit of the basic turtle design—a body covered by a shell. The shell consists of fused bony plates. The **carapace** is the top, or dorsal, part of the shell, and the **plastron** is the lower, or ventral, portion. In most species, the vertebrae and ribs are fused to the inner surface of the carapace. Because the ribs are fused to the carapace, the pelvic and pectoral girdles lie within the ribs instead of outside the ribs, as they do in all other terrestrial vertebrates. Unlike other reptiles, turtles have a sharp beak instead of teeth.

Turtles and tortoises live in a variety of habitats. Some species are permanently aquatic, some are permanently terrestrial, and some spend time both on land and in the water. The differing demands of these habitats are reflected in the shells and limbs of turtles. For example, water-dwelling turtles usually have a streamlined, disk-shaped shell that permits rapid turning in water, and their feet are webbed for swimming. The limbs of marine turtles, which spend their entire lives in the ocean, have evolved into flippers for swimming and maneuvering. Many tortoises have a domed carapace into which they can retract their head, legs, and tail as a means of protection from predators. Their limbs are sturdy and covered with thick scales.

Reproduction

All turtles and tortoises lay eggs. The female selects an appropriate site on land, scoops out a hole with her hind limbs, deposits the eggs, and covers the nest. She provides no further care for the eggs or the hatchlings. Marine turtles often migrate long distances to lay their eggs on the same beach where they hatched. For example, Atlantic green sea turtles travel from their feeding grounds off the coast of Brazil to Ascension Island in the South Atlantic—a distance of more than 2,000 km (1,242 mi). These turtles probably rely on several environmental cues, possibly even the Earth's magnetic field and the direction of currents, to find this tiny island.

ORDER CROCODILIA

The living reptiles most closely related to the dinosaurs are the crocodilians, order Crocrodilia. This group is composed of about 21 species of large, heavy-bodied, aquatic reptiles. In addition to crocodiles and alligators, the order includes the caimans and the gavial. Figure 41-14 shows some examples of crocodilians.

Crocrodilians live in many tropical and subtropical regions of the world. Alligators live in China and the southern United States. Caimans are native to Central America and South America, and they have been introduced into Florida.

FIGURE 41-14

(a) Crocodiles, such as genus *Crocodylus*, are found in Africa, Asia, Australia, and the Americas. (b) The gavial, *Gavialis gangeticus*, is a crocodilian with an extremely long and slender snout adapted for seizing and eating fish. Gavials live only in India and Burma.



(a)



(b)



All crocodilians are carnivorous. They feed on fish and turtles and on land animals that come to the water to feed or drink. Crocodilians capture their prey by lying in wait until an animal approaches and then attacking swiftly. A crocodilian can see and breathe while lying quietly submerged in water. A valve at the back of the throat prevents water from entering the air passage when a crocodilian feeds underwater.

ORDER SQUAMATA

The order Squamata consists of about 5,500 species of lizards and snakes. A distinguishing characteristic of this order is an upper jaw that is loosely joined to the skull. Squamates are the most structurally diverse of the living reptiles, and they are found worldwide.

Lizards

There are about 3,000 species of living lizards. Common lizards include iguanas, chameleons, and geckos. Lizards live on every continent except Antarctica. Figure 41-15 shows some examples of lizards. Most lizards prey on insects or on other small animals. A few of the larger species, such as the chuckwalla and desert iguana of the southwestern United States, feed on plants. The Komodo dragon feeds on prey as large as goats and deer. Only two species of lizards are venomous. They are the Gila monster of the southwestern United States and northern Mexico and the related beaded lizard of southern Mexico.

Most lizards rely on agility, speed, and camouflage to elude predators. If threatened by a predator, some lizards have the ability to detach their tail. This ability is called **autotomy**. The tail continues to twitch and squirm after it detaches, drawing the predator's attention while the lizard escapes. The lizard grows a new tail in several weeks to several months, depending on the species.

Most lizards are small, measuring less than 30 cm (12 in.) in length. The largest lizards belong to the monitor family (Varanidae). Like snakes, monitors have deeply forked tongues that pick up airborne particles and transfer them to the Jacobson's organ in the roof of the mouth.

FIGURE 41-15

(a) The largest of all monitors is the Komodo dragon, *Varanus komodoensis*, of Indonesia. The Komodo dragon can grow to 3 m long (10 ft). (b) A colorful gecko of the genus *Phelsuma* has specialized structures on the pads of its fingers and toes that allow it to cling to almost any surface.



(a)



(b)



(a)



(b)

Snakes

There are about 2,500 species of snakes, and like lizards, they are distributed worldwide. Figure 41-16 shows some examples of snakes. The most obvious characteristic of snakes is the lack of legs, which affects all other aspects of their biology. What was the selective pressure that caused snakes to evolve leglessness? One possibility is that the ancestors of snakes were terrestrial but lived in thick vegetation, where legs were a hindrance to rapid movement.

The graceful movements of snakes are made possible by their unique anatomy. A snake has a backbone of 100 to 400 vertebrae, and a pair of ribs is attached to each vertebra. These bones provide the framework for thousands of muscles. The muscles manipulate not only the skeleton but also the snake's skin, causing the overlapping scales to extend and contract, propelling the snake.

Capturing and Consuming Prey

A snake may just seize and swallow its prey. However, many snakes employ one of two methods for killing: constriction or injection of venom. Snakes that are **constrictors** wrap their bodies around prey. A constrictor suffocates its prey by gradually increasing the tension in its coils, squeezing a little tighter each time the prey breathes out. This technique is used both by large snakes, such as boas, pythons, and anacondas, and by smaller snakes, such as gopher snakes and king snakes.

Some snakes inject their prey with a toxic venom in one of three different ways. The snakes with fangs in the back of the mouth, such as the boomslang and twig snakes of Africa, bite the prey and use grooved teeth in the back of the mouth to guide the venom into the puncture. Cobras, kraits, and coral snakes are elapids. **Elapid** snakes inject poisons through two small, fixed fangs in the front of the mouth. **Vipers** inject venom through large, mobile fangs in the front of the mouth. Rattlesnakes, copperheads, and water moccasins are examples of vipers. When a viper strikes, these hinged fangs swing forward from the roof of the mouth and inject venom more deeply than can the fangs of elapids.

FIGURE 41-16

(a) The Gaboon viper, *Bitis gabonica*, injects a toxic venom to kill its prey before it begins the process of swallowing. (b) The boa constrictor, *Constrictor constrictor*, suffocates its prey.



Quick Lab

Modeling Snake Swallowing

Materials rubber tubing, small marble

Procedure Find a way to get the marble into the middle of the rubber tubing.

Analysis How is this model similar to the feeding mechanism of a snake? If you used a marble that was larger than the opening of the tubing, what problems would you encounter? Why is the size of larger prey not a problem for snakes?



FIGURE 41-17

This series of photographs shows a snake, *Dasypeltis scabra*, swallowing a bird's egg. Prey is often larger than the diameter of the snake's head, so the process of swallowing can take an hour or more.

Once killed, the prey must be swallowed whole because a snake's curved, needlelike teeth are not suited for cutting or chewing. Several features of a snake's skull enable it to swallow an animal larger in diameter than its head, as shown in Figure 41-17. The upper and lower jaws are loosely hinged, move independently, and can open to an angle of 130 degrees. In addition, a snake's lower jaw, palate, and parts of its skull are joined by a flexible, elastic ligament that allows the snake's head to stretch around its prey.

FIGURE 41-18

Unlike most reptiles, the endangered tuataras, such as the one shown, are most active at low temperatures.



ORDER RHYNCHOCEPHALIA

The order Rhynchocephalia (RING-koe-suh-FAY-lee-uh) is an ancient one that contains only the tuataras of the genus *Sphenodon*. Tuataras inhabit only a few small islands of New Zealand. The Maoris of New Zealand named the tuataras for the conspicuous spiny crest that runs down the animal's back, seen in Figure 41-18. The word *tuatara* means "spiny crest". Tuataras grow to about 60 cm (24 in.) in length. They usually hide in a burrow during the day and feed on insects, worms, and other small animals at night.

Since arriving in New Zealand about 1,000 years ago, humans have radically changed the landscape and introduced predators such as rats and cats, which feed on tuataras and their eggs. As a result, tuataras have disappeared from most of their original range.

SECTION 3 REVIEW

1. Compare the characteristics of aquatic turtles to the characteristics of land tortoises.
2. Identify characteristics of crocodilians that allow them to feed efficiently in water.
3. Identify three strategies employed by lizards to avoid predators.
4. Describe two methods snakes use for killing prey.
5. Explain why tuataras are rarely seen in the wild.

CRITICAL THINKING

6. **Making Comparisons** Compare the costs versus the benefits of autotomy for lizards.
7. **Inferring Relationships** Why do you think turtles migrate long distances to lay eggs on the same shore on which they hatched?
8. **Applying Information** Why do you think snakes have many more vertebrae than any other group of reptiles does?

CHAPTER HIGHLIGHTS

SECTION 1

Origin and Evolution of Reptiles

- An abundance of food, a dry climate, and a mass extinction of other species led to the dominance of reptiles. The Mesozoic era is often called the age of reptiles.
- Several factors contributed to the success of dinosaurs. Legs positioned under their bodies made dinosaurs faster and more agile than other reptiles. Dinosaurs were well adapted to dry conditions. At the end of the Triassic period, a mass extinction reduced competition and allowed dinosaurs to flourish.
- Dinosaurs became extinct about 65 million years ago. Evidence suggests that environmental conditions over many years endangered dinosaur survival, and either a single asteroid impact or multiple asteroid impacts might have triggered a mass extinction event.
- There are four modern orders of reptiles: Chelonia (turtles and tortoises), Crocodilia (alligators and crocodiles), Squamata (lizards and snakes), and Rhynchocephalia (tuataras).
- Three characteristics that contribute to the success of reptiles on land are the amniotic egg, watertight skin, and efficient respiration and excretion.

Vocabulary

dinosaur (p. 819)
asteroid impact
hypothesis (p. 821)

amniotic egg (p. 823)
amnion (p. 823)
yolk sac (p. 823)

allantois (p. 823)
chorion (p. 823)
albumen (p. 823)

keratin (p. 824)

SECTION 2

Characteristics of Reptiles

- Most reptiles have a three-chambered heart, but crocodiles have a four-chambered heart. Reptiles can divert blood from the lungs to conserve energy and warm their bodies.
- Reptiles inflate their lungs by expanding the ribs, which lowers air pressure in the chest cavity and draws in air. When the ribs relax, air is forced out.
- A reptile's brain is about the same size as the brain of an amphibian but has a much larger cerebrum. A reptile's senses include sight, hearing, smell, and heat detection.
- All living reptiles are ectotherms. Ectotherms warm their bodies mainly by absorbing heat from their surroundings.
- Many reptiles lay shelled eggs, which is called oviparity. Some species retain shelled eggs inside the female's body, which is ovoviviparity. Other species have eggs with placentas rather than shells, which develop within the female's body. This is viviparity.
- Of the reptiles, crocodiles and alligators provide the greatest amount of parental care.

Vocabulary

septum (p. 825)
alveolus (p. 826)
Jacobson's organ (p. 827)

thermoregulation (p. 828)
ectotherm (p. 828)
endotherm (p. 828)

oviparity (p. 829)
ovoviviparity (p. 829)
viviparity (p. 829)

placenta (p. 829)

SECTION 3

Modern Reptiles

- The order Chelonia, which includes turtles and tortoises, have a shell composed of bony plates. The vertebrae and ribs are fused to the interior surface of the shell.
- Members of the order Crocodilia are large aquatic or semiaquatic carnivores. They include crocodiles, alligators, caimans, and gavials. A valve in the oral cavity of crocodilians covers the esophagus and windpipe while the mouth is submerged. This allows crocodilians to capture and swallow prey underwater.
- The order Squamata consists of lizards and snakes. Most lizards rely on speed, agility, and camouflage to elude predators. Some can detach their tail to distract predators.
- Some snakes kill their prey by constriction. Others kill their prey by injecting venom.
- The order Rhynchocephalia contains only the tuataras. Tuataras hide in burrows during the day and feed at night. The arrival of humans and new predators in New Zealand has reduced the number of tuataras.

Vocabulary

carapace (p. 830)
plastron (p. 830)

autotomy (p. 832)
constrictor (p. 833)

elapid (p. 833)
viper (p. 833)

CHAPTER REVIEW

USING VOCABULARY

1. Choose the term that does not belong in the following group, and explain why it does not belong: *elapids*, *vipers*, *constrictors*, and *dinosaurs*.
2. Explain the relationship between amnion and chorion.
3. Use each of the following terms in a separate sentence: *oviparity*, *ovoviviparity*, and *viviparity*.
4. **Word Roots and Origins** The word *ectotherm* comes from the Greek word *ecto*, meaning “outside” and *therm*, meaning “heat”. The word *endotherm* comes from the Greek word *endo*, meaning “within” and *therm*, meaning “heat”. Explain why each term is a good name for the biological process that the term describes.

UNDERSTANDING KEY CONCEPTS

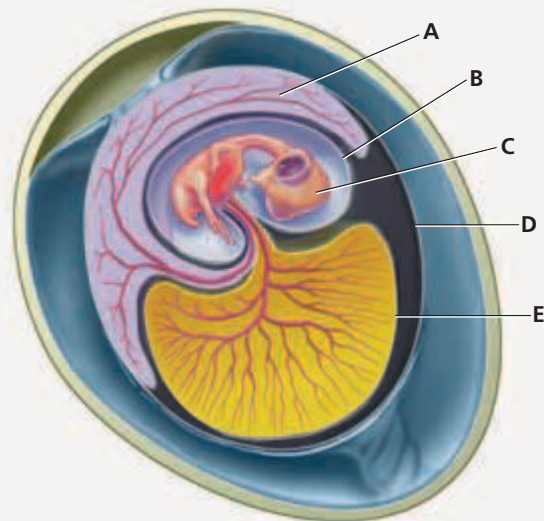
5. **Summarize** the events that led to the rise of reptiles during the Permian period.
6. **Identify** three factors that contributed to the success of dinosaurs during the Triassic period.
7. **Explain** why scientists advocate the asteroid impact hypothesis for dinosaur extinction.
8. **Name** one example of each of the four modern orders of reptiles.
9. **Explain** how the evolution of the amniotic egg is an adaptation to life on land.
10. **Explain** how the respiratory and excretory systems in reptiles show adaptations for life on land.
11. **Contrast** a turtle’s heart and a crocodile’s heart.
12. **Describe** how the structure of a turtle’s heart allows for flexibility in blood circulation.
13. **Identify** the advantage of alveoli in respiration.
14. **Describe** two senses other than vision and hearing that reptiles use to find prey.
15. **Identify** the major benefit of ectothermy.
16. **Compare** the reproductive strategies of oviparity, ovoviviparity, and viviparity.
17. **Describe** the features that make turtles unique among reptiles.
18. **Explain** how crocodilians swallow prey underwater.
19. **Explain** the purpose of a lizard’s ability to lose its tail and grow a new one.
20. **List** three ways in which snakes can inject venom into their prey.

21. **Identify** the reason populations of tuataras have decreased in the last 1,000 years.

22. **CONCEPT MAPPING** Use the following terms to create a concept map that relates various adaptations of reptiles to life on land: *reptiles*, *water loss*, *amniotic eggs*, *keratin*, *scales*, *watertight*, and *skin*.

CRITICAL THINKING

23. **Analyzing Information** If birds did not evolve from the pterosaurs, what can you conclude about the evolution of the ability to fly?
24. **Analyzing Concepts** Why is it advantageous for a snake to kill its prey, either through constriction or venom, before trying to eat the prey?
25. **Recognizing Relationships** The skin of a basking lizard is usually dark. As the lizard warms, the skin lightens. Suggest a functional explanation for this change. (Hint: Consider how this change might affect the lizard’s absorption of heat.)
26. **Analyzing Data** Fossil evidence collected in Alaska suggests that some dinosaurs were year-round residents of areas subject to freezing temperatures and long periods of darkness. Does this evidence of arctic dinosaurs support or contradict the hypothesis that the extinction of dinosaurs was due to the intense cold produced by a cloud of debris in the atmosphere? Explain your answer.
27. **Relating Concepts** Some viviparous snakes and lizards live in cold climates. Why might viviparity be advantageous in such environments?
28. **Interpreting Graphics** The following diagram shows five parts of the amniotic egg, indicated by the letters A, B, C, D, and E. Name and identify the function of each part.



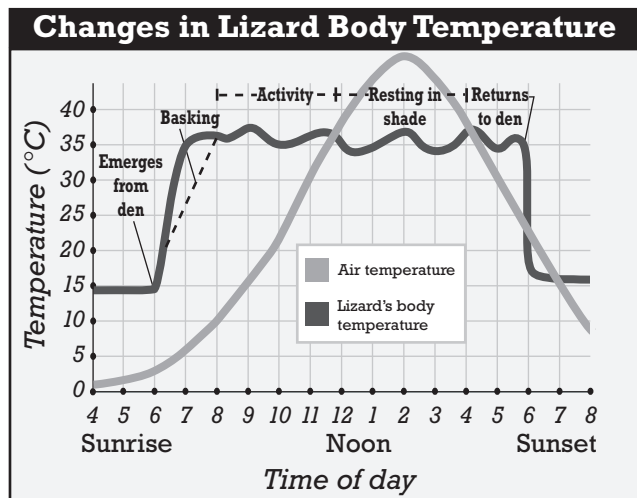


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which is an adaptation that made reptiles the dominant species in the Mesozoic era?
 - They were endotherms.
 - They could live in Antarctica.
 - They were all large predators.
 - They were well adapted to dry areas.
- What are the two basic parts of a turtle's shell?
 - septum and amnion
 - chorion and allantois
 - keratin and columella
 - carapace and plastron
- What is the purpose of a lizard's ability to lose its tail and grow a new one?
 - to capture prey
 - to hide from predators
 - to escape from predators
 - to reduce its need for food
- Long legless bodies may have arisen as an adaptation that helped snakes do what?
 - catch prey
 - swallow large animals
 - absorb oxygen through their skin
 - burrow and move through thick vegetation

INTERPRETING GRAPHICS: The graph below shows changes in air temperature and changes in the body temperature of a lizard. Use the graph below to answer the question that follows.

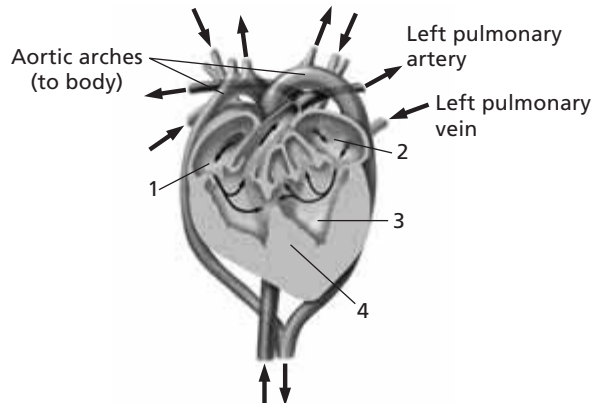


- In order for the lizard to raise its internal temperature it must do which of the following?
 - rest in the shade
 - bask in sunshine
 - increase its internal temperature through activity
 - decrease its internal temperature through activity

DIRECTIONS: Complete the following analogy.

- Ectotherm : reptiles :: endotherm :
 - fishes
 - insects
 - mammals
 - amphibians

INTERPRETING GRAPHICS: The illustration below shows a cross section of a turtle's heart. Use the illustration to answer the question that follows.



- Which feature of a turtle's heart structure is different from that of a crocodile?
 - 1
 - 2
 - 3
 - 4

SHORT RESPONSE

A reptile can redirect blood flow through the heart to send some deoxygenated blood back to the body instead of to the lungs.

What is the advantage to the lizard of redirecting blood flow in this manner?

EXTENDED RESPONSE

There are three patterns of reproduction among reptiles. They differ in how long eggs remain in the female and how the developing young are provided with nutrition.

Part A With these differences in mind, compare oviparity, ovoviviparity, and viviparity.

Part B In which pattern of reproduction is a placenta present? Explain the function of the placenta.

Test TIP

When analyzing a graph, pay attention to its title. It should tell you what the graph is about and provide a context for the data.

Observing Color Adaptation in Anoles

OBJECTIVES

- Observe live anoles.
- Test whether background color stimulates color change in anoles.

PROCESS SKILLS

- observing
- hypothesizing
- experimenting
- organizing data
- analyzing data

MATERIALS

- glass-marking pencil
- 2 large clear jars with wide mouths and lids with air holes
- 2 live anoles
- 6 shades each of brown and green construction paper, ranging from light to dark (2 swatches of each shade)

Background

1. Anoles include 250–300 species of lizards in the genus *Anolis*.
2. Anoles can change color, ranging from brown to green, and are sometimes mottled.

3. Anoles live in shrubs, grasses, and trees. Describe some ways in which the ability to change color might be an advantage to anoles.
4. Light level, temperature, and other factors, such as whether the anole is frightened or whether it has eaten recently, can affect color. When anoles are frightened, they usually turn a dark grey or brown and are unlikely to respond to other stimuli.
5. Anoles can change color within a few minutes.

Procedure

1. Observe the anoles in two terraria, and discuss the purpose of this investigation with your partners. Develop a hypothesis that describes a relationship between anole skin color and background color. Write your hypothesis in your lab report.
2. Obtain swatches of construction paper in at least six different shades of green and brown. You will need two swatches of paper in each color.
3. Obtain two clear jars. Label one jar “Anole 1,” and label the other jar “Anole 2.”
4.  **CAUTION** You will be working with live animals. Handle them gently and follow instructions carefully. Select two anoles of the same color from the terraria. Plan your actions and cooperate with a partner to transfer one anole into each labeled jar. Anoles will run fast and are easily frightened. Carefully pick them up and place the animals in separate jars. Do not pick up anoles by their tails. Grasp them gently behind the head. Quickly and carefully place a lid with air holes on each jar.
5. Gently place the jar with Anole 1 on a swatch of construction paper that most closely matches the anole’s color. Try not to jostle the anole in the jar, and move the jar as little and as gently as possible. Repeat this procedure for Anole 2. Both anoles should closely match the color of the swatch.



6. When you have obtained and matched two anoles to closely matching colors, label the back of the pieces of paper "Initial Color of Anole 1" and "Initial Color of Anole 2," respectively. Replace the swatches underneath the jars after you have labeled them. The anoles should stay in their respective jars until the end of this investigation.
7. Using the given setup and the remaining swatches of colored construction paper, devise a control experiment to test whether background color stimulates color change in the anole.
8. In your lab report, list the independent variable and the dependent variables that you intend to use in your experiment. Describe how you will vary the independent variable and how you will measure changes in your dependent variable.
9. In your lab report, describe the control you will use in your experiment.
10. Create a data table similar to the one below to record your experimental observations for your lab report. For example, the table below is designed to record any change in anole skin color on four different background colors and the time it took for each change to take place. Design your data table to fit your own experiment. Remember to allow plenty of space to write your observations.
11. Have your experiment approved by your teacher before conducting it. As you conduct your experiment, be sure to record all of your data and observations in your lab report.
12. Attach your color swatches to your lab report, or include a color-coded key so that others reading your report will be able to understand how you measured initial color and color changes in your anoles. Be sure the color that most closely represents the initial color of both anoles is clearly indicated in your lab report.
13.   Clean up your materials and wash your hands before leaving the lab.

Analysis and Conclusions

1. What effect, if any, did changes in the independent variable have on the dependent variable in your experiment?
2. Do your data support your hypothesis? Explain.
3. Can you think of any sources of error in your experiment?
4. Was your experiment a controlled experiment? If yes, describe your control and why you think a control is necessary for your experiment.
5. Were there any uncontrolled variables in your experiment, such as loud noises, bright light, or sudden movements, that could have affected your experiment? Describe how you might be able to improve your methods.

Further Inquiry

Design an experiment that tests the effects of temperature on anole skin color.

DATA TABLE OBSERVING ANOLES

	Color 1		Color 2		Color 3		Color 4	
	Change	Time	Change	Time	Change	Time	Change	Time
Anole 1								
Anole 2								



Young birds, such as this owl, depend on their parents for food and protection.

SECTION 1 *Origin and Evolution of Birds*

SECTION 2 *Characteristics of Birds*

SECTION 3 *Classification*



For project ideas from *Scientific American*, visit go.hrw.com and type in the keyword **HM6SAK**.

ORIGIN AND EVOLUTION OF BIRDS

Birds belong to the class Aves, which, with nearly 10,000 species, is the largest class of terrestrial vertebrates. Birds are also the most recently evolved group of vertebrates, having appeared only about 150 million years ago. Among living vertebrates, only birds and bats can fly. The bodies of most birds are well adapted to flight.

OBJECTIVES

- **Identify** and describe seven major characteristics of birds.
- **List** three similarities between birds and dinosaurs.
- **Describe** the characteristics of *Archaeopteryx*.
- **Summarize** the two main hypotheses for the evolution of flight.

VOCABULARY

furcula

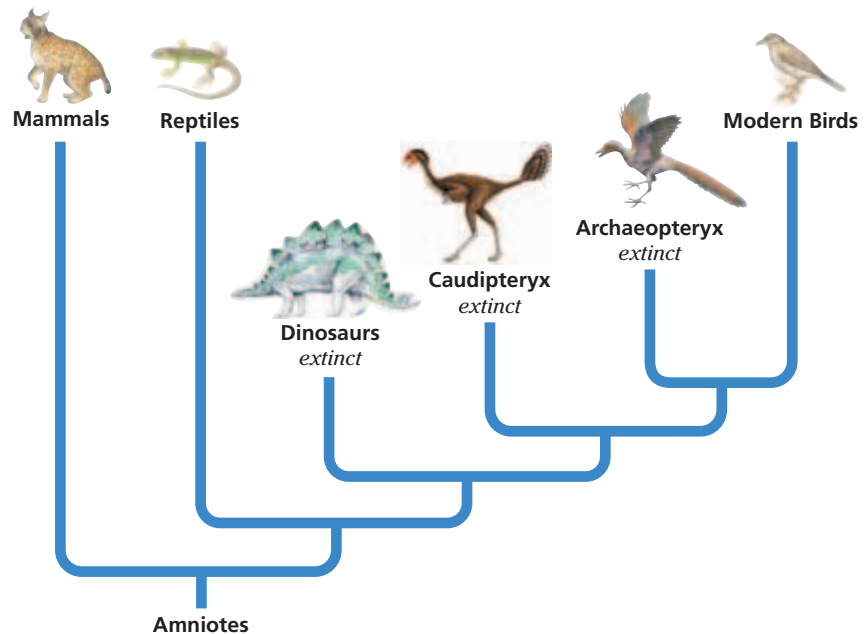
CHARACTERISTICS

Although there are many kinds of birds, birds are so distinctive that it is difficult to mistake one for any other kind of vertebrate. All birds—even those that cannot fly—share the seven important characteristics described below.

- **Feathers**—Feathers are unique to birds, and all birds have them. Like hair, feathers are composed mainly of the versatile protein keratin. Feathers are essential for flight, and they insulate a bird's body against heat loss.
- **Wings**—A bird's forelimbs are modified into a pair of wings. Feathers cover most of the surface area of the wing.
- **Lightweight, rigid skeleton**—The skeleton of a bird reflects the requirements of flight. Many of the bones are thin-walled and hollow, making them lighter than the bones of nonflying animals. Air sacs from the respiratory system penetrate some of the bones. Because many bones are fused, the skeleton is rigid and can resist the forces produced by the strong flight muscles.
- **Endothermic metabolism**—A bird's rapid metabolism supplies the energy needed for flight. Birds maintain a high body temperature of 40–41°C (104–106°F). The body temperature of humans, by contrast, is about 37°C, or 98.6°F.
- **Unique respiratory system**—A rapid metabolism requires an abundant supply of oxygen, and birds have the most efficient respiratory system of any terrestrial vertebrates. The lungs are connected to several sets of air sacs, an arrangement that ensures that oxygen-rich air is always in the lungs.
- **Beak**—No modern bird has teeth, but the jaws are covered by a tough, horny sheath called a *beak*.
- **Oviparity**—All birds lay amniotic eggs encased in a hard, calcium-containing shell. In most species, the eggs are incubated in a nest by one or both parents.

FIGURE 42-1

This phylogenetic diagram represents a hypothesis for the relationship among birds, reptiles, mammals, and some extinct relatives of birds. For updates on phylogenetic information, visit go.hrw.com. Enter the keyword HM6 Phylo.



EVOLUTION

There are many similarities between birds and some dinosaurs, such as *Caudipteryx* show in Figure 42-1. Three of these similarities include a flexible S-shaped neck, a unique ankle joint, and hollow bones. Birds are thought to have evolved from small, fast-running carnivorous dinosaurs during the Jurassic period (200–146 million years ago). Figure 42-1 shows the likely relationships between birds and other terrestrial vertebrates.

The oldest known bird fossils are classified in the genus *Archaeopteryx* and date from the late Jurassic period, about 150 million years ago. In the fossil in Figure 42-2a, the shapes of feathers are clearly visible. Feathers covered *Archaeopteryx*'s forelimbs, forming wings, and covered its body and tail as well. Like modern birds, *Archaeopteryx* had hollow bones and a **furcula** (FUHR-kyoo-luh), the fused pair of collarbones commonly called a *wishbone*. The furcula plays an important role in flight by helping to stabilize the shoulder joint. Based on such similarities with modern birds, scientists think that *Archaeopteryx* could fly. However, *Archaeopteryx* also had several characteristics of its dinosaur ancestors, including teeth, claws on its forelimbs, and a long, bony tail. Figure 42-2b shows an artist's conception of what an *Archaeopteryx* might have looked like.

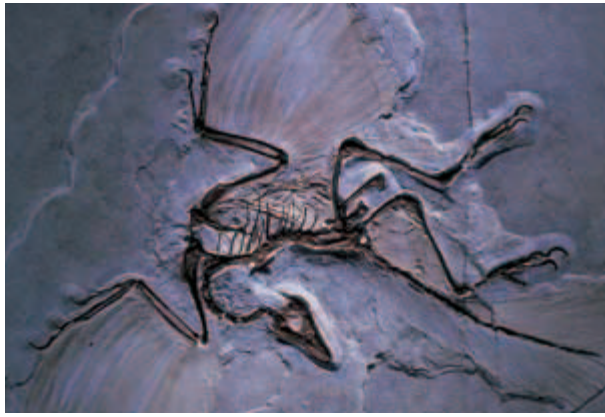
Word Roots and Origins

archaeopteryx

from the Greek *archaios*, meaning "ancient," and *pteryx*, meaning "wing"

Origin of Flight

The evolution of a flying animal from nonflying ancestors entails many changes in anatomy, physiology, and behavior. According to one hypothesis, the ancestors of birds were tree dwellers that ran along branches and occasionally jumped between branches and trees. Wings that allowed these animals to glide evolved. Once gliding was possible, the ability to fly by flapping the wings evolved.



(a)

Another hypothesis draws on the fact that the dinosaurs most closely related to birds were terrestrial and states that the evolution of birds must have occurred on the ground, not in the trees. Wings may have originally served to stabilize the animals as they leapt after prey. Or they may have been used for trapping or knocking down insect prey. Over generations, the wings became large enough to allow the animal to become airborne.

Evolution After *Archaeopteryx*

A number of recent discoveries show that by the early Cretaceous period (146–66 million years ago), birds had begun diversifying. *Sinornis*, a 140-million-year-old specimen discovered in China in 1987, had some key features of modern birds, including a shortened, fused tail and a wrist joint that allowed the wings to be folded against the body. The diversification of birds continued throughout the Cretaceous period. Figures 42-2c and 42-2d show two birds from the late Cretaceous period.

Only two of the modern orders of birds had appeared by the end of the Cretaceous period. Birds survived the global catastrophe that is thought to have wiped out the dinosaurs, and then underwent a dramatic and rapid evolutionary radiation. By about 40 million years ago, most of the modern orders of birds had originated.



(b) *Archaeopteryx*



(c) *Ichthyornis*



(d) *Hesperornis*

FIGURE 42-2

In this fossil of *Archaeopteryx* (a), one can find characteristics of both birds and dinosaurs. These artist's renderings of three extinct birds are based on fossil evidence. *Archaeopteryx* (b) is the oldest bird; it still had claws on its wings. *Ichthyornis* (c) had strongly developed wings and was about 21–26 cm (8–10 in.) in length. *Hesperornis* (d) was considered flightless, but its well-developed legs made it a strong swimmer.

SECTION 1 REVIEW

1. List two unique features of a bird's skeleton.
2. What are two functions of feathers?
3. Identify the characteristics that birds have in common with dinosaurs.
4. Describe two characteristics shared by *Archaeopteryx* and modern birds.
5. Name two differences between *Archaeopteryx* and modern birds.
6. Summarize the two major hypotheses for the evolution of flight.

CRITICAL THINKING

7. **Forming Hypotheses** Modern birds lack teeth. Form a hypothesis to explain how birds might have evolved to lack teeth.
8. **Evaluating Hypotheses** After studying the fossil of *Archaeopteryx*, evaluate the two hypotheses for the evolution of flight. Identify strengths and weaknesses of each.
9. **Applying Information** Identify three adaptations that help birds satisfy their high need for oxygen.

SECTION 2

OBJECTIVES

- **Describe** the structure of a contour feather.
- **Identify** two modifications for flight seen in a bird's skeletal system.
- **Contrast** the function of the gizzard with that of the crop.
- **Trace** the movement of air through the respiratory system of a bird.
- **Explain** the differences between altricial and precocial young.

VOCABULARY

feather
follicle
shaft
vane
barb
barbule
preen gland
sternum
pygostyle
crop
proventriculus
gizzard
vas deferens
oviduct
brood patch
precocial
altricial
ornithologist

CHARACTERISTICS OF BIRDS

A number of unique anatomical, physiological, and behavioral adaptations enable birds to meet the aerodynamic requirements of flight. Natural selection has favored a lightweight body and powerful wing muscles that give birds their strength.

FEATHERS

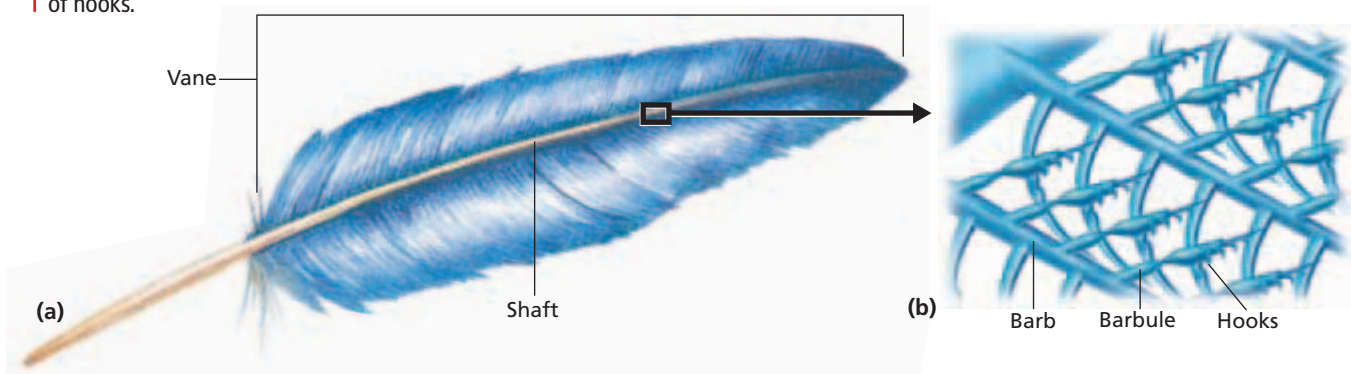
Feathers are modified scales that serve two primary functions: providing lift for flight and conserving body heat. Soft, fluffy *down feathers* cover the body of very young birds and provide an insulating undercoat in adults. *Contour feathers* give adult birds their streamlined shape and provide coloration and additional insulation. *Flight feathers* are specialized contour feathers on the wings and tail. Birds also have dust-filtering bristles near their nostrils.

The structure of a feather combines maximum strength with minimum weight. Feathers develop from tiny pits in the skin called **follicles**. A **shaft** emerges from the follicle, and two **vanes**, pictured in Figure 42-3a, develop on opposite sides of the shaft. At maturity, each vane has many branches, called **barbs**. The barbs, in turn, have many projections, called **barbules**, equipped with microscopic hooks, as shown in Figure 42-3b. The hooks interlock and give the feather its sturdy but flexible shape. Feathers are made of keratin, an insoluble protein that is highly resistant to decomposition. Keratin is also the protein that makes up fingernails, claws, hair, and scales in animals.

Feathers need care. In a process called *preening*, birds use their beaks to rub their feathers with oil secreted by a **preen gland**, located at the base of the tail. Birds periodically *molt*, or shed and regrow their feathers. Birds living in temperate climates usually replace their flight feathers during the late summer.

FIGURE 42-3

Bird feathers, such as this contour feather (a), usually have a shaft, with two vanes growing out either side of the shaft. The vanes (b) consist of barbs and barbules that interlock by means of hooks.



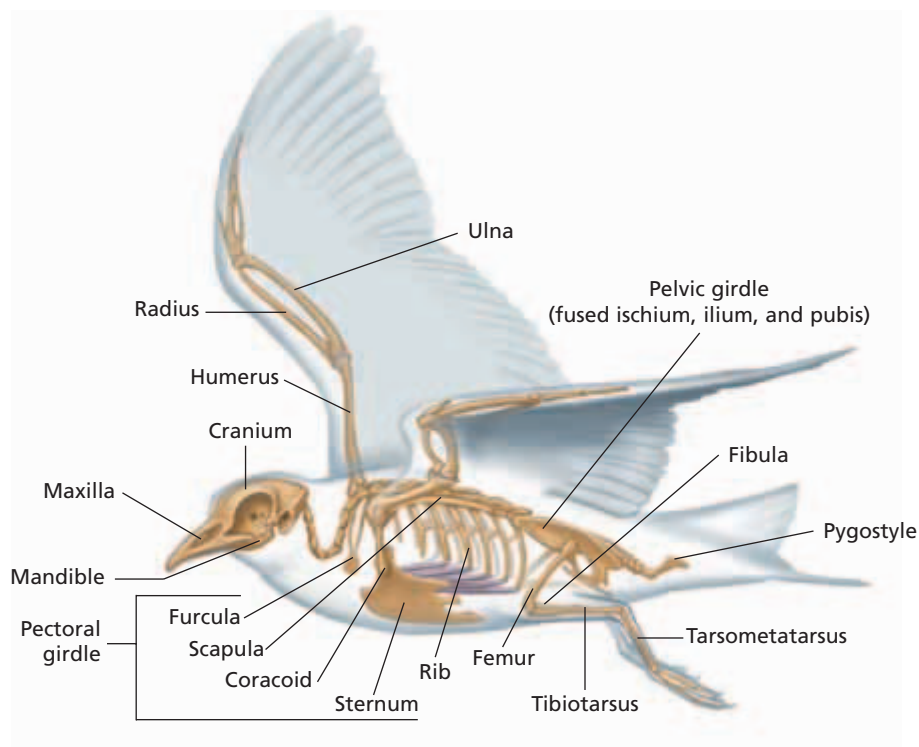


FIGURE 42-4

The avian skeleton is well adapted for flight. The bones are air filled, making them light but strong. The skeleton is arranged in such a way that it supports the large muscles necessary for flight.

SKELETON AND MUSCLES

The avian skeleton combines lightness with strength. The bones are thin and hollow. Many bones are fused, so the skeleton is more rigid than the skeleton of a reptile or mammal. The rigid skeleton provides stability during flight. Note in Figure 42-4 that the bones of the trunk and hip vertebrae and the pectoral and pelvic girdles are highly fused. Along with the furcula, the large, keel-shaped **sternum**, or breastbone, is an attachment point for flight muscles. The humerus, ulna, and radius, along with the pectoral girdle and the sternum, support the wing. The **pygostyle** (PIEG-uh-stiel), the fused terminal vertebrae of the spine, supports the tail feathers. The tail provides additional lift and aids in steering and braking.

Flight involves a series of complex wing movements, each one using a different set of muscles. On the downstroke, the wings cut forward and downward through the air. During upstroke, they move upward and backward. These movements are made possible by large, powerful flight muscles in the breast and wings. In some birds, flight muscles account for up to 50 percent of the body weight.

METABOLISM

Birds are endothermic; that is, they generate heat to warm the body internally. Rapid breathing and digestion of large quantities of food support the high metabolic rate necessary to generate this heat. Birds, unlike reptiles, cannot go for long periods without eating. To help conserve body heat, birds may fluff out their feathers. Aquatic birds have a thin layer of fat that provides additional insulation.



Quick Lab

Comparing Wing Structures

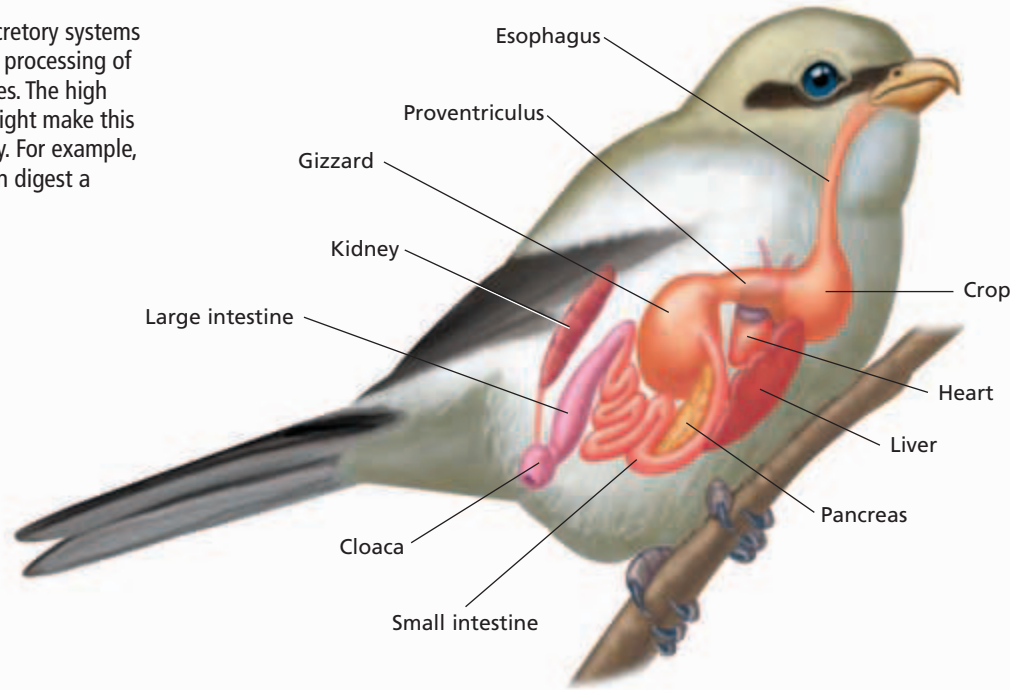
Materials pictures of different kinds of birds, ruler

Procedure Examine each sheet of birds and their wings. Compare the structure and shape of the wings. Measure the wingspan relative to the bird's body length. Record your observations.

Analysis Predict the type of habitat in which each bird lives. How does the shape of the wing relate to the bird's niche? Explain why the type of wings each bird has might make the bird unsuccessful if it were introduced into a much different environment.

FIGURE 42-5

A bird's digestive and excretory systems are adapted for the rapid processing of food and metabolic wastes. The high energy requirements of flight make this efficient system necessary. For example, the carnivorous shrike can digest a mouse in three hours.



Word Roots and Origins

proventriculus

from the Greek *pro*, meaning "before," and the Latin *venter*, meaning "belly"

Digestive and Excretory Systems

The high amount of energy required to fly and regulate body heat is obtained by a quick and efficient digestive system, as illustrated in Figure 42-5. Because birds do not have teeth, they are not able to chew their food. Instead, food passes from the mouth cavity straight to the esophagus. An enlargement of the esophagus called the **crop** stores and moistens food. Food then passes to the two-part stomach. In the first chamber, the **proventriculus** (PROH-ven-TRIK-yoo-luhs), acid and digestive enzymes begin breaking down the food. Food then passes to the **gizzard**, the muscular portion of the stomach, which kneads and crushes the food. The gizzard often contains small stones that the bird has swallowed. These aid in the grinding process. Thus, the gizzard performs a function similar to that of teeth and jaws. Seed-eating birds usually have a larger crop and gizzard—relative to body size—than meat-eating birds do.

From the stomach, food passes into the small intestine. There, bile from the liver and enzymes from the pancreas and intestine further break down the food. The nutrients are then absorbed into the bird's bloodstream. Passage of food through the digestive system of a bird is usually very rapid. For instance, a thrush can eat blackberries, digest them, and excrete the seeds 45 minutes later.

The avian excretory system is efficient and lightweight. Unlike other vertebrates, most birds do not store liquid waste in a urinary bladder. The two kidneys filter a nitrogenous waste called *uric acid* from the blood. Concentrated uric acid travels through ducts called *ureters* to the cloaca, where it mixes with feces and is then excreted. This system is adaptive for flight because birds do not need to carry much water in their bodies.

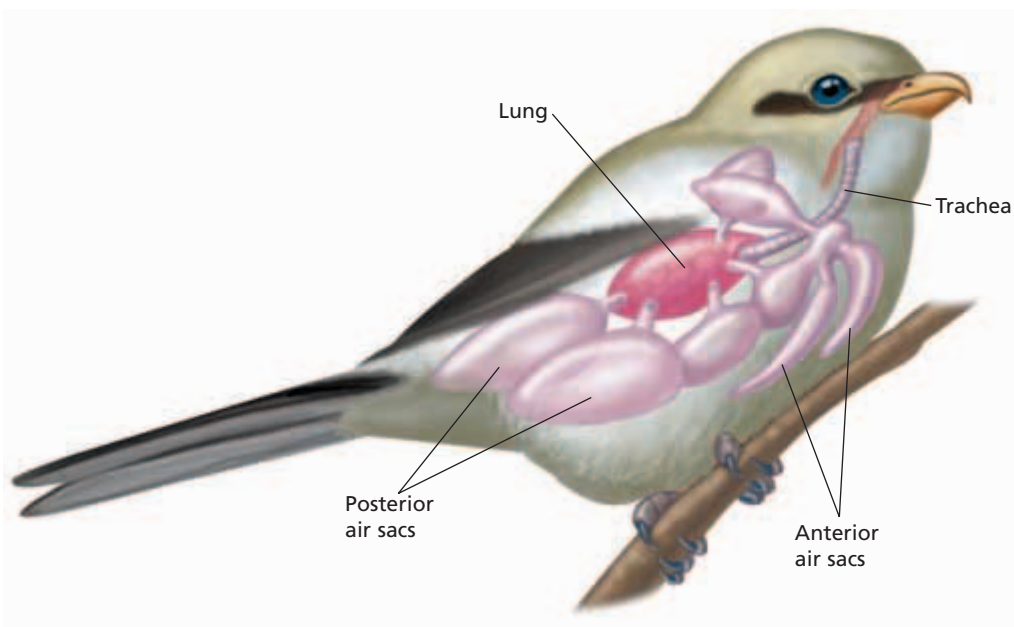


FIGURE 42-6

The unique architecture of the bird's respiratory system provides a constant flow of oxygenated air to the lungs. This highly efficient system allows birds to maintain the high metabolic rate necessary for flight. It also enables birds to function at high altitudes, where other animals would suffer from the low availability of oxygen.

Respiratory System

The high metabolic rate of birds requires large amounts of oxygen. Yet some birds migrate thousands of miles at altitudes as high as 7,000 m (23,000 ft), where air pressure is very low. So, birds have an elaborate and highly efficient respiratory system. Air enters the bird's body through paired nostrils located near the base of the beak. The air passes down the trachea and enters the two primary bronchi. From the bronchi, some of the air moves to the lungs. However, about 75 percent of the air bypasses the lungs and flows directly to posterior air sacs, shown in Figure 42-6. In most birds, nine sacs extend from the lungs and occupy a large portion of the bird's chest and abdominal cavity. These sacs also extend into some of the long bones. Thus, the air sacs not only function in respiration but also greatly reduce the bird's density.

Gas exchange does not occur in the air sacs. Their function is to store and redirect air. When the bird exhales, the oxygen-poor air from its lungs is forced into the anterior air sacs, and the oxygen-rich air in the posterior air sacs is forced into the lungs. This way, the bird has oxygenated air in its lungs during both inhalation and exhalation.

Circulatory System

The avian circulatory system has characteristics that are similar to those of either reptiles or mammals or both. Like crocodiles and mammals, birds have a four-chambered heart with two separate ventricles. Deoxygenated blood is always kept separate from oxygenated blood. In comparison with most other vertebrates, most birds have a rapid heartbeat. A hummingbird's heart beats about 600 times a minute. An active chickadee's heart beats 1,000 times a minute. In contrast, the heart of the larger, less active ostrich averages 70 beats per minute, or about the same rate as a human heart. Avian red blood cells have nuclei.

Nervous System and Sense Organs

Relative to their body size, birds have large brains. The most highly developed areas of the bird's brain are those that control flight-related functions, such as the cerebellum, which coordinates movement. The cerebrum is also large. It controls complex behavior patterns, such as navigation, mating, nest building, and caring for the young. The large optic lobes receive and interpret visual stimuli.

Keen vision is necessary for taking off, landing, spotting landmarks, hunting, and feeding. Most birds have strong, color vision that aids them in finding food. In most species, the eyes are large and are located near the sides of the head, giving the bird a wide field of vision. Birds that have eyes located near the front of the head have better binocular vision, meaning they can perceive depth in the area where the visual fields of the two eyes overlap.

Hearing is important to songbirds and to nocturnal species, such as owls, which rely on sounds to help them locate their prey. Though birds lack external ears, owls have feathers around their ear openings that direct sound into the ear. The sense of smell is also strong in many birds.

REPRODUCTION

In the male bird, sperm is produced in two testes that lie anterior to the kidneys. Sperm passes through small tubes called **vasa deferentia** (singular, *vas deferens*) into the male's cloaca. During mating, the male presses his cloaca to the female's cloaca and releases sperm. Most females have a single ovary located on the left side of the body. The ovary releases eggs into a long funnel-shaped **oviduct**, where the eggs are fertilized by sperm. Fertilized eggs move down the oviduct, where they are encased in a protective covering and a shell. The egg passes out of the oviduct and into the cloaca. From the cloaca, it is expelled from the bird.

Eco Connection

DDT and Bird Eggs

DDT is a pesticide that was widely used on crops until the 1970s. DDT was banned in several countries because of the harm it was causing to birds. DDT causes some birds to produce thin egg shells, decreasing survival rates of the birds' offspring. The result was a significant drop in the populations of several species of raptors and pelicans. The thinning of the eggs was so significant that even the weight of an incubating parent could crush the eggs. With the banning of DDT in the United States, populations of the affected birds have increased.

Nest Building and Parental Care

Birds usually lay their eggs in a nest. Nests hold the eggs, conceal young birds from predators, provide shelter from the elements, and sometimes serve to attract a mate. Most birds build nests in sheltered, well-hidden spots—ranging from holes in the ground to treetops. Woodpeckers, for example, nest in a hole they have drilled in a tree. Orioles suspend their nests from branches, well beyond the reach of predators. And barn swallows build a saucer of mud on the beam of a building. Birds construct their nests from almost any available material. Twigs, bark, grasses, feathers, and mud are common materials used.

One or both parents warm, or *incubate*, the eggs by sitting on them and covering them with a thickened, featherless patch of skin on the abdomen called a **brood patch**. Once the eggs hatch, the young usually receive extensive parental care.

Birds have two general patterns of rearing young. Some birds lay many eggs and incubate them for long periods. These birds hatch **precocial** (pree-KOH-shuhl) young, which can walk, swim, and feed themselves as soon as they hatch. Ducks, quail, chickens, and other ground-nesting species produce precocial offspring. Other birds lay only a few eggs that hatch quickly and produce **altricial** (al-TRISH-uhl) young, which are blind, naked, and helpless, as shown in Figure 42-7. These young depend on both parents for several weeks. The young of woodpeckers, hawks, pigeons, parrots, warblers, and many aquatic birds are altricial.

MIGRATION

Each year, thousands of bird species exploit the spring and summer food resources of temperate regions. Then, when temperatures drop and the food supply dwindles, they travel to warmer climates. The seasonal movement of animals from one habitat to another habitat is called *migration*. Many of the birds that nest in the United States and Canada during the spring and summer fly south in the fall to spend the winter in Mexico, Central America, the Caribbean, or South America.

How do birds manage to navigate thousands of kilometers across varied terrains and return to the same places each year? **Ornithologists**—biologists who study birds—have learned that birds rely on a variety of cues to help them navigate. Some species monitor the position of the stars or the sun. Others rely on topographical landmarks, such as mountains. The Earth's magnetic field, changes in air pressure due to altitude, and low-frequency sounds may also provide information to migrating birds.

Many species migrate thousands of kilometers and must rely on their fat reserves in order to complete the journey. To prepare for their migration, some birds, such as blackpoll warblers, eat so much food before their journey that their weight nearly doubles.



FIGURE 42-7

This yellow warbler cares for its young by feeding and protecting them.

SECTION 2 REVIEW

1. Distinguish between vanes, barbs, and barbules.
2. What is the function of the keel-shaped sternum?
3. In what way does the gizzard compensate for the lack of teeth in birds?
4. What are the functions of the anterior and posterior air sacs?
5. Contrast altricial and precocial young.
6. Identify the cues birds use to help them navigate during a migration.

CRITICAL THINKING

7. **Analyzing Information** Why do you think the brood patch is featherless?
8. **Applying Information** Ground-nesting birds often produce precocial young. How might this improve the odds of survival for the young?
9. **Forming Reasoned Opinions** Some biologists have proposed that birds and reptiles should be grouped in the same class of vertebrates. Do you agree? Support your answer.

MIGRATING BIRDS IN DANGER: Conservation Issues and Strategies

Since Europeans first settled in the Americas, more than half of the Western hemisphere's wetlands have been destroyed. Today, in spite of regulations, wetland destruction continues.

Crucial Roadside Park

More than 40 North American shorebird species—such as sandpipers, plovers, and curlews—breed in the Arctic and migrate to wintering sites in Central and South America. To complete these long-distance flights, shorebirds must accumulate large fuel reserves. The birds prepare for their journeys at a few food-rich staging areas, which are usually wetlands. In some cases, between 50 and 80 percent of the entire population of a species may visit a single site. The loss of such a staging area could devastate a population of shorebirds.

Studies show that the numbers of shorebirds are declining worldwide. In order to preserve and protect these valuable species, researchers must first determine shorebird populations and map their migratory routes.

Technology to the Rescue

Today, the work of preserving shorebirds has been greatly advanced through the use of computer technology. One group that uses computers in this work is the International Shorebird Survey (ISS). The ISS was established in 1974 and consists of both volunteers and professional researchers. The volunteers record information on populations, habitat characteristics, weather conditions, and human activity. Researchers use computer programs to identify relationships among these variables. Complex methods of analysis have revealed much about the migratory habits of shorebirds—

their velocity, altitude, flight path, and gathering sites, for example. Computer analyses have also helped researchers identify significant trends, such as a 70 to 80 percent drop in the population of sanderlings.

Sanctuary Without Borders

Another organization that works to protect shorebirds is the Western Hemisphere Shorebird Reserve Network (WHSRN). The WHSRN has identified more than 90 sites where significant numbers of shorebirds breed, feed, and rest. The organization has proposed that these sites be recognized internationally as a single, vast wildlife reserve whose boundaries are, in effect, defined by the migrating birds.

Because most of the staging sites for shorebirds are wetlands, humans must develop a strategy for the conservation of these sites. The hemispheric wildlife reserve is one example of such a strategy.



Delaware Bay, located between Delaware and New Jersey, is a staging site for shorebirds that migrate between South America and the Arctic. As much as 80 percent of the population of red knots (shown here) feeds and rests at this site.

REVIEW

1. What are some threats to shorebirds posed by humans?
2. How is technology aiding the conservation of shorebirds?
3. **Critical Thinking** How do the migration behaviors of shorebirds make them especially vulnerable to habitat loss?



CLASSIFICATION

Birds are the most widespread terrestrial vertebrates. Their ability to navigate over long distances and their many adaptations for flight enable them to migrate to and inhabit virtually any environment. Their anatomical diversity reflects the diversity of places they inhabit.

DIVERSITY

By looking closely at a bird's beak and feet, you can infer many things about where it lives and how it feeds. Hawks and eagles have powerful beaks and clawed talons that help them capture and tear apart their prey. Swifts have a tiny beak that opens wide like a catcher's mitt and snares insects in midair. Because swifts spend most of their lives in flight, their feet are small and adapted for infrequent perching. The feet of flightless birds, on the other hand, are modified for walking and running. Some examples of the variety of bird beaks and feet are shown in Figure 42-8.



(a)



(b)



(c)



(d)

OBJECTIVES

- **Describe** the relationship between beak shape and diet in birds.
- **List** 10 major orders of living birds, and name an example of each order.
- **Describe** the function of the syrinx.

VOCABULARY

syrinx
crop milk

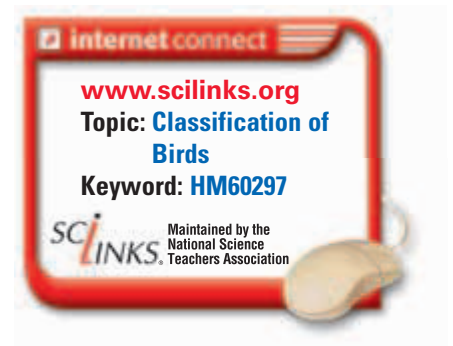


FIGURE 42-8

This cardinal (a), *Cardinalis cardinalis*, has a short, strong beak for cracking seeds and feet that enable it to perch on small tree branches. This kestrel (b), *Falco sparverius*, has a beak that enables it to tear flesh and talons that enable it to grip and kill prey. This calliope hummingbird (c), *Stellula calliope*, has a long, thin beak that enables it to extract nectar from flowers. This northern shoveler (d), *Anas clypeata*, has a flat beak that enables it to shovel mud while searching for food.



FIGURE 42-9

This mute swan, *Cygnus olor*, is able to take off from water and fly at very high speeds despite its great weight. Weighing up to 23 kg (50 lb), mute swans are the heaviest flying birds. Swans are monogamous, meaning they mate for life. While the female incubates the eggs, the male helps guard the nest.

FIGURE 42-10

The parrot pictured below is a lesser sulfur-crested cockatoo, *Cacatua sulphurea*. Parrots range in length from 8 cm (3 in.) to over 91 cm (3 ft). The earliest fossils of parrots indicate that they have existed as a group for at least 20 million years. Like most parrots, these cockatoos nest in holes in trees, and they usually lay only two eggs per year. Because of human activities, many species have become extinct or endangered.



Most taxonomists divide about 10,000 species of living birds into 23 orders. Taxonomists have traditionally used morphological evidence from beaks, feet, plumage, bone structure, and musculature to classify birds. Technological advances in the analysis of blood proteins, chromosomes, and DNA have also been used more recently. Despite the introduction of these new methods, the relationships among the 23 orders of birds are still not well resolved. Ten of the most familiar orders of living birds are described below.

Order Anseriformes

Swans, geese, and ducks—commonly called *waterfowl*—belong to this order of 160 species. Found worldwide, members of this order are usually aquatic and have webbed feet for paddling and swimming. Waterfowl feed on a variety of aquatic and terrestrial foods, ranging from small invertebrates and fish to grass. The bill is typically flattened. The young are precocial, and parental care is usually provided by the female. A mute swan is shown in Figure 42-9.

Order Strigiformes

This order includes the owls, the nocturnal counterparts to the raptors. Owls are predators that have a sharp, curved beak and sharp talons or claws. As shown in the chapter opener photo, owls also have large, forward-facing eyes that provide improved vision at night. Owls rely on their keen sense of hearing to help locate prey in the dark. There are about 180 species of owls, and they are found throughout the world.

Order Apodiformes

Hummingbirds and swifts belong to this order. All of the roughly 430 species are small, fast-flying, nimble birds with tiny feet. Swifts pursue insects and capture them in flight. Hummingbirds eat some insects but also feed on nectar, which they lap up with a very long tongue. The long, narrow bill of a hummingbird can reach deep into a flower to locate nectar. Swifts have a worldwide distribution, but hummingbirds live only in the Western Hemisphere.

Order Psittaciformes

This order includes the parrots and their relatives, the parakeets, macaws, cockatoos, and cockatiels. Most of the roughly 360 species in this order live in the tropics. Parrots are characterized by a strong, hooked beak that is often used for opening seeds or slicing fruits. Their upper mandible is hinged on the skull and movable. Unlike most birds, parrots have two toes that point forward and two toes that point toward the rear, an adaptation for perching and climbing. They are vocal birds, and many species gather in large, noisy flocks. Parrots have long been prized as pets because of their colorful plumage and intelligence and because some species can be taught to mimic human speech. However, habitat destruction and excessive collecting for the pet trade now threaten many parrot species with extinction. Figure 42-10 shows a cockatoo.

Order Piciformes

This diverse group of tree-dwelling birds contains woodpeckers, honeyguides, and toucans. All members of this order nest in tree cavities. Like parrots, they have two forward-pointing toes and two that point to the rear. There are about 350 species found throughout the world except in Australia. The diversity of foods consumed by these birds is reflected in the diversity of their bills. Woodpeckers, which drill holes into trees to capture insects, have strong, sharp, chisel-like bills. Toucans feed mainly on fruit, which they pluck with a long bill, as shown in Figure 42-11.

Order Passeriformes

This large order contains about 5,900 species—more than half the total number of bird species—and includes most of the familiar North American birds. Robins, warblers, blue jays, and wrens are just some of the birds belonging to this group.

Passerines are sometimes called *perching birds*. In most birds, three toes point forward and one points backward. Passerines, too, have this arrangement of toes, but the rear toe is enlarged and particularly flexible to provide a better grip on branches. Passerines feed on a variety of foods, including nectar, seeds, fruit, and insects.

Many passerines are called *songbirds* because the males produce long, elaborate, and melodious songs. Male birds sing to warn away other males and to attract females. The song is produced in the structure known as the **syrinx** (SIR-ingks), which is located at the base of the bird's trachea. By regulating the flow of air through the syrinx, birds can generate songs of great range and complexity.

Order Columbiformes

This globally distributed group contains about 320 species of pigeons and doves. Figure 42-12 shows a mourning dove. These birds usually are plump-breasted and have relatively small heads; short necks, legs, and beaks; and short, slender bills. Most feed on fruit or grain.

The crop, which in most other birds is used to store food, secretes a nutritious milklike fluid called **crop milk**. Both sexes produce crop milk to feed their young. Columbiform birds usually lay a clutch of two eggs, which hatch after a two-week incubation period. The young usually leave the nest two weeks after hatching. Another member of this order is the now-extinct dodo of Mauritius, an island in the Indian Ocean.

Order Ciconiiformes

The order Ciconiiformes is highly diverse, and has a worldwide distribution. This order includes about 1040 species of herons, storks, ibises, egrets, raptors, and penguins. Many have a long, flexible neck, long legs, and a long bill. Many are wading birds, and they feed on fish, frogs, and other small prey in shallow water. Many species of Ciconiiformes are diurnal (daytime) hunters with keen vision. Vultures, however, feed on dead animals and use their sense of smell to detect the odor of decomposing flesh.



FIGURE 42-11

Toucans, such as this keel-billed toucan, *Ramphastos sulfuratus*, mate once per year, usually laying two to four eggs. The male and female toucans take turns sitting on the eggs. The eggs usually hatch after about 15 days of incubation.

FIGURE 42-12

The adult mourning dove, *Zenaida macroura*, stands about 30 cm (12 in.) tall and nests in trees or bushes. Mourning doves breed throughout North America. They winter as far south as Panama.





FIGURE 42-13

The great blue heron, *Ardea herodias*, uses its spearlike beak to stab fish, frogs, and other prey. Young herons must be taught how to hunt. Scientists have learned that young herons often miss their intended prey and must also learn what is and is not food.

Word Roots and Origins

syrinx

from the Greek *syrinx*, meaning "reed" or "pipe"

Raptors have a sharp, curved beak and sharp talons and include ospreys, hawks, falcons, vultures, and eagles. About 310 species of raptors are distributed throughout the world. Some members of this order grow to be quite large. Figure 42-13 shows a great blue heron, a large species that is in North America. The marabou stork of Australia, for example, can be more than 1.5 m (59 in.) in height.

Penguins are a unique group of flightless marine birds. All 17 species live in the Southern Hemisphere. The penguin's wedge-shaped wings have been modified into flippers, and the feet are webbed. Underwater, penguins flap their flippers to propel themselves forward—they "fly" through the water. Most penguins have a thick coat of insulating feathers and a layer of fat beneath the skin, enabling them to live in polar conditions. They maintain this fat layer by consuming large quantities of fish and krill.

Order Galliformes

Members of this group, which includes turkeys, pheasants, chickens, grouse, and quails, are commonly called *fowl*. These terrestrial birds are usually plump-bodied and may have limited flying ability. Grains form a large part of the diet of many fowl, and all species have a large, strong gizzard. Some are also an important part of the human diet. The young are precocial. There are about 220 species distributed worldwide.

Order Struthioniformes

Some of the world's largest birds belong to this order. They include ostriches, rheas, emus, and cassowaries. Ostriches are native to Africa and can attain a height of nearly 3 m and weigh 150 kg. Ostriches cannot fly, but they are specialized as high-speed runners. Propelled by their long, strong legs, ostriches can reach speeds of 55 km per hour. Each large foot has only two toes. Reduction in the number of toes is common in running animals.

Rheas are a South American version of the ostrich. Emus are the second largest of the world's birds, originally found in Australia. Cassowaries, from New Guinea, are the most colorful of this order, with black bodies and blue heads.

SECTION 3 REVIEW

1. Explain how a bird's beak and feet can provide information about the bird's lifestyle.
2. Identify the order to which each of the following birds belongs: dove, robin, goose, and penguin.
3. Identify similarities and differences between raptors and owls.
4. Identify the function of the syrinx.
5. Describe the source and function of crop milk.

CRITICAL THINKING

6. **Recognizing Relationships** Penguins have a large, keel-shaped sternum, but ostriches do not. Provide an explanation for this difference.
7. **Analyzing Information** Why can birds inhabit more diverse environments than reptiles can?
8. **Applying Information** Why might crops and gizzards be less common in carnivorous birds than in seed-eating birds?

CHAPTER HIGHLIGHTS

SECTION 1

Origin and Evolution of Birds

- Seven major characteristics of birds are feathers; wings; a lightweight, rigid flight skeleton; a respiratory system involving air sacs; endothermy; a beak instead of teeth; and oviparity.
- Three similarities between birds and dinosaurs include an S-shaped neck, a unique ankle joint, and hollow bones.
- *Archaeopteryx* had feathers covering its body, tail, and forelimbs. It had hollow bones and a fused collar bone, called a furcula.
- There are two hypotheses for the origin of flight in birds. One hypothesis is that flight evolved in tree-dwellers. The second hypothesis is that flight evolved in ground-dwellers.

Vocabulary

furcula (p. 842)

SECTION 2

Characteristics of Birds

- Contour feathers are made of a central shaft composed of two vanes with branches, called barbs, that are connected by interlocking hooked barbules.
- Two modifications to the bird skeletal system include an enlarged sternum and hollow bones.
- The crop stores food. The two parts of the stomach are the proventriculus and the gizzard, which crushes food.
- The lungs of a bird are connected to several air sacs that store and move air but do not participate in gas exchange.
- Birds lack a urinary bladder and excrete their nitrogenous waste as uric acid mixed with feces.
- The cerebellum, cerebrum, and optic lobes of the bird brain are large.
- All birds lay hard-shelled eggs. Precocial young are active as soon as they hatch. Altricial young are born helpless and require parental care for several weeks.
- Many birds migrate using a variety of environmental cues to guide their migration.

Vocabulary

feather (p. 844)
follicle (p. 844)
shaft (p. 844)
vane (p. 844)
barb (p. 844)

barbule (p. 844)
preen gland (p. 844)
sternum (p. 845)
pygostyle (p. 845)
crop (p. 846)

proventriculus (p. 846)
gizzard (p. 846)
vas deferens (p. 848)
oviduct (p. 848)
brood patch (p. 848)

precocial (p. 849)
altricial (p. 849)
ornithologist (p. 849)

SECTION 3

Classification

- The feet and beak of a bird reflect its way of life.
- There are currently 23 commonly recognized orders of living birds, but technological advances add to our knowledge of relationships among animals, and taxonomy is subject to change with new information.
- Ducks, geese, and swans belong to the order Anseriformes.
- Owls are nocturnal hunters and belong to the order Strigiformes.
- Swifts and hummingbirds belong to the order Apodiformes.
- Parrots and relatives belong to the order Psittaciformes.
- Woodpeckers and toucans belong to the order Piciformes.
- Perching birds and songbirds belong to the order Passeriformes. Passeriformes have a structure called the syrinx, which they use to produce songs.
- Pigeons and doves belong to the order Columbiformes.
- Raptors, long-legged water birds, and penguins belong to the order Ciconiiformes.
- Chickens and turkeys belong to the order Galliformes.
- Large birds such as the ostrich belong to the order Struthioniformes.

Vocabulary

syrinx (p. 853)

crop milk (p. 853)

CHAPTER REVIEW

USING VOCABULARY

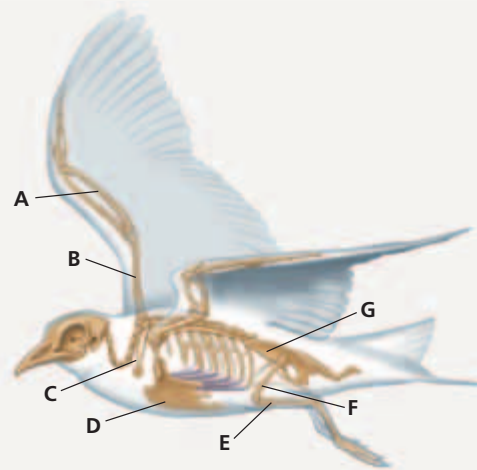
- For each pair of terms, explain how the meanings of the terms differ.
 - furcula* and *sternum*
 - flight feathers* and *contour feathers*
 - crop* and *gizzard*
 - altricial* and *precocial*
- Explain the relationship between vas deferentia, oviducts, and cloaca.
- Use the following key terms in the same sentence: *shaft*, *follicle*, *vane*, *barb*, *barbule*, *preening*, and *preen gland*.
- Word Roots and Origins** The word *precocial* is derived from the Latin prefix *pre-*, which means “before,” and the term *coquere*, which means “to mature.” Using this information, explain why the term *precocial* is a good word for the type of young birds it describes.

UNDERSTANDING KEY CONCEPTS

- List** the seven defining characteristics of birds.
- Identify** characteristics that *Archaeopteryx* shared with its dinosaur ancestors.
- Summarize** the evidence indicating that *Archaeopteryx* could fly.
- Compare** two possible explanations for the evolution of flight.
- Contrast** the function of down feathers with that of contour feathers.
- Explain** how a bird’s skeleton maximizes strength for flight while minimizing weight.
- Describe** the functions of two of the organs in a bird’s digestive system.
- Name** the structure that grinds food, aided by stones that a bird swallows.
- Describe** the role that air sacs play in increasing respiratory efficiency in birds.
- Summarize** the differences between newly hatched ducks and newly hatched pigeons.
- List** several cues that birds might use to guide their movements when migrating long distances.
- Explain** the relationship between beak and feet shapes and the lifestyles of raptors, seed-eating birds, and flightless birds.
- Identify** the order to which each of the following birds belongs: ducks, owls, hummingbirds, parrots.
- Name** the characteristic that is unique to song-producing members of the order Passeriformes.
- Explain** how some members of the order Columbiformes are similar to mammals.
- CONCEPT MAPPING** Use the following terms to create a concept map that describes the adaptations of birds for flight: *contour feathers*, *flight feathers*, *sternum*, *keel*, *pygostyle*, *hollow*, *urinary bladder*, and *air sacs*.

CRITICAL THINKING

- Analyzing Concepts** The right and left sides of a bird’s heart are completely separated. Thus, the oxygenated blood is never mixed with the deoxygenated blood. Why is this complete separation in the heart necessary?
- Interpreting Graphics** Look at the diagram below of a bird’s skeleton. Identify the following structures: pelvic girdle, furcula, sternum, femur, humerus, ulna, and tibiotarsus.



- Applying Information** Why is binocular vision important to some birds?
- Inferring Relationships** Cowbirds lay their eggs in the nests of other birds. The young cowbirds hatch slightly earlier than the other birds do, and the cowbird hatchlings are slightly larger. Why might these characteristics be advantageous for the young cowbirds?
- Making Comparisons** Although many species of temperate-zone birds migrate to the tropics to escape winter, some species remain behind. What benefits might these birds gain from not migrating?
- Justifying Conclusions** In terms of evolutionary adaptations, explain why the young of some birds, such as ducks and quail, are precocial.

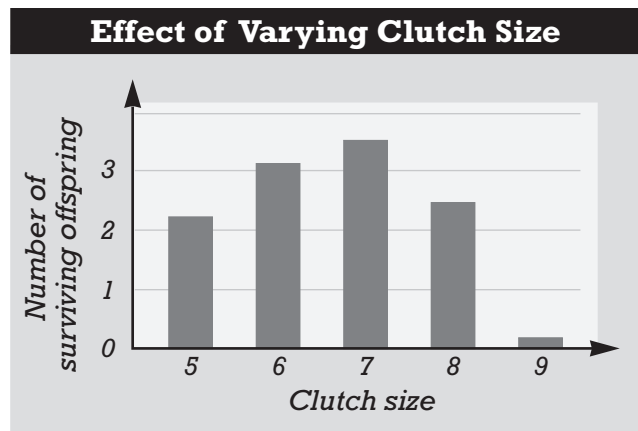


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following characteristics of *Archaeopteryx* is not shared by modern birds?
A. tail
B. teeth
C. furcula
D. feathers
- Which of the following characteristics do birds share with dinosaurs?
F. crop
G. lack of teeth
H. presence of feathers
J. structure of the ankle joint
- What is the function of the preen gland?
A. to produce digestive enzymes
B. to control salt balance in the body
C. to release scents that help attract mates
D. to produce an oily substance used to condition the feathers
- Which bone supports the tail feathers?
F. ulna
G. furcula
H. pygostyle
J. pelvic girdle

INTERPRETING GRAPHICS: The graph below shows the effect of varying clutch size on the number of surviving offspring in one bird species. Use the graph to answer the question that follows.

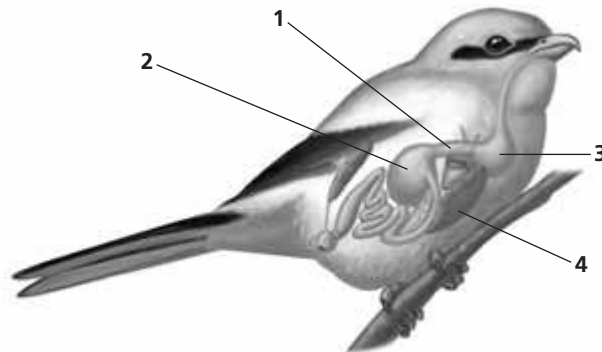


- Based on these data, which of the following statements is true for this species?
A. The optimal number of eggs in a clutch is seven.
B. The greater the clutch size is, the greater the number of surviving offspring.
C. Nests with five eggs produced the fewest number of surviving offspring.
D. More offspring died in nests containing eight eggs than in nests containing nine eggs.

DIRECTIONS: Complete the following analogy.

6. crop milk : crop :: song :
F. syrinx
G. trachea
H. proventriculus
J. anterior air sacs

INTERPRETING GRAPHICS: The diagram below shows the digestive system of a bird. Use the diagram to answer the question that follows.



7. Which digestive structure grinds food, aided by stones swallowed by the bird?
A. 1
B. 2
C. 3
D. 4

SHORT RESPONSE

Each type of feather on a bird serves a specific purpose.

What are the functions of contour feathers and down feathers in birds?

EXTENDED RESPONSE

Imagine that a museum display of bird skeletons became mixed up, and all of the labels were lost.

Part A How could you separate the skeletons of flightless birds from those of birds that fly?

Part B How could you tell which birds flew rapidly and which birds could soar?

Test TIP

When a question refers to a graph, study the data plotted on the graph to determine any trends before you try to answer the question.

Comparing Feather Structure and Function

OBJECTIVES

- Observe a flight feather, a contour feather, and a down feather.
- Compare the structure and function of different kinds of feathers.

PROCESS SKILLS

- observing
- relating structure to function
- comparing and contrasting

MATERIALS

- 1 quill feather (large flight feather from wing or tail)
- 1 contour feather
- 1 down feather
- unlined paper
- prepared slide of a contour feather
- compound light microscope
- prepared slide of a down feather

Background

1. List several distinguishing characteristics of birds.
2. How do birds differ from other vertebrates?
3. What are the functions of feathers?

Procedure

1. In your lab report, make a table like the one on the next page. Record your observations of each kind of feather in your data table.
2. Examine a quill feather. Hold the base of the central shaft with one hand, and gently bend the tip of the feather with your other hand. Be careful not to break the feather. Next, hold the shaft, and wave the feather in the air. Record your observations concerning the structure of the quill feather. Relate your observations to the feather's possible function. Describe the function of the feather under "Function of feather" in your data table.



Quill feather, a large flight feather

3. Examine the vane of the feather. Does the vane appear to be a solid structure? Include a description of the quill feather's vane structure under "Structure of feather" in your data table.
4. Make a drawing of the quill feather. Label the shaft, vanes, and barbs. Compare your feather with the figure above.
5. Examine a contour feather. Make a sketch of the contour feather in your data table. Label the shaft, vanes, and barbs on your sketch. Does the feather resemble the one in the figure on the next page?
6. Describe the structure of the contour feather under "Structure of feather" in your data table.
7. Examine a prepared slide of a contour feather under low power. Note the smaller barbs, called *barbules*, extending from each of the barbs.
8. How might you observe the region between the barbs? Locate the tiny hooks at the end of each barbule. Note the arrangement of the hooks on adjacent barbs. Why do you think the hooks are so small? Make a separate, labeled drawing of the hooks in your lab report.
9. Examine the down feather, and sketch it in your data table. How does your down feather compare with the figure on the next page?
10. Describe the structure of the down feather in your data table. Do you notice a difference in the structure of the contour and down feathers?

COMPARISON OF FEATHERS

Type of feather	Sketch of feather	Structure of feather	Function of feather
Quill feather			
Contour feather			
Down feather			



Contour feather



Down feather

- Examine the prepared slide of the down feather under low power. Locate the barbs and barbules. Switch your microscope to high power, and make a separate, labeled drawing of the down feather in your lab report. Does it resemble the one in the figure above?

- 
 Clean up your materials, and wash your hands before leaving the lab.

Analysis and Conclusions

- What is the function of the shaft? What is the function of the vanes and barbs?
- How do hooks increase the strength and air resistance of a feather?
- How is the structure of the quill feather related to its function of aiding flight?
- Based on your observations, why might down feathers be more effective at keeping a bird warm than the other two feather types you observed?
- Based on your observations, how do you expect to see these feathers arranged on a bird? Explain how position of the feather affects the function of the feather.
- What evolutionary pressure(s) would have caused the evolution of these different types of feathers?

Further Inquiry

Each of the feather types you have examined has a specific structure and function. Review your observations, and try to think of features that account for the efficiency of the three types of feathers.



These young opossums depend on their mother's body to supply milk, a nourishing fluid from her mammary glands. The production of milk is one of the key characteristics of mammals.

SECTION 1 *Origin and Evolution of Mammals*

SECTION 2 *Characteristics of Mammals*

SECTION 3 *Diversity of Mammals*

SECTION 4 *Primates and Human Origins*

ORIGIN AND EVOLUTION OF MAMMALS

Mammals (class Mammalia) are a highly diverse group. The opossums on the previous page are mammals. What makes them recognizable as mammals? What traits do they share with more than 4,000 other mammalian species?

MAJOR CHARACTERISTICS

All mammals, such as the lions shown in Figure 43-1, have the following six major characteristics:

- **Endothermy**—Mammals, like birds, regulate body heat internally through metabolism and externally through insulation. A mammal's body temperature stays high and nearly constant because of adjustments in metabolic rate and regulation of heat loss through the body surface. This manner of controlling body temperature is called **endothermy**.
- **Hair**—All mammals have some hair. Most mammals are covered with a thick coat of hair, which insulates the body against heat loss. Hair is made of filaments of the protein keratin.
- **Completely divided heart**—Mammals have a four-chambered heart whose two ventricles are completely separated by a muscular wall. This division keeps deoxygenated blood from mixing with oxygenated blood and allows efficient pumping of blood through the circulatory system.
- **Milk**—Female mammals produce milk to feed their offspring. Milk is a nutritious fluid that contains fats, protein, and sugars. Milk is produced by **mammary glands**, which are modified sweat glands located on the thorax or abdomen.
- **Single jawbone**—A mammal's lower jaw is made up of a single bone. By comparison, a reptile's lower jaw is made up of several bones. Mammalian fossils are often identified by the jawbone.
- **Specialized teeth**—Mammals have various types of teeth modified for different functions. Teeth at the front of the jaw bite, cut, or hold prey. Teeth along the sides of the jaw crush, grind, or slice. By comparison, most reptiles' teeth are uniformly sharp and cone-like throughout the mouth.

OBJECTIVES

- **Describe** the major characteristics of mammals.
- **Compare** the characteristics of early synapsids, early therapsids, and modern mammals.
- **Relate** the adaptive radiation of mammals to the history of dinosaurs.
- **Differentiate** between monotremes, marsupials, and placental mammals.

VOCABULARY

endothermy
mammary gland
synapsid
therapsid
monotreme
oviparous
marsupial
viviparous
placental mammal
placenta

FIGURE 43-1

Lions have the six key characteristics of mammals.



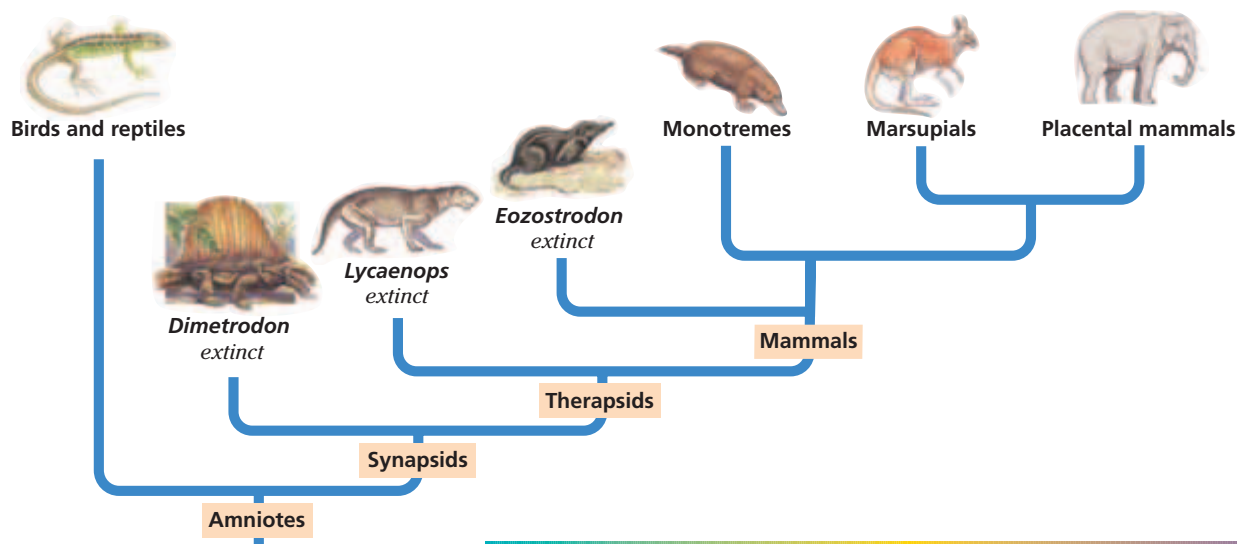


FIGURE 43-2

This phylogenetic diagram reflects hypotheses about the evolutionary relationships among mammals and other vertebrates. Mammals are thought to have descended from four-legged, land-dwelling vertebrates known as *therapsids*, which are sometimes referred to as *mammal-like reptiles*.

ANCESTORS OF MAMMALS

Scientists think that the ancestors of all mammals appeared more than 300 million years ago. Around that time, amniotes seem to have split into two groups. One group gave rise to dinosaurs, birds, and modern reptiles. The other group, known as **synapsids**, gave rise to mammals and their extinct relatives, as shown in Figure 43-2. Paleontologists recognize early synapsids by the structure of their skull, which has a single opening in a bone just behind the eye socket. This same type of skull is found in all later synapsids, including mammals, although often in a modified form.

The first synapsids were small and looked like modern lizards. By the early Permian period (272 million to 298 million years ago), various large synapsids had appeared. Some reached 4 m (13 ft) in length and weighed more than 200 kg (440 lb). Figure 43-3 shows an illustration of a carnivorous synapsid of the genus *Dimetrodon* (die-MET-ruh-DON). Unlike most other reptiles, which have uniformly shaped teeth, these early synapsids had specialized teeth—long bladelike front teeth and smaller back teeth.

FIGURE 43-3

This synapsid had a sail-like structure on its back. Scientists think that this structure was filled with blood vessels that may have helped regulate the animal's body temperature. This illustration was made based on information from the fossil record.

Therapsids

A subset of synapsids, called **therapsids**, appeared later in the Permian period and gave rise to mammals. Therapsids were the most abundant terrestrial vertebrates during the late Permian period. They survived through the Triassic period (251 million to 203 million years ago) and into the Jurassic period (203 million to 144 million years ago).

A rich fossil record of transitional forms between therapsids and mammals exists. By studying these fossils, scientists can trace the anatomical changes that occurred during this transition and infer additional physiological, ecological, and behavioral changes. Several features we associate with mammals evolved first among early therapsids. For example, like the limbs of many early therapsids, mammals' limbs are directly beneath the body. Evidence suggests that some early therapsids were endothermic and may have had hair.



Dimetrodon sp.

Early Mammals

Both the first mammals and the first dinosaurs appeared during the Triassic period. So, dinosaurs coexisted with mammals for more than 150 million years. Figure 43-4 shows a hypothesized image of an early mammal. Early mammals were about the size of mice. Fossil skulls with large eye sockets suggest that these mammals were active at night. Also, their teeth were adapted for feeding on insects. Hiding by day and specializing on insects allowed the mammals to avoid threats from dinosaurs or competition with them. Similarities in the mammary tissues of several kinds of mammals suggest that milk production had evolved by the end of the Triassic.



FIGURE 43-4

The earliest mammals, such as this species of the genus *Eozostrodon*, had to compete with the larger, dominant dinosaurs. These early mammals were active at night, when dinosaurs were probably less active.

DIVERSIFICATION OF MAMMALS

Dinosaurs dominated most terrestrial habitats while populations of small mammals continued to evolve. By the middle of the Cretaceous period, about 100 million years ago, three different kinds of mammals had appeared. Modern mammals belong to one of these three groups. The first group is made up of **monotremes**. They are **oviparous**, meaning that they lay eggs. The second group is made up of **marsupials**. They are **viviparous**, which means that they give birth to live young. In marsupials, the young develop within a pouch on the mother's body for some time after birth. The third group is made up of **placental mammals**. They are also viviparous, but in this group, the fetus typically develops within the mother's reproductive system for a longer time than it does in marsupials. Also, in this group, the developing fetus receives nourishment through a blood-rich structure called the **placenta**.

Many scientists think that some sort of natural disaster changed Earth's climate and forced the dinosaurs into extinction about 65 million years ago, at the end of the Cretaceous period. This change opened up many new habitats and resources to mammals. So, mammals took over many of the ecological roles that dinosaurs previously had. Today, nearly all large terrestrial animals are mammals.



SECTION 1 REVIEW

1. What is the function of hair in mammals?
2. Identify two differences between the skull of a mammal and the skull of a reptile.
3. Compare the characteristics of early synapsids, early therapsids, and modern mammals.
4. Differentiate between monotremes, marsupials, and placental mammals.

CRITICAL THINKING

5. **Analyzing Theories** Describe a possible species that would be a transitional form between therapsids and mammals.
6. **Examining Evidence** What factors limit biologists' knowledge of early mammals?
7. **Making Comparisons** Which trait(s) do monotremes share with reptiles?

SECTION 2

OBJECTIVES

- **Explain** the advantage of endothermy in mammals.
- **Identify** features of the mammalian respiratory and circulatory systems that help sustain a rapid metabolism.
- **Describe** mammalian adaptations for obtaining food.
- **Compare** the nervous system of mammals to that of other groups of animals.
- **Differentiate** among the patterns of development in monotremes, marsupials, and placental mammals.

VOCABULARY

diaphragm
incisor
canine
premolar
molar
baleen
rumen
cecum
echolocation

CHARACTERISTICS OF MAMMALS

Mammals live in many different kinds of habitats and climates all over the world. Several adaptations enable mammals to live in such diverse environments.

ENDOTHERMY

All animals produce heat internally when they metabolize, or produce energy from food. Mammals are *endotherms*, meaning they conserve and regulate this body heat. Most other animals, such as insects and lizards, are *ectotherms*, animals that are heated or cooled by their surroundings. Endothermy allows mammals to live in cold climates yet remain active. Moreover, the type of metabolism needed for endothermy also provides energy to perform strenuous activities for extended periods, such as migrating long distances.

Mammalian organ systems are uniquely adapted for endothermy. Figure 43-5 shows the internal anatomy of a mammal. Because of its faster metabolism, a mammal uses more oxygen and food than does a reptile of the same size. So, mammals have unique circulatory and digestive systems. Also, the body temperature of a mammal is often above that of its environment, so heat constantly escapes through the animal's skin and breath. The energy used to heat a mammal's body would be wasted without adaptations such as body insulation. Often, mammals that live in very cold climates have heavy coats of fur and, or thick layers of fat, called *blubber*.

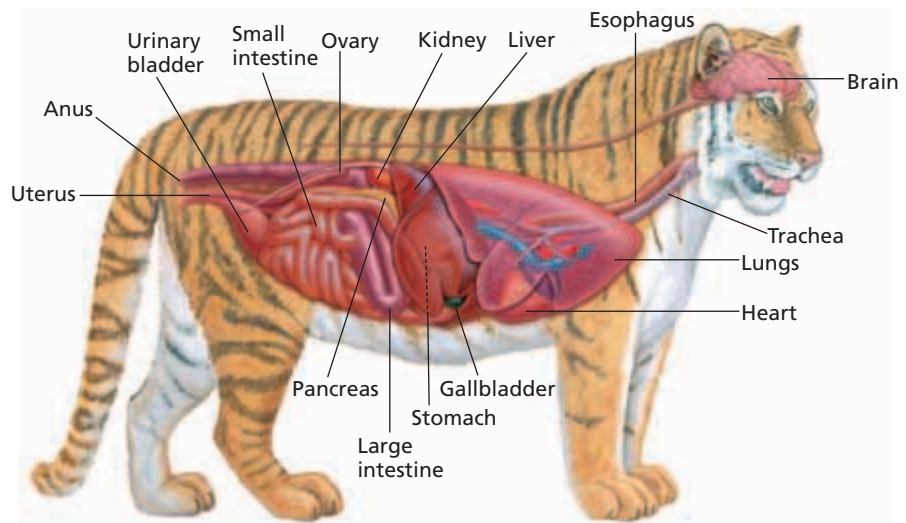


FIGURE 43-5

The organs and organ systems of modern mammals have a high demand for energy. The mammalian heart efficiently pumps oxygenated blood throughout the body, delivering nutrients that fuel the high-energy requirements of endothermy.

Circulatory System

The structure of the mammalian heart allows efficient pumping of blood throughout the body, as shown in Figure 43-6. The mammalian heart has two atria and two ventricles. A *septum*, or wall of tissue, completely separates the ventricles. The septum prevents oxygenated and deoxygenated blood from mixing. Recall that in the hearts of lizards and turtles, an incomplete septum allows the oxygen-rich blood and oxygen-poor blood to mix when the animal is inactive. The complete septum is an adaptation that allows mammals' bodies to transport oxygen more efficiently.

Respiratory System

A mammal's respiratory system is adapted for efficient gas exchange. The lungs are large and contain millions of *alveoli*, the small sacs in which gas exchange occurs. As a result, mammalian lungs have a much larger surface area available for gas exchange than reptilian lungs do.

Mammals breathe using two mechanisms: one they share with some reptiles and one that is unique. Lizards and snakes inhale by using their rib-cage muscles to expand the thoracic cavity, the body cavity that holds the lungs. Mammals inherited this breathing mechanism but use it mostly under conditions of strenuous activity. The second mammalian breathing mechanism uses a sheet of muscle below the rib cage called the **diaphragm**. Contraction of the diaphragm enlarges the thorax and thus expands the thoracic cavity. At rest, mammals breathe primarily with the diaphragm.

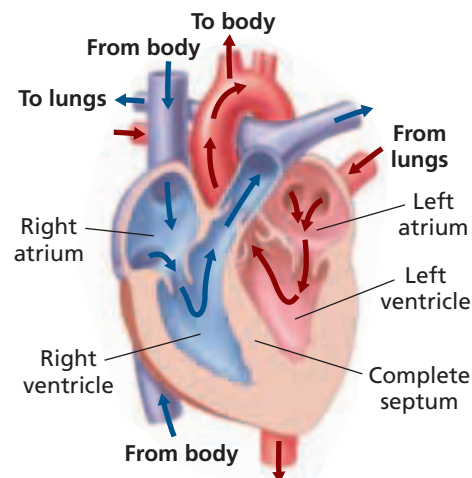


FIGURE 43-6

The red arrows show the path of oxygenated blood through the heart, and the blue arrows show the path of deoxygenated blood. Notice that the ventricles are completely separated by a septum. As a result, the blood pumped to the body contains a higher percentage of oxygen than a reptile's or fish's pumping heart can circulate.

FEEDING AND DIGESTION

For most mammals, the breakdown of food begins with chewing. Other vertebrates simply swallow their food whole or in large pieces. Chewing speeds up digestion by breaking food into small pieces that provide a large surface area for attack by enzymes. Variations in the size and shape of teeth among different mammalian species reflect differences in diet. Chisel-like **incisors** cut. Pointed **canines** grip, puncture, and tear. **Premolars** shear, shred, cut, or grind. **Molars** grind, crush, or cut. For example, predatory carnivores, such as the bear in Figure 43-7, have a set of teeth specialized for gripping, holding, tearing, and crushing food. Mammalian carnivores are recognizable by their sharp incisors and long canines.

Baleen whales, such as the blue whale, lack teeth. Instead, they have **baleen**, thin plates of keratin that hang from the skin of the upper jaw like a curtain. As a baleen whale swims, it gulps water, then closes its mouth, and pushes the water out through the baleen. Shrimp and other invertebrates get trapped behind the baleen and then are swallowed.

FIGURE 43-7

Carnivores, such as this bear, have large, sharp incisor and canine teeth that can cut and tear flesh. Bears and many other carnivores also have strong, crushing molars.





FIGURE 43-8

Many herbivores, such as this zebra, have flat teeth that can efficiently grind grasses, grains, or leaves. Herbivores also have digestive systems that harbor symbiotic microbes that help digest plant materials.

Special Adaptations for Digesting Plants

Meat is simple to digest, so most carnivores have short, simple digestive systems. Plants, however, can be difficult to digest because plants contain *cellulose*, a polymer of the sugar glucose. Animals do not produce enzymes that can break down cellulose. However, the long digestive tracts of herbivorous mammals, such as the zebra shown in Figure 43-8, contain microorganisms that can break down cellulose.

In some herbivorous mammals, the structure that is called a *stomach* is actually made up of four chambers. One of these chambers is the true stomach. Another chamber, known as the **rumen** (ROO-muhn), contains symbiotic microorganisms. Plant material that has been chewed and swallowed enters the rumen, where microorganisms begin to break the cellulose into smaller molecules that can be absorbed into the animal's bloodstream. The material is partly digested in the rumen, then regurgitated, chewed again, and swallowed again. The animal may regurgitate and swallow the same food several times. Mammals that have a rumen are called *ruminants* and include cows, sheep, goats, giraffes, and deer.

In horses, zebras, rodents, rabbits, and elephants, microorganisms that live in the cecum (SEE-kuhm) complete digestion of the food. The **cecum** is a large sac that branches from the small intestine and acts as a fermentation chamber. Food passes through the stomach and small intestine before entering the cecum. Mammals with a cecum do not chew cud.

NERVOUS SYSTEM

A mammal's brain is about 15 times heavier than the brain of a similarly sized fish, amphibian, or reptile. Of all animals, humans have one of the highest ratios of brain size to body size. Whales, dolphins, and some primates also have high ratios. These differences are due mostly to the size of the cerebrum. The *cerebrum* is the outer region of the brain and the largest part of the brain in mammals. The cerebrum's surface is usually folded and fissured, which greatly increases its surface area without increasing its volume. The cerebrum evaluates input from the sense organs, controls movement, initiates and regulates behavior, and functions in memory and learning.

As with other terrestrial vertebrates, a mammal's survival depends on five major senses: vision, hearing, smell, touch, and taste. The importance of each sense depends on the mammal's environment. For example, most bats, which are active at night, rely largely on sound rather than vision for navigating and finding food. Using a process called **echolocation** (EK-oh-loh-KAY-shuhn), these bats emit high-frequency sound waves, which bounce off objects, including potential prey. The bat then analyzes the returning echoes to determine the size, distance, direction, and speed of the objects.

Word Roots and Origins

cecum

from the Latin *intestinum caecum*, meaning "blind intestine"

DEVELOPMENT

In all living mammals, milk from the mother's mammary glands nourishes newborns. However, the pattern of development of the offspring differs from group to group.

Monotremes

A female monotreme typically lays one or two large eggs encased in thin, leathery shells and then incubates them. Her body heat keeps the eggs warm. The yolk nourishes the developing embryo within the egg. At hatching, a monotreme is very small and only partially developed. Its mother protects it and feeds it milk from her mammary glands until it is ready to survive on its own.

Marsupials

In marsupials, such as opossums and kangaroos, embryos develop for just a short period within the mother's uterus and then emerge from the uterus and crawl into the mother's pouch, a skin-lined pocket on her abdomen. The newborn offspring of a kangaroo at this stage is only 2 to 3 cm (1 in.) long. In the mother's pouch, the newborn attaches to a nipple to feed. The newborn's development and growth then continue inside the pouch for several months.

Placental Mammals

Placental mammals, such as the horse in Figure 43-9, give birth to well-developed young after a long period of development inside the uterus. During this period, the placenta provides nourishment and oxygen to the developing offspring. The placenta begins to form shortly after fertilization, when the fertilized egg attaches to the lining of the uterus. Extensions from the *chorion*, the outer membrane of the embryo, grow into the lining of the uterus. Blood vessels from the uterus surround these extensions. Nutrients and oxygen diffuse from the mother's blood into the blood of the offspring, and carbon dioxide and other wastes diffuse from the offspring into the mother's blood. After birth, infants feed on milk for several weeks or months.



FIGURE 43-9

Placental mammals, such as horses, carry their developing fetuses for a long time and give birth to infants that are relatively large and well developed, but still need parental care.

Word Roots and Origins

placenta

from the Greek *plakos*, meaning "flat object" or "flat cake"

SECTION 2 REVIEW

1. A mammal eats about 10 times as much food as a lizard of the same size. Explain this difference.
2. Compare a mammalian heart to a reptilian heart.
3. Describe the function of a rumen.
4. Compare a mammalian brain to a reptilian brain.
5. Compare the developmental patterns of monotremes, marsupials, and placental mammals.

CRITICAL THINKING

6. **Analyzing Information** What is the advantage of large ears to a bat that is active at night?
7. **Applying Concepts** What ecological relationship does a cow have with the bacteria in its rumen?
8. **Recognizing Relationships** Propose a functional reason why endothermy is associated with an enlarged cerebrum.

SECTION 3

OBJECTIVES

- **Identify** an example from each of the 12 major orders of mammals.
- **Distinguish** between monotremes, marsupials, and placental mammals.
- **Compare** the characteristics of artiodactyls and perissodactyls.
- **Compare** the adaptations for aquatic life in cetaceans, pinnipeds, and sirenians.

VOCABULARY

pinniped
ungulate

DIVERSITY OF MAMMALS

Mammals can be found in a diverse range of habitats, including the open ocean, underground, and mountaintops. Each mammalian species is uniquely adapted for its particular way of life.

MAMMALIAN ORDERS

Mammals are commonly classified into a single order of monotremes, 7 orders of marsupials, and about 18 orders of placental mammals. Possible evolutionary relationships between the mammalian orders are shown in Figure 43-10.

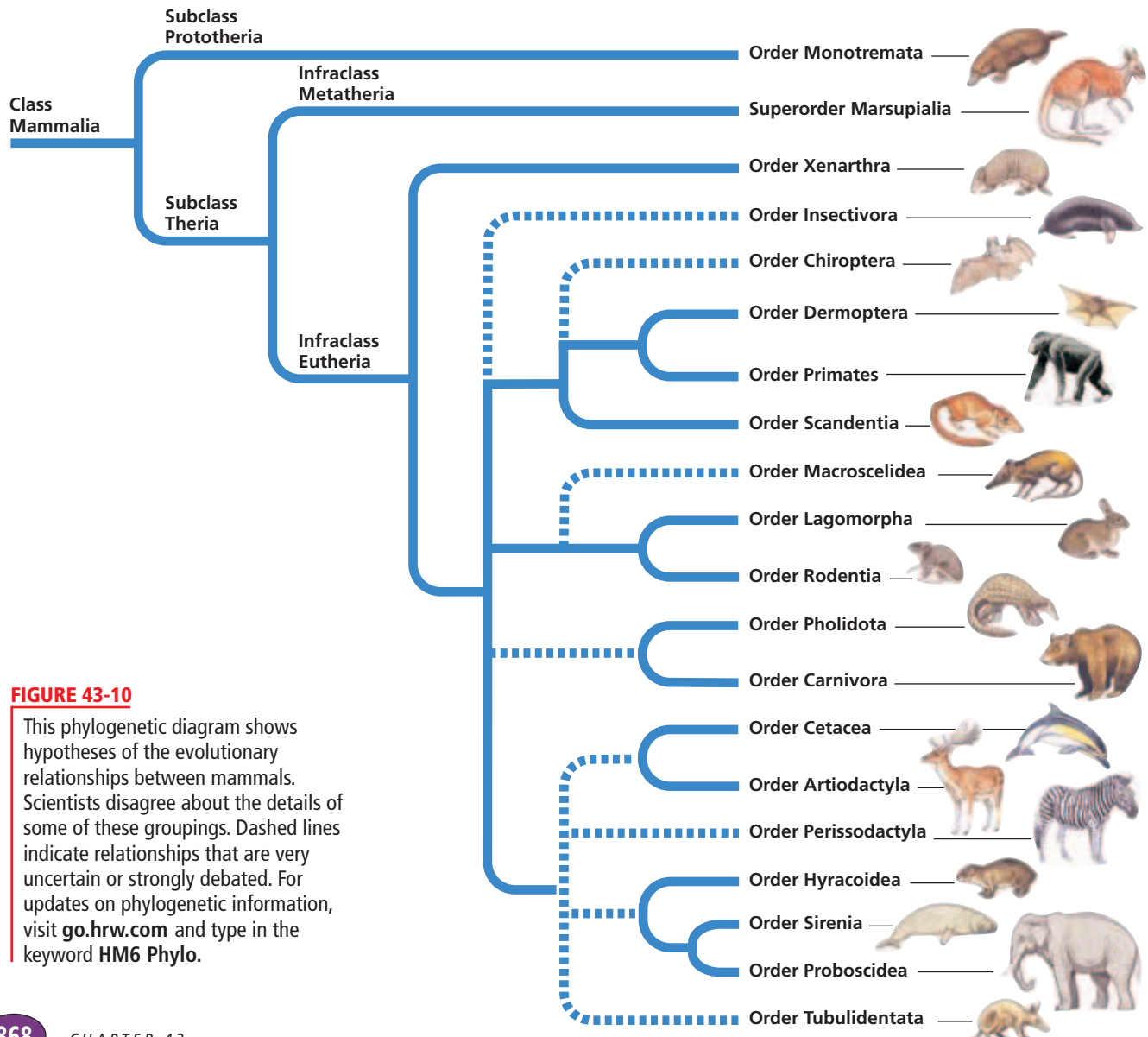


FIGURE 43-10

This phylogenetic diagram shows hypotheses of the evolutionary relationships between mammals. Scientists disagree about the details of some of these groupings. Dashed lines indicate relationships that are very uncertain or strongly debated. For updates on phylogenetic information, visit go.hrw.com and type in the keyword **HM6 Phylo**.



(a) Duckbill platypus, *Ornithorhynchus anatinus*



(b) Short-beaked echidna, *Tachyglossus aculeatus*

MONOTREMES

The order Monotremata (MAHN-oh-truh-MAHT-uh), the *monotremes*, is the only order in the subclass Prototheria. Because monotremes lay eggs, biologists consider this order to be very ancient, meaning that monotremes existed before other kinds of mammals did. Just three species exist today: the platypus and two echidna species.

The duckbill platypus, shown in Figure 43-11a, is adapted to life around rivers or streams in Australia. It has waterproof fur, webbed feet, and a flattened tail that aids in swimming. It uses its wide, flat, leathery bill to find worms, crayfish, and other invertebrates in soft mud. The female platypus digs a den in a riverbank to lay her eggs and curls around the eggs to protect and warm them. The babies lick milk from mammary glands on the mother's abdomen.

The two echidna species live in dry woodlands or deserts in Australia and New Guinea. As shown in Figure 43-11b, they have protective spines, a long snout, and a sticky tongue used to feed on ants and other insects (though they are unrelated to other mammal anteaters). Echidnas incubate their eggs in a pouch on the belly.

FIGURE 43-11

Two of the three species of the order Monotremata are shown here: the duckbill platypus (a) and an echidna that is also called a *spiny anteater* (b).

MARSUPIALS

The marsupials had previously been classified in one order, Marsupialia (mahr-SOO-pee-AY-lee-uh), but are now divided into at least seven orders within the super order Marsupialia. The majority of about 280 species of marsupials live in Australia, but some live in New Guinea and the Americas. The Virginia opossum is the only marsupial native to the United States.

Scientists think that marsupials began to evolve in isolation when Australia and New Guinea drifted away from the other continents more than 40 million years ago. At that time, placental mammals were rare in the Australian region, so marsupials evolved to take advantage of many ecological opportunities. Marsupials were once common in South America as well but were displaced by placental mammals that migrated in from the north.

PLACENTAL MAMMALS

Nearly 95 percent of all mammalian species are placental mammals, making up the infraclass Eutheria of the subclass Theria. They are classified into about 20 orders.

Order Xenarthra

The order Xenarthra (zuh-NAHR-thruh) includes about 30 living species of anteaters, armadillos, and sloths living in southern North America, Central America, and South America. Biologists think that, based on fossil and molecular evidence, Xenarthra evolved as a unique lineage in what is now South America. The name *xenarthra* means “strange joints” and refers to the unique structure of the lumbar vertebrae of members of this order.

This order was once named Edentata (EE-den-TAH-duh), meaning “toothless,” because many members of this order do not have prominent teeth. Anteaters completely lack teeth. Armadillos and sloths have peglike teeth that lack enamel. Most edentates feed on insects, which they capture with a long, sticky tongue. With their powerful front paws and large, sharp claws, they rip open anthills and termite nests. Armadillos supplement their insect diet with small reptiles, frogs, mollusks, and scavenged meat. Sloths, on the other hand, are herbivores; their continuously growing teeth are adapted to grinding plants.

Order Lagomorpha

The order Lagomorpha (LAG-uh-MAWR-fuh), members of which are called *lagomorphs*, includes about 70 species of rabbits, hares, and pikas. A pika (PIE-kuh) is shown in Figure 43-12a. Lagomorphs are native to many continents. They differ from rodents in that lagomorphs have a double row of upper incisors, with two large front teeth backed by two smaller ones. The teeth of lagomorphs continue to grow throughout their lifetime. Such teeth are an adaptation to a herbivorous diet.

Order Rodentia

Related to Lagomorpha is the order Rodentia (roh-DEN-chuh), members of which are called *rodents*. Rodentia is the largest mammalian order, which includes more than 1,800 species, or about 40 percent of all placental mammals. Rodents flourish on every continent except Antarctica and are adapted to a wide range of habitats. They tend to produce many young in each litter. Squirrels, marmots, chipmunks, gophers, muskrats, mice, and rats are rodents. The porcupine in Figure 43-12b is also a rodent.

A rodent’s teeth consist of a few molars or premolars and two pairs of incisors that continue to grow as long as the rodent lives. The sharp incisors are an adaptation to gnaw on seeds, twigs, roots, and bark. As a rodent gnaws, the back surface of the tooth wears away faster than the front surface, maintaining the tooth’s edge.

FIGURE 43-12

The North American pika (a) is a lagomorph, related to rabbits and hares. The North American porcupine (b), a rodent, ranges from Canada to northern Mexico.



(a) North American pika,
Ochotona princeps



(b) North American porcupine,
Erethizon dorsatum

Order Primates

The order Primates is made up of 235 living species, including lemurs, tarsiers, lorises, monkeys, gibbons, apes, and humans. Most primates are omnivores and have teeth suited for a varied diet. Primates have brains that have a relatively large cortex, which make possible the complex behaviors characteristic of this group.

A wide range of body sizes and adaptations allow primates to live in a variety of terrains. The smallest known primate, the pygmy mouse lemur, weighs only about 30 g and was discovered in 2000 in Madagascar, where it lives mostly in trees. In contrast, the largest primate, the mountain gorilla, can weigh 140–180 kg (300–400 lb) and lives on the ground in dense African mountain forests.

Most primates have forward-facing eyes, a feature that enables depth perception. Many primates are active at night and have large eyes adapted for night vision. All primates have grasping hands and, with the exception of humans, grasping feet. Some primates also have a grasping tail. Many primates live in trees, where grasping feet, hands, and tails are essential adaptations. In humans, grasping hands serve many purposes.

Order Chiroptera

The only mammals that truly fly, the bats, make up the order Chiroptera (kie-RAHP-tuh-ruh). There are more than 900 species of bats, and they live throughout the world, except in polar environments. A bat's wing is a modified front limb with a membrane of skin that stretches between extremely long finger bones to the hind limb, as shown in Figure 43-13a. A bat's wingspan can measure up to 1.5 m (4.5 ft). The bat's clawed thumb sticks out from the top edge of the wing. Bats use their thumbs for walking, climbing, and grasping.

Most bats have small eyes and large ears and navigate by echolocation. Most bats are active at night and feed on insects. However, some tropical bats are active in the day and feed on fruit or flower nectar. These bats locate food by using their large eyes and keen sense of smell. A few species of bats feed on meat or blood.

Order Insectivora

The order Insectivora (in-sek-TIV-uh-ruh) includes about 390 species of shrews, hedgehogs, and moles living in North America, Africa, and Europe. Figure 43-13b shows a shrew. Most members of this order are *insectivores*, which means “animals that eat insects.” However, not all insectivores are members of the order Insectivora. Furthermore, some Insectivora eat meat.

Scientists disagree about whether to include certain families of mammals in this order. For example, a family of mammals called *colugos*, which are commonly called *flying lemurs*, was once placed in Insectivora. This family is now usually classified as another order, Dermoptera. The Insectivora are usually small animals with a high metabolic rate. Most have long, pointed noses that enable them to probe in the soil for insects, worms, and other invertebrates. Their sharp teeth are adapted for grasping and piercing prey.

FIGURE 43-13

(a) Bats, in the order Chiroptera, are the only mammals that truly fly. The physics of the bat's wing in flight gives the bat more lift in relation to its body weight than most birds have. Thus, bats can remain airborne at slower speeds than birds can. (b) Shrews, in the order Insectivora, must eat more than twice their own body weight daily to fuel their high metabolic rate.



(a) Peter's epauletted fruit bat, *Epomophorus crypturus*



(b) Least shrew, *Cryptotis barua*

Word Roots and Origins

ungulate

from the Latin *ungula*,
meaning "hoof"

FIGURE 43-14

This caribou (a) is an artiodactyl. Artiodactyls are native to all continents except Australia and Antarctica. Although this tapir (b) looks like a pig (Order Artiodactyla), it is a perissodactyl. The similarity to a pig is an example of convergent evolution.



(a) Caribou, *Rangifer tarandus*



(b) Baird's tapir, *Tapirus bairdi*

Order Carnivora

The 274 living species of the order Carnivora (kahr-NIV-uh-ruh) are distributed worldwide. Most members of this order are called *carnivores*, which means “animals that eat meat.” Dogs, cats, raccoons, bears, hyenas, otters, seals, and sea lions are some well-known carnivores. Most are skilled hunters with strong senses of sight and smell. Other adaptations of carnivores include strong jaws, long canine teeth, and clawed toes to seize and hold prey. Many terrestrial carnivores have skeletal adaptations, such as long limbs, to run quickly.

Aquatic carnivores, known as **pinnipeds**, include the sea lions, seals, and walruses. They are efficient at swimming, with streamlined bodies and four limbs adapted as flippers. Although pinnipeds spend much of their time in the sea feeding, they return to land to sleep and to give birth. They are generally larger than land carnivores, and their large size helps them maintain body temperature. Most pinnipeds can dive to depths of 400 m (1,313 ft) and remain underwater for up to five minutes, but some can remain submerged for as long as one hour. Some scientists once placed pinnipeds in their own order—Pinnipedia.

Order Artiodactyla

Mammals with hoofs are **ungulates** (UHNG-yoo-lits). Two main groups of ungulates are characterized by their foot structure and by the presence of either a rumen or a cecum.

Ungulates with an even number of toes are *artiodactyls*, in the order Artiodactyla (AHRD-ee-oh-DAK-tuh-luh). This order includes about 210 species of deer, cattle, giraffes, pigs, and camels. Artiodactyls are native to every continent except Antarctica and Australia. Figure 43-14a shows a common artiodactyl, a caribou. Most artiodactyls can run quickly to escape predators.

Most artiodactyls are herbivores, although pigs are omnivores. Their molars are usually large and flat, for grinding plant material. Most artiodactyls are ruminants, or animals that have a rumen.

Order Perissodactyla

Ungulates with an odd number of toes are *perissodactyls*, in the order Perissodactyla (PUH-ris-oh-DAK-tuh-luh). This order includes about 17 living species, such as horses, zebras, rhinoceroses, and tapirs. Most species are native to Africa and Asia. However, some species of tapirs, such as the one in Figure 43-14b, live in Central and South America. Perissodactyls have a cecum.

Order Cetacea

Closely related to Artiodactyla is the order Cetacea (see-TAY-shuh), members of which are called *cetaceans*. Cetaceans include about 90 species of whales, dolphins, and porpoises worldwide. The orca in Figure 43-15a is an example. Cetaceans have fish-shaped bodies with forelimbs modified as flippers. They lack hind limbs and have broad, flat tails that help propel them through the water.



(a) Killer whale, *Orcinus orca*



(b) Manatee, *Trichechus manatus*

Cetaceans are totally aquatic but evolved from land-dwelling mammals. They breathe through modified nostrils called *blowholes*. Adult cetaceans lack hair except for a few bristles on the snout. A thick layer of blubber below the skin provides insulation. Cetaceans use echolocation to navigate, communicate, and find prey.

Two subgroups of cetaceans are the toothed whales and baleen whales. Toothed whales include sperm whales, narwhals, dolphins, porpoises, and orcas. Toothed whales can have up to 100 teeth. They prey on fish, squid, seals, and other whales. Baleen whales, such as blue whales, lack teeth and filter food from the water with the baleen attached to the roof of the mouth.

Order Sirenia

Four species of manatees and dugongs (DOO-gawngz) make up the order Sirenia (sie-REE-nee-uh), commonly called the *sirenians*. These large torpedo-shaped herbivores live in tropical seas, estuaries, and rivers. Their front limbs are flippers modified for swimming. Like whales (order Cetacea), sirenians lack hind limbs and have a flattened tail for propulsion. Although manatees and dugongs look like whales, they are more closely related to elephants. The similarities between whales and sirenians came about through convergent evolution. Figure 43-15b shows the only sirenian found in North America, the manatee.

FIGURE 43-15

This orca (a), a cetacean, looks very different from most artiodactyls, but scientists think cetaceans are closely related to artiodactyls. These manatees, or sea cows (b), belong to one of four species of Sirenia. Sirenia may share recent ancestry with Proboscidea.



Quick Lab

Comparing Gestation Periods

Materials paper, pencil

Procedure

1. Make a table of gestation periods for different mammals. Make three columns labeled "Mammal," "Gestation period," and "Offspring per pregnancy."
2. Fill in your table with the following data:

Bat, 210 days, 1 offspring
 Gerbil, 19–21 days, 4–7 offspring
 Horse, 332–342 days, 1 offspring
 Monkey, 226–232 days, 1 offspring
 Rabbit, 31 days, 3–6 offspring
 Squirrel, 44 days, 3 offspring
 Whale, 420–430 days, 1 offspring
 Wolf, 63 days, 4–5 offspring

Analysis Make a graph that relates gestation length and the number of offspring per pregnancy. Then, propose a hypothesis to explain this relationship.

TABLE 43-1 Minor Orders of Mammals

Order	Description	Examples
Macroscelidea	ground-dwelling insectivores with long, flexible snouts; 15 species found only in Africa	elephant shrews (not true shrews)
Pholidota	insectivores with protective scales composed of fused hair; resemble reptiles; found in Africa and southern Asia	pangolins or scaly anteaters (not true anteaters)
Tubulidentata	nearly hairless insectivores with piglike bodies and long snouts; found in southern Africa	aardvarks
Scandentia	squirrel-like omnivores that live on ground and in trees; feed on fruit and small animals; found in tropical Asia	tree shrews (not true shrews)
Dermoptera	only two species exist; glide in air using a thin membrane stretched between their limbs; found only in parts of Asia	colugos or flying lemurs (not true lemurs)
Hyracoidea	small rabbitlike herbivores; 7 species found mostly in Africa	hyrax

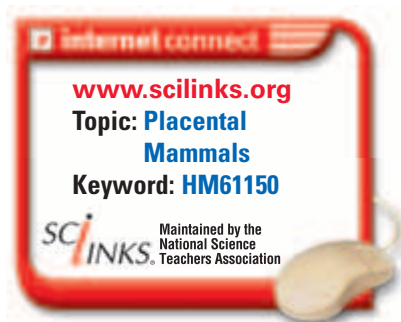
Order Proboscidea

Members of the order Proboscidea (PROH-buh-SID-ee-uh) have a nose that is modified into a long, boneless trunk, or *proboscis*. The only living species of this order are the Asian elephant and the African elephant, which is the largest living land mammal. Mammoths and mastodons are extinct members of this order. The African elephant can reach 6,000 kg (13,200 lb). To sustain such a large body, an elephant feeds on plants for up to 18 hours a day. The trunk allows an elephant to gather water or gather leaves from high branches. Modified incisors, called *tusks*, efficiently dig up roots and strip bark from branches. Large, jagged molars at the back of the jaw can grow up to 30 cm (1 ft) long and grind plant material.

Elephants have long gestation periods. A calf takes 20 to 22 months to develop. Female elephants can continue to give birth until the age of 70, and elephants can live to be 80 years old.

Other Orders of Placental Mammals

The 12 orders just described include most of the familiar placental mammals. The 6 remaining orders contain just 1 percent of the mammalian species and are summarized in Table 43-1.



SECTION 3 REVIEW

1. Give an example of each from the major orders of mammals.
2. Which continent is a natural home of both monotremes and marsupials?
3. What is unusual about the incisors of rodents?
4. Compare artiodactyls to perissodactyls.
5. Compare manatees to toothed whales.

CRITICAL THINKING

6. **Inferring Relationships** Give an adaptive reason why the pouch of the marsupial mole, a burrowing animal, opens toward the rear of its body.
7. **Applying Information** Why is *flying lemurs* a poor name for members of the order Dermoptera?
8. **Making Comparisons** Cetaceans live in cold ocean waters yet lack fur. Explain this.

PRIMATES AND HUMAN ORIGINS

To understand human origins, one must understand the relationships of humans to other primates.

PRIMATE CHARACTERISTICS

Some people speak of primates as the “highest” mammalian order. However, as Figure 43-16 shows, many primate characteristics are generalized rather than specialized and are similar to features possessed by ancestral mammals. Primate limbs, for example, are most similar to the limbs of the earliest mammals.

Many primate traits are adaptations for living in groups in trees. Examples include strong three-dimensional vision and **prehensile appendages**, or hands, feet, and tails that can grasp. The primate brain, with its large cerebrum, is able to interpret complex visual information and keep track of subtle shifts in social organization.

Large brain parts relative to size

Primate brains support complex skills, such as using hands, interpreting visual information, interacting socially, and caring for offspring.

Acute color vision

Forward-facing eyes allow binocular vision, depth perception, and skilled movement in three-dimensional space.

Generalist teeth

The variety of teeth permits herbivorous and omnivorous diets.

Communication

Facial and vocal structure enables broad range of expressions and sounds.

Infant care

Infants require prolonged care; reduced litter size permits greater mobility and attention to each young; there is usually one pair of mammary glands on the chest.



OBJECTIVES

- **Identify** traits that distinguish primates from other mammals.
- **Describe** fossil evidence relating humans to primate ancestors.
- **Compare** hypotheses concerning hominid evolution.

VOCABULARY

prehensile appendage
anthropoid primate
opposable thumb
great ape
bipedalism
hominid
australopithecine
human

FIGURE 43-16

Primate characteristics are mostly adaptations for a social life in the trees.

Manual dexterity

Opposable thumbs can touch other four fingers; fingers can grip or manipulate objects; flattened nails protect finger pads.

Social organization

Many primates live in social groups with complex behaviors among members.

Characteristic skeletal structure

Primates can sit upright, cling to trees, or hang from branches; major bones of limbs are like those of earliest mammals, with one upper and two lower bones.

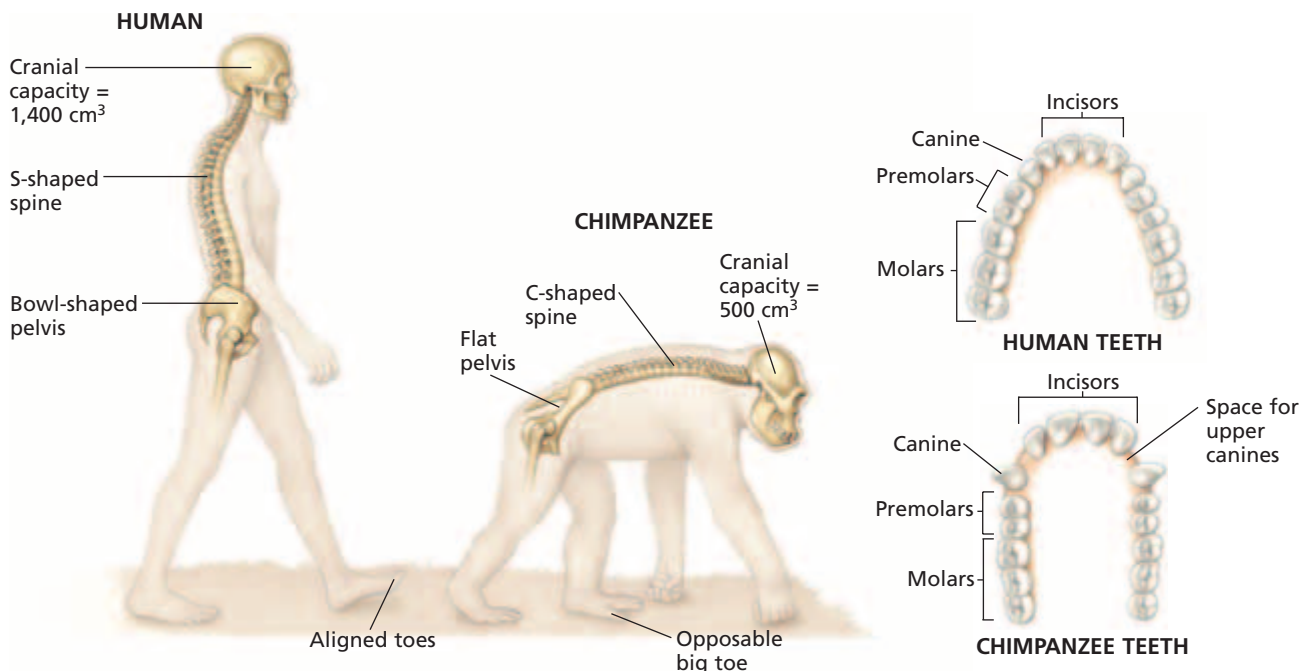


FIGURE 43-17

Anthropoids such as this white-handed gibbon, *Hylobates lar*, have rotating shoulder and elbow joints. This adaptation enables anthropoids to swing by their arms through trees.

FIGURE 43-18

Humans have certain physical traits that differ markedly from those of the chimpanzee, a modern ape: the jaw, pelvis, spine, feet, and toes.



Anthropoids

The primate lineages that evolved the earliest include lemurs, lorises, and tarsiers. These groups are sometimes referred to as *prosimians*. The gibbon in Figure 43-17 is one of the **anthropoid primates**, a group that also includes New World monkeys, Old World monkeys, apes, and humans. Anthropoid adaptations include rotating shoulder and elbow joints and an **opposable thumb**, which can touch the other fingers. Anthropoids can hold and manipulate objects precisely, as when a chimpanzee peels a banana or when a student holds a pencil. Nonhuman anthropoids also have grasping feet with an opposable big toe.

Humans, apes, and Old World monkeys have a similar *dental formula*, or number and arrangement of teeth. Each half of the upper and lower mouth includes two incisors, one canine, two premolars, and three molars, as shown in Figure 43-18. Compared to other primates, anthropoids have a more complex brain structure and a larger brain relative to body size.

Orangutans, gorillas, chimpanzees, bonobos, and humans make up the **great apes**. Chimpanzee and human DNA is so similar that humans are thought to be more closely related to chimpanzees than to any other living primates. DNA and fossil evidence suggests that humans and chimpanzees share a common ancestor that lived about 6 million years ago. Humans, however, did not descend from chimpanzees. Rather, modern apes and humans both descended from an ancestral apelike species.

Modern Humans

Among living mammals, only we humans, *Homo sapiens*, have the trait of **bipedalism** (bie-PED'l-iz-uhm), the tendency to walk upright on two legs. The human skeleton is adapted for bipedalism in several ways, as shown in Figure 43-18.

The bowl-shaped human pelvis supports internal organs during upright walking. The human spine curves in an S shape that allows for upright posture. Human toes are aligned with each other and are much shorter than ape toes. Because humans are the only primates that have this foot structure, the shape of the human foot is likely an adaptation for bipedalism.

The larger brain and smaller jaw in modern humans result in a flatter face than that found in apes. The modern human brain has an average size of about 1,400 cm³ and the chimpanzee, about 400 cm³. Among other unique structures, the human brain has extensive areas that function in the production and understanding of speech. Apes have similar areas of their brains that function in communication, and apes can learn to mimic certain forms of human sign language. Apes living in the wild, however, do not use the complexity of signals found in human language.

Hominids

Hominids include humans and extinct humanlike anthropoid species. Bipedalism is the distinguishing characteristic of this group. All other living anthropoid primates are *quadrupedal*, meaning they tend to walk on all four limbs. Apelike ancestors of the first hominids were probably also quadrupedal. How long ago did the first bipedal hominid evolve? And did human traits such as upright walking and a larger brain evolve together or at different times? Fossil evidence has provided some clues to the answers.

FOSSIL HOMINIDS

Paleontologists and anthropologists (scientists who study humans) have unearthed sufficient fossil evidence to conclude that a variety of humanlike species lived on Earth within the past 10 million years. However, scientists continue to investigate and debate hypotheses about the evolutionary relationships among all known hominids. Even the species name of some fossils is a matter of debate.

One important fossil discovery was made in 1974 in the Afar Valley region of Africa by Donald Johanson and colleagues. The 3.2 million-year-old fossilized skeleton, shown in Figure 43-19, was of an anthropoid primate with the brain size of a chimpanzee. But the skeletal structure clearly showed that this organism was bipedal.

Australopithecines

Johanson and colleagues gave the new fossil the species name *Australopithecus afarensis* and the nickname “Lucy.” Additional fossils of the same species have been discovered in other parts of Africa and date from about 2.5 million to 4 million years ago. A number of fossil organisms similar to Lucy have been classified as species of the genus *Australopithecus* within the subfamily of **australopithecines** (AW-struh-loh-PITH-uh-SEENZ), which may include other genera.

Word Roots and Origins

anthropoid

from the Greek *anthropos*, meaning “man,” and *eidos*, meaning “shape”



FIGURE 43-19

The original fossil find of *Australopithecus afarensis* consisted of a partial skeleton. The team of investigators who discovered it nicknamed the fossil “Lucy.”

MANY HOMINID SPECIES

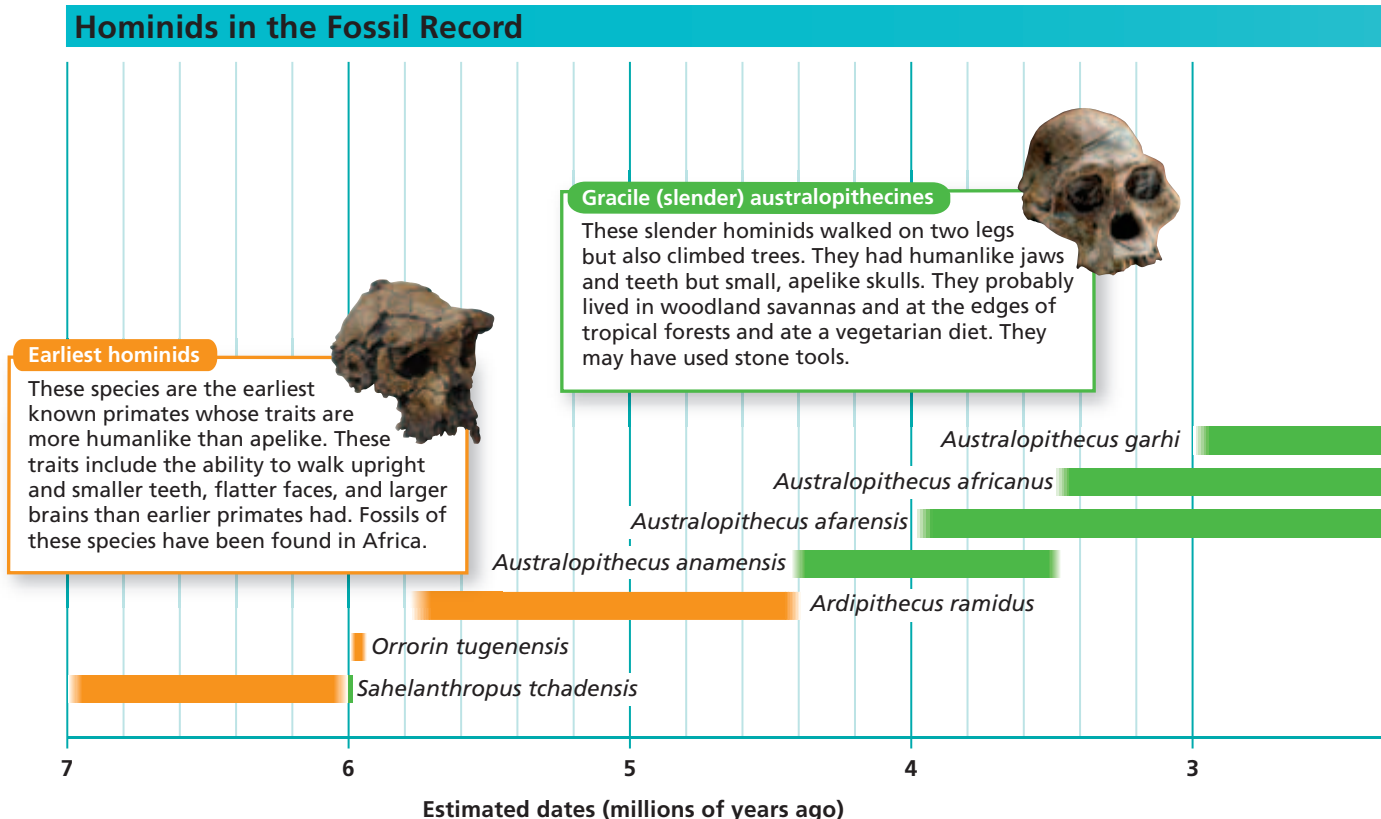
Paleontologists continue to find new hominid fossils. Although the exact classification of some fossils is strongly debated, it is clear that human evolution did not proceed as a single lineage of increasingly humanlike forms. Rather, several hominid forms arose, thrived, and became extinct over the past 7 million years, as shown in Figure 43-20. Furthermore, different species of hominids may have coexisted in time and possibly interacted.

In 1995, Mary Leakey and colleagues at the National Museums of Kenya announced the finding of a new hominid species, *Australopithecus anamensis*, that predated *A. afarensis* (Lucy's species) by about 300,000 years. Like Lucy's species, this species was also similar to a chimpanzee but probably bipedal. A possible descendant of Lucy's species was *A. africanus*, which lived about 2.3 to 3 million years ago. It was taller and heavier than Lucy's species and had a slightly larger brain capacity (430 to 550 cm³).

Three more-recent species, *A. aethiopicus*, *A. robustus*, and *A. boisei*, date from about 2.6 million to 1 million years ago. The physical characteristics of these later species suggest that they were a different lineage from *A. afarensis*. For example, they had heavier skulls, larger molars, and generally thicker bodies than Lucy's species did. Their brain capacity ranged from 450 to 600 cm³. Some scientists call these later hominids *robust australopithecines*, and some scientists place them in the genus *Paranthropus* instead of in the genus *Australopithecus*.

FIGURE 43-20

Many different species of hominids may have coexisted in time and possibly interacted during the past 7 million years. Scientists continue to find new hominid fossils and debate the exact classification of some fossils. This diagram represents one interpretation of the fossil record.



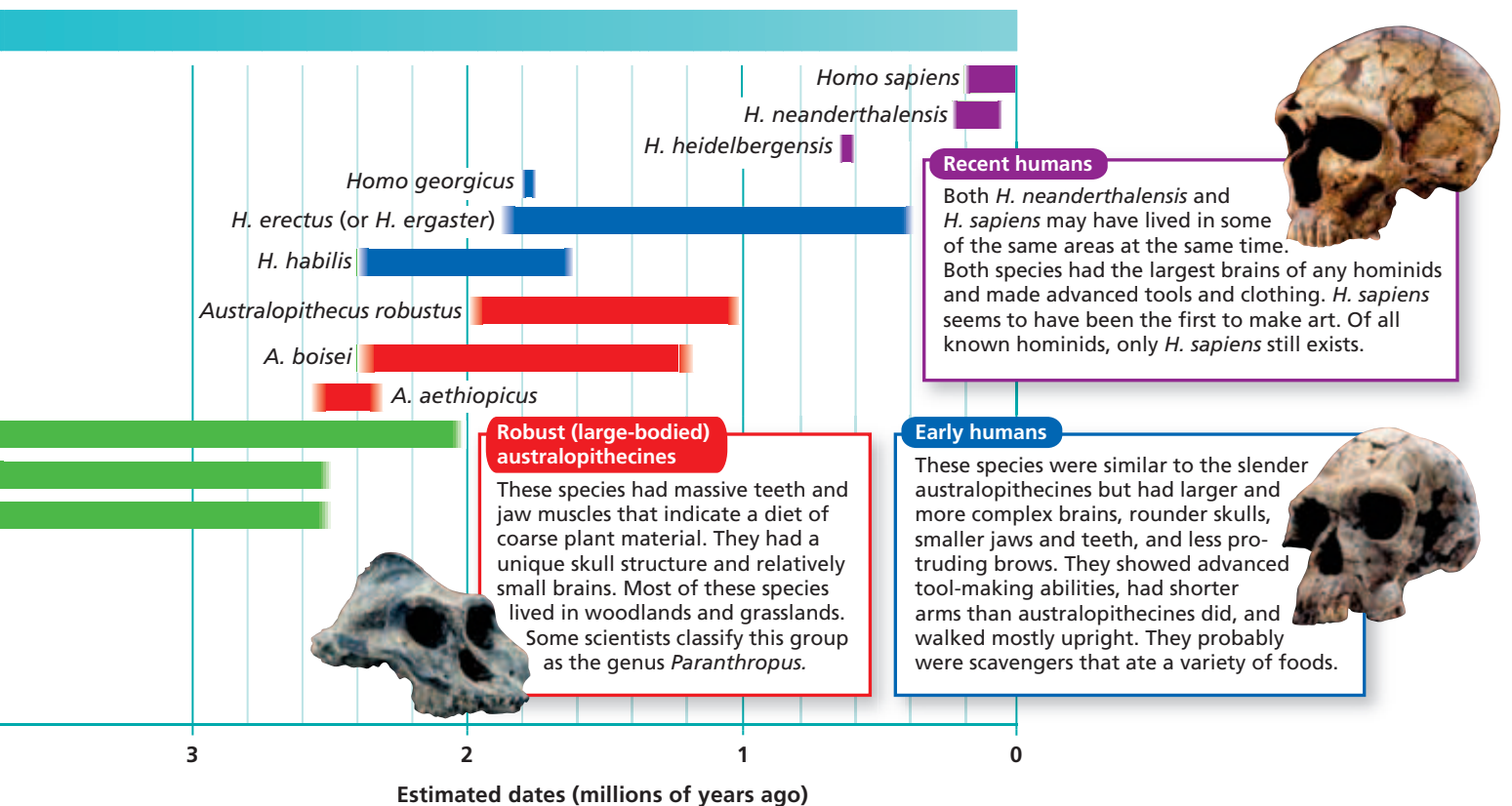
HUMANS

Sometime after the appearance of the australopithecines, new hominids appeared that are classified in the genus *Homo*. Extinct and living members of this genus are called **humans**. Many fossil humans have physical structures that are transitions between those of australopithecines and of modern humans.

Homo habilis and *Homo erectus*

In the early 1960s, scientists in East Africa found a hominid skull whose brain capacity was much larger than the brain capacity of Lucy's species but whose body was not much taller than Lucy's body. Importantly, the new fossils were found along with stone tools. Scientists named the new species *Homo habilis*, the "handy human." Fossils of *H. habilis* are between 1.6 million and 2.5 million years old and have a brain capacity of 590 to 690 cm³.

Later species, *Homo erectus* (meaning "upright human"), had a brain capacity of 800 to 1,250 cm³, or about two-thirds that of a modern human. *H. erectus* had a thicker skull, larger brow ridges, a lower forehead, and larger, protruding teeth than modern humans have. Some individuals were as tall as modern humans. Because *H. erectus* fossils have been found on several continents, scientists think that this hominid was the first to travel out of Africa. Charred animal bones indicate that *H. erectus* hunted and cooked its food.



Word Roots and Origins

sapiens

from the Latin *sapere*, meaning
"to taste" or "to know"

Homo sapiens and *Homo neanderthalensis*

Neanderthals, a distinctive type of human, lived in Europe and Asia from about 230,000 to 30,000 years ago. They had heavy bones, thick brows, protruding jaws, and brains of about the same size as the brains of modern humans. They lived in caves and made stone scraper tools. The reason for their extinction is an ongoing scientific question. This species had once been classified as a subspecies of *Homo sapiens* but is now mainly classified as *H. neanderthalensis*. Neanderthals may have interacted with *H. sapiens* in some places.

The first humans classified as *H. sapiens* appeared in Africa about 160,000 years ago. The first discovery of *H. sapiens* fossils was in Cro-Magnon cave in France, so some members of *H. sapiens* are referred to as *Cro-Magnons* (KROH-man-YAWNS). Other fossils are known from several continents. The earliest members of *H. sapiens* differed only slightly from modern humans. Their average brain and body size were about the same as modern human's.

As hominid fossils are discovered and studied, scientists revise the classification of some hominid species and debate hypotheses about the possible evolutionary relationship between these species. The ancestry of *H. sapiens* is one such topic of debate.

Modern Humans

How did modern humans come to occupy the entire globe? In one hypothesis, local populations of *H. erectus* gave rise to local populations of *H. sapiens* all over the world. According to this *multiregional hypothesis*, interbreeding among populations was sufficient to keep all of humanity as a single species. In contrast, the *recent-African-origin hypothesis* suggests that *H. sapiens* evolved from *H. erectus* uniquely in Africa about 100,000 to 200,000 years ago, then migrated out of Africa, and populated the globe. An analysis of mitochondrial DNA from people around the world suggests that humans did arise in Africa. It is possible that humans migrated out of Africa more than once. Also, interbreeding among populations around the world would have been possible during and after these migrations.

SECTION 4 REVIEW

1. Identify which characteristics humans share with primates and which are unique to humans.
2. What kind of evidence shows that chimpanzees are the closest living relatives of humans?
3. Identify several traits that differ among the variety of hominids known from fossils.
4. Describe how members of *Homo habilis* differ from australopithecines.
5. Contrast the multiregional hypothesis and the recent-African-origin hypothesis.

CRITICAL THINKING

6. **Analyzing Processes** Propose an explanation of the adaptive value of bipedalism in hominids.
7. **Analyzing Models** Using Figure 43-20, name the hominids that would have coexisted 2 million, 1 million, and 50,000 years ago.
8. **Calculating Information** From the information about hominid brain size and fossil ages in this section, construct a graph of how hominid brain sizes changed with time.

CHAPTER HIGHLIGHTS

SECTION 1

Origin and Evolution of Mammals

- Six key characteristics of mammals are endothermy, a fully divided heart, hair, milk production by females, a single jawbone, and a diversity of complex teeth.
- Mammals belong to an ancient group of animals called *synapsids*. Synapsids have a skull with one opening in a bone behind the eye socket.
- Mammals are probably descended from a subgroup of early synapsids called *therapsids*. Therapsids had legs that were positioned beneath their body and had complex teeth.
- Small mammals first appeared in the Triassic period. Mammals underwent adaptive radiation after the Cretaceous mass extinction of dinosaurs.

Vocabulary

endothermy (p. 861)
mammary gland (p. 861)
synapsid (p. 862)

therapsid (p. 862)
monotreme (p. 863)
oviparous (p. 863)

marsupial (p. 863)
viviparous (p. 863)
placental mammal (p. 863)

placenta (p. 863)

SECTION 2

Characteristics of Mammals

- Mammals are endothermic. Endothermy enables mammals to occupy a range of habitats and sustain strenuous activity. Endothermy requires large amounts of food and oxygen.
- Mammals have a heart with two atria and two ventricles and have large lungs with a large internal surface area.
- Mammals, unlike most vertebrates, chew their food. Mammal teeth include incisors, canines, premolars, and molars.
- Some mammals digest cellulose with the aid of microorganisms. These mammals have either a rumen or a cecum that contains the symbiotic microorganisms.
- Monotremes lay eggs. Marsupials give birth to young that continue development in the mother's pouch. Placental mammals typically develop young within the uterus for longer periods than marsupials do before giving birth.

Vocabulary

diaphragm (p. 865)
incisor (p. 865)
canine (p. 865)

premolar (p. 865)
molar (p. 865)
baleen (p. 865)

rumen (p. 866)
cecum (p. 866)
echolocation (p. 866)

SECTION 3

Diversity of Mammals

- There are at least 26 living orders of mammals. The most ancient orders are Monotremata and seven orders of marsupials. The remaining orders are placental mammals.
- The major placental mammal orders are Xenarthra, Lagomorpha, Rodentia, Primates, Chiroptera, Insectivora, Carnivora, Artiodactyla, Cetacea, Perissodactyla, Sirenia, and Proboscidea.

Vocabulary

pinniped (p. 872)

ungulate (p. 872)

SECTION 4

Primates and Human Origins

- Most primates share several characteristics, including generalized teeth; dental formula; three-dimensional vision; large brains; and grasping hands, feet, and tails.
- Of all living primates, only humans are bipedal. Bipedalism is the defining characteristic of hominids.
- Paleontologists have found many hominid fossils, including a variety of now-extinct australopithecines and humans.
- The multiregional and recent-African-origin hypotheses seek to explain how humans came to occupy the entire globe.

Vocabulary

prehensile appendage (p. 875)
anthropoid primate (p. 876)

opposable thumb (p. 876)
great ape (p. 876)

bipedalism (p. 876)
hominid (p. 877)

australopithecine (p. 877)
human (p. 879)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - viviparous* and *oviparous*
 - septum* and *diaphragm*
 - canine* and *molar*
 - rumen* and *cecum*
 - placenta* and *charion*
 - anthropoid primate* and *hominid*
- Use the following terms in the same sentence: *mammals*, *synapsids*, *therapsids*, and *endothermy*.
- Word Roots and Origins** The word *incisor* is derived from the Latin *incidere*, which means “to cut into.” Using this information, explain why the term *incisor* is a good name for the type of tooth that the term describes.

UNDERSTANDING KEY CONCEPTS

- Identify** six key characteristics of mammals.
- Relate** the characteristics of therapsids and synapsids to those of mammals.
- Relate** the evolution of mammals to the mass extinction of Cretaceous reptiles.
- Compare** the costs and benefits of endothermy for mammals.
- Explain** how the structure of the mammalian heart supports endothermy.
- Describe** the important adaptations of the teeth of mammals.
- Compare** the teeth of rodents to those of lagomorphs.
- Describe** two mammalian adaptations for digesting plants.
- Identify** the part of the brain that is relatively larger in mammals than in other animal groups.
- Describe** the distinguishing characteristic of placental mammals.
- Give** an example of an animal from each of the 12 orders of placental mammals.
- Compare** the characteristics of artiodactyls and perissodactyls.
- Compare** sirenians, pinnipeds, and cetaceans.
- Describe** the distinguishing characteristics of primates.
- Compare** fossil australopithecines with fossil humans.

-  **CONCEPT MAPPING** Use the following terms to create a concept map that compares the process of reproduction in different animal groups: *reptiles*, *monotremes*, *marsupials*, *placental mammals*, *oviparous*, *viviparous*, and *placenta*.

CRITICAL THINKING

- Relating Structure and Function** Mammalian species that live in very cold environments are usually larger than species of the same genus that live in warmer climates. Propose an explanation for this. (Hint: Consider the effect that increasing size has on volume and surface area.)
- Analyzing Processes** Some kinds of mice give birth about 21 days after mating. Why might this characteristic make mice ideal laboratory animals for experiments dealing with mammalian development and heredity?
- Forming Hypotheses** Sloths are arboreal xenarthrans that spend most of their lives hanging upside down from tree branches as they feed in tropical forests. Most sloths have green algae growing in tiny pits in their hair. What advantage might each species gain from this relationship?
- Applying Concepts** In recent years, surgeons have tried transplanting baboon and pig hearts into humans. Explain why surgeons tried these hearts rather than a large turtle's heart.
- Analyzing Results** In a famous study conducted 200 years ago, the Italian scientist Lazzaro Spallanzani showed that a blinded bat could still fly and capture insects. However, a bat whose ears had been plugged with wax could neither fly nor hunt. Explain these results.
- Interpreting Graphics** Study the two skulls below. Which is more like a mammal skull? Justify your answer.





Standardized Test Preparation

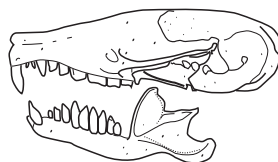
DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following structures is found in all modern mammals and birds?
 - hair
 - skull with teeth
 - lungs with air sacs
 - heart with four chambers
- What is the function of a mammalian diaphragm?
 - enables efficient breathing
 - provides nourishment for young
 - carries the young inside the uterus
 - keeps oxygenated blood separate in heart
- Which of these animals is a marsupial?
 - lion
 - echidna
 - opossum
 - duckbill platypus
- Which of these animals is a monotreme?
 - zebra
 - opossum
 - kangaroo
 - duckbill platypus
- Which of the following structures is found in cats but not in opossums?
 - hair
 - uterus
 - placenta
 - mammary gland

INTERPRETING GRAPHICS: The illustration below shows the skulls of two different mammals. Use the illustration to answer the question that follows.



A



B

- What can be inferred about these mammals?
 - Mammal A has more fat than mammal B.
 - Mammal A has more hair than mammal B.
 - Mammal A eats more meat than mammal B.
 - Mammal A eats more grass than mammal B.

DIRECTIONS: Complete the following analogy.

- hair : insulation :: milk :
 - armor
 - circulation
 - endothermy
 - nourishment

INTERPRETING GRAPHICS: The illustration below shows the skulls of two different mammals. Use the illustration to answer the question that follows.



A



B

- Which of the following accurately describes the differences between these skulls?
 - Skull A has more teeth than skull B does.
 - Skull A has more brain capacity than skull B does.
 - Skull A is the skull of a primate, and skull B is not the skull of a primate.
 - Skull A is the skull of an ape, and skull B is the skull of a human.

SHORT RESPONSE

Mammals and birds are endothermic vertebrates. Describe the functional costs and benefits of endothermy.

EXTENDED RESPONSE

Scientists classify amniotes from 300 million years ago into two major groups: the diapsids and the synapsids. A unique subset of synapsids from 245 million years ago is made up of therapsids. Modern mammals are grouped into the monotremes, the marsupials, and the placental mammals.

Part A Describe modern scientific hypotheses about the evolutionary relationships among these groups of amniotes.

Part B Describe the kinds of evidence that scientists examine to test these hypotheses.

Test TIP

For questions requiring an extended response, make an outline listing the key points of your response before you begin writing.

Examining Mammalian Characteristics

OBJECTIVES

- Observe examples of mammals.
- Examine the distinguishing characteristics of mammals.

PROCESS SKILLS

- observing
- inferring

MATERIALS

- hand lens or stereomicroscope
- microscope slide of mammalian skin
- compound light microscope
- mirror
- selection of vertebrate skulls (some mammalian, some nonmammalian)
- field guide to mammals

Background

1. List the distinguishing characteristics of mammals.
2. Define the term *endothermy*.
3. Mammalian skin is characterized by cutaneous glands, such as sebaceous glands and sweat glands, that develop as ingrowths from the epidermis into the dermis.



PART A Mammalian Hair and Skin

1. Use a hand lens to examine several areas of your skin that appear to be hairless. Record your observations in your lab report.
2. Compare the amount of hair on humans with the amount on other mammals that you have seen or read about. What role does hair or fur play in endothermy? What other roles does hair (or whiskers) play in mammals?
3. Examine a slide of mammalian skin under low power. Notice the glands in the skin.
4. Identify the sebaceous glands and the sweat glands in the skin. Sweat glands are found only in mammals, but some mammals do not have them or have few of them. What mechanism for cooling might these other animals have? Which other glands are unique to mammals?

PART B Mammalian Reproduction

5. Look at the photographs of mammals on this page. Which characteristics do these animals share?
6. Name the two orders of mammals represented in the photographs on this page.



PART C Mammalian Mouth and Teeth

7. Use a mirror to look in your mouth, and identify the four kinds of mammalian teeth. Count how many of each you have on one side of your lower jaw.
8. Look at the skulls of several mammals. Identify the four kinds of teeth in each skull, and count them as you counted your own. How are the four types of teeth different from yours?
9. Look at the skulls of several nonmammalian vertebrates. Describe the teeth in each one, and compare them with mammalian teeth.
10. Breathe through your nose with your mouth closed. Do you feel a flow of air into your mouth? You have a hard palate (the roof of your mouth) that separates your mouth from your nose.
11. Look again at the different skulls. In which vertebrates do you see a hard palate? What is an advantage of having a hard palate?
12. Compare the jaws of the mammalian skulls with those of the nonmammalian skulls. Notice how the upper jawbone and the lower jawbone connect in each skull. Is there a similarity in the mammalian jaws that distinguishes them from the nonmammalian jaws? Explain.
13. Create a data table, similar to the model below, to record your observations for your lab report. For example, the table below is designed to record observations of differences that you will find among the animal skulls. Remember to allow plenty of space to record your observations.

PART D Vertebrate Diversity

14. Use a field guide to find out more about the following mammalian orders: Cetacea, Xenarthra, Pholidota, and

Chiroptera. Answer the following questions about these mammals in your lab report:

- a. Cetaceans, such as whales and dolphins, are marine mammals. Cetaceans are hairless except for a few bristles. Why are cetaceans classified as mammals?
- b. Some mammals—including some members of Xenarthra (anteaters and armadillos) and Pholidota (pangolins)—lack teeth. Which characteristics do these animals share with other mammals?
- c. Like many birds, chiropterans (bats) have wings, fly, and are endotherms. Which characteristics distinguish these mammals from birds?

Analysis and Conclusions

1. List the characteristics that distinguish mammals from other vertebrates.
2. List several characteristics you observed that most mammals share.
3. Birds are also endotherms. Which structure in birds serves the same function as hair in mammals? Explain.
4. Compare the data you collected on the teeth from different animal skulls with the diet of each of those animals. How does the type of teeth that it has help with the particular diet that each animal has?

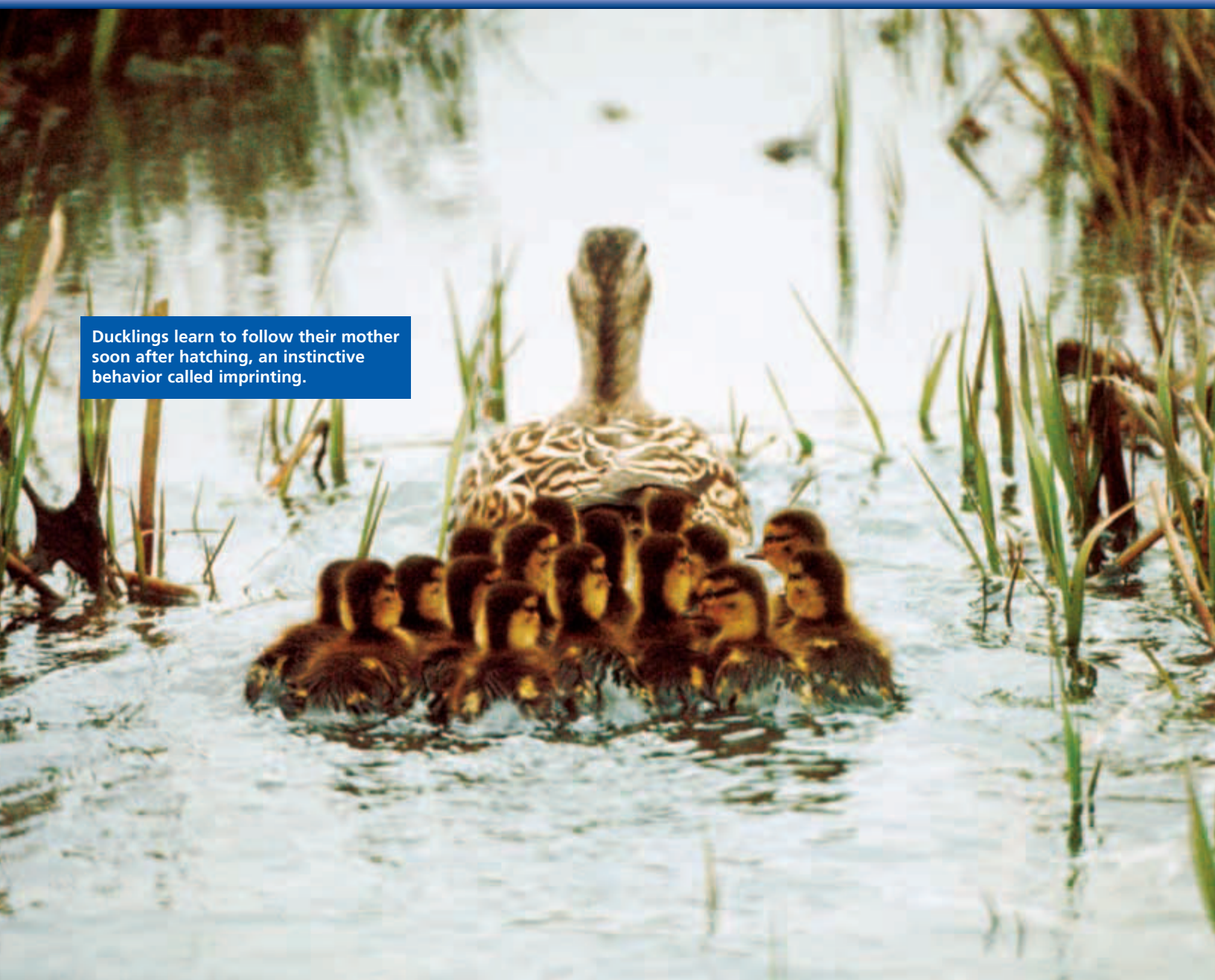
Further Inquiry

Find out how mammalian brains are different from the brains of other vertebrates. What adaptive advantage might these differences provide mammals?

OBSERVATIONS OF ANIMAL SKULLS

Animal	Mammal?	Number of incisors	Number of canines	Number of premolars	Number of molars	Hard palate?	Jaw

ANIMAL BEHAVIOR

A photograph of a mother duck with brown and white mottled feathers swimming in a pond. She is surrounded by a large group of her ducklings, which are small and fluffy with brown and yellow feathers. They are in a pond with tall green reeds and other aquatic plants in the background.

Ducklings learn to follow their mother soon after hatching, an instinctive behavior called imprinting.

SECTION 1 *Development of Behavior*

SECTION 2 *Types of Animal Behavior*



Unit 7—Ecosystem Dynamics
Topic 2



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DEVELOPMENT OF BEHAVIOR

A snake plays dead. A mouse presses a lever. A chimpanzee gathers termites on a stick. This section discusses how these and other behaviors are developed.

THE STUDY OF BEHAVIOR

Leaf-cutter ants cut sections of leaves with their sharp mouthparts and head back to the nest with their load of leaf bits. In the nest, other individuals tend and harvest the fungi that grow on the leaves as food for the colony. **Ethologists** (ee-THAHL-uh-JISTS), biologists who study behavior, seek to learn about such varied behaviors. **Behavior** is an action or series of actions performed by an organism, usually in response to a stimulus. In order to learn about behavior, ethologists ask questions about how and why a behavior occurs. For example, how do ants find their way to food and back to the colony? Why do ants share leaves with other colony members?

When considering an animal's behavior, ethologists ask four main questions: (1) What causes the behavior? What are the mechanisms that respond to stimuli? (2) What is the role of genes in the behavior; and how does it develop during an individual's lifetime? (3) What is the behavior's evolutionary history? (4) How does the behavior affect the organism's survival and reproduction?

Genes and Behavior

When studying behavior, ethologists often ask how much of an animal's behavior is determined by genetics and how much of the behavior is based on the unique developmental environment of the animal? Ethologists have studied bees to learn more about the role of genes in behavior. Some adult bees can detect which young in the hive have bacterial infections. These adult "hygienic" bees pull the diseased young from their cells and throw them out of the nest, as in Figure 44-1. In contrast, "nonhygienic" adult bees ignore diseased young.

American and Australian biologists mated hygienic queen bees to nonhygienic males and studied the inheritance pattern of this trait. They found that certain genes with set inheritance patterns control young-removal behaviors. This study showed that genes can underlie animal behavior. The triggers for the behavior, however, come from the environment. In this case, diseased young present in the hive trigger the removal behavior.

OBJECTIVES

- **Identify** four questions asked by biologists who study behavior.
- **Describe** an example of an innate behavior.
- **Compare** four types of learned behavior.
- **Explain** how learning and genes can interact to affect behavior.

VOCABULARY

ethologist
behavior
innate behavior
fixed action pattern
learning
habituation
operant conditioning
classical conditioning
reasoning
imprinting
sensitive period

FIGURE 44-1

"Hygienic" bees carry diseased or dead young from the hive. Studies have shown that this behavior is highly heritable.





FIGURE 44-2

A male lion who has taken over a pride will often kill cubs fathered by rival males.

Natural Selection and Behavior

Ethologists have hypothesized that animals usually behave in ways that promote their survival and offspring production. Because genes control some behaviors, natural selection can affect genetic variation that involves behavioral genes. Lions present a good example of this principle. The males living in a particular pride (family group) father all of the cubs. Eventually new males come along and force out the previous males. After the new males take over, they often kill cubs fathered by the previous males, as shown in Figure 44-2.

Observing this, ethologists wondered how natural selection could favor killing young of the same species? Further observation revealed that females will not breed with males as long as those females are caring for young. If the infants die, however, the female will mate again. By killing the existing offspring, the new male is likely to produce more offspring than males who do not kill cubs. Ethologists hypothesize that natural selection has favored genes that cause male lions to kill cubs that are not their offspring. It is important to remember that this behavior is instinctive. It is neither conscious nor deliberately aimed at other males or their offspring.

INNATE BEHAVIOR

Innate behaviors, more commonly called instincts, are inherited actions that are performed effectively the first time without being taught. An orb spider, for example, builds her web the same way every time. There is very little variation in what she does, and all her female offspring will build their webs in a similar manner without being taught.

Fixed Action Pattern

A **fixed action pattern** is a rigid innate behavior that all members of a species perform the same way each time they perform it. Figure 44-3 shows an Eastern hognose snake displaying a fixed action pattern in response to a predator. The snake spreads its jaws, hisses, and rolls on its back when threatened. Individuals that perform this behavior are less likely to get eaten and more likely to reproduce than individuals that do not perform this behavior.

Fixed action patterns continue from start to finish without modification once an environmental stimulus triggers them. However, there are still factors that influence whether an animal will perform this behavior or not. For example, Greylag geese retrieve eggs that have rolled out of the nest the same way every time. They will also retrieve other objects that are similar in shape or size to an egg. However, only mother geese retrieve eggs that have rolled out of the nest, and they only perform this behavior between the time of egg laying and hatching.

FIGURE 44-3

This Eastern hognose snake (*Heterodon platyrhinos*) is “playing dead.” The snake spreads its jaw, hisses, and rolls on its back.



LEARNED BEHAVIOR

Some aspects of behavior are influenced by genes, but to what degree can behaviors be modified by experience? Learned behaviors are actions that change with experience. **Learning** is the modification of a behavior based on experience. Learning can influence the expression of behaviors that are innate and also behaviors that are not innate. The study of learned behavior is central to much of ethology, and learning types can vary from simple to complex.

Habituation

The simplest type of learning, **habituation** (huh-bi-choo-AY-shuhn), occurs when an animal learns to ignore a frequent, harmless stimulus. For example, when an object passes overhead, a young gull chick tries to hide. As the chick grows older, and as parents, other common birds, or falling leaves pass over the chick's head without consequence, the youngster learns not to react. When a predatory bird flies over, however, the gull stops feeding to hide. Because the passing of a predator's shape is rare, the chicks never habituate to it. Habituation in gulls saves energy and allows feeding, yet preserves defenses for the rare emergency.

Operant Conditioning

A more complex type of learning occurs by trial and error. A dog, for example, learns to associate a cat's hiss and arched back with a painful scratch on the nose. When trial-and-error learning occurs under highly controlled conditions, it is called **operant** (AH-puhr-uhnt) **conditioning**. An animal associates some action or operation (the "operant") with a punishment or reward.

American psychologist B. F. Skinner investigated operant conditioning by placing a rat in a box with a lever, such as the one shown in Figure 44-4. As the rat explored the box, the rat eventually pressed the lever, which delivered a food pellet. After several accidental pressings, the rat learned to press the lever deliberately for food. Skinner thought that nearly any behavior could be "conditioned," or trained.

Further research by others, however, showed that although rats can easily learn to press a lever to receive food, they have trouble learning to press a lever to avoid electric shocks to their feet. Rats can learn quite quickly to avoid such foot shocks by jumping. Yet, it's much harder for them to learn to jump to get a food reward. These seeming contradictions do make sense: Rats in nature get food by manipulating objects with their hands, not their feet. Like all animals, rats most easily learn those things that are related to natural skills for survival and reproduction.



Quick Lab

Recognizing Learned Behavior

Materials small wads of paper towel (one moist; one dry), T maze made of five pieces of cardboard taped in a T shape inside a cardboard box, sow bug, blunt probe

Procedure

1. Place the moist paper in the left side of the T. Place the dry paper wad on the right side.
2. Place the sow bug at the bottom of the T. If it does not start to crawl, gently prod it with a blunt probe. Observe what the sow bug does when it reaches the T section.
3. Retrieve the sow bug. Perform as many trials as time allows. Record the results of each trial.

Analysis Summarize the sow bug's behavior. Determine if the sow bug modified its behavior through learning. Use evidence to support your answer.

FIGURE 44-4

A rat placed in the "Skinner box" shown here will learn to press a lever to obtain food. This behavior is an example of operant conditioning.





Classical Conditioning

Famed Russian biologist Ivan Pavlov studied another complex type of learning and published his results in 1903. Pavlov observed that dogs salivate at the sight and smell of meat. Was this response learned? To find out, he presented meat to a group of dogs. At the same time that the dogs received the meat, Pavlov would ring a bell. The dogs learned to associate the bell with a meat reward and eventually would salivate in response to the bell tone.

This kind of learning is called **classical conditioning**. The animal learns to associate a response with a predictive stimulus. The animal responds (salivation) after the natural stimulus (the meat) and learns to associate the natural stimulus with the predictive stimulus (the bell). This type of conditioning differs from operant conditioning in which the animal learns to respond (pushing the lever) *before* the reward appears (the food pellet).

Classical conditioning occurs in nature as well as in artificial conditions. For example, a crow learns to associate the sight of shiny, broken eggshells on the beach with the presence of newly hatched gull chicks and swoops down for a tasty meal. Advertising agencies capitalize on classical conditioning in humans by associating their product with positive imagery. They associate a certain car brand with financial success, or a certain cola with healthy good looks, all to trigger the desired response: purchase of their product.

Problem-Solving and Reasoning

A more complex type of learning is called *problem-solving learning*. Figure 44-5 shows an example of problem solving in which a chimpanzee uses a tool to get termites out of a nest. This behavior may be learned from watching a parent, may be a result of trial-and-error, or a combination of several learning mechanisms.

One type of problem-solving, **reasoning**, involves the ability to solve a problem not previously encountered by the individual in a

way that is not dictated by instinct. For example, if a chimpanzee enters a room with boxes scattered on the floor and sees a bunch of bananas tied to the ceiling, the animal will arrange the boxes to form a platform in order to retrieve the bananas.

In another example, researchers placed a jar containing a fish in a tank with an octopus. The octopus used its arms to unscrew the lid, removed the fish, and then discarded the jar and lid. This type of behavior cannot be considered instinctive, because boxes and jars are not in the evolutionary history of these species. The behavior occurred without trial-and-error, as if the animal used reasoning to develop an insight into how to solve the problem.

FIGURE 44-5

A chimpanzee will look for a stick and use it as a tool to retrieve termites from the nest for food.



GENES, LEARNING, AND BEHAVIOR

Biologists have learned from studying bees and other organisms that genes can influence behavior. Some scientists have argued that most behaviors are genetically programmed because different individuals in the same species act in the same ways. Other scientists assert that behaviors are shaped by an animal's experiences. Most ethologists today have come to agree that animal behavior, especially complex behavior such as that seen in primates, is affected both by genes and by experience.

Imprinting

One class of behavior that is determined by both genes and learning is called imprinting. **Imprinting** is a form of learning in which a young animal forms permanent associations with its environment. The most common imprinting occurs when new-born animals learn to identify a mother figure. Nobel Prize-winning ethologist Konrad Lorenz studied how geese come to identify their mothers and follow them to ponds to feed. Lorenz found that goslings follow the first large object they see moving away from the nest. The goslings would follow wagons, boxes, balloons, and even Lorenz himself, as seen in Figure 44-6.

Another example is found in sea turtles. Sea turtles imprint on characteristics of the beach where they hatch. Years later they are able to find their way back to the same beach to breed.

Imprinting occurs during a specific phase in an animal's development, called a **sensitive period**. During this time, certain types of learning take place that are later very difficult to change. For example, it is much easier for young children to learn multiple languages. This is much more difficult later in life. Scientists hypothesize that this change in the ability to learn as young animals mature is related to genes that control processes of development.



FIGURE 44-6

These young geese are following Nobel Prize winner Konrad Lorenz. Lorenz cared for the goslings the first day or so after they hatched, and the goslings imprinted on him.

SECTION 1 REVIEW

1. What are four questions that an ethologist might ask when studying behavior?
2. How is it possible for natural selection to affect an innate behavior?
3. Describe four types of learned behavior.
4. Explain the difference between the learning styles involved with Skinner's box and Pavlov's dogs.
5. Why is imprinting an example of interaction between learned and innate behavior?

CRITICAL THINKING

6. **Applying Information** If you trained bears for a stage show, which principles of behavior would you use?
7. **Forming Reasoned Opinions** A 6'10" man has a 5-year-old son. When the son goes through a doorway, he lowers his head as the father does. Do you think this behavior is likely to be learned or innate? Why?
8. **Justifying Conclusions** What do you think is the adaptive value of imprinting? Explain.

Science in Action

Does Sonar Cause Whale Strandings?

Every year, thousands of whales and dolphins strand themselves on beaches and often die. Scientists think that the whales may strand themselves individually or in groups for a variety of reasons. In recent years, beaked whales, which are not usually known for this behavior, have been beaching themselves with alarming frequency. Are human actions related to this increase?



Kenneth Balcomb

HYPOTHESIS: Sonar Causes Whale Beachings

The ocean is filled with sounds generated during human activities: shipping, fishing, oil drilling, oceanography research, and military operations. The U.S. Navy, for instance, uses a sonar system to detect submarine movements. The blasts of low frequency sound used by the military are so loud they can travel hundreds of miles at decibel levels equivalent to the sound of jet engines.

In March of 2000, 16 whales beached themselves on the Bahamas within 24 hours of a Naval sonar test. Six died. Kenneth Balcomb heads the Bahamas Marine Mammal Survey research station. Balcomb was aware of several past incidents in which mass strandings occurred around the time of naval maneuvers. It was unknown whether these events were connected. The closeness and timing of the Naval testing in the Bahamas seemed to suggest a sound-related stranding. Balcomb and co-researchers hypothesized that the loud sonar damaged the whales' hearing and in some way led to the beachings.

METHODS: Autopsy Whales and Analyze Tissue

Balcomb quickly removed the heads of two dead whales, froze the 200-pound specimens, and had them flown to the Woods Hole Oceanographic Institute in Massachusetts. There, he and others performed CT scans and microanalysis of inner ear and brain tissues.



Scientists are currently studying why whales are beaching themselves at increasing rates. Members of the Marine Animal Rescue Society are performing an autopsy on this beached whale.

RESULTS: Tissues Show Signs of Damage

The whales that were examined all showed similar lesions, including internal bleeding, known as hemorrhaging, in the acoustic regions of the cranium and mandible and in tissues next to airspaces around the ear bones. One specimen that was examined by ultra high-resolution computerized tomography showed a brain hemorrhage. In addition, dissection of this same specimen showed lung hemorrhage and laryngeal hemorrhage.

CONCLUSION: Evidence Supports Sonar-Induced Tissue Damage and Death

Based on the results of specimen examination, the National Marine Fisheries Service and the Navy came to the same conclusion: the injuries were all consistent with an intense acoustic or pressure event.

In September of 2002, 14 more whales stranded themselves in the Canary Islands following military maneuvers by NATO using sonar. Paul Jepson of the Zoological Society of London and colleagues found that the whales suffered tissue and organ damage similar to that found in the whales beached in the Bahamas.

In June 2003, a federal judge-magistrate in San Francisco ruled that the Navy must limit its plans for low-frequency sonar exercises. The Navy is currently working to establish those limits.

REVIEW

1. What evidence supports Balcomb's hypothesis?
2. Evaluate the conclusion reached by Balcomb. Is it a valid conclusion? Explain your reasoning.
3. **Critical Thinking** Are there situations in which the use of sonar by the military is justified? Explain.



TYPES OF ANIMAL BEHAVIOR

Animals all need to obtain food, find places to live, protect themselves, attract mates, reproduce, and care for young. This section discusses types of behaviors that help animals accomplish these actions.

FEEDING BEHAVIOR

Most animals spend the majority of their waking hours searching for, catching, or eating their food. How do they balance the energy they expend versus the energy they gain? One possible explanation is described by the **optimality hypothesis**, the idea that animals tend to behave in a way that maximizes food gathering while minimizing effort and exposure to predators.

For example, researchers have watched crows choose a large whelk (a snail-like mollusk) fly with it to a height of about 5 m (16 ft), and then drop it onto a rock. If the shell breaks, the crow eats the whelk. If the shell fails to break, the crow picks it up and drops it again. The optimality hypothesis would predict that dropping a whelk a second time is more likely to provide a meal than is searching for another large whelk. Experiments confirm that crow behavior optimizes nutrition versus effort expended.

COMPETITIVE BEHAVIOR

Each species needs a place to live that provides shelter from bad weather, adequate food and water, and access to mates. Because such resources are limited, competition is often the result. Competition between animals of the same species can be seen in several types of behavior.

Aggressive Behavior

Physical conflict or threatening behavior between animals is called **aggressive behavior**. This type of behavior includes displays and contests of strength that determine which individual is larger or stronger. The male impalas in Figure 44-7 display aggressive behavior when competing for mates. Most often, these contests result in one animal “surrendering” to the other. Usually, the larger or healthier male wins, and both leave unhurt.

OBJECTIVES

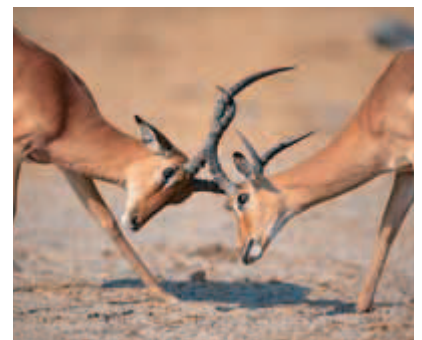
- **Discuss** the optimality hypothesis and feeding behavior.
- **List** three types of competitive behavior.
- **Describe** three different types of reproductive behavior.
- **Name** five kinds of communication.
- **Identify** costs and benefits of social behavior.
- **Describe** four types of cyclic behavior.

VOCABULARY

optimality hypothesis
aggressive behavior
territory
dominance hierarchy
courtship
communication
aposematic coloration
mimicry
pheromone
social behavior
circadian rhythm
hibernation
migration

FIGURE 44-7

Male impalas defend their territories against intrusion by other males.



Territorial Behavior

One way to ensure that an animal is able to obtain sufficient resources for itself, its mate, and its offspring is for the animal to select and establish a territory. A **territory** is an area that an animal or a group of animals occupies and defends from other members of the same species. An animal establishes a territory in many ways, including marking the boundaries with urine or visual cues, and claiming an area with vocal signals. Territorial animals will threaten or attack intruders.

The male bowerbird shown in Figure 44-8 has established a small territory which includes his nest and the area around it. He has marked the territory by building and decorating a bower and will defend it from other males. Animals may be territorial under certain circumstances. For example, the bowerbird only builds and defends his territory during breeding season. Natural selection can reinforce territorial behavior. When animals space themselves out, they do not compete for the same resources. This behavior increases the likelihood that the young of territorial animals will survive and inherit traits that promote territoriality.

Dominance Hierarchies

Competition can lead to a clear ranking of individuals within the group, from most dominant to most subordinate. This type of ranking, called a **dominance hierarchy** (HIE-rahr-kee), reduces the need for competition and aggressive behavior as subordinates learn to submit to avoid conflict. One example of a dominance hierarchy is that formed by chickens. The dominant chicken can peck all other chickens. The most subordinate chicken gets pecked by all other chickens in the group. It is from this behavior in chickens that the term “pecking order” is derived.

FIGURE 44-8

Male bowerbirds (*Amblyornis inornatus*) build elaborate nests of various shapes and with different decorations. Females choose a mate based on his bower-building abilities.



REPRODUCTIVE BEHAVIOR

Elaborate behaviors have evolved around the process of reproduction in many animals. These behaviors often differ between males and females. Differences generally center around attracting or competing for a mate. Reproductive behaviors may help animals recognize members of the same species, or members of the opposite sex. They may also be indicators of good health.

Sexual Selection

Animals generally choose mates based on certain traits or behaviors. This tendency creates a process called *sexual selection*. Traits or behaviors that increase an individual's ability to acquire a mate will appear with increased frequency in a population. The female bowerbird in Figure 44-8 chose this male based on his ability to build and decorate an attractive bower. His offspring will probably build bowers in a similar way.

Another means of attracting a mate involves certain behaviors, called **courtship**. In some species, courtship can include a complex series of behaviors called *rituals*. Courtship rituals are instinctive behaviors that are performed the same way by all members of a population and that may help animals identify receptive mates of the same species. Most courtship rituals consist of specific signals and responses that indicate willingness to mate.

Mating Systems

Mating systems increase the likelihood that young will survive. Male polygamy (more than one female), monogamy, and female polygamy (more than one male) are reproductive strategies that are determined primarily by the amount and type of parental care required by the young. Monogamy is favored in situations in which there are advantages to both parents raising the young. In birds, for example, it would be difficult for one parent to protect the nest and the hatchlings while also providing enough food for the young. This type of situation may explain why birds tend to be monogamous.

Parental Behavior

Parental investment is the time and energy an individual must spend to produce and nurture offspring. The benefit of parental care is that it increases the likelihood that young will survive to adulthood. The costs are that parental care can generally only be provided for a small number of young because of the large energy investment by the parent. Usually, females invest more in parental care than males do. In mammals like the whales in Figure 44-9, the female carries the young within her body during development, and after birth the young must nurse. In some species, the male provides the majority of parental investment. Male seahorses, for example, carry the eggs until they hatch.

Word Roots and Origins

monogamy

from the Greek *mono*, meaning "one," and *gamos*, meaning "marriage"

FIGURE 44-9

Whales provide extensive parental care for their young. Baby mammals are often born rather helpless and learn about their environments from parents.





FIGURE 44-10

Ants use chemical communication in the form of a pheromone trail, which other ants will follow. The ants above are following a trail on a leaf.

COMMUNICATION

Ant pheromones, bird songs, whale song; these behaviors are all examples of **communication**, signals produced by one animal that result in some type of response in another. There are many ways animals can communicate, including sight, sound, chemicals, touch, and possibly even language.

Sight and Sound

Species living in open environments often use visual signals to provide rapid communication. Behavioral displays communicate within and between species. Bright colors often serve as a warning that an animal is poisonous. This is called **aposematic** (A-poh-suh-MA-tic) **coloration**. After several encounters, predators learn to associate this color or pattern with a bad experience. Some animals gain protection by looking like a dangerous animal. This strategy is called **mimicry**.

Nocturnal animals, and animals in habitats with restricted visibility, often use sound to communicate. Bullfrogs and crickets, for example, use sound to attract a mate. Elephants communicate at a frequency that is too low for humans to hear.

Chemicals

Chemical communication can convey information over greater distance and time than can communication by sight or sound. Some animals release chemicals called **pheromones** (FER-uh-mohns) that cause individuals of the same species to react in a predictable way. For example, ants leave a pheromone trail that other individuals can follow, as shown in Figure 44-10. Female moths release a pheromone that attracts a male of her species from miles away.

Touch

Species that inhabit dark hives or dens often communicate by touch in addition to using sound or chemicals. Honeybees use a display that includes sight, touch, and sound, shown in Figure 44-11, to communicate the direction and distance to a food source.

Language

Most scientists have regarded language as a uniquely human behavior. In order for communication to be considered language, there are certain criteria that must be met. Among these are *phonemes* (sounds that can be combined to form words), *productivity* (many combinations of phonemes to produce different meanings), and *grammar* (rules for combining words that affect the meaning). Most animal communication lacks at least one of the characteristics of true language. Although animals do not use language systems for communication in the wild, it is possible that they can learn to use them. Research on language is being done with gorillas, chimpanzees, bonobos, parrots, dogs, and dolphins.

FIGURE 44-11

The blurred image in the center of the hive is a bee vigorously performing a waggle dance. Her sisters are receiving information about the direction and distance of a food source in part by touch communication.



SOCIAL BEHAVIOR

Social behavior can be defined as any kind of interaction between two or more animals, usually of the same species. Some species, like the bonobos in Figure 44-12, spend the majority of their lives in social groups.

Social Groups

Social groups have evolved in the animal kingdom because there are benefits to living in a group. These benefits can include protection from predators and more success in foraging. For example, fish on the outer edges of a school assume most of the danger of predation. The fish swimming on the edges shift constantly, so most individuals are not exposed to danger for a long time. Lions hunting cooperatively can bring down large prey much more efficiently than lions hunting alone.

There are also disadvantages to living in a social group. For example, there is often increased competition for food, mates, and other resources. The risk of spreading disease is also higher within a social group than it would be among a population of nonsocial animals. One species that exemplifies both the benefits and disadvantages of social groups is the blue-gill sunfish. During mating season, blue-gills nest close together. The larger number of individuals helps provide protection against predators. However, disadvantages include competition during courtship, theft of eggs by non-breeding males, and possibly transmission of disease.

Altruism

Occasionally, one member of a social group acts in a way that benefits other members of the group while putting the individual at a disadvantage. This type of behavior is called *altruism* (AL-troo-ism). One example of altruism can be seen in the ground squirrel found in North America. These animals live in large colonies. If one member of the colony sees a predator, it will give a high-pitched alarm call, as shown in Figure 44-13. This call warns other members of the colony to hide while putting the animal that calls at risk.

Another example of altruism occurs in animal societies in which the workers are sterile. Bees, ants, termites, and naked mole rats are all examples of animals that live in this type of society. Usually, members of a society are related to each other and share a large proportion of their genes. Therefore, helping a relative survive increases the chance that the genes an individual shares with that relative will be passed to the next generation. This type of natural selection is called *kin selection*.

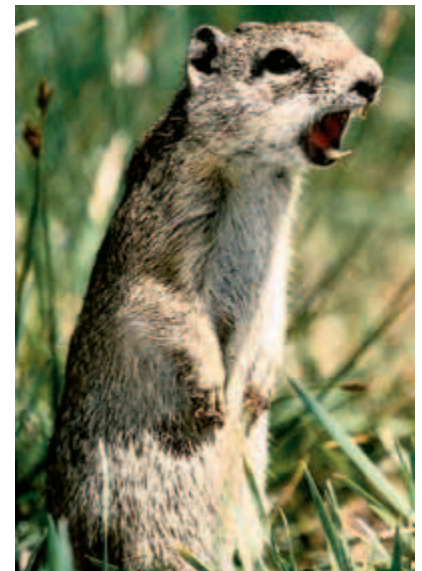


FIGURE 44-12

Bonobos are a species of ape closely related to chimpanzees. Bonobos have a highly complex and unusually peaceful social system.

FIGURE 44-13

A ground squirrel gives a piercing warning call that signals the presence of a predator.



Word Roots and Origins

circadian

From the Latin *circa*, meaning "about," and the Latin *dian*, meaning "day"

FIGURE 44-14

Monarch butterflies migrate hundreds of miles to spend their winters in central Mexico.



CYCLIC BEHAVIOR

Animals display a variety of cyclic behaviors that are synchronized with changes in their environment. These behaviors generally develop as a result of temperature changes, variations in availability of food, or likelihood of predation.

Biological Rhythms

During the day, gorillas are busy foraging but during the night, they rest. A daily biological cycle is called a **circadian** (suh-KAY-dee-uhn) **rhythm**. Predators have evolved biological rhythms in response to the activity of their prey. For example, mice are active at night, and as a result, owls are also active at night.

Many marine animals that live along the shore have biological cycles related to the tides. Tidal cycles are also called *lunar cycles* because the tide is determined by the phases of the moon. For example, fiddler crabs on the eastern coast of the United States emerge from their burrows at every low tide (twice every 24 hours).

There are also annual biological cycles. Some animals go into a period of inactivity and lowered body temperature during the winter when food is scarce. This type of behavior is called **hibernation**. Some hibernating animals, such as ground squirrels, drop their body temperature to a few degrees above freezing and do not wake for weeks at a time.

Migratory Behavior

Migration is a periodic group movement that is characteristic of a population or species. There are many types of migration. The monarch butterflies, shown in Figure 44-14, migrate hundreds of miles each year to winter in Mexico. Salmon migrate from their ocean habitat into streams to breed, then back to the ocean. Migration is exhausting and risky yet it allows animals to find habitats with plentiful seasonal foods and provides nesting sites safe from predators.

SECTION 2 REVIEW

1. Summarize the optimality hypothesis.
2. Give an example of three types of competitive behavior seen in animals.
3. How can sexual selection affect reproductive behaviors?
4. Describe an example of four types of communication used by animals.
5. Identify advantages and disadvantages of living in a social group.
6. Describe four types of cyclic behavior.

CRITICAL THINKING

7. **Applying Information** Male widowbirds have very long tails. What might be some advantages and disadvantages of having long tails?
8. **Forming Hypotheses** If a species of squirrel eats only soft acorns, what would the optimality hypothesis predict about the energy required to open hard acorns?
9. **Analyzing Information** How does defending a territory benefit an animal when aggressive competition may cause physical injury?

CHAPTER HIGHLIGHTS

SECTION 1

Development of Behavior

- Behavior is an action or series of actions performed by an animal in response to a stimulus.
- Ethologists seek to learn the cause, genetic development, evolutionary history, and reproductive advantage of a behavior.
- Innate behaviors are instinctive behaviors that do not vary despite an animal's environment or experience. A fixed action pattern (FAP) is an example of an innate behavior.
- Learned behavior is a behavior that is based on experience.
- The four types of learned behavior are habituation, operant conditioning, classical conditioning, and problem solving.
- In some cases, such as with imprinting, genes and learning can interact to affect behavior.

Vocabulary

ethologist (p. 887)
behavior (p. 887)
innate behavior (p. 888)

fixed action pattern (p. 888)
learning (p. 889)
habituation (p. 889)

operant conditioning (p. 889)
classical conditioning (p. 890)
reasoning (p. 890)

imprinting (p. 891)
sensitive period (p. 891)

SECTION 2

Types of Animal Behavior

- According to the optimality hypothesis, animals tend to maximize food gathering while minimizing effort and exposure to predators.
- Three types of competitive behavior include aggressive behavior, territorial behavior, and formation of dominance hierarchies.
- Sexual selection happens when certain mates are chosen over others based on a certain trait or set of traits.
- Animals show variation in mating systems and parental care, based on the amount of care the young require at birth.
- Five forms of communication are visual, sound, chemical, touch, and language.
- The benefits of social behavior include predator protection and successful foraging. The costs of social behavior include increased competition and risk of spreading disease.
- Some behaviors occur in synchrony with cyclical changes in the environment. These cyclic behaviors include circadian rhythms, lunar cycles, hibernation, and migration.

Vocabulary

optimality hypothesis (p. 893)
aggressive behavior (p. 893)
territory (p. 894)
dominance hierarchy (p. 894)

courtship (p. 895)
communication (p. 896)
aposematic coloration (p. 896)
mimicry (p. 896)

pheromone (p. 896)
social behavior (p. 897)
circadian rhythm (p. 898)
hibernation (p. 898)

migration (p. 898)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - innate behavior* and *learned behavior*
 - classical conditioning* and *operant conditioning*
 - hibernation* and *migration*
- Explain the relationship between *competitive behavior* and *dominance hierarchy*.
- Use the following terms in the same sentence: *social behavior*, *altruism*, and *kin selection*.
- Word Roots and Origins** The word *ethology* comes from the Greek *etho*, which means “habit” and the suffix *ology* which means “the study of.” Using this information, explain why the term *ethology* is a good name for the study of behavior.

UNDERSTANDING KEY CONCEPTS

- Identify** four questions an ethologist would ask when studying a behavior.
- Explain** whether nest building in bowerbirds is innate or learned.
- Name** the behavior a mother goose is displaying when she rolls a baseball back to her nest.
- List** examples of each of the four types of learning.
- Describe** the learning experiment involving rats and Skinner’s box.
- Describe** the experiments of Konrad Lorenz.
- Explain** the reason a dog tied to a tree would probably wrap the rope around the trunk while a chimpanzee would not.
- Identify** how genes and learning are involved in imprinting.
- State** the hypothesis that explains why a cheetah quits chasing prey after 100 yards.
- Describe** three types of competitive behavior.
- Compare** the costs versus the benefits of territoriality for impalas.
- Name** the type of selection that would result in peacocks with larger tails.
- Explain** the relationship between parental care and the type of mating system found in a species.
- Identify** an example for each of the four different types of communication found in animals.
- Identify** the types of communication that are displayed during a waggle dance.
- Relate** the costs and benefits of social behavior to life in a prairie dog colony.
- Explain** the biological reason for each of the four types of cyclic behaviors.
- Describe** the biological cycle that explains jet lag in people who travel across time zones.
- CONCEPT MAPPING** Use the following  terms to create a concept map that describes animal behavior: *behavior*, *stimulus*, *innate behavior*, *fixed action pattern*, *learned behavior*, *conditioning*, *reasoning*, *imprinting*, and *sexual selection*.

CRITICAL THINKING

- Analyzing Information** Imagine that a mutation resulted in an animal that could not experience habituation. Explain the effect on the individual and how natural selection might act on this mutation.
- Forming Reasoned Opinions** A friend insists that animals can only communicate with members of their own species. Is this true? Explain and support your opinion.
- Critiquing a Scientific Explanation** Biologists cut the tails of male widowbirds and found that they mated with only half as many females as males with uncut tails. Should the scientists conclude that female widowbirds prefer long tails? Why or why not?
- RECOGNIZING RELATIONSHIPS** In many bird species, the male is more brightly colored than the female. Explain how natural selection and sexual selection might result in these characteristics.
- INTERPRETING GRAPHICS** Identify the type of behavior displayed in the following image. Explain how natural selection might play a role in developing this behavior.





Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

1. An orb spider builds her web in exactly the same way every time. This is an example of what type of behavior?
A. random behavior
B. learned behavior
C. abnormal behavior
D. fixed action pattern behavior
2. What type of learning can only occur during a specific period early in an animal's life?
F. reasoning
G. assuming
H. imprinting
J. conditioning
3. A male lion kills all the young cubs of rival males. The genes of which of the following would be favored by this behavior?
A. the pride
B. the female
C. the male lion
D. the male lion's siblings
4. Which of the following represents classical conditioning?
F. rats feeding in Skinner boxes
G. a dog salivating at the sound of a bell
H. a male bullfrog croaking loudly in a pond
J. a primate giving a warning signal to troop members

INTERPRETING GRAPHICS: The photo below shows ants following a pheromone trail. Use the photo to answer the questions that follow.



5. The behavior of the ants is most likely to be what type of behavior?
A. innate
B. learned
C. habituation
D. classical conditioning
6. What type of communication are the ants using?
F. visual communication
G. sound communication
H. communication by touch
J. chemical communication

DIRECTIONS: Complete the following analogy.

7. aggression : competitive behavior :: altruism :
A. conditioning
B. social behavior
C. parental behavior
D. fixed action pattern behavior

INTERPRETING GRAPHICS: The photo below shows Konrad Lorenz with goslings that imprinted on him. Use the photo to answer the question that follows.



8. What is the most likely advantage to the behavior illustrated above?
F. Goslings who follow buckets are more successful in finding food.
G. Adult wild geese that associate with humans reproduce more successfully.
H. Traveling in single file is the most successful way for geese to avoid predators.
J. Goslings who follow their mother are more likely to find food and safety from predators.

SHORT RESPONSE

A male lion entering a pride kills all the young cubs. What are the benefits of the male's behavior?

EXTENDED RESPONSE

You have been hired to invent a humane method for fighting household ants. Use the behavior pictured in Questions 5–6 to accomplish this job and explain your invention.

Test TIP

Do not try to rush to finish a test. Many mistakes are made as a result of carelessness.

Studying Nonverbal Communication

OBJECTIVES

- Recognize that posture is a type of nonverbal communication.
- Observe how human posture changes during a conversation.
- Determine the relationship of gender to the postural changes that occur during a conversation.

PROCESS SKILLS

- observing
- analyzing
- graphing
- collecting data

MATERIALS

- paper
- pencil
- stopwatch or clock with a second hand

Background

People communicate nonverbally with their **posture**, or body position. The position of the body while standing is called the **stance**. In an **equal stance**, the body weight is supported equally by both legs. In an **unequal stance**, more weight is supported by one leg than by the other. In this lab, you will observe and analyze how stance changes during conversations between pairs of people who are standing.

1. Write a definition for each boldface term in the paragraph above.
2. Make a data table similar to the one on the next page. The sample data entered in row 1 show how to enter data. Do not copy these data.
3. Based on the objectives for this lab, write a question you would like to explore about nonverbal communication.



PART A Observing Behavior

1. Work in a group of two or three to observe conversations between pairs of people. Each conversation must last between 45 seconds and 5 minutes. One person in your group should be the timekeeper and the other group members should record data. Be sure that your subjects are unaware they are being observed.
2. Observe at least three conversations. Record the genders of the two participants in each conversation and the gender of the one person whose posture you observe. Be sure that the timekeeper accurately clocks the passage of each 15-second interval.
3. For each 15-second interval, record all of the changes in stance by the person you are observing. For example, note every time your subject shifts from an equal stance to an unequal stance, or vice versa.

To record the stance simply, you may write *E* to identify an equal stance and *U* to identify an unequal stance.

4. If the subject assumes an unequal stance, also record the number of weight shifts from one foot to the other. Indicate a weight shift simply by writing *W*.
5. When a conversation ends, write down whether the pair departed together or separately. To record this, write *T* to indicate departing together, or *S* to indicate departing separately.
6. After you have completed each observation, tally the total number of weight shifts within each 15-second block. **IMPORTANT!** Retain data only for conversations that last 45 seconds. If a conversation ends before you have collected data for 45 seconds, observe another conversation.

PART B Analyzing Behavior

7. After all observations have been completed, combine the data from all of the groups in your class. Analyze the data, without regard to gender.
 - a. Determine the most common stance during the first 15 seconds of a conversation, the middle 15 seconds, and the last 15 seconds. Make a bar graph to summarize the class data.
 - b. Find the average number of weight shifts in the beginning, middle, and end intervals. Make a bar graph to summarize the class data.

8. Repeat step 7, but analyze the data according to gender this time.
9. Compile the data and make bar graphs for each of the following: males talking with a male, males talking with a female, females talking with a male, and females talking with a female. Compare these graphs with the ones you made in step 7.

Analysis and Conclusions

1. Which stance was used most often during a conversation?
2. Based on your observations, which behavior most often signals that a conversation is about to end: stance change or weight shift?
3. Do males and females differ in their departure signals? If so, describe the differences you observed. Justify your conclusion.
4. What do you think might be an adaptive significance of a departure signal?
5. What other behaviors did you observe that were forms of nonverbal communication? Propose a reason for each type of behavior. Justify your answer.

Further Inquiry

Write a new question about animal behavior. Propose an experiment that you could use to answer your question. Identify the animal that you think would be best for your study and justify your reason for choosing it.

TABLE 1 OBSERVING NONVERBAL COMMUNICATION

Pairs	Gender		15-s intervals		
	Involved	Observed	15 s	30 s	45 s
1	F, M	M	U, W	E	E
2					
3					

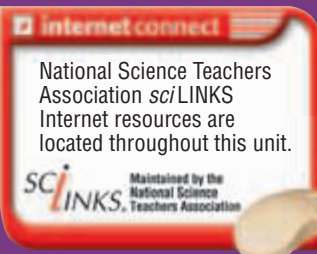
UNIT 10

CHAPTERS

- 45** *Skeletal, Muscular, and Integumentary Systems*
- 46** *Circulatory and Respiratory Systems*
- 47** *The Body's Defense Systems*
- 48** *Digestive and Excretory Systems*
- 49** *Nervous System and Sense Organs*
- 50** *Endocrine System*
- 51** *Reproductive System*



References to *Scientific American* project ideas are located throughout this unit.



HUMAN BIOLOGY

“The human body is marvelous. It can move freely, act deliberately, and survive under the most variable conditions. Its construction is complex and its requirements many.”

From “Exploring Man,” from *Behold Man: A Photographic Journey of Discovery Inside the Body*, by Lennart Nilsson in collaboration with Jan Lindberg. English translation copyright © 1974 by Albert Bonniers, Förlag, Stockholm. Reprinted by permission of *Little, Brown and Company*.



Near-perfect coordination of the many organ systems enables humans to play soccer and carry out daily activities.

At six weeks old, this developing human embryo (right) weighs less than 1 g. By eight weeks, all of the major organ systems will be recognizable.

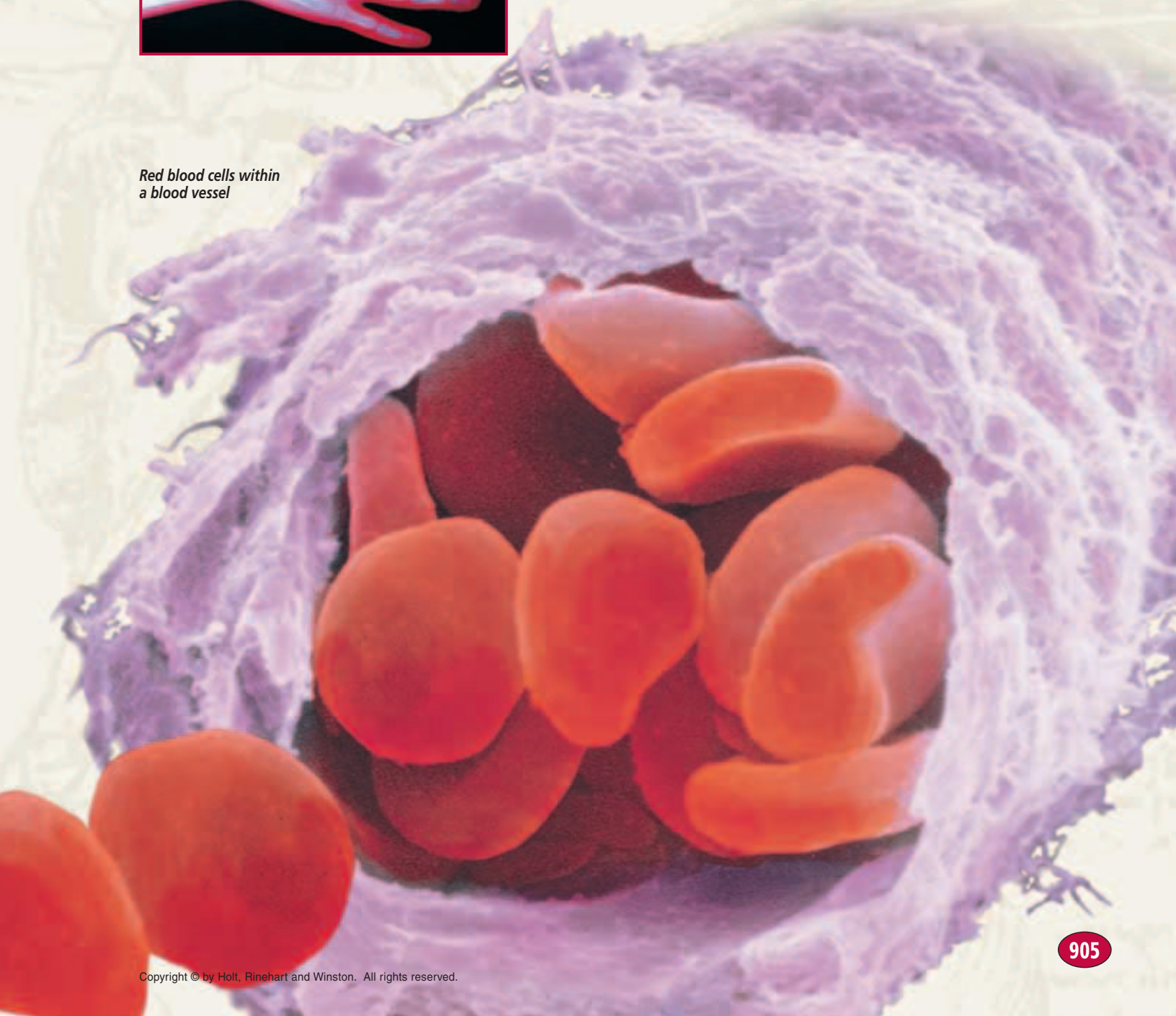


This X ray of a child's hand (below) reveals the hand's many bones.



This researcher, like other scientists around the world, spends many hours in the lab each day searching for safe drugs that can be used to treat human ailments.

Red blood cells within a blood vessel



SKELETAL, MUSCULAR, AND INTEGUMENTARY SYSTEMS



This X-ray shows a color-enhanced image of the human skull, mandible, teeth, and neck.

SECTION 1 *The Human Body Plan*

SECTION 2 *Skeletal System*

SECTION 3 *Muscular System*

SECTION 4 *Integumentary System*

THE HUMAN BODY PLAN

The human body begins to take shape during the earliest stages of embryonic development. While the embryo is a tiny ball of dividing cells, it begins forming the tissues and organs that compose the human body. By the end of its third week, the human embryo has bilateral symmetry and is developing vertebrate characteristics that will support an upright body position.

BODY TISSUES

A tissue is a collection of cells that are similar in structure and that work together to perform a particular function. The human body has four main types of tissues: muscle, nervous, epithelial, and connective.

Muscle Tissue

Muscle tissue is composed of cells that can contract. Every function that muscle tissue performs—from creating a facial expression to keeping the eyes in focus—is carried out by groups of muscle cells that contract in a coordinated fashion. The human body has three types of muscle tissue: skeletal, smooth, and cardiac. **Skeletal muscle** moves the bones in your trunk, limbs, and face. **Smooth muscle** handles body functions that you cannot control consciously, such as the movement of food through your digestive system. **Cardiac muscle**, found in your heart, pumps blood through your body. Figure 45-1a, on the following page, shows an illustration of cells of skeletal muscle tissue.

Nervous Tissue

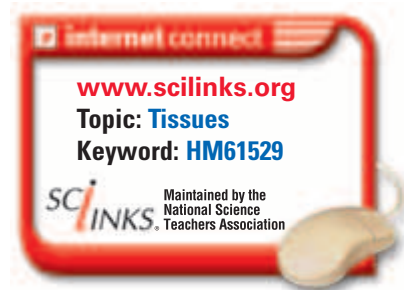
Nervous tissue contains cells that receive and transmit messages in the form of electrical impulses. These cells, called **neurons** (NOO-rahnz), are specialized to send and receive messages throughout the body. Nervous tissue makes up your brain, spinal cord, and nerves. It is also found in parts of sensory organs, such as the retina in your eye. Some nervous tissue senses changes in the internal and external environment. Other nervous tissue interprets the meaning of sensory information. Still other types of nervous tissue cause the body to move in response to sensory information. Coordination of voluntary and involuntary activities and regulation of some body processes are also accomplished by nervous tissue. Figure 45-1b, on the following page, shows an illustration of cells of nervous tissue.

OBJECTIVES

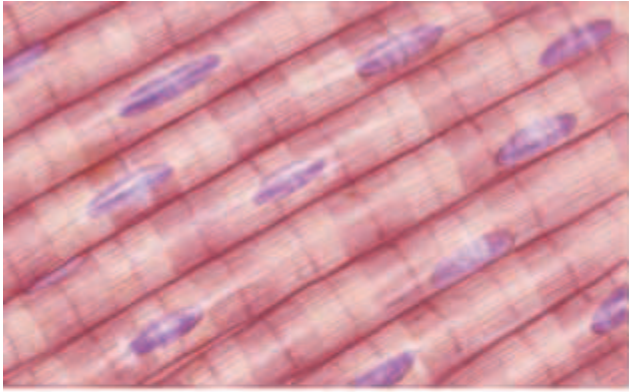
- **Describe** four types of tissues that make up the human body.
- **Explain** how tissues, organs, and organ systems are organized.
- **Summarize** the functions of the primary organ systems in the human body.
- **Identify** the five human body cavities and the organs that each contains.

VOCABULARY

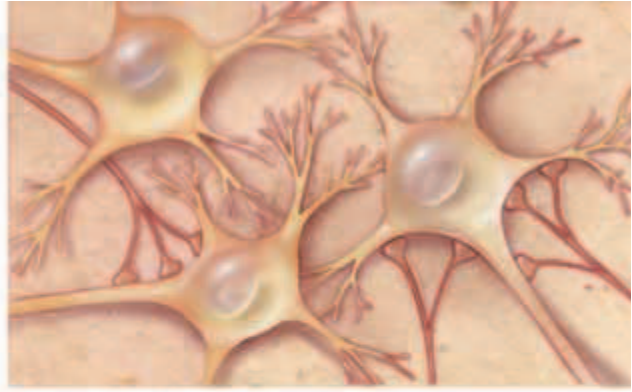
muscle tissue
skeletal muscle
smooth muscle
cardiac muscle
nervous tissue
neurons
epithelial tissue
connective tissue
matrix
organ
cranial cavity
spinal cavity
diaphragm
thoracic cavity
abdominal cavity
pelvic cavity



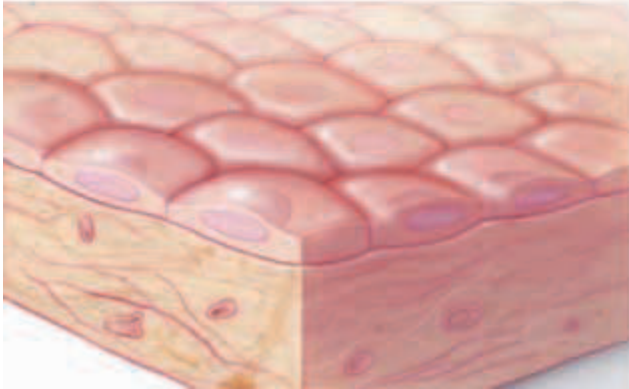
(a) MUSCLE TISSUE



(b) NERVOUS TISSUE



(c) EPITHELIAL TISSUE (top layer of cells)



(d) CONNECTIVE TISSUE

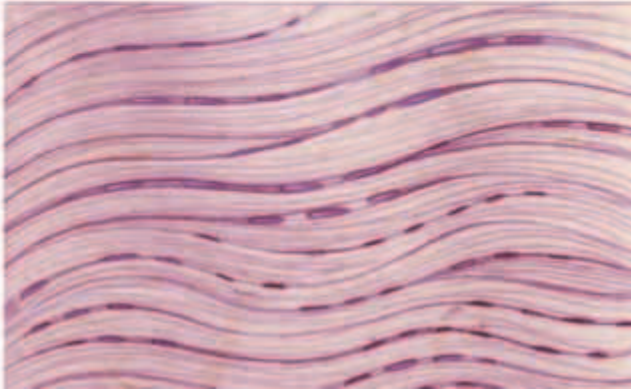


FIGURE 45-1

These four drawings show representative cells of the four main types of tissues in the human body: (a) muscle tissue, (b) nervous tissue, (c) epithelial tissue, and (d) connective tissue.

Epithelial Tissue

Epithelial (ep-uh-THEE-lee-uhl) **tissue** consists of layers of cells that line or cover all internal and external body surfaces. Each epithelial layer is formed from cells that are tightly bound together, often providing a protective barrier for these surfaces. Epithelial tissue is found in various thicknesses and arrangements, depending on where it is located. For example, the epithelial tissue that lines blood vessels is a single layer of flattened cells through which substances can easily pass. But the epithelial tissue that lines the trachea consists of a layer of cilia-bearing cells and mucus-secreting cells that act together to trap inhaled particles. The most easily observed epithelial tissue, the body's outer layer of skin, consists of sheets of dead, flattened cells that cover and protect the underlying living layer of skin. Figure 45-1c shows an illustration of cells of epithelial tissue.

Connective Tissue

Connective tissue binds, supports, and protects structures in the body. Connective tissues are the most abundant and diverse of the four types of tissue, and include bone, cartilage, tendons, fat, and blood. These tissues are characterized by cells that are embedded in large amounts of an intercellular substance called **matrix**. Matrix can be solid, semisolid, or liquid. Bone cells are surrounded by a hard, crystalline matrix containing calcium. The cells in cartilage, tendons, and fat are surrounded by a semisolid fibrous matrix. Blood cells are suspended in a liquid matrix. Figure 45-1d shows an illustration of cells of connective tissue.

ORGANS AND ORGAN SYSTEMS

An **organ** consists of various tissues that work together to carry out a specific function. The stomach, a saclike organ in which food is mixed with digestive enzymes, is composed of the four types of tissues. A single organ, such as the stomach, usually does not function in isolation. Rather, groups of organs interact in an organ system. For example, in the digestive system, the stomach, small intestine, liver, and pancreas all work together to break down food into molecules the body can use for energy. Table 45-1 lists the body's organ systems and names their major structures and functions. As you study the table, think about the ways in which the different organ systems work together to function in an efficient, integrated manner.

TABLE 45-1 *Summary of Organ Systems*

System	Major structures	Functions
Skeletal	bones	provides structure; supports and protects internal organs
Muscular	muscles (skeletal, cardiac, and smooth)	provides structure; supports and moves trunk and limbs; moves substances through body
Integumentary	skin, hair, nails	protects against pathogens; helps regulate body temperature
Cardiovascular	heart, blood vessels, blood	transports nutrients and wastes to and from all body tissues
Respiratory	air passages, lungs	carries air into and out of lungs, where gases (oxygen and carbon dioxide) are exchanged
Immune	lymph nodes and vessels, white blood cells	provides protection against infection and disease
Digestive	mouth, esophagus, stomach, liver, pancreas, small and large intestines	stores and breaks down food; absorbs nutrients; eliminates waste
Excretory	kidneys, ureters, bladder, urethra, skin, lungs	eliminates waste; maintains water and chemical balance
Nervous	brain, spinal cord, nerves, sense organs, receptors	controls and coordinates body movements and senses; controls consciousness and creativity; helps monitor and maintain other body systems
Endocrine	glands (such as adrenal, thyroid, pituitary, and pancreas); hypothalamus and specialized cells in the brain, heart, stomach, and other organs	maintains homeostasis; regulates metabolism, water and mineral balance, growth, behavior, development, and reproduction
Reproductive	ovaries, uterus, mammary glands (in females), testes (in males)	produces eggs and milk in females, sperm in males, and offspring after fertilization

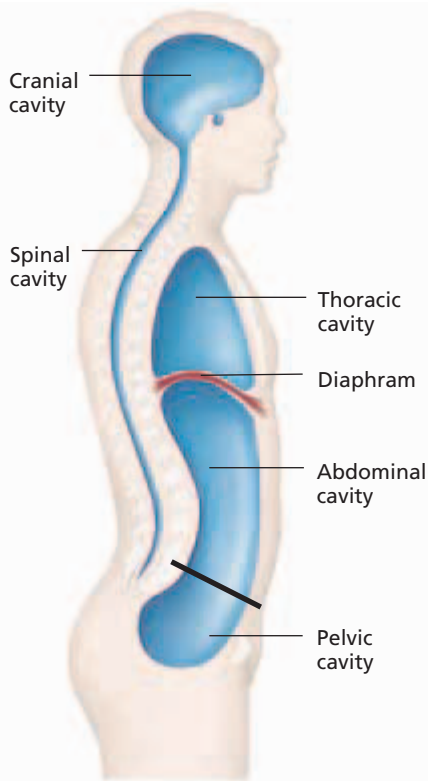


FIGURE 45-2

The human body has five main cavities that house and protect delicate internal organs.

Integration of Organ Systems

An even higher level of organization is the integration of organ systems. Each organ system has organs associated with it according to the organ's primary function. However, the boundaries are not always well defined. For example, nearly all of the juices produced by the pancreas are designed to aid in digestion. But because the pancreas produces vitally important hormones, it is also considered a component of the endocrine system. Each organ system carries out its own specific function, but for the organism to survive, the organ systems must work together. For example, nutrients from the digestive system are distributed by the cardiovascular system. The efficiency of the cardiovascular system depends on nutrients from the digestive system and oxygen from the respiratory system.

BODY CAVITIES

Many organs and organ systems in the human body are housed in compartments called body cavities. These cavities protect internal organs from injuries and permit organs such as the lungs to expand and contract while remaining securely supported. As shown in Figure 45-2, the human body has five main body cavities. Each cavity contains one or more organs. The **cranial cavity** contains the brain. The **spinal cavity** surrounds the spinal cord.

The two main cavities in the trunk of the human body are separated by a wall of muscle called the **diaphragm** (DIE-uh-FRAM). The upper compartment, or **thoracic** (toh-RAS-ik) **cavity**, contains the heart, the esophagus, and the organs of the respiratory system. The lower compartment, or **abdominal** (ab-DAHM-uh-nuhl) **cavity**, contains organs of the digestive system. The **pelvic cavity** contains the organs of the reproductive and excretory systems.

SECTION 1 REVIEW

1. Name the four types of tissues in the human body, and give an example of each.
2. Explain the difference between muscle tissue and nervous tissue.
3. How are tissues, organs, and organ systems organized in the body?
4. How do the organ systems function together in the human body?
5. Give an example of interaction between the endocrine system and another organ system.
6. Identify the organs each body cavity contains.

CRITICAL THINKING

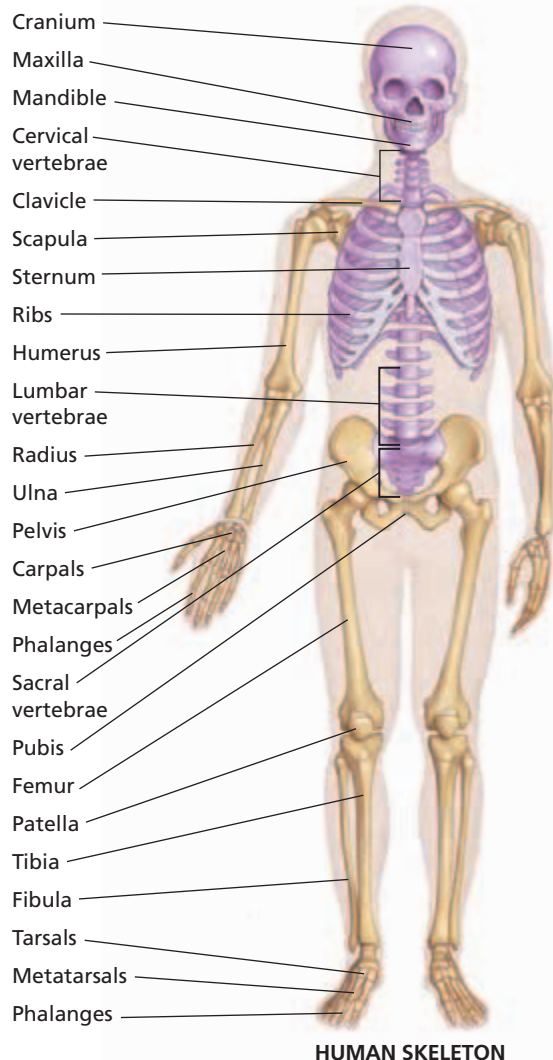
7. **Applying Information** Describe how the skeletal, muscular, nervous, respiratory, and circulatory systems function in a person swimming in a pool.
8. **Analyzing Concepts** Explain how the function of the body's organs might be affected if the body were not divided into cavities?
9. **Forming Reasoned Opinions** The body cavity that protects the brain is encased in bone. Why do you think the abdominal cavity is not encased in bone?

SKELETAL SYSTEM

*The adult human body consists of approximately 206 bones, which are organized into an internal framework called the **skeleton**. The variation in size and shape among the bones that make up the skeleton reflects their different roles in the body.*

THE SKELETON

As shown in Figure 45-3, the human skeleton is composed of two parts—the axial skeleton and the appendicular (AP-uhn-DIK-yuh-luhr) skeleton. The bones of the skull, ribs, spine, and sternum form the **axial skeleton**. The bones of the arms and legs, along with the scapula, clavicle, and pelvis, make up the **appendicular skeleton**.



OBJECTIVES

- **Distinguish** between the axial skeleton and the appendicular skeleton.
- **Explain** the function and structure of bones.
- **Summarize** how bones develop and elongate.
- **List** three types of joints, and give an example of each.
- **Describe** a common disorder that affects the skeletal system.

VOCABULARY

skeleton
axial skeleton
appendicular skeleton
periosteum
compact bone
Haversian canal
osteocyte
spongy bone
bone marrow
fracture
ossification
epiphyseal plate
joint
fixed joint
semimovable joint
movable joint
ligament
synovial fluid
rheumatoid arthritis
osteoarthritis

FIGURE 45-3

The skeleton is the framework that supports and protects the body. The bones of the axial skeleton are colored purple. The bones of the appendicular skeleton are colored yellow.

Word Roots and Origins

periosteum

from the Greek *peri*, meaning "around," and *osteon*, meaning "bone"

BONE FUNCTION AND STRUCTURE

The bones that make up the skeleton function in a variety of ways. Bones provide a rigid framework against which muscles can pull, give shape and structure to the body, and support and protect delicate internal organs. Notice, for example, that the ribs curve to form a cage that contains the heart and lungs. Similarly, bones in the skull form the cranium, a dome-shaped case that protects the brain. Bones also store minerals, such as calcium and phosphorus, which play vital roles in important metabolic processes. In addition, the internal portion of many bones produces red blood cells, platelets, and white blood cells.

Despite their number and size, bones make up less than 20 percent of the body's mass. The reason for their having relatively little mass can be better understood by looking at bone structure. Bones are not dry, rigid structures, as they may appear in a museum exhibit. They are moist, living tissues.

Long Bone Structure

As shown in Figure 45-4, a long bone consists of a porous central cavity surrounded by a ring of dense material. The bone's surface is covered by a tough membrane called the **periosteum** (PER-ee-AHS-tee-uhm). This membrane contains a network of blood vessels, which supply nutrients, and nerves, which signal pain.

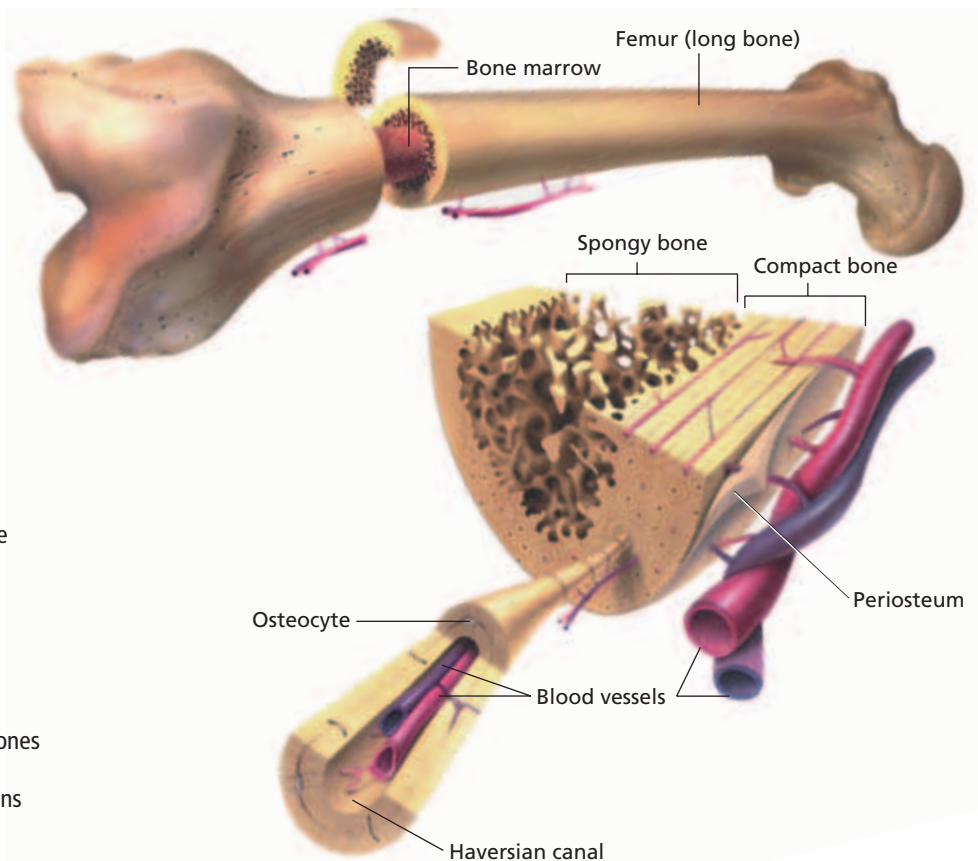


FIGURE 45-4

Long bones, found in the limbs of the body, are hollow and cylindrical. The outer shell of hard compact bone consists of closely packed rings of minerals and protein fibers. Narrow canals running through these rings contain blood vessels and nerves. Spongy bone is found in small flat bones and in the ends of long bones. The central shaft of the long bone contains marrow and blood vessels.

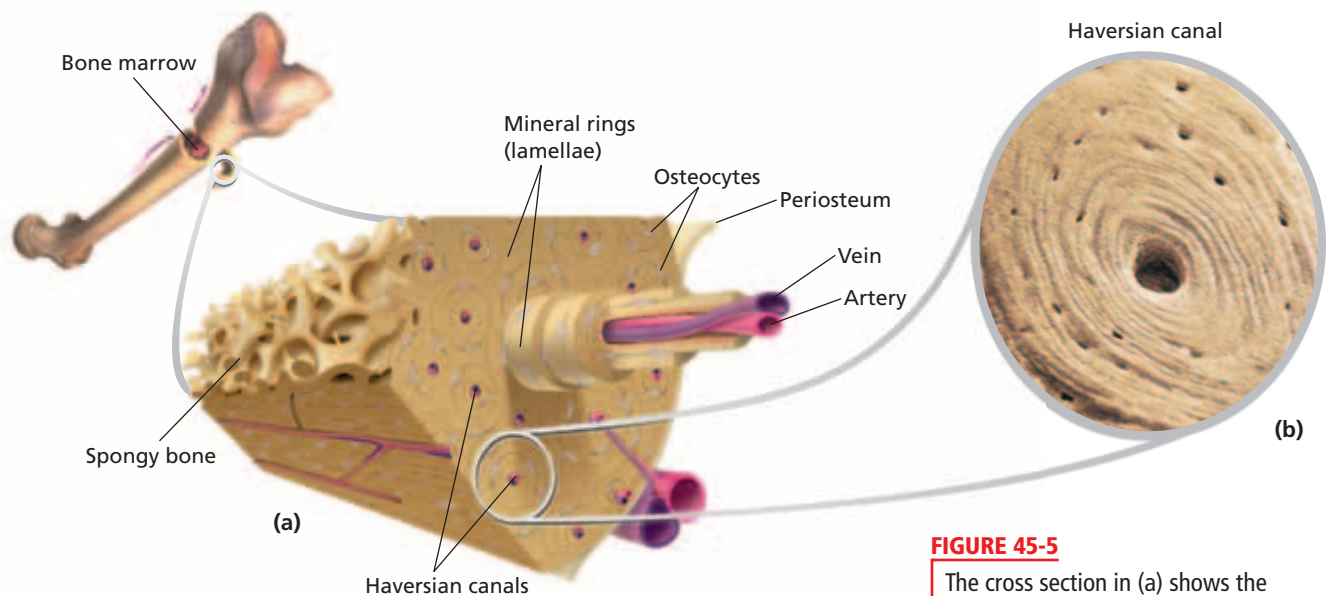


FIGURE 45-5

The cross section in (a) shows the internal structure of compact bone. A micrograph of a Haversian canal ($380\times$) surrounded by lamellae in compact bone is shown in (b).

Under the periosteum is a hard material called **compact bone**. A thick layer of compact bone enables the shaft of the long bone to endure the large amount of stress it receives during activities such as jumping. In the cross section shown in Figure 45-5a, notice that compact bone is composed of cylinders of mineral crystals and protein fibers called *lamellae*. In the center of each cylinder is a narrow channel called a **Haversian** (huh-VER-shuhn) **canal**, as shown in Figure 45-5b. Blood vessels run through interconnected Haversian canals, creating a network that carries nourishment to the living bone tissue. Several layers of protein fibers wrap around each Haversian canal. Embedded within the gaps between the protein layers are bone cells called **osteocytes** (AH-tee-uh-SIETS).

Beneath some compact bone is a network of connective tissue called **spongy bone**. Although its name suggests that it is soft, this tissue is hard and strong. As shown in Figure 45-4, spongy bone has a latticework structure that consists of bony spikes arranged along points of pressure or stress, making bones both light and strong.

Bone Marrow

Many bones also contain a soft tissue called **bone marrow**, which can be either red or yellow. Red bone marrow—found in spongy bone, the ends of long bones, ribs, vertebrae, the sternum, and the pelvis—produces red blood cells, platelets, and white blood cells. Yellow bone marrow fills the shafts of long bones. It consists mostly of fat cells and serves as an energy reserve. It can also be converted to red bone marrow and produce blood cells when severe blood loss occurs.

Injury and Repair

Despite their strength, bones will crack or even break if they are subjected to extreme loads, sudden impacts, or stresses from unusual directions. The crack or break is referred to as a **fracture**. If circulation is maintained and the periosteum survives, healing will occur even if the damage to the bone is severe.

Eco Connection

Bones of Lead

Millions of Americans have been exposed to lead in the environment. Following exposure to lead, the kidneys excrete most of the metal. But 7 to 10 percent of the remaining lead in the body is stored in bone and can stay there for a lifetime. The rapid bone uptake of lead acts as a detoxifying mechanism. But lead may not be permanently locked in bone. As people age, bone degeneration may occur, releasing lead into the bloodstream. Even very small concentrations of lead in the bloodstream can cause damage to kidneys, and high blood pressure.

The United States has outlawed the addition of lead to gasoline, water pipes, and paint. As a result, people who are now under age 25 may not accumulate as much lead in their bones as people from earlier generations.

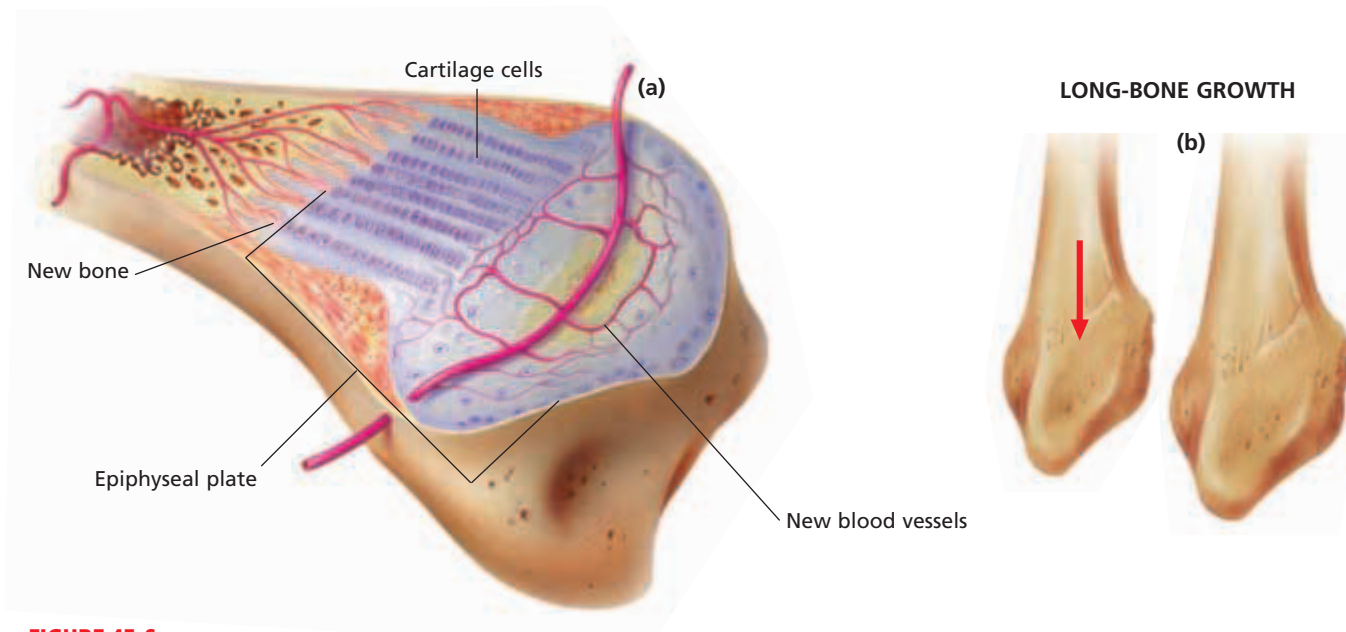


FIGURE 45-6

The epiphyseal plate, found at the ends of immature long bones, such as the fibula shown above, is the site of bone elongation. This region is rich with cartilage cells, which divide, enlarge, and push older cells toward the middle of the bone shaft. As older cells move back, they are replaced by new bone cells, forming new regions of bone. A long bone (a) will grow in length, circumference, and density in this manner, as shown in (b).

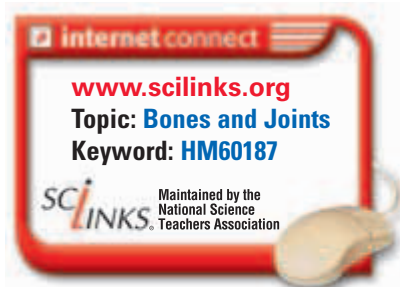
BONE DEVELOPMENT

Most bones develop from cartilage, a tough but flexible connective tissue. In the second month of fetal development, much of the skeleton is made of cartilage. During the third month, osteocytes begin to develop and release minerals that lodge in the spaces between the cartilage cells, turning the cartilage to bone. The process by which cartilage is slowly replaced by bone as a result of the deposition of minerals is called **ossification** (AHS-uh-fuh-KAY-shuhn). Most fetal cartilage is eventually replaced by bone. However, some cartilage remains, lending flexibility to the areas between bones, at the end of the nose, in the outer ear, and along the inside of the trachea.

A few bones, such as some parts of the skull, develop directly into hard bone without forming cartilage first. In these cases, the osteocytes are initially scattered randomly throughout the embryonic connective tissue but soon fuse into layers and become flat plates of bone. In the skull, suture lines can be seen where the plates of bone meet.

Bone Elongation

Bones continue to develop after a person's birth. Between early childhood and late adolescence, bone cells gradually replace the cartilage in long bones of limbs, such as the arms and legs. Bone elongation takes place near the ends of long bones in an area known as the **epiphyseal** (EP-uh-FIZ-ee-uhl) **plate**. As shown in Figure 45-6a, the epiphyseal plate is composed of cartilage cells that divide and form columns, pushing older cells toward the middle of the bone. As these older cells die, they are replaced by new bone cells. Growth continues, as shown in Figure 45-6b, until bone has replaced all the cartilage in the epiphyseal plate. At this point, bones no longer elongate and a person has usually reached full height. The epiphyseal plates then become *epiphyseal lines*.



JOINTS

The place where two bones meet is known as a **joint**. Three major kinds of joints are found in the human body—fixed, semimovable, and movable. Examples of these joints are shown in Figure 45-7.

Fixed Joints

Fixed joints prevent movement. They are found in the skull, where they securely connect the bony plates and permit no movement of those bones. A small amount of connective tissue in a fixed joint helps absorb impact to prevent the bones from breaking.

Semimovable Joints

Semimovable joints permit limited movement. For example, semimovable joints hold the bones of the vertebral column in place and allow the body to bend and twist. The vertebrae of the spine are separated by disks of cartilaginous tissue. These tough, springy disks compress and absorb shocks that could damage the fragile spinal cord. Semimovable joints are also found in the rib cage, where long strands of cartilage connect the upper ten pairs of ribs to the sternum, allowing the chest to expand during breathing.

Movable Joints

All other joints in the body are **movable joints**. These joints enable the body to perform a wide range of movements and activities. Movable joints include hinge, ball-and-socket, pivot, saddle, and gliding joints. An example of a *hinge joint* is found in the elbow, which allows you to move your forearm upward and downward, like a hinged door. An example of a *ball-and-socket joint* is the shoulder joint, which enables you to move your arm up, down, forward, and backward, as well as to rotate it in a complete circle. The joint formed by the top two vertebrae of your spine is an example of a *pivot joint*; it allows you to turn your head from side to side, as when shaking your head “no.” The *saddle joint*, found at the base of each thumb, allows you to rotate your thumbs and helps you grasp objects with your hand. Finally, *gliding joints* allow bones to slide over one another. Examples are the joints between the small bones of your foot, which allow your foot to flex when you walk.

Joint Structure

Joints, such as the knee, are often subjected to a great deal of pressure and stress, but their structure is well suited to meet these demands. As in all movable joints, the parts of the bones that come in contact with each other are covered with cartilage, which protects the bones’ surface from friction. Tough bands of connective tissue, called **ligaments**, hold the bones of the joint in place. The surfaces of the joints that are subjected to a great deal of pressure are lined with tissue that secretes a lubricating substance called **synovial** (sih-NOH-vee-uhl) **fluid**. Synovial fluid helps protect the ends of bones from damage by friction.

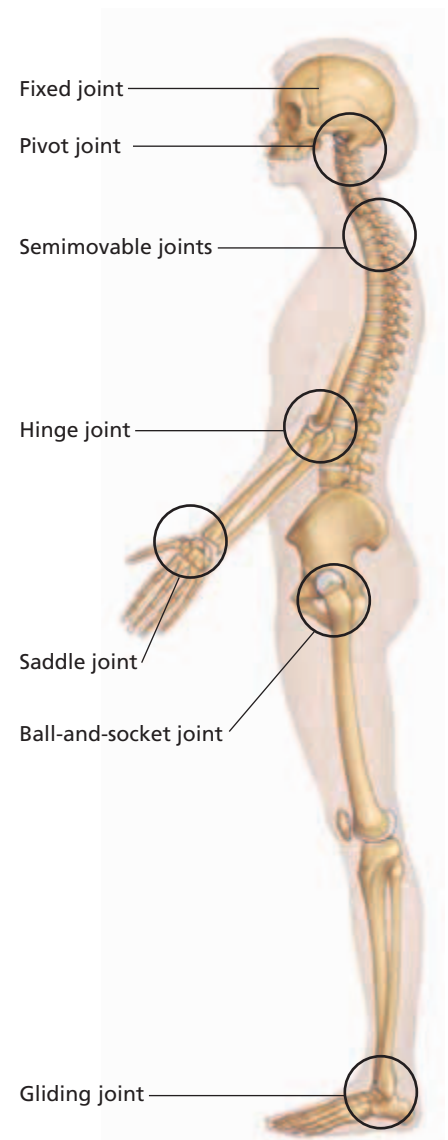
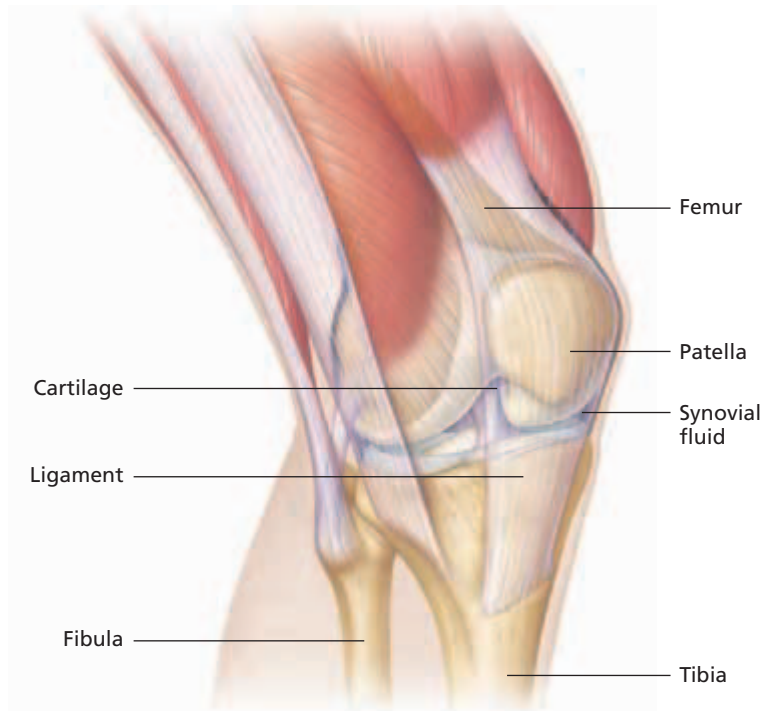


FIGURE 45-7

In addition to fixed joints and semimovable joints, the human body has five types of movable joints: pivot, hinge, saddle, ball-and-socket, and gliding.

FIGURE 45-8

The knee is a movable joint formed by the ends of the femur, the tibia, and the patella. Many cordlike ligaments stabilize the joint, especially during movement. Pads of cartilage protect the ends of bones and act as shock absorbers. Like many joints in the body, the knee is a synovial joint. It contains membranes that secrete synovial fluid, which lubricates and nourishes the tissues inside the joint.



Sometimes these protective structures are not enough to prevent a joint from becoming injured. Of all the joints in the body, the knee joint is the most susceptible to injury because it carries the body's weight and relies on many ligaments for stability. Damage to the knee joint can cause swelling in the compartment that contains the synovial fluid. Figure 45-8 shows the internal structures of the knee joint.

The term *arthritis* is used to describe disorders that cause painful, swollen joints. There are two forms of arthritis that affect joints. **Rheumatoid arthritis** develops when the immune system begins to attack body tissues. The joints become inflamed, swollen, stiff, and deformed. **Osteoarthritis** is a degenerative joint disease in which the cartilage covering the surface of bone becomes thinner and rougher. As a result, bone surfaces rub against each other, which is sensed by the nerves in the periosteum, and causes severe discomfort.

SECTION 2 REVIEW

1. List the major parts of the axial skeleton and the major parts of the appendicular skeleton.
2. Name five functions of bones.
3. Illustrate the structure of a long bone.
4. When does the ossification of most of the bones in the body begin and end?
5. Describe the function of the three major types of joints, and give an example of each.
6. Differentiate between the two types of arthritis.

CRITICAL THINKING

7. **Applying Information** What is the advantage of a cartilaginous skeleton during prenatal development?
8. **Analyzing Information** Which type of arthritis is not related to age?
9. **Relating Concepts** How are the structures of cartilage and bone related to the function each performs in the body?

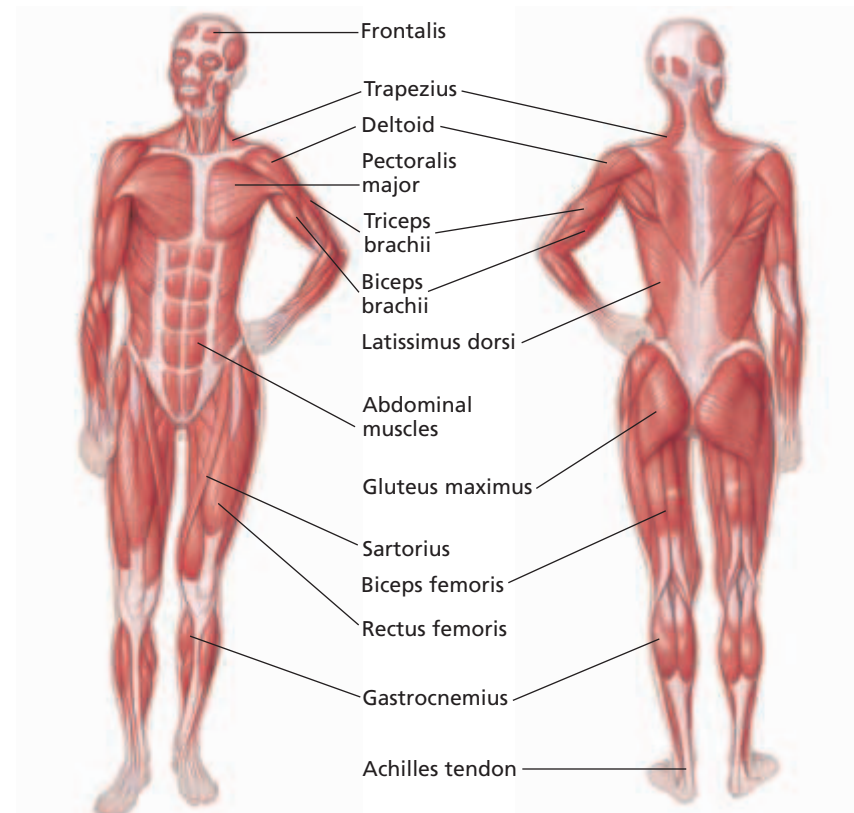
MUSCULAR SYSTEM

Muscles make up the bulk of the body and account for about one-third of its weight. Their ability to contract and relax not only enables the body to move, but also provides the force that pushes substances, such as blood and food, through the body.

MUSCLE TYPES

A muscle is an organ that can contract in a coordinated fashion and includes muscle tissue, blood vessels, nerves, and connective tissue. Some of the major muscles of the human body are shown in Figure 45-9. Recall that the human body has three types of muscle tissues: skeletal, smooth, and cardiac.

Skeletal muscle is responsible for moving parts of the body, such as the limbs, trunk, and face. Skeletal muscle tissue is made up of elongated cells called **muscle fibers**. Each muscle fiber contains many nuclei and is crossed by light and dark stripes, called **striations**, as shown on the following page in Figure 45-10a. Skeletal muscle fibers are grouped into dense bundles called **fascicles**. A group of fascicles are bound together by connective tissue to form a muscle. Because their contractions can usually be consciously controlled, skeletal muscles are described as **voluntary muscles**.



OBJECTIVES

- **Distinguish** between the three types of muscle tissues.
- **Describe** the structure of skeletal muscle fibers.
- **Explain** how skeletal muscles contract.
- **Describe** how muscles move bones.
- **Explain** the process in which a muscle becomes fatigued.

VOCABULARY

muscle fiber
striation
fascicle
voluntary muscle
involuntary muscle
myofibril
myosin
actin
Z line
sarcomere
tendon
origin
insertion
flexor
extensor
muscle fatigue
oxygen debt

FIGURE 45-9

Skeletal muscle tissue is shown in these diagrams of some of the major muscles in the human body.

Certified Athletic Trainer

Job Description A certified athletic trainer (ATC) is a highly educated and skilled professional who specializes in athletic health care. Certified athletic trainers must have a bachelor's degree, usually in athletic training, health, physical education, or exercise science. In addition, an ATC must pass a certification exam. The job includes the prevention, identification, evaluation, treatment, referral, and rehabilitation of sports-related injuries.

Focus on Certified Athletic Trainer

In high school, Veronica Ampey was interested in science, medicine, rehabilitation, and community service. Today, she has found a career that brings

together all of these interests and her love for sports. "I like the fact that I am not tied to a desk," says Ampey. "I get paid to watch athletes practice and compete."

Ampey works as an ATC at a small high school in Washington, D.C. Ampey says "The bonus is using my education and experience to address incidents when they occur. Additionally, there is a great deal of satisfaction in seeing someone you've worked with return to full athletic participation."

Education and Skills

- **High School**—at least three years of science courses and four years of math courses.



- **College**—bachelor's degree from a college with an accredited athletic training curriculum, including course work in biology; a master's degree for some jobs; and certification.
- **Skills**—patience, good organizational skills, and ability to work as a member of a team.



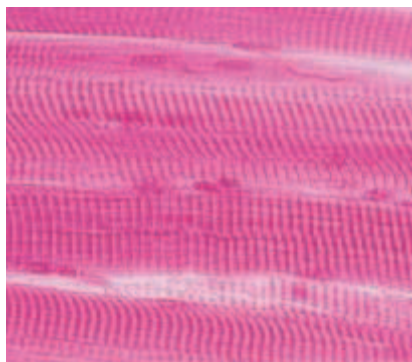
For more about careers, visit go.hrw.com and type in the keyword **HM6 Careers**.

FIGURE 45-10

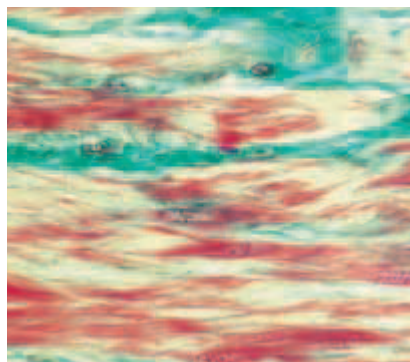
These light micrographs show the three types of muscle tissue. Skeletal muscle tissue (a) has a striped appearance when viewed under a microscope (430 $\times$). Smooth muscle tissue (b) is found in the digestive tract, the uterus, the bladder, and the blood vessels (400 $\times$). Cardiac muscle tissue (c) is found only in the heart (270 $\times$).

Smooth muscle forms the walls of the stomach, intestines, blood vessels, and other internal organs. Individual smooth muscle cells are spindle-shaped, have a single nucleus, and interlace to form sheets, as shown in Figure 45-10b. Notice that smooth muscle lacks the striations found in skeletal muscle tissue. Smooth muscle fibers are surrounded by connective tissue, but the connective tissue does not unite to form tendons as it does in skeletal muscles. Because most of its movements cannot be consciously controlled, smooth muscle is referred to as **involuntary muscle**.

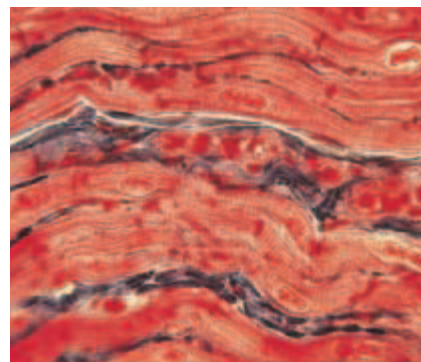
Cardiac muscle, shown in Figure 45-10c, makes up the walls of the heart. Cardiac muscle shares some characteristics with both skeletal muscle and smooth muscle. As with skeletal muscle, cardiac muscle tissue is striated; as with smooth muscle, it is involuntary and each cell has one nucleus.



(a) SKELETAL MUSCLE TISSUE



(b) SMOOTH MUSCLE TISSUE



(c) CARDIAC MUSCLE TISSUE

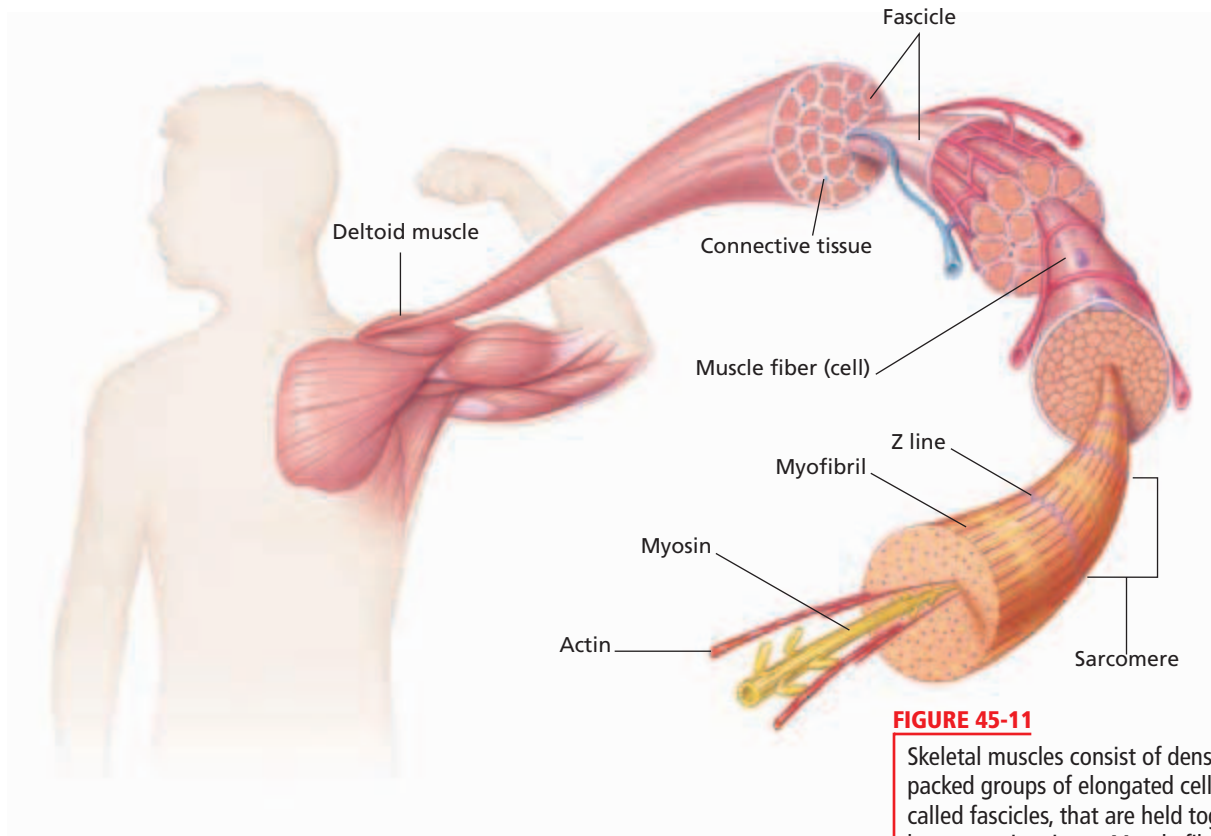


FIGURE 45-11

Skeletal muscles consist of densely packed groups of elongated cells, called fascicles, that are held together by connective tissue. Muscle fibers consist of protein filaments called myofibrils. Two types of filaments are found in muscle fibers—actin and myosin. The complementary structures of actin and myosin interact to contract and relax muscles.

MUSCLE STRUCTURE

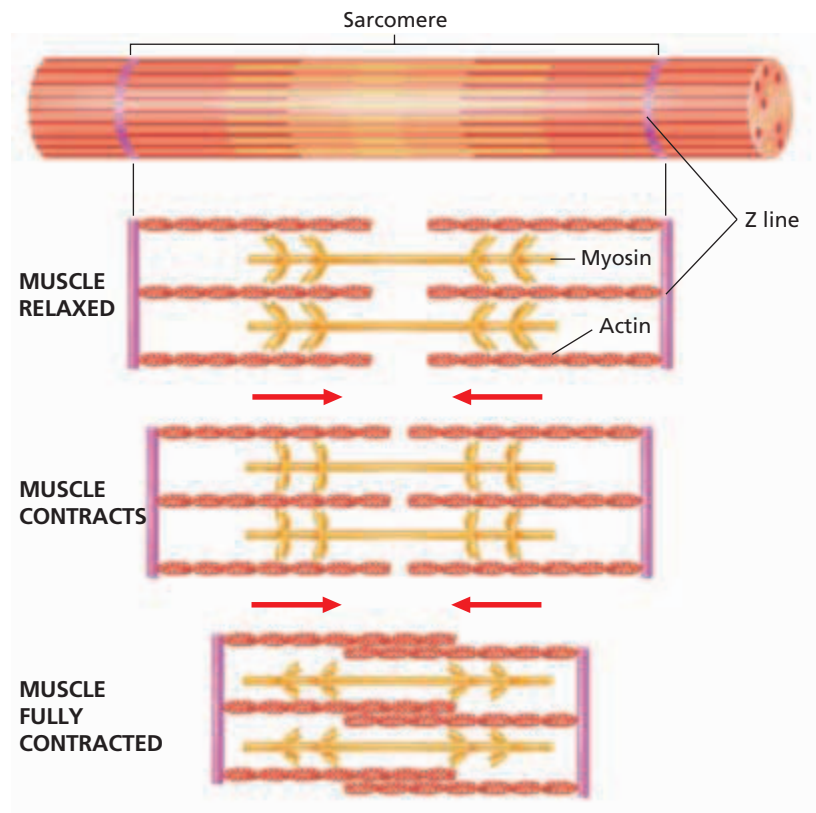
A skeletal muscle fiber is a single, multinucleated muscle cell. A skeletal muscle may be made up of hundreds or even thousands of muscle fibers, depending on the muscle's size. Although muscle fibers make up most of the muscle tissue, a large amount of connective tissue, blood vessels and nerves are also present. Like all body cells, muscle cells are soft and easy to injure. Connective tissue covers and supports each muscle fiber and reinforces the muscle as a whole.

The health of a muscle depends on a sufficient nerve and blood supply. Each skeletal muscle fiber has a nerve ending that controls its activity. Active muscles use a lot of energy and therefore require a continuous supply of oxygen and nutrients, which are supplied by arteries. Muscles produce large amounts of metabolic waste that must be removed through veins.

A skeletal muscle fiber, such as the one shown in Figure 45-11, contains bundles of threadlike structures called **myofibrils** (MIE-oh-FIE-bruhlz). Each myofibril is made up of two types of protein filaments—thick ones and thin ones. Thick filaments are made of the protein **myosin** (MIE-uh-suhn), and thin filaments are made of the protein **actin**. Myosin and actin filaments are arranged to form an overlapping pattern, which gives striated muscle tissue its striped appearance. Thin actin filaments are anchored at their endpoints to a structure called the **Z line**. The region from one Z line to the next is called a **sarcomere** (SAHR-kuh-MIR).

FIGURE 45-12

In a relaxed muscle, the actin and myosin filaments overlap. During a muscle contraction, the filaments slide past each other and the zone of overlap increases. As a result, the length of the sarcomere shortens.



Quick Lab

Testing Muscle Stamina and Strength

Materials bathroom scale, notepad, pencil

Procedure

1. Create a chart that compares the strength of pectoral muscles at four different time intervals, each 1 minute apart.
2. Press the scale between the palms of your hands. Have your partner record the amount of pressure applied by your pectoral muscles.
3. Set the scale down and press your hands together in front of you for 1 minute. Press the scale between your hands again and have your partner record the pressure.
4. Repeat steps 2 and 3 two more times. Then repeat the experiment with your partner pressing the scale while you record the pressure.

Analysis What trends did you notice, if any, in the amount of pressure recorded? What might be a reason for this trend?

MUSCULAR CONTRACTION

The sarcomere is the functional unit of muscle contraction. When a muscle contracts, myosin filaments and actin filaments interact to shorten the sarcomere. Myosin filaments have extensions shaped like oval “heads.” Actin filaments look like a twisted strand of beads. When a nerve impulse stimulates a muscle to contract, the myosin filaments’ heads attach to points between the beads of the actin filaments. The myosin heads then bend inward, pulling the actin with them. The myosin heads then let go, bend back into their original position, attach to a new point on the actin filament, and pull again. This action shortens the sarcomere. The synchronized shortening of sarcomeres along the length of a muscle fiber causes the whole fiber, and hence the muscle, to contract. Figure 45-12 shows a sarcomere’s structures.

Muscle contraction requires energy, which is supplied by ATP. This energy is used to detach the myosin heads from the actin filaments. Because myosin heads must attach and detach a number of times during a single muscle contraction, muscle cells must have a continuous supply of ATP. Without ATP, the myosin would remain attached to the actin, keeping a muscle permanently contracted.

Muscle contraction is an all-or-none response—either the fibers contract or they remain relaxed. How, then, are you able to contract your muscles tightly enough to lift a dumbbell or gently enough to lift a pen? The force of a muscle contraction is determined by the number of muscle fibers that are stimulated. As more fibers are activated, the force of the contraction increases.

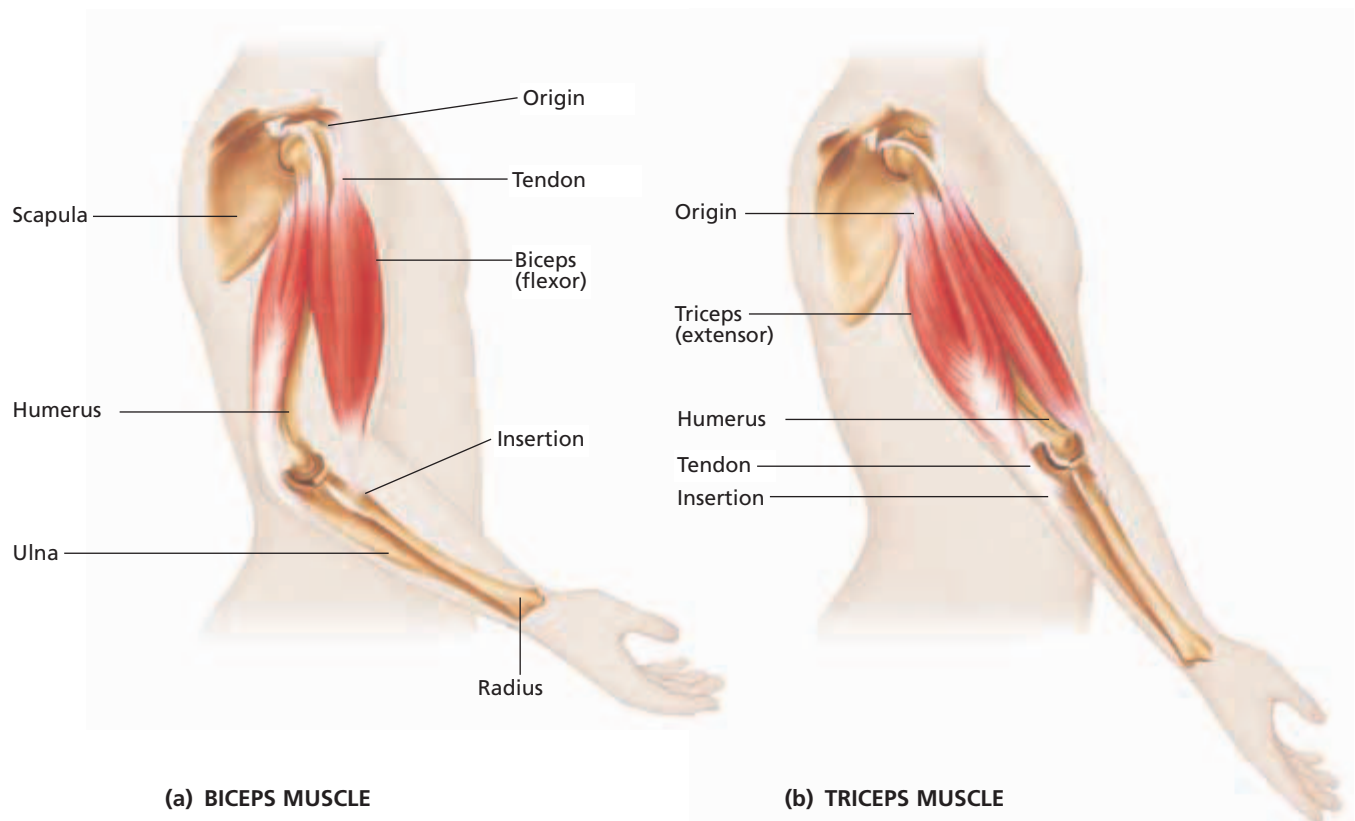
MUSCULAR MOVEMENT OF BONES

Generally, skeletal muscles are attached to one end of a bone, stretch across a joint, and are fastened to a point on another bone. Muscles are attached to the outer membrane of bone, the **periosteum**, either directly or by a tough fibrous cord of connective tissue called a **tendon**. For example, as shown in Figure 45-13, one end of the large biceps muscle in the arm is connected by tendons to the radius in the forearm, while the other end of the muscle is connected to the scapula in the shoulder. When the biceps muscle contracts, the forearm flexes upward while the scapula remains stationary. The point where the muscle attaches to the stationary bone—in this case, the scapula—is called the **origin**. The point where the muscle attaches to the moving bone—in this case the radius—is called the **insertion**.

Most skeletal muscles are arranged in opposing pairs. One muscle in a pair moves a limb in one direction; the other muscle moves it in the opposite direction. Muscles move bones by pulling them, not by pushing them. For example, when the biceps muscle contracts, the elbow bends. The biceps muscle is known as a **flexor**, a muscle that bends a joint. Contraction of the triceps muscle in the upper arm straightens the limb. The triceps muscle is an example of an **extensor**, a muscle that straightens a joint. To bring about a smooth movement, one muscle in a pair must contract while the opposing muscle relaxes.

FIGURE 45-13

Skeletal muscles, such as the biceps and triceps muscles in the upper arm, are connected to bones by tendons. (a) When the biceps muscle contracts, the elbow bends. (b) When the triceps muscle contracts, the elbow straightens.





MUSCLE FATIGUE

Muscle cells store glycogen, which is used as a source of energy when the blood cannot deliver adequate amounts of glucose. The breakdown of glycogen releases large amounts of energy, but sometimes even those reserves are used up. During prolonged and vigorous exertion, fat molecules are utilized for energy. Fat molecules contain a greater concentration of potential energy than any other molecule in the body. When energy availability fails to keep pace with its use, muscle fatigue sets in and controlled muscle activity ceases, even though the muscle may still receive nerve stimulation to move. **Muscle fatigue** is the physiological inability of a muscle to contract. Muscle fatigue is a result of a relative depletion of ATP. When ATP is absent, a state of continuous contraction occurs. An example of depletion of ATP is when a marathon runner collapses during a race, suffering from severe muscle cramps.

FIGURE 45-14

These athletes are in the process of repaying their oxygen debts. Oxygen debt occurs frequently after strenuous, sustained exertion.



Oxygen Debt

Oxygen is used during cellular respiration in the synthesis of ATP. Large amounts of oxygen are needed to maintain the rate of maximum ATP production required to sustain strenuous exercise. However, after several minutes of heavy exertion, the circulatory system and the respiratory system are not able to bring in enough oxygen to meet the demands of energy production. Oxygen levels in the body become depleted. This temporary lack of oxygen availability is called **oxygen debt**. Oxygen debt leads to an accumulation of lactic acid as metabolic waste in the muscle fibers. The presence of lactic acid produces the soreness you may experience after prolonged exercise. Oxygen debt causes a person to spend time in rapid, deep breathing after strenuous exercise, as the athletes shown in Figure 45-14 are doing. The oxygen debt is repaid quickly as additional oxygen becomes available, but muscle soreness may persist until all of the metabolic wastes that have accumulated in the muscle fibers are carried away or converted.

SECTION 3 REVIEW

1. Compare the three main types of muscle tissues found in the body.
2. Why is smooth muscle referred to as involuntary muscle?
3. Why do skeletal muscle fibers appear striated?
4. How do skeletal muscles contract?
5. How do muscles work together to move bones?
6. Contrast the functions of flexors and extensors.
7. What causes muscles to become fatigued?

CRITICAL THINKING

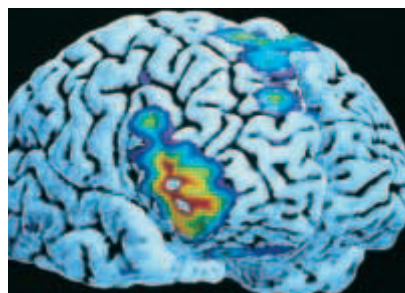
8. **Analyzing Information** Rigor mortis is a condition in which all of the body muscles become rigid shortly after a person dies. Why does rigor mortis develop?
9. **Applying Information** What causes muscle cramping after vigorous exercise or repeated movement?
10. **Applying Information** Why do you think the heart muscle never suffers from fatigue?

Looking Inside the Human Body

In 1895, the development of X-ray equipment provided physicians a way to look at images of dense tissue in the body, such as bones. Modern imaging techniques rely on computers.

Computerized Tomography

Computerized tomography (CT) uses a focused beam of low-dose X-rays to obtain cross-sectional images of structures in the body. Tomography is the technique used to take images of a specific "slice" or plane of tissue. Computerized tomography can differentiate tissues of various densities.



(a)



(b)

(a) A three dimensional PET scan shows the surface detail and cellular activity of the brain. (b) This X-ray shows the bones of the skull, mandible, and neck.

Positrons and Brain Imaging

Still newer imaging technology is positron emission technology, or PET. Positrons are positively charged particles with the same mass as electrons that result from the disintegration of radioisotopes. Michael M. TerPogossian and his colleagues at Johns Hopkins suggested using short-lived radioisotopes in medical research. They developed scanners to detect the positrons released by radioisotopes that had been injected into a patient's bloodstream. As technology improved, biomedical engineers redesigned scanning equipment to create three-dimensional images from the positron emissions. Positron emission technology is used to map areas of the brain that are involved in memory, sensation, perception, speech, and information processing. In addition, positron emission technology provides clues about the causes of psychiatric disorders, such as depression.

Holographic Imaging

A new holographic imaging system combines images obtained by computerized tomography or magnetic resonance scanners and displays an accurate three-dimensional image of anatomical structures. Magnetic resonance imaging (MRI) creates images of soft tissues. MRI uses radio waves emitted by the nuclei of hydrogen atoms that are activated by a magnetic scanner. A holograph is a method of photography that uses laser light to produce

a three-dimensional image. The transparent but solid looking image floats in front of the holographic film and can be studied from all sides.

Holograms in Medicine

Physicians can make surgical plans by studying the true appearance of a patient's organs, such as the three dimensional PET scan that highlights the verbal center in the brain (a). Compare the three dimensional PET scan with the X-ray of the same part of the body (b). The X-ray may not be as helpful as the three dimensional PET scan when a physician must diagnose an illness or injury of the brain.

REVIEW

1. How do X-rays differ from PET scans?
2. How might a holograph be more useful than a PET scan?
3. **Justifying Conclusions** If a surgeon had to remove a piece of metal lodged in a patient's skull, which type of imaging would you choose? Support your answer, and give reasons why you did not choose the other imaging techniques.



SECTION 4

OBJECTIVES

- **Describe** the functions of the skin.
- **Distinguish** between the two layers that form the skin.
- **Identify** two types of glands found in the skin, and describe their functions.
- **Describe** the structure of nails.
- **Describe** the structure of hair.

VOCABULARY

epidermis
keratin
melanin
dermis
exocrine gland
sweat gland
oil gland
sebum

Word Roots and Origins

Epidermis

from the Greek *epi*, meaning "on" or "upon," and *derma*, meaning "skin"

INTEGUMENTARY SYSTEM

The integumentary system, consisting of the skin, hair, and nails, acts as a barrier to protect the body from the outside world. It also functions to retain body fluids, protect against disease, eliminate waste products, and regulate body temperature.

SKIN

The skin is one of the human body's largest organs. Subjected to a lifetime of wear and tear, the layers of skin are capable of repairing themselves. Skin contains sensory receptors that monitor the external environment, and mechanisms that rid the body of wastes. The skin is composed of two layers—the epidermis and the dermis.

Epidermis

The **epidermis**, or outer layer of skin, is composed of many sheets of flattened, scaly epithelial cells. Its top layers are made of mostly dead cells. These cells are exposed to the dangers of the external environment. Scraped or rubbed away on a daily basis, they are replaced by new cells made in the rapidly dividing lower layers. The cells of the epidermis are filled with a protein called **keratin**, which gives skin its rough, leathery texture and its waterproof quality.

There is a great variety in skin color among humans. The color of skin is mainly determined by a brown pigment called **melanin** (MEL-uh-nin), which is produced by cells in the lower layers of the epidermis. Melanin absorbs harmful ultraviolet radiation. The amount of melanin produced in skin depends on two factors: heredity and the length of time the skin is exposed to ultraviolet radiation. Increased amounts of melanin are produced in a person's skin in response to ultraviolet radiation. All people, but especially people with light skin, need to minimize exposure to the sun and protect themselves from its ultraviolet radiation, which can damage the DNA in skin cells and lead to deadly forms of skin cancer.

Dermis

The **dermis**, the inner layer of skin, is composed of living cells and specialized structures, such as sensory neurons, blood vessels, muscle fibers, hair follicles, and glands. Sensory neurons make it possible for you to sense many kinds of conditions and signals from the environment, such as heat and pressure. Blood vessels provide nourishment to the living cells and help regulate body temperature.

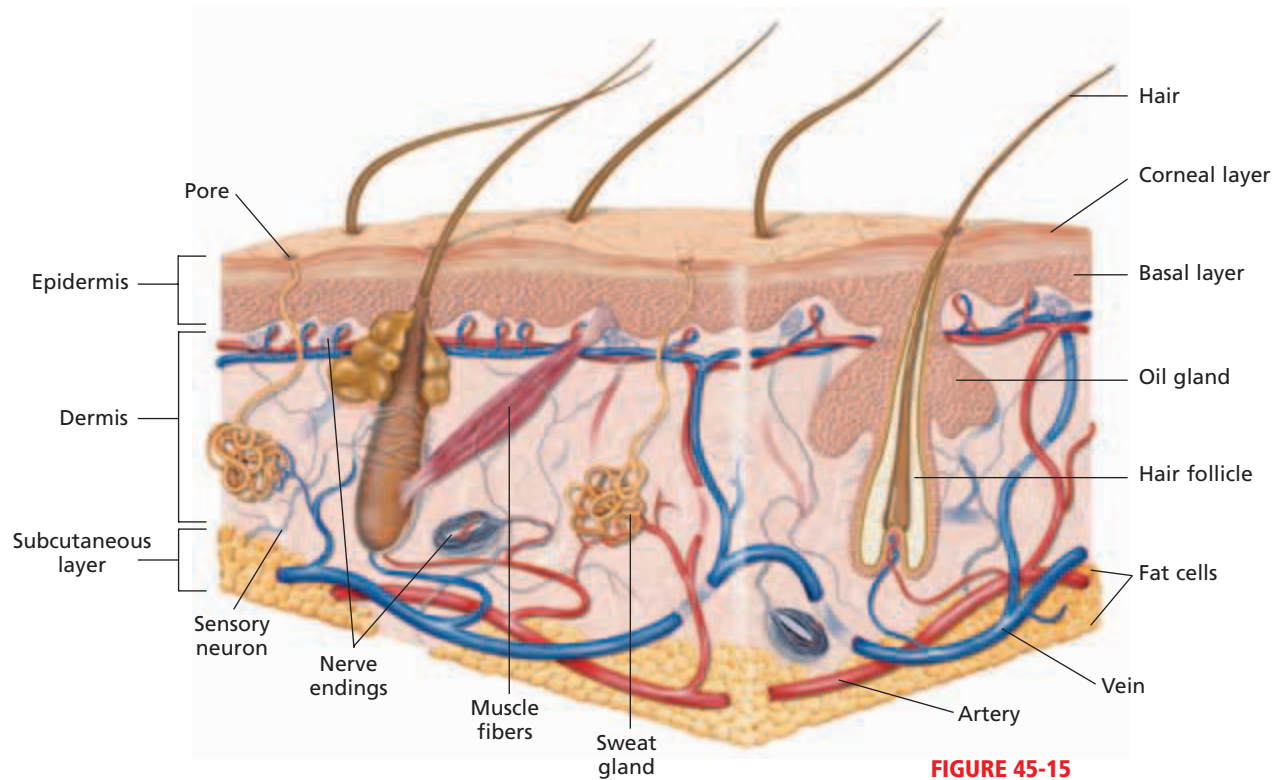


FIGURE 45-15

Skin is composed of two layers: the epidermis and the dermis. The top of the epidermis consists of dead, flattened cells that are shed and replaced every day. The dermis contains specialized structures that protect the body from infectious diseases, regulate body temperature, sense the environment, and secrete oil, sweat, and some wastes.

Tiny muscle fibers attached to hair follicles contract and pull hair upright when you are cold or afraid, producing what are commonly called goose bumps. Glands produce sweat, which helps cool your body, and oil, which helps soften your skin. A layer of fat cells lies below the dermis. These cells act as energy reserves; add a protective, shock-absorbing layer; and insulate the body against heat loss. Study the structures of the skin in Figure 45-15.

Glands

The skin contains **exocrine glands**, glands that release secretions through ducts. The main exocrine glands of the skin are the sweat glands and the oil glands.

The skin functions as an excretory organ by releasing excess water, salts, and urea through the **sweat glands**. By releasing excess water, the skin also helps regulate body temperature. When the body's temperature rises, circulation increases, and the skin becomes warm and flushed, as shown in Figure 45-16. The sweat glands then release more sweat. As the water in sweat evaporates, the skin is cooled.

Oil glands, found in large numbers on the face and scalp, secrete the fatty substance **sebum**. Oil glands are usually connected by tiny ducts to hair follicles. Sebum coats the surface of the skin and the shafts of hairs, preventing excess water loss and softening the skin and hair. Sebum is also mildly toxic to some bacteria. The production of sebum is controlled by hormones. During adolescence, high levels of sex hormones increase the activity of the oil glands. If the ducts of oil glands become clogged with excessive amounts of sebum, dead cells, and bacteria, the skin disorder *acne* can result.

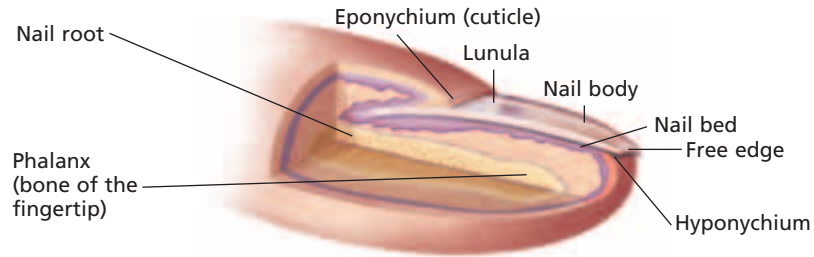
FIGURE 45-16

Skin acts as a temperature-controlling device. It contains millions of sweat glands that secrete microscopic droplets of water. The water droplets help cool the body when its temperature rises, such as after a rigorous workout.



FIGURE 45-17

This illustration of the structure of a fingernail shows that the nail root, from which the nail is constantly regenerated, is protected well beneath the surface of the finger, next to the bone of the fingertip.



NAILS

Nails, which protect the ends of the fingers and toes, form from nail roots under skin folds at the base and sides of the nail. As new cells form, the nail grows longer. Like hair, nails are composed primarily of keratin. The nail body is about 0.5 mm (0.02 in.) thick. Nails grow at about 1 mm (0.04 in.) per week. Nails rest on a bed of tissue filled with blood vessels, giving the nails a pinkish color. The structure of a fingernail can be seen in Figure 45-17.

Changes in the shape, structure, and appearance of the nails may be an indicator of a disease somewhere in the body. They may turn yellow in patients with chronic respiratory disorders, or they may grow concave in certain blood disorders.

HAIR

Hair, which protects and insulates the body, is produced by a cluster of cells at the base of deep dermal pits called *hair follicles*. The hair shaft is composed of dead, keratin-filled cells that overlap like roof shingles. Oil glands associated with hair follicles prevent hair from drying out. Most individual hairs grow for several years and then fall out.

Hair color is the result of the presence of the pigment melanin in the hair shaft. Black, brown, and yellow variants of melanin combine to determine an individual's hair color. Hair color is influenced by hereditary factors.

SECTION 4 REVIEW

1. What are the names and functions of the two layers of skin?
2. Identify the reason the dermis is considered the living layer of skin.
3. What are the functions of the two types of exocrine glands found in the dermis?
4. Illustrate and label the structure of a fingernail.
5. Describe the structure of hair.

CRITICAL THINKING

6. **Relating Concepts** Why can sunbathing be considered dangerous?
7. **Analyzing Information** A third-degree burn may be surrounded by painful areas of second- and first-degree burns. However, a third-degree burn is often painless. Why?
8. **Recognizing Relationships** How might muscles in the dermis benefit mammals in cold weather?

CHAPTER HIGHLIGHTS

SECTION 1

The Human Body Plan

- The human body has four main types of tissue: muscle, nervous, epithelial, and connective.
- A tissue is a collection of cells, an organ is a collection of tissues, and an organ system is a collection of organs.
- Some of the primary organ systems in the body include the integumentary, nervous, and cardiovascular systems.
- Many organs are located in the body's five main cavities: abdominal, cranial, spinal, thoracic, and pelvic.

Vocabulary

muscle tissue (p. 907)
skeletal muscle (p. 907)
smooth muscle (p. 907)
cardiac muscle (p. 907)

nervous tissue (p. 907)
neurons (p. 907)
epithelial tissue (p. 908)
connective tissue (p. 908)

matrix (p. 908)
organ (p. 909)
cranial cavity (p. 910)
spinal cavity (p. 910)

diaphragm (p. 910)
thoracic cavity (p. 910)
abdominal cavity (p. 910)
pelvic cavity (p. 910)

SECTION 2

Skeletal System

- The human skeleton is composed of the axial skeleton (skull, ribs, spine, and sternum) and the appendicular skeleton (arms and legs, scapula, clavicle, and pelvis).
- Bones support muscles, give structure to the body, protect organs, store minerals, and make blood cells.
- Bones are made up of minerals, protein fibers, and cells.
- Most bones develop from cartilage through a process called *ossification*.
- The human body has three types of joints: fixed joints, semimovable joints, and movable joints. The joints can be affected by a disease called arthritis.

Vocabulary

skeleton (p. 911)
axial skeleton (p. 911)
appendicular skeleton (p. 911)
periosteum (p. 912)
compact bone (p. 913)

Haversian canal (p. 913)
osteocyte (p. 913)
spongy bone (p. 913)
bone marrow (p. 913)
fracture (p. 913)

ossification (p. 914)
epiphyseal plate (p. 914)
joint (p. 915)
fixed joint (p. 915)
semimovable joint (p. 915)

movable joint (p. 915)
ligament (p. 915)
synovial fluid (p. 915)
rheumatoid arthritis (p. 916)
osteoarthritis (p. 916)

SECTION 3

Muscular System

- The human body has three types of muscle tissues: skeletal, smooth, and cardiac.
- Skeletal muscles consist of groups of fibers. Muscle fibers contain myofibrils made up of protein filaments.
- During a muscle contraction, myosin and actin filaments interact to shorten the length of a sarcomere.
- Most skeletal muscles are arranged in opposing pairs.

Vocabulary

muscle fiber (p. 917)
striation (p. 917)
voluntary muscle (p. 917)
involuntary muscle (p. 918)

myofibril (p. 919)
myosin (p. 919)
actin (p. 919)
Z line (p. 919)

sarcomere (p. 919)
tendon (p. 921)
origin (p. 921)
insertion (p. 921)

flexor (p. 921)
extensor (p. 921)
muscle fatigue (p. 922)
oxygen debt (p. 922)

SECTION 4

Integumentary System

- Skin, hair, and nails act as barriers that protect the body from the environment.
- Skin is composed of two layers, which are the epidermis and the dermis.
- Hair and nails are composed of the protein keratin; they grow from a root of rapidly dividing cells.
- Sweat glands produce sweat, which cools the body. Oil glands secrete sebum, which softens the skin.

Vocabulary

epidermis (p. 924)
keratin (p. 924)

melanin (p. 924)
dermis (p. 924)

exocrine gland (p. 925)
sweat gland (p. 925)

oil gland (p. 925)
sebum (p. 925)

CHAPTER REVIEW

USING VOCABULARY

1. Choose the term that does not belong in the following group, and explain why it does not belong: *saddle joint*, *pivot joint*, *fixed joint*, and *hinge joint*, and *ball-and-socket joint*.
2. Distinguish between *compact bone* and *spongy bone*.
3. Use the following key terms in the same sentence: *actin*, *muscle fiber*, *myofibrils*, and *myosin*.
4. **Word Roots and Origins** The word *epidermis* is derived from the Greek *derma*, which means “skin.” The prefix *epi* means “on.” Using this information, explain why the term *epidermis* is a good name for the anatomical structure it describes.

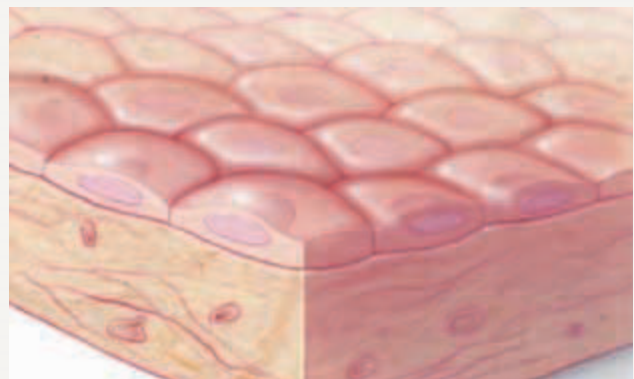
UNDERSTANDING KEY CONCEPTS

5. **Define** epithelial tissue.
6. **Explain** the relationship between cells, tissues, organs, and organ systems.
7. **Summarize** the functions of the primary organ systems in the human body.
8. **Describe** the organs that can be found in the abdominal cavity.
9. **List** all of the bones in the axial skeleton.
10. **Identify** the five functions of the skeletal system.
11. **Explain** the role the Haversian canals play in compact bone.
12. **Define** red bone marrow. Where is it produced, and what is its function?
13. **Summarize** how bones develop and elongate.
14. **State** three types of joints, and give examples of each type.
15. **Describe** the cause and symptoms of the disease rheumatoid arthritis.
16. **Explain** the difference between skeletal muscle, smooth muscle, and cardiac muscle.
17. **Describe** the components of a sarcomere.
18. **Illustrate** how a skeletal muscle contracts.
19. **Explain** how muscles move bones.
20. **Name** the functions of tendons and ligaments.
21. **List** four functions of the skin.
22. **Identify** the difference between the epidermis and the dermis.
23. **Define** melanin. What is its role in the body?

24. **Explain** the similarities and differences between nails and hair.
25. **Identify** the substance that prevents the hair and skin from drying out, and the gland where this substance is produced.
26.  **CONCEPT MAPPING** Use the following terms to create a concept map that illustrates the body's four levels of structural organization: *muscle tissue*, *connective tissue*, *epithelial tissue*, *nervous tissue*, *organ*, and *organ system*.

CRITICAL THINKING

27. **Inferring Relationships** Young thoroughbred horses that are raced too early in life have an increased risk of breaking the bones in their legs. What can you infer about the process of ossification in horses?
28. **Evaluating Information** During a normal birth, a baby passes through the mother's pelvis. A woman's pelvis has a larger diameter and is more oval-shaped than a man's pelvis. In addition, a newborn's skull bones are not completely ossified. How are these skeletal properties advantageous to the birthing process?
29. **Analyzing Concepts** Oil glands secrete an oily substance that helps keep the skin soft and flexible. They also secrete fatty acids, which help kill bacteria. How can the function of oil glands be affected if you wash your skin too frequently?
30. **Interpreting Graphics** Examine the drawing of epithelial cells below. The flat epithelial cells of the skin overlap each other much like shingles on a roof do. How does this arrangement enable these cells to perform their protective function?





Standardized Test Preparation

DIRECTIONS: Choose the letter that best answers the question or completes the sentence.

- The thoracic cavity contains which organs?
 - brain
 - spine
 - organs of the digestive system
 - organs of the respiratory system
- The cells of connective tissue are embedded in what substance?
 - matrix
 - keratin
 - marrow
 - synovial fluid
- The periosteum is a membrane that does which of the following?
 - covers the bone
 - contains marrow
 - produces red blood cells
 - increases the length of long bones
- Which of the following is true about the dermis?
 - It is the top layer of skin.
 - It contains cardiac muscle.
 - It is made up of dead cells.
 - It contains nerves and blood vessels.

INTERPRETING GRAPHICS: The graph below shows the relationship between skin type, UV index, and sunburns. Use the table to answer the question that follows.

Relationship of UV Index and Sunburns

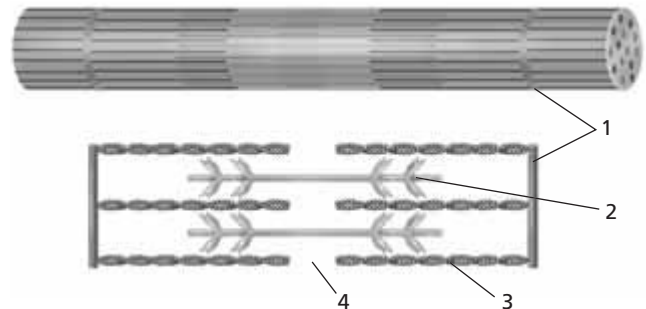
UV index	Minutes before Skin Type 1 burns	Minutes before Skin Type 4 burns
0–2	30	> 120
3	20	90
5	12	60
7	8.5	40
9	7	33

- Which of the following statements about skin type 1 is true?
 - Skin type 4 will never sunburn.
 - Skin type 1 will always burn in less than 20 minutes.
 - Skin type 1 is less sensitive to UV exposure than skin type 4 is.
 - Skin type 1 is more sensitive to UV exposure than skin type 4 is.

DIRECTIONS: Complete the following analogy.

- nerve : neuron :: bone :
 - marrow
 - skeleton
 - osteocyte
 - Haversian canal

INTERPRETING GRAPHICS: The figure below shows a sarcomere and an enlargement of actin and myosin filaments. Use the figure to answer the question that follows.



- Which part of the sarcomere represents the Z line?
 - feature 1
 - feature 2
 - feature 3
 - feature 4

SHORT RESPONSE

Red bone marrow inside spongy bone produces red blood cells, which are specialized cells used to carry oxygen throughout the body.

How are red blood cells transported around the body?

EXTENDED RESPONSE

A single layer of smooth muscle encircles the walls of blood vessels. The walls of the stomach and small intestine have a layer of circular smooth muscle and a layer of longitudinal smooth muscle.

Part A How does the muscle arrangement of blood vessels reflect the function of this structure?

Part B How does the muscle arrangement of the stomach and small intestine reflect the function of these structures?

Test TIP

For a question involving experimental data, determine the constants, variables, and control **before** answering the questions.

Dehydrating and Demineralizing Bone

OBJECTIVES

- Determine the amount of water and minerals in bone.
- Identify structures in bone cells.

PROCESS SKILLS

- observing
- identifying
- calculating

MATERIALS

- balance
- beaker, 250 mL
- beakers, 500 mL (2)
- bones (2)
- bone slides, prepared
- drying oven
- gauze, circular piece
- glass plate or parafilm
- hot pad
- hydrochloric acid, 1 M (300 mL)
- lens paper
- marker, permanent
- microscope, compound
- pencil, wax
- plastic bag, resealable
- specimen tag
- tongs

Background

1. Dehydration is the process of removing the water from a substance.
2. Demineralization is the process of removing the minerals from a substance.

PART A Dehydrating a Bone

1. In your lab report, prepare a data table similar to Table A.
2.    Put on safety goggles, a lab apron, and gloves. Wear this protective gear during all parts of this investigation.

TABLE A DEHYDRATION OF BONE

Mass before drying	Mass after drying	Percentage of bone mass lost

3. Obtain a bone from your teacher. Test the flexibility of the bone by trying to bend and twist it.
4. Place the bone on a balance. Measure the mass of the bone to the nearest 0.1 g, and record it in your data table. Then, use a permanent marker to write the initials of each member of your group on a specimen tag, and tie the tag to the bone.
5. Place the bone in a drying oven at 100°C for 30 minutes. While the bone is in the oven, complete Part C.
6.  **CAUTION** Do not touch hot objects with your bare hands. Use insulated gloves and tongs as appropriate. Using tongs, remove the bone from the oven and place it on a heat-resistant pad to cool for 10 minutes.
7. Use tongs to place the cooled bone on the balance. Measure the mass of the bone to the nearest 0.1 g, and record it in your data table.
8. Use the equation below to calculate the percentage of the bone's mass that was lost during heating.

Percentage mass lost =

$$\frac{\text{mass before heating} - \text{mass after heating}}{\text{mass before heating}} \times 100$$

PART B Demineralizing a Bone

9. In your lab report, prepare a data table similar to Table B.
10. Obtain a second bone from your teacher. Test the flexibility of the bone by trying to bend and twist it.

TABLE B DEMINERALIZATION OF A BONE

Mass before demineralizing	Mass after demineralizing and drying	Percentage of bone mass lost

11. Place the bone on a balance. Measure the mass of the bone, and record it in your data table.
12.  **CAUTION** Glassware is fragile. Notify your teacher promptly of any broken glass or cuts. Do not clean up broken glass or spills unless your teacher tells you to do so. Using a wax pencil, label a 500 mL beaker "1 M HCl." Also label the beaker with the initials of all group members. Place a piece of gauze in the bottom of the beaker.
13.  **CAUTION** If you get an acid on your skin or clothing, wash it off at the sink immediately while calling to your teacher. Place the bone on top of the gauze in the beaker, and add enough 1 M HCl to cover the bone. Use a glass plate or parafilm to cover the beaker.
14. Place the beaker under a fume hood, and allow the bone to soak in the acid until it softens and becomes spongy. This should take 5 to 7 days. Periodically use tongs to test the hardness of the bone. *Note: Do not touch the bone with your fingers while it is soaking in acid. Rinse the tongs with water thoroughly each time you finish testing the bone.*
15. When the bone becomes spongy, use tongs to carefully remove it from the beaker, and rinse it under running water for two minutes.
16. After the bone has been thoroughly rinsed, test the bone for hardness by twisting and bending it with your fingers. *Note: Be sure you are wearing gloves.*
17. Then, use a permanent marker to write the initials of each member of your group on a specimen tag, and tie the tag to the bone. Place the bone in a drying oven at 100°C for 30 minutes.
18.  **CAUTION** Do not touch hot objects with your bare hands. Use insulated gloves and tongs as appropriate. Using tongs, remove the bone from the oven and place it on a heat-resistant pad. Allow the bone to cool for 10 minutes.

19. Use tongs to place the cooled bone on the balance. Measure the mass of the bone to the nearest 0.1 g, and record the measurement in your data table.
20. Use the equation below to calculate the percentage of the bone's mass that was lost through demineralization and dehydration.

Percentage of mass lost =

$$\frac{\text{mass before demineralizing} - \text{mass after demineralizing and drying}}{\text{mass before demineralizing}} \times 100$$

PART C Observing Prepared Slides of Bone

21.  **CAUTION** Do not use electrical equipment near water or with wet hands or clothing. Using a compound light microscope, focus on a prepared slide of bone by using low power, and then switch to high power. Locate a Haversian canal, the darkly stained circle in the center of a set of lamellae. Find the darkly stained osteocytes between the lamellae.
22. In your lab report, draw and label the following bone structures: Haversian canal, lamella, and osteocyte.

Analysis and Conclusions

1. What effect did water loss have on the bone? What effect did mineral loss have on the bone?
2. Why did you have to dehydrate the bone before measuring its mass in Part B?
3. What percentage of bone is water? What percentage of bone is made of minerals?
4. If you were to prepare a slide using the dehydrated and demineralized bone, what do you think the bone would look like?
5. What happened when the demineralized bone was dried? Why do you think this happened?
6. If a person's diet lacked calcium, how could this affect his or her bones?

Further Inquiry

Research the differences in the amount and distribution of different bone types from various parts of the human skeleton.

CIRCULATORY AND RESPIRATORY SYSTEMS



This photograph shows the air sacs of a human lung. (SEM 780×)

SECTION 1 *The Circulatory System*

SECTION 2 *Blood*

SECTION 3 *The Respiratory System*

THE CIRCULATORY SYSTEM

Most of the cells in the human body are not in direct contact with the external environment. The circulatory system acts as a transport service for these cells. Two fluids move through the circulatory system: blood and lymph.

THE HEART

The blood, heart, and blood vessels form the **cardiovascular system**. The lymph, lymph nodes, and lymph vessels form the **lymphatic system**. The cardiovascular system and lymphatic system collectively make up the *circulatory system*. The circulatory system transports nutrients, hormones, and gases; gets rid of wastes; and helps maintain a constant body temperature.

The central organ of the cardiovascular system is the heart, the muscular organ that pumps blood through a network of blood vessels. The heart beats more than 2.5 billion times in an average life span. Yet this organ is slightly larger than a fist. The heart lies within the thoracic (chest) cavity, behind the sternum (breastbone) and between the two lungs. A tough, saclike membrane called the *pericardium* surrounds the heart and secretes a fluid that reduces friction as the heart beats.

Notice in Figure 46-1 that a *septum* (wall) vertically divides the heart into two sides. The right side pumps blood to the lungs, and the left side pumps blood to the other parts of the body. Each side of the heart is divided into an upper and lower chamber. Each upper chamber is called an **atrium**, and each lower chamber is called a **ventricle**.

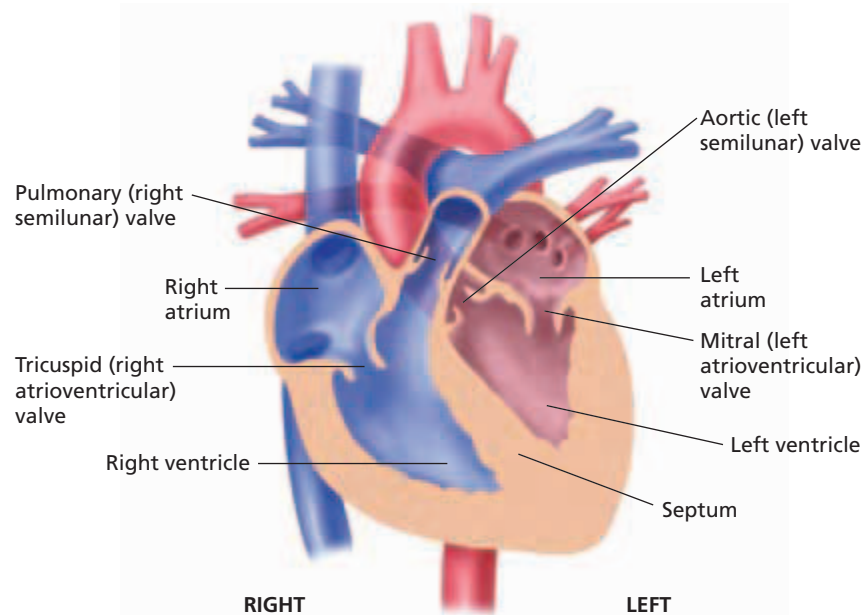


FIGURE 46-1

The septum prevents mixing of blood from the two sides of the heart, and the valves ensure that blood flows in only one direction.

OBJECTIVES

- **Describe** the structure and function of the human heart.
- **Trace** the flow of blood through the heart and body.
- **Distinguish** between arteries, veins, and capillaries in terms of their structure and function.
- **Distinguish** between pulmonary circulation and systemic circulation.
- **Summarize** the functions of the lymphatic system.

VOCABULARY

cardiovascular system

lymphatic system

atrium

ventricle

valve

aorta

sinoatrial node

atrioventricular node

pulse

artery

blood pressure

hypertension

capillary

vein

pulmonary circulation

systemic circulation

atherosclerosis

lymph



Quick Lab

Determining Heart Rate

Materials stopwatch or clock with second hand

Procedure

1. Have your partner find the pulse in your wrist and count your heartbeats for 15 seconds while you are seated. Calculate your resting heart rate in beats per minute.
2. Have your partner count your heartbeats for 15 seconds while you are standing. Calculate your heart rate in beats per minute.
3. Have your partner count your heartbeats for 15 seconds after you jog or march in place for 1 minute. Calculate your heart rate in beats per minute.

Analysis What causes your pulse? What causes the change in your heart rate in each situation?

Valves are flaps of tissue that open in only one direction. The *atrioventricular* (AY-tree-oh-ven-TRIH-kyuh-luhr) *valve* (AV valve) on the right side of the heart is called the *tricuspid valve*. The *mitral valve*, also called the *bicuspid valve*, is on the left. As the ventricles pump, blood pressure closes the AV valves to prevent blood from flowing backward into the atria. From the ventricles, blood is pumped out of the heart into large vessels. A *semilunar* (SEM-ee-LOON-uhr) *valve* (SL valve) separates the ventricles from these large vessels on each side of the heart. The SL valve on the right side is known as the *pulmonary valve*, and the SL valve on the left side is known as the *aortic valve*. The SL valves prevent blood from flowing back into the ventricles when the heart relaxes.

Circulation in the Heart

Refer to Figure 46-2 to trace the path of the blood as it circulates through the heart. Blood returning to the heart from parts of the body other than the lungs has a high concentration of carbon dioxide and a low concentration of oxygen. **1** Deoxygenated (O_2 -poor) blood enters the right atrium.

2 The right atrium sends deoxygenated blood into the right ventricle. **3** The muscles of the right ventricle contract and force the blood into the pulmonary arteries. **4** The pulmonary artery sends the blood to the lungs. In the lungs, the carbon dioxide diffuses out of the blood, and oxygen diffuses into the blood. **5** The oxygenated blood returns to the left atrium of the heart. Notice in Figure 46-2 that the flow of blood on the left side of the heart is illustrated with a red arrow representing oxygenated blood, which has a bright red color.

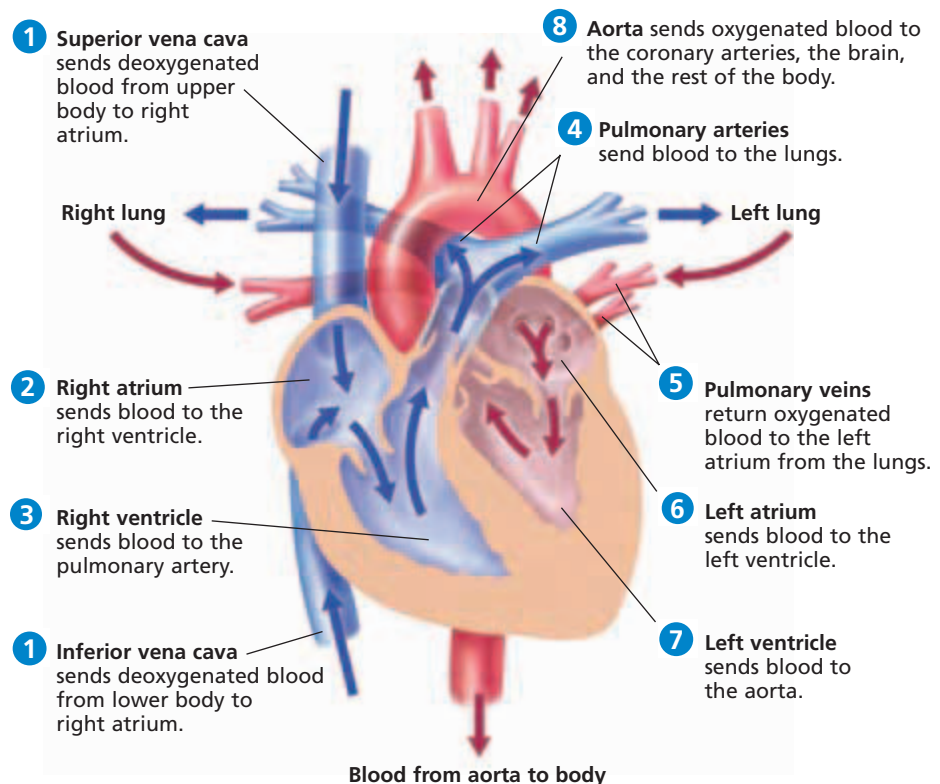


FIGURE 46-2

Trace the path of blood through the heart. Notice that illustrations of a heart are drawn as if the heart were in a person facing you. That is, the left side of the heart is shown on the right as you face the heart, and the right side of the heart is on the left as you face the heart.

6 The oxygenated blood is then pumped into the left ventricle. 7 Contraction of the muscular walls of the left ventricle forces the blood into a large blood vessel called the **aorta**. 8 From the aorta, blood is transported to all parts of the body. The left ventricle is the thickest chamber of the heart because it has to do the most work to pump blood to all parts of the body.

Deoxygenated blood is commonly represented with the color blue. However, it is a misconception that deoxygenated blood is blue. When oxygen is attached to hemoglobin, the blood is bright red. Without oxygen, blood is dark red. The dark red blood in veins appears blue when it shows through the vein walls and skin.

Control of the Heartbeat

The heart consists of muscle cells that contract in waves. When the first group of cells are stimulated, they in turn stimulate neighboring cells. Those cells then stimulate more cells. This chain reaction continues until all the cells contract. The wave of activity spreads in such a way that the atria and the ventricles contract in a steady rhythm. The first group of heart-muscle cells that get stimulated lie in an area of the heart known as the sinoatrial node, shown in Figure 46-3.

The **sinoatrial** (SIEN-oh-AY-tree-uhl) (**SA**) **node** is a group of specialized heart-muscle cells located in the right atrium. These muscle cells spontaneously initiate their own electrical impulse and contract. The SA node is often called the *pacemaker* because it regulates the rate of contraction of the entire heart. The electrical impulse initiated by the SA node subsequently reaches another special area of the heart, known as the **atrioventricular (AV) node**. The AV node is located in the septum between the atria, as shown in Figure 46-3. The AV node relays the electrical impulse to the muscle cells that make up the ventricles. As a result, the ventricles contract a fraction of a second after the atria, completing one full heartbeat. In an average adult at rest, the heart beats about 70 times each minute.

A heartbeat has two phases. Phase one, called *systole* (SIS-tohl), occurs when the ventricles contract, closing the AV valves and opening the SL valves to pump blood into the two major vessels that exit the heart. Phase two, called *diastole* (DIE-a-stohl), occurs when the ventricles relax, allowing the back pressure of the blood to close the SL valves and opening the AV valves. The closing of these two heart valves results in the characteristic *lub dup* sound we call a heartbeat. If one of the valves fails to close properly, some blood may flow backward, creating a different sound, which is known as a heart murmur.

A person's **pulse** is a series of pressure waves within an artery caused by the contractions of the left ventricle. When the ventricle contracts, blood surges through the arteries, and the elastic walls in the vessels expand and stretch. The most common site for taking a pulse is at a radial artery, on the thumb side of each wrist. The average pulse rate ranges from 70 to 90 beats per minute for adults.

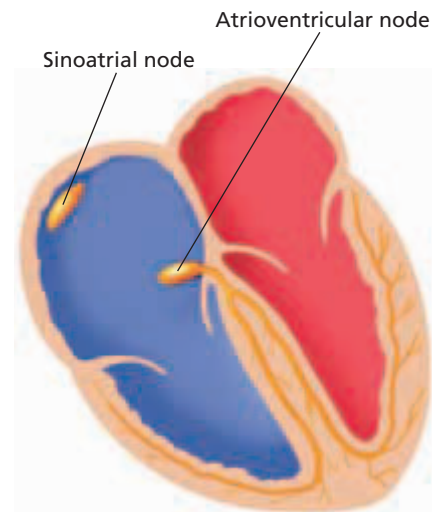
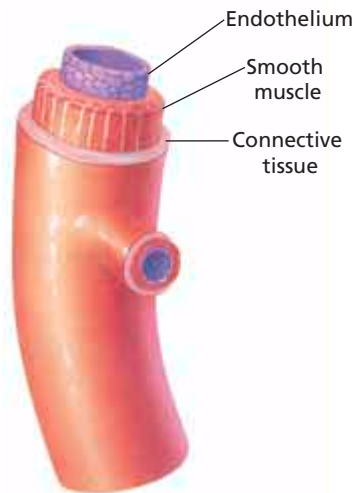


FIGURE 46-3

Two areas of specialized tissue, known as nodes, control the heartbeat. A person whose SA node is defective can have an operation to implant an artificial pacemaker. An artificial pacemaker can also help a defective AV node.



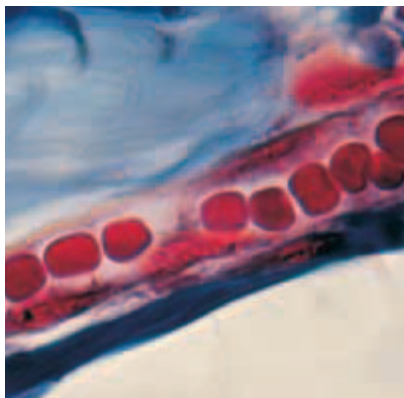
Artery
(carries blood away from the heart)

FIGURE 46-4

Notice the thick muscular layer of an artery. The layers of the artery wall are separated by elastic tissue. This tissue provides strength, preventing systolic pressure from bursting the artery.

FIGURE 46-5

The diameter of a capillary is so small that red blood cells must move single file through these vessels, as shown in this photograph (1,200 $\times$). All exchange of nutrients and waste between blood and cells occurs across the thin walls of the capillaries.



BLOOD VESSELS

The circulatory system is known as a closed system because the blood is contained within either the heart or the blood vessels at all times. This type of system differs from an open system, in which blood leaves the vessels and circulates within tissues throughout the organism's body. The blood vessels that are part of the closed circulatory system of humans form a vast network to help keep the blood flowing in one direction.

Arteries and Blood Pressure

The large, muscular vessels that carry blood away from the heart are called **arteries**. As shown in Figure 46-4, the thick walls of the arteries have three layers: an inner endothelial layer, a middle layer of smooth muscle, and an outer layer of connective tissue. This structure gives arteries a combination of strength and elasticity, which allows them to stretch as pressurized blood enters from the heart. You can feel this stretching of arteries—it is your pulse.

Contraction of the heart moves the blood through the arteries with great force. The force that blood exerts against the inside walls of a blood vessel is known as **blood pressure**. Blood pressure is highest in the two main arteries that leave the heart. It is usually measured in the artery that supplies blood to the arm. To measure blood pressure, a trained person inflates a cuff that is placed around a patient's arm, temporarily stopping the flow of blood through the artery. Connected to the cuff is a gauge containing a column of mercury (Hg) that rises as the pressure in the cuff increases. The trained person then releases the air in the cuff slowly while listening to the artery with a stethoscope and watching the column of mercury. The first sounds of blood passing through the artery indicates the *systolic pressure*, or the pressure of the blood when the ventricles contract. In a normal adult, the systolic pressure is about 120 mm of Hg for males and 110 mm of Hg for females. Continuing to release the air in the cuff, the trained person next listens for the disappearance of sound, which indicates a steady flow of blood through the artery in the arm. This indicates the *diastolic pressure*. In a normal adult, the diastolic pressure is about 80 mm of Hg for males and 70 mm of Hg for females.

High blood pressure, or **hypertension**, is a leading cause of death in many countries. Blood pressure that is higher than normal places a strain on the walls of the arteries and increases the chance that a vessel will burst.

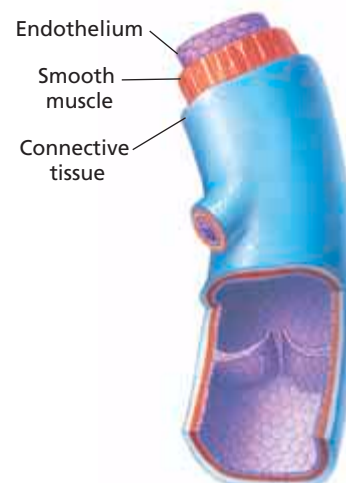
Capillaries and Veins

Recall that when the left ventricle contracts, it forces blood into the aorta, the body's largest artery. From the aorta, blood travels through a network of smaller arteries, which in turn divide and form even smaller vessels, called *arterioles*. The arterioles branch into a network of tiny vessels, called **capillaries**. A capillary is shown in Figure 46-5.

The network formed by capillaries is so extensive that all of the approximately 100 trillion cells in the body lie within about 125 μm of a capillary. This close association between capillaries and cells allows for rapid exchange of materials. Capillary walls are only one cell thick; gases and nutrients can diffuse through these thin walls. Wherever the concentration of oxygen or nutrients is higher in the blood than in the surrounding cells, the substance diffuses from the blood into the cells. Wherever the concentrations of carbon dioxide and wastes are higher in the cells than in the blood, these substances diffuse from the cells into the blood.

Blood flows through capillaries that merge to form larger vessels called *venules* (VEN-yoolz). Several venules in turn unite to form a **vein**, a large blood vessel that carries blood to the heart. Veins returning deoxygenated blood from the lower parts of the body merge to form the *inferior vena cava*. Veins returning deoxygenated blood from the upper parts of the body merge to form the *superior vena cava*. Refer back to Figure 46-2, and locate the inferior vena cava and the superior vena cava.

As you can see in Figure 46-6, although the walls of the veins are composed of three layers, like those of the arteries, they are thinner and less muscular. By the time blood reaches the veins, it is under much less pressure than it was in the arteries. With less pressure being exerted in the veins, the blood could flow backward and disrupt the pattern of circulation. To prevent that, valves in the veins help keep the blood flowing in one direction. Many veins pass through skeletal muscle. When these muscles contract, they are able to squeeze the blood through the veins. When these muscles relax, the valves can close, thus preventing the blood from flowing backward. Figure 46-6 shows the structure of a valve in a vein.



Vein
(returns blood to the heart)

FIGURE 46-6

Like an artery, a vein has three layers: the outer layer of connective tissue, the middle layer of smooth muscle, and the inner layer of endothelial tissue.

PATTERNS OF CIRCULATION

The English scientist William Harvey (1578–1657) first showed that the heart and the blood vessels form one continuous, closed system of circulation, as shown in Figure 46-7. He also reasoned that this system consists of two primary subsystems: **pulmonary circulation**, in which the blood travels between the heart and lungs, and **systemic circulation**, in which the blood travels between the heart and all other body tissues.

Pulmonary Circulation

Deoxygenated blood returning from all parts of the body except the lungs enters the right atrium, where it is then pumped into the right ventricle. When the right ventricle contracts, the deoxygenated blood is sent through the pulmonary artery to the lungs. The pulmonary artery is the only artery that carries deoxygenated blood. The pulmonary artery branches into two smaller arteries, with one artery going to each lung. These arteries branch into arterioles and then into capillaries in the lungs.



FIGURE 46-7

The cardiovascular system transports materials throughout the body and distributes heat.

FIGURE 46-8

The pulmonary circulation between the heart and the lungs involves the pulmonary arteries and the pulmonary veins. Deoxygenated blood flows from the right side of the heart to the lungs. Oxygenated blood is returned to the left side of the heart from the lungs. This is the opposite of systemic and coronary blood flow, in which oxygen-rich blood flows from the heart and oxygen-poor blood is returned to the heart.

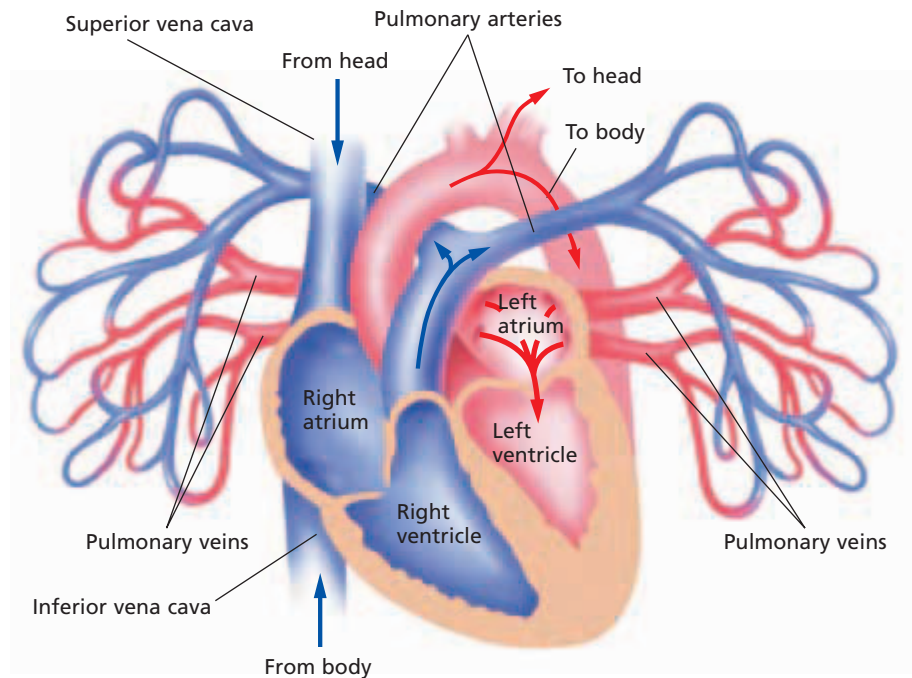
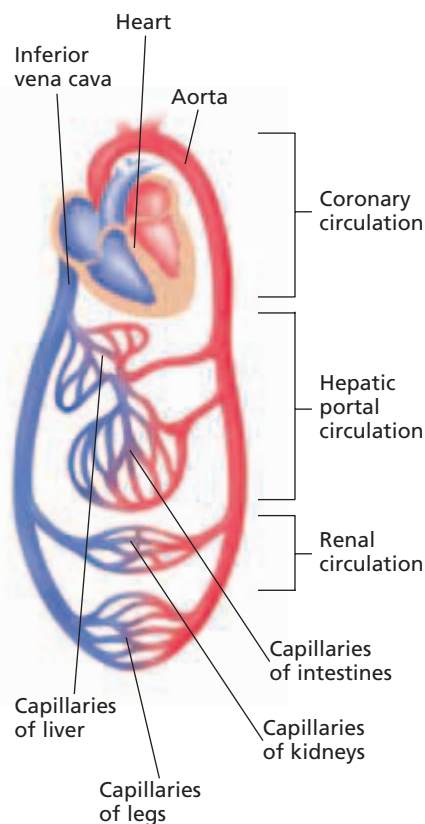


FIGURE 46-9

Notice three subsystems of systemic circulation. Other subsystems transport blood between the heart and the head, arms, and other organs.



In the lungs, carbon dioxide diffuses out of the capillaries and oxygen diffuses into the capillaries. The oxygenated blood then flows into venules, which merge into the *pulmonary veins* that lead to the left atrium of the heart. From the left atrium, blood is pumped into the left ventricle and then to the body through the aorta. In Figure 46-8, trace the path blood takes as it passes through pulmonary circulation.

Systemic Circulation

Systemic circulation is the movement of blood between the heart and all parts of the body except the lungs. Trace the path blood follows in systemic circulation in Figure 46-9. Notice that oxygenated blood is pumped out of the left ventricle and into the aorta. From the aorta, blood flows into other subsystems of systemic circulation.

Coronary circulation is the subsystem of systemic circulation that supplies blood to the heart itself. If blood flow in the *coronary arteries*, which supply blood to the heart, is reduced or cut off, muscle cells will die. This can happen when an artery is blocked by a blood clot or by **atherosclerosis** (ATH-uhr-oh-skler-OH-sis), a disease characterized by the buildup of fatty materials on the interior walls of the coronary arteries. Either type of blockage can lead to a *heart attack*.

Hepatic portal circulation is a subsystem of systemic circulation. Nutrients are picked up by capillaries in the small intestine and are transported by the blood to the liver. Excess nutrients are stored in the liver for future needs. The liver receives oxygenated blood from a large artery that branches from the aorta.

Renal circulation, another subsystem of systemic circulation, supplies blood to the kidneys. Nearly one-fourth of the blood that is pumped into the aorta by the left ventricle flows to the kidneys. The kidneys filter waste from the blood.

LYMPHATIC SYSTEM

In addition to the cardiovascular system, the circulatory system also includes the lymphatic system. One function of the lymphatic system is to return fluids that have collected in the tissues to the bloodstream. Fluids diffuse through the capillary walls just as oxygen and nutrients do. Some of these fluids pass into cells, some return to the capillaries, and some remain in the intercellular spaces.

Excess fluid in the tissues moves into the tiny vessels of the lymphatic system; this fluid is called **lymph**. Lymph vessels merge to form larger vessels. The lymph vessels are similar in structure to capillaries, and the larger lymph vessels are similar in structure to veins. However, an important difference exists between blood vessels and lymph vessels. Blood vessels form a complete circuit so that blood passes from the heart to all parts of the body and then back again to the heart. In contrast, lymph vessels form a one-way system that returns fluids collected in the tissues back to the bloodstream. In addition, the lymphatic system has no pump. Like the blood in veins, lymph must be moved through the vessels by the squeezing of skeletal muscles. Like veins, the larger lymph vessels have valves to prevent the fluid from moving backward.

Notice in Figure 46-10 that lymph vessels form a vast network that extends throughout the body. The lymph that travels in these vessels is a transparent yellowish fluid, much like the liquid part of the blood. As the lymph travels through these vessels on its way to the heart, it passes through small organs known as lymph nodes. Notice in Figure 46-10 that lymph nodes are like beads on a string. These nodes filter the lymph as it passes, trapping foreign particles, microorganisms, and other tissue debris. Lymph nodes also store *lymphocytes*, white blood cells that are specialized to fight disease. When a person has an infection, the nodes may become inflamed, swollen, and tender because of the increased number of lymphocytes.

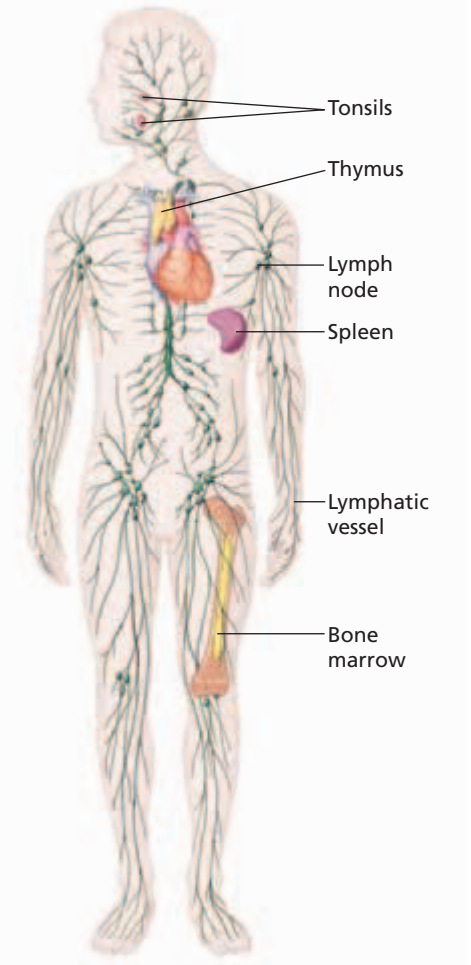


FIGURE 46-10

Like the cardiovascular system, the lymphatic system forms a vast network. Concentrated in certain regions of this network are lymph nodes that contain some of the disease-fighting cells of the immune system.

SECTION 1 REVIEW

1. Describe the structure of the heart.
2. Outline the path that blood follows through the heart and body, starting at the superior vena cava.
3. Describe the process by which the heartbeat is regulated.
4. How are the structures of arteries, veins, and capillaries related to their function?
5. Compare oxygenation levels in pulmonary circulation and systemic circulation.
6. Explain how the lymphatic system works with the cardiovascular system.

CRITICAL THINKING

7. **Analyzing Information** Some babies are born with a hole in the septum between the two atria. Based on what you know about blood flow through the heart, explain why this condition would be harmful to the baby.
8. **Forming Reasoned Opinions** In which blood vessels would you expect to find the lowest average blood pressure? Explain your answer.
9. **Applying Information** A man's arm is cut by a piece of glass. Blood comes out of the wound in rapid spurts. Which type of vessel was cut?

SECTION 2

OBJECTIVES

- **List** the components of blood.
- **Distinguish** between red blood cells, white blood cells, and platelets in terms of their structure and function.
- **Summarize** the process of blood clotting.
- **Explain** what determines the compatibility of blood types for transfusion.

VOCABULARY

plasma
red blood cell (erythrocyte)
hemoglobin
white blood cell (leukocyte)
phagocyte
antibody
platelet
fibrin
blood type
antigen
Rh factor

FIGURE 46-11

Notice that a mature red blood cell (RBC) is disk-shaped and is concave on both sides. A red blood cell is little more than a cell membrane filled with hemoglobin. How is this structure related to its function?



BLOOD

Blood is a liquid connective tissue. The two main functions of the blood are to transport nutrients and oxygen to the cells and to carry carbon dioxide and other waste materials away from the cells. Blood also transfers heat to the body surface and plays a major role in defending the body against disease.

COMPOSITION OF BLOOD

Blood is composed of a liquid medium and blood solids (formed elements). Formed elements include red blood cells, white blood cells, and platelets. The liquid makes up about 55 percent of the blood, and formed elements make up the remaining 45 percent. A healthy adult has about 4 to 5 L of blood in his or her body.

Plasma

Plasma, the liquid medium, is a sticky, straw-colored fluid that is about 90 percent water and includes metabolites, nutrients, wastes, salts, and proteins. Cells receive nourishment from dissolved substances carried in the plasma. These substances, which may include vitamins, minerals, amino acids, and glucose, are absorbed from the digestive system and transported to the cells. Plasma also carries hormones and brings wastes from the cells to the kidneys or the lungs to be removed from the body.

A variety of proteins are carried in the plasma. The plasma proteins have various functions. Some of the proteins in the plasma are essential for the formation of blood clots. Another protein, called albumin, plays an important role in the regulation of osmotic pressure between plasma and blood cells and between plasma and tissues. Other proteins, called antibodies, help the body fight disease.

Red Blood Cells

Red blood cells, or **erythrocytes** (uh-RITH-ruh-siets), shown in Figure 46-11, transport oxygen to cells in all parts of the body. Red blood cells are formed in the red marrow of bones. Immature red blood cells synthesize large amounts of an iron-containing protein called **hemoglobin**. Hemoglobin is the molecule that actually transports oxygen and, to a lesser degree, carbon dioxide. During the formation of a red blood cell, its cell nucleus and organelles disintegrate. The mature red blood cell is little more than a membrane containing hemoglobin.

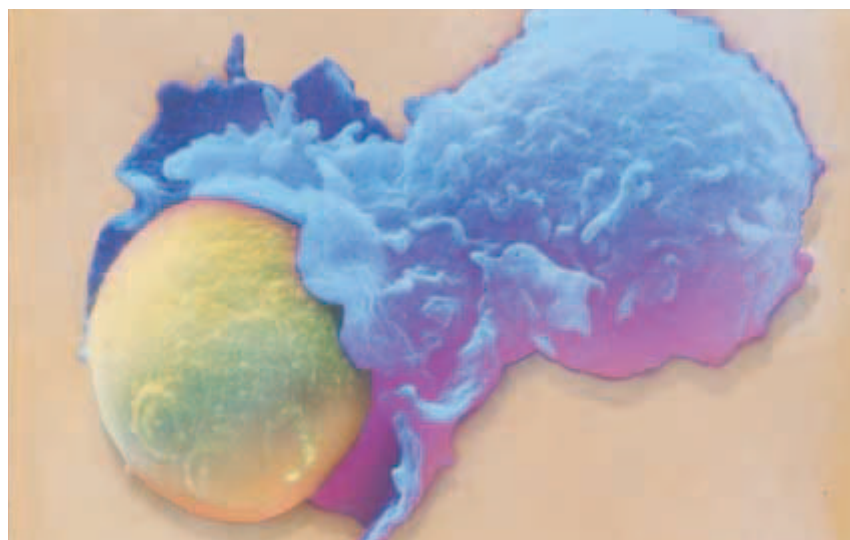
Because red blood cells lack nuclei, they cannot divide and they have a limited survival period, usually 120 to 130 days. Of the more than 30 trillion red blood cells circulating throughout the body at one time, 2 million disintegrate every second. To replace them, new ones form at the same rate in the red marrow of bones. Some parts of the disintegrated red blood cells are recycled. For example, the iron portion of the hemoglobin molecule is carried in the blood to the marrow, where it is reused in new red blood cells.

White Blood Cells

White blood cells, or **leukocytes** (LOO-kuh-siets), help defend the body against disease. They are formed in the red marrow, but some must travel to lymph nodes, tonsils, the thymus, or the spleen to mature. White blood cells are larger than red blood cells and significantly less plentiful. Each cubic millimeter of blood normally contains about 4 million red blood cells and 7,000 white blood cells. White blood cells can squeeze their way through openings in the walls of blood vessels and into the intercellular fluid. In that way, white blood cells can reach the site of infection and help destroy invading microorganisms.

Notice in Figure 46-12 that a white blood cell has a very different structure from that of a red blood cell. For instance, a white blood cell may be irregularly shaped and may have a rough outer surface. There are other differences between red blood cells and white blood cells as well. In contrast with the short-lived red blood cells, white blood cells may function for years. And while there is only one type of red blood cell, there are several types of white blood cells.

The white blood cell shown in Figure 46-12 is the type of white blood cell known as a **phagocyte** (FA-guh-siet). Phagocytes are cells that engulf invading microorganisms. Locate the microorganisms that are being engulfed by the phagocyte in Figure 46-12. Another type of white blood cell produces **antibodies**. Antibodies are proteins that help destroy substances, such as bacteria and viruses, that enter the body and can cause disease. When a person has an infection, the number of white blood cells can double.



Word Roots and Origins

leukocyte

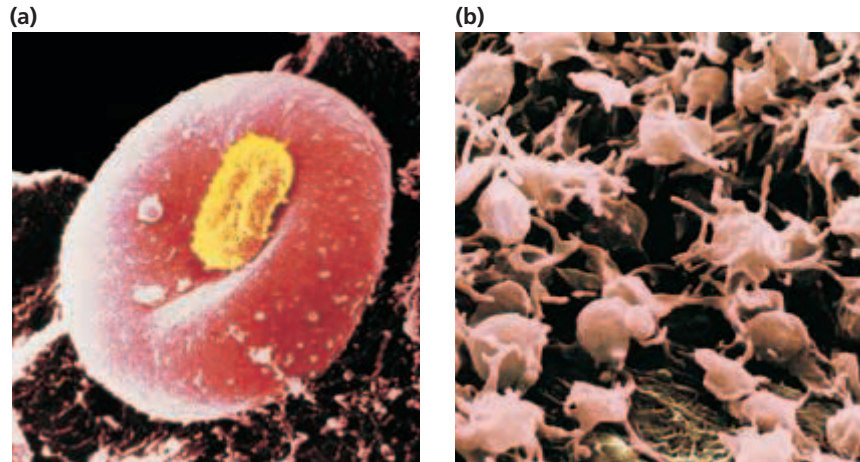
from the Greek *leuco*, meaning "white," and *cyte*, meaning "cell"

FIGURE 46-12

Some white blood cells, like the phagocyte shown in blue, engulf and destroy invading microorganisms.

FIGURE 46-13

Inactive platelets, such as the yellow object shown in (a), derive their name from the fact that they look like little plates. Platelets are colorless and contain chemicals that are involved in the clotting process. (b) The platelets change shape during the clotting process. When activated, the platelets settle and spread on the substrate.



Eco Connection

Vampire Bats Help Save Stroke Victims

Vampire bats have an anticoagulant in their saliva that prevents blood clotting when it flows from a wound. In 1995, this enzyme was isolated and named *Draculin*.

Researchers have developed a clot-dissolving agent called *Desmodus rotundus salivary plasminogen activator* (DSPA), which is based on the salivary enzyme Draculin. DSPA targets and destroys fibrin. The current treatment must be given within three hours of the onset of a stroke (a sudden loss of consciousness or paralysis that occurs when the blood flow to the brain is interrupted). Otherwise, brain-cell death and brain damage may occur. According to research, DSPA could be a safe treatment for longer periods of time and appears to have no detrimental effect on brain cells.

Platelets

Platelets are essential to the formation of a blood clot. A blood clot is a mass of interwoven fibers and blood cells that prevents excess loss of blood from a wound. Platelets are not whole cells. They are fragments of very large cells that were formed in the bone marrow. As you can see in Figure 46-13a, platelets get their name from their platelike structure. Platelets lack a nucleus and have a life span of 7 to 12 days. A cubic micrometer of blood may contain as many as half a million platelets.

When a blood vessel tears or rips, platelets congregate at the damaged site, sticking together and forming a small plug. The vessel constricts, slowing blood flow to the area. Then, special clotting factors are released from the platelets and the damaged tissue. These factors begin a series of chemical reactions that occur at the site of the bleeding. The last step in this series brings about the production of a protein called **fibrin**. Fibrin molecules consist of long, sticky chains. As you can see in Figure 46-14, these chains form a net that traps red blood cells, and the mass of fibrin and red blood cells hardens into a clot, or scab.

Hemophilia is a disorder caused by the absence of one or more of the proteins required for blood clotting. When a person with hemophilia is injured, bleeding continues for much longer than it would in a person without hemophilia. Large cuts or internal injuries can be life threatening. Today, people with hemophilia are treated with injections of the missing clotting factors.

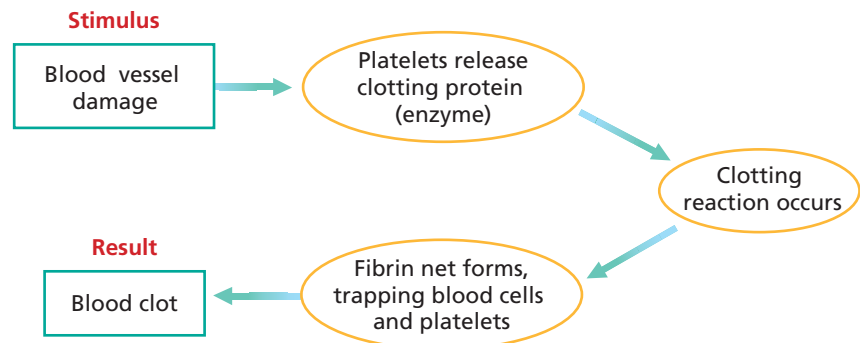


FIGURE 46-14

The release of enzymes from platelets at the site of a damaged blood vessel initiates a "clotting cascade."

BLOOD TYPES

Blood type is determined by the type of antigen present on the surface of the red blood cells. An **antigen** is a substance that stimulates an immune response. Antigens that are normally present in a person's body provoke no response. However, when foreign antigens enter the body, cells respond by producing antibodies. In fact, the word *antigen* is an abbreviation for “antibody-generating substance.”

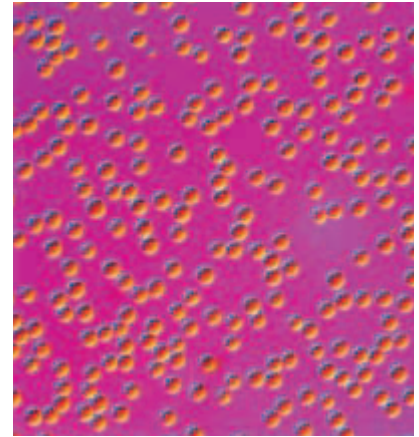
In the early 1900s, Karl Landsteiner used blood taken from his laboratory workers and made observations similar to those you see in Figure 46-15. He noticed that mixing blood samples from two people sometimes resulted in the cells clumping together, or *agglutinating*. When samples of two different blood types are mixed together, reactions occur between the antigens on the red blood cells and the antibodies in the plasma, causing the cells to agglutinate. When samples of the same blood type are mixed, no reaction occurs, and the blood cells do not agglutinate.

Landsteiner's observations led to the classification of human blood by blood types. Three of the most important human antigens are called A, B, and Rh. The A-B-O system of blood typing, described below, is based on the A and B antigens.

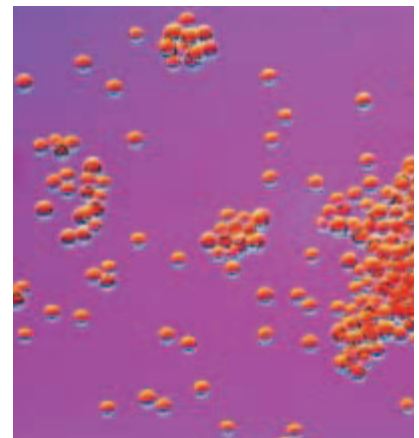
A-B-O System

The A-B-O system is a means of classifying blood by the antigens located on the surface of the red blood cells and the antibodies circulating in the plasma. As shown in Table 46-1, an individual's red blood cells may carry an A antigen, a B antigen, both A and B antigens, or no antigen at all. These antigen patterns are called blood types A, B, AB, and O, respectively.

Notice in Table 46-1 that an individual with type A blood also has anti-B antibodies against type B blood. If type B blood is given to a recipient with type A blood, the recipient's anti-B antibodies will react with the B antigens on the donated red blood cells and the blood will agglutinate. In addition, the donor's type B blood has anti-A antibodies. Their presence will compound the antigen-antibody reaction. The result will be agglutinated blood that will block the flow of blood through the vessels. For this reason, transfusion recipients must receive blood that is compatible with their own.



(a)



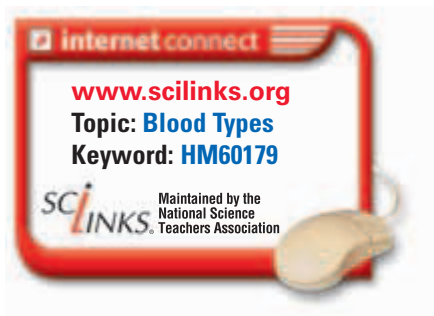
(b)

FIGURE 46-15

Notice that there is no agglutination of red blood cells in the slide in (a), where blood samples from two people with the same blood type were mixed. Compare this with the slide in (b), where blood samples from two people with different blood types were mixed.

TABLE 46-1 Blood Types, Antigens, and Antibodies

Blood types	Antigen on the red blood cells	Antibodies in the plasma	Can get blood from	Can give blood to
A	A	anti-B	O, A	A, AB
B	B	anti-A	O, B	B, AB
AB	A and B	none	A, B, AB, O	AB
O	none	anti-A, anti-B	O	A, B, AB, O



People who have type AB blood are *universal recipients*. They can receive A, B, AB, or O blood because they do not have anti-A or anti-B antibodies. People who have type O blood are *universal donors*. They can donate blood to people who have A, B, AB, or O blood because the blood cells of people who have type O blood do not have A or B antigens.

Rh System

An antigen that is sometimes present on the surface of red blood cells is the **Rh factor**, named after the rhesus monkey in which it was first discovered. Eighty-five percent of the United States' population is Rh-positive (Rh^+), meaning that Rh antigens are present. People who do not have Rh antigens are called Rh-negative (Rh^-).

If an Rh^- person receives a transfusion of blood that has Rh^+ antigens, antibodies may react with the antigen and agglutination will occur. The most serious problem with Rh incompatibility occurs during pregnancy. If the mother is Rh^- and the father is Rh^+ , the child may inherit the dominant Rh^+ allele from the father. During delivery, a small amount of the fetus's Rh^+ blood may reach the mother's bloodstream. If this happens, the mother will develop antibodies to the Rh factor. If a second Rh^+ child is conceived later, the mother's antibodies can cross the placenta and attack the blood of the fetus. This condition is called *erythroblastosis fetalis*. The fetus may die as a result of this condition, or if the child is born alive, he or she may need an immediate transfusion of Rh^+ blood.

To prevent this condition, an Rh^- mother of an Rh^+ child can be given antibodies to destroy any Rh^+ cells that have entered her bloodstream from the fetus. The mother is, in effect, immunized against the Rh antigen before her immune system has a chance to develop its own antibodies. The antibody treatment prevents Rh sensitization in Rh^- women only if their bodies have not already produced Rh antibodies. If an Rh^- mother has not yet been sensitized, she receives the antibody treatment in the 28th week of pregnancy and again immediately after delivery.

SECTION 2 REVIEW

1. Identify the four main components of blood.
2. Explain how the structure of red blood cells, white blood cells, and platelets relates to the function of these cells.
3. Identify the stages and structures involved in the clotting process.
4. What factors determine the compatibility of blood types for transfusion?
5. Which blood types, in terms of the A-B-O and Rh antigens, can be donated to somebody with type AB^- blood?

CRITICAL THINKING

6. **Analyzing Information** Hemophilia is a disorder in which there is a failure in one of the steps of clot formation. What might be some advantages and disadvantages of this disorder?
7. **Evaluating Results** A patient's blood has an elevated count of leukocytes. What does this most likely indicate?
8. **Relating Concepts** Explain why a pregnant woman should know her blood type and the blood type of her baby's father.

MILESTONES

IN Blood Transfusions

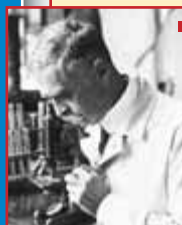
Timeline

1628 William Harvey describes the circulation of blood.



1667 Jean-Baptiste Denis successfully transfuses blood from a lamb to a human.

1818 First human-to-human blood transfusion.



1901 Karl Landsteiner determined three of four blood types (A, B, and O).

1914 Blood is stored for the first time.



1937 Bernard Fantus formed the first blood bank.

1939 Charles Drew set up collection centers for blood to fill World War II needs.



1950 Carl Walter and W.P. Murphy, Jr. introduced the plastic bag to collect blood.

1985 Blood is screened for HIV and other diseases.

Every three seconds someone needs a blood transfusion—blood, plasma, or saline is introduced into the body. Blood and blood products are used to treat accident and burn victims, cancer patients, and patients undergoing surgeries and medical treatments. The development of safe blood transfusion techniques was an important achievement in modern medicine.

In 1665, Richard Lower, an English physician, transferred blood between two dogs. Jean-Baptiste Denis, a French physician and astrologer, transfused blood from a lamb to a human in 1667. The first successful human-to-human blood transfusions were performed in 1818 by James Blundell, a British physician, but these were followed by many failures.

In 1901, Austrian-born American physiologist Karl Landsteiner determined the first three blood types—A, B, and O—and that incompatible types will clot. This discovery explained why the outcome of transfusions had been so unpredictable in the past. For his work, Landsteiner received the Nobel Prize in medicine or physiology in 1930.

During World War I, scientists began to study how to preserve and transport blood for wounded soldiers. They found that the addition of sodium citrate prevented clotting, making the storage of blood possible for the first time.

In 1937, Bernard Fantus, a physician, established the first nonprofit blood bank at Cook County Hospital in Chicago. The next year, Charles Drew, a medical doctor, found that blood plasma (the liquid component of blood) could substitute for whole blood in emergency situations.

Germany's attack on France in 1940 created a great need for blood. Based on Drew's findings, the United States began to provide liquid plasma and whole blood for France. Working with the National Research

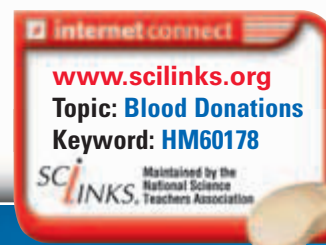
Council and the American Red Cross, Drew set up blood collection centers.

Early in the 1980s, some transfused blood was found to carry the human immunodeficiency virus (HIV), the virus that causes acquired immune deficiency syndrome (AIDS). Since 1985, careful screening for HIV, hepatitis, and other diseases has almost entirely removed the risk of receiving contaminated blood.

Today, many people bank their own blood for later use in surgery. Blood can also be collected during surgery and returned to the patient later. The AIDS epidemic has triggered a race to create artificial blood. Several companies have begun testing blood substitutes from chemically treated animal blood and outdated human blood.

Review

1. What type of patients can benefit from blood transfusions?
2. **Critical Thinking** How might safe blood transfusion techniques affect you?
3. **Critical Thinking** Use the Web site below to research the latest progress on creating artificial blood. Write a short report to describe your findings.



SECTION 3

OBJECTIVES

- **Differentiate** external respiration from internal respiration.
- **Trace** the path of air from the atmosphere to the bloodstream.
- **Describe** how gases are exchanged in the lungs and transported in the bloodstream.
- **Summarize** the skeletal and muscular changes that occur during breathing.
- **Describe** how the rate of breathing is controlled.

VOCABULARY

respiratory system
external respiration
internal respiration
lung
pharynx
epiglottis
trachea
larynx
bronchus
bronchiole
alveolus
inspiration
diaphragm
expiration

THE RESPIRATORY SYSTEM

*The blood transports oxygen from the lungs to cells and carries carbon dioxide from the cells to the lungs. It is the function of the **respiratory system** to exchange gases with the cardiovascular system.*

RESPIRATION

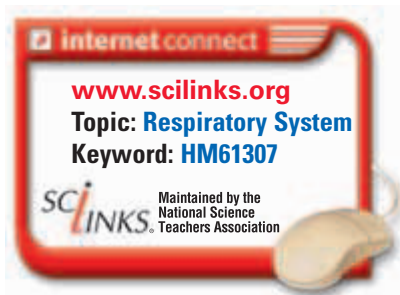
The respiratory system involves both external respiration and internal respiration. **External respiration** is the exchange of gases between the atmosphere and the blood. **Internal respiration** is the exchange of gases between the blood and the cells of the body. Once oxygen is in the cells, the cells use it to break down glucose and make ATP by the process of aerobic respiration. Without oxygen, the body could not obtain enough energy from food to survive. Excess carbon dioxide produced as a waste product of aerobic respiration is toxic to cells and is removed from the cells by internal respiration.

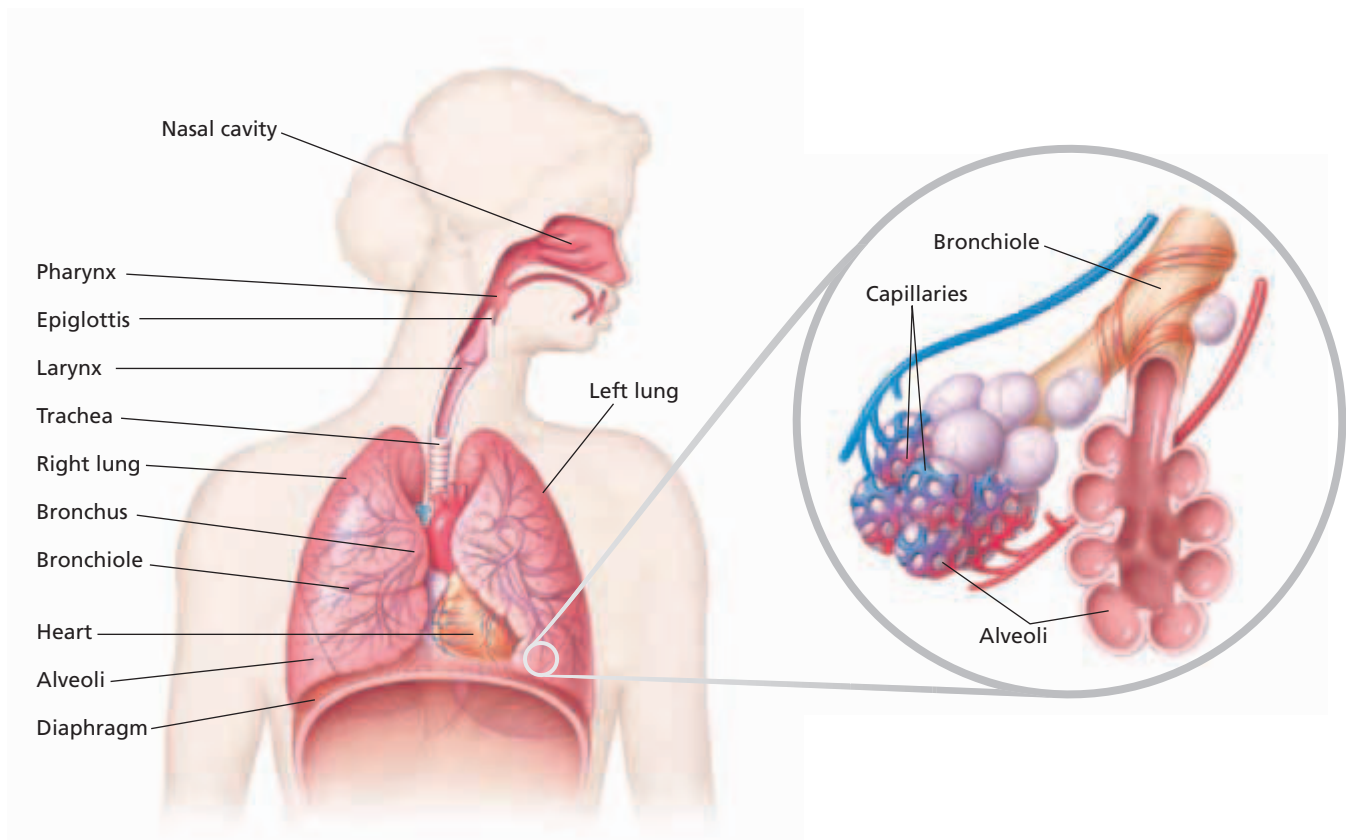
THE LUNGS

The **lungs** are the site of gas exchange between the atmosphere and the blood. Notice in Figure 46-16 that the right lung has three divisions, or lobes. It is slightly heavier than the two-lobed left lung. The lungs are located inside the *thoracic cavity*, bounded by the rib cage and the diaphragm. Lining the entire cavity and covering the lungs are *pleura*, membranes that secrete a slippery fluid that decreases friction from the movement of the lungs during breathing.

The Path of Air

Refer to Figure 46-16 to trace the path air follows from the atmosphere to the capillaries in the lungs. External respiration begins at the mouth and at the nose. Air filters through the small hairs of the nose and passes into the nasal cavity, located above the roof of the mouth. In the nasal cavity, mucous membranes warm and moisten the air, which helps prevent damage to the delicate tissues that form the respiratory system. The walls of the nasal cavity are also lined with cilia. These cilia trap particles that are inhaled and are eventually swept into the throat, where they are swallowed.





The moistened, filtered air then moves into the throat, or **pharynx** (FER-inks), a tube at the back of the nasal cavities and the mouth. The pharynx contains passageways for both food and air. When food is swallowed, a flap of cartilage, called the **epiglottis**, presses down and covers the opening to the air passage. When air is being taken in, the epiglottis is in an upright position, allowing air to pass into a cartilaginous tube called the windpipe, or **trachea** (TRAY-kee-uh). The trachea is about 10 to 12 cm long and has walls lined with ciliated cells that trap inhaled particles. The cilia sweep the particles and mucus away from the lungs toward the throat.

At the upper end of the trachea is the voicebox, or **larynx** (LER-inks). Sounds are produced when air is forced past two ligaments—the *vocal cords*—that stretch across the larynx. The pitch and volume of the sound produced varies with the amount of tension on the vocal cords and on the amount of air being forced past them.

The trachea then branches into two **bronchi** (BRAHN-kie) (singular, bronchus), each of which leads to a lung. The walls of the bronchi consist of smooth muscle and cartilage and are lined with cilia and mucus. Within the lungs, the bronchi branch into smaller and smaller tubes. The smallest of these tubes are known as **bronchioles**, which are also lined with cilia and mucus. Eventually the bronchioles end in clusters of tiny air sacs called **alveoli** (al-VEE-oh-LIE) (singular, alveolus). A network of capillaries surrounds each alveolus, as you can see in the detailed view shown in Figure 46-16. All exchange of gases in the lungs occurs in the alveoli. To facilitate this exchange, the surface area of the lungs is enormous. A healthy lung contains nearly 300 million alveoli and has a total surface area of about 70 m²—about 40 times the surface area of the skin.

FIGURE 46-16

Trace the passage of air from the atmosphere to the lungs. Oxygen in the air finally reaches the alveoli, the functional units of the respiratory system. All exchange of gases between the respiratory system and the cardiovascular system occurs in the alveoli.

GAS EXCHANGE AND TRANSPORT

In the lungs, gases are exchanged between the alveoli and the blood in the capillaries. Oxygen (O_2) to be transported throughout the body moves into the bloodstream, and carbon dioxide (CO_2) to be eliminated from the body moves into the alveoli.

Gas Exchange in the Lungs

Figure 46-17 illustrates the direction in which oxygen and carbon dioxide move in the alveoli. When air moves into the lungs, the oxygen in the air crosses the thin alveolar membranes as well as the capillary walls and dissolves in the blood. Carbon dioxide moves in the opposite direction, crossing the capillary walls and thin alveolar membranes and entering the alveoli.

Air moving into the alveoli is rich in oxygen and contains little carbon dioxide. In contrast, blood in the capillaries surrounding the alveoli is low in oxygen and contains high levels of carbon dioxide. Substances diffuse from an area of higher concentration to an area of lower concentration. Consequently, oxygen diffuses from the alveoli into the blood, and carbon dioxide diffuses from the blood into the alveoli. The enormous surface area of the alveoli increases the rate of diffusion of these two gases.

Transport of Oxygen

When oxygen diffuses into the blood, only a small amount remains dissolved in the plasma. Most of the oxygen—95 to 98 percent—moves into the red blood cells, where it combines with hemoglobin, an iron-containing protein. Each hemoglobin molecule contains four iron atoms. Each iron atom can bind to one oxygen molecule. Thus, one hemoglobin molecule can carry up to four molecules of oxygen. There are about 250 million hemoglobin molecules in each red blood cell. When oxygenated blood reaches body tissues, the oxygen concentration is higher in the blood than in the body tissues. Thus, oxygen is released from hemoglobin and diffuses out of the capillaries and into surrounding cells.

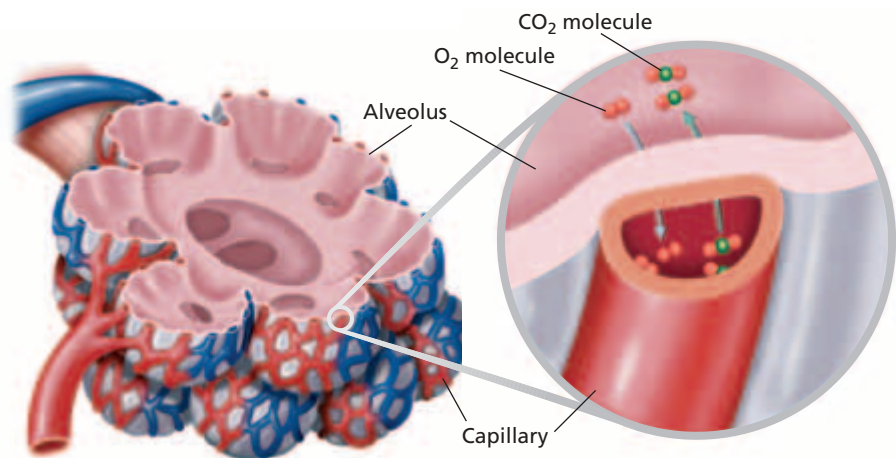


FIGURE 46-17

Because of concentration gradients, oxygen and carbon dioxide diffuse across the alveoli and capillary walls.

Transport of Carbon Dioxide

Because the concentration of carbon dioxide (CO_2) is higher in the cells, it diffuses out of the cells and into the blood. Only about 7 percent of the carbon dioxide dissolves in the plasma. Approximately 23 percent binds to hemoglobin. The remaining 70 percent is carried in the blood as bicarbonate ions (HCO_3^-). As shown in the equation below, CO_2 reacts with water in the plasma to form carbonic acid (H_2CO_3). In turn, the carbonic acid disassociates into bicarbonate ions and hydrogen ions (H^+):



Thus, most of the CO_2 travels in the blood as bicarbonate ions. When the blood reaches the lungs, the reactions are reversed:



Bicarbonate ions combine with hydrogen ions to form carbonic acid, which in turn forms carbon dioxide and water. The carbon dioxide diffuses out of the capillaries into the alveoli and is exhaled into the atmosphere.

MECHANISM OF BREATHING

Breathing is the process of moving air into and out of the lungs. **Inspiration**, shown in Figure 46-18, is the process of taking air into the lungs. When you take a deep breath, your chest expands as muscles contract to move the ribs up and outward. At the same time, your **diaphragm**, a large skeletal muscle that separates the thoracic cavity from the abdominal cavity, flattens and pushes down on the abdomen. Muscles in the abdominal wall in turn relax. This action provides room for the flattened diaphragm.

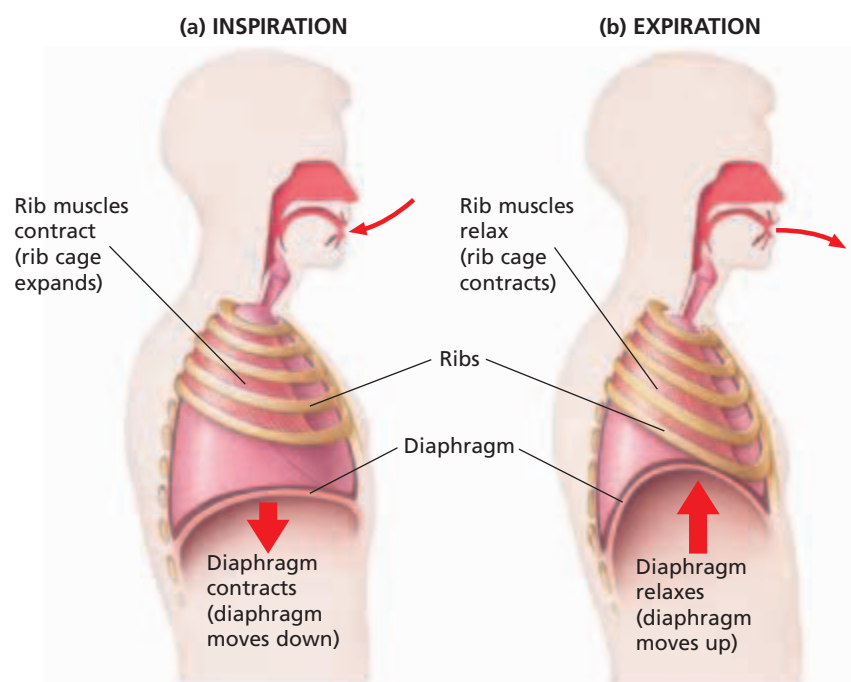


FIGURE 46-18

The diaphragm, a large skeletal muscle that separates the thoracic cavity from the abdominal cavity, and the muscles between the ribs control the movement of the thoracic cavity during breathing. If these muscles were paralyzed, then inspiration and expiration would not occur.

Word Roots and Origins

expiration

from the Latin *expir*, which means
"to breathe out"

When the diaphragm flattens and the ribs are lifted up and out, the volume of the lungs increases. An increased volume reduces the air pressure within the lungs. At this point, the air pressure inside the lungs is lower than the air pressure outside the body. As a result, air from the atmosphere moves into the lungs.

During **expiration**, the process of releasing air from the lungs, the reverse movements take place, as shown in Figure 46-18. As the diaphragm and rib muscles relax, the elastic tissues of the lungs recoil, deflating the lungs. The volume of the lungs decreases. Because the volume is smaller, the air pressure inside the cavity becomes greater than the air pressure outside the body. This pressure difference forces air out of the lungs until the pressures are again equal.

Regulation of Breathing

The rate at which oxygen is used depends on the activity of the cells. The greater their activity, the more oxygen they need and the faster the body needs to breathe. The slower their activity, the slower the body needs to breathe. Both rate and depth of breathing change in order to provide oxygen and eliminate carbon dioxide.

The rate of breathing is controlled by the brain and brain stem, which monitors the concentration of carbon dioxide in the blood. As activity increases, high levels of carbon dioxide in the blood stimulate nerve cells in the brain. The brain stem in turn stimulates the diaphragm to increase the breathing rate and depth. When the carbon dioxide concentration in the blood returns to lower levels, the sensors in the brain send a message to the respiratory muscles to return to a slower breathing rate. All this is controlled subconsciously by control centers in the brain. However, a person can temporarily override the respiratory control system at any time, holding his or her breath until losing consciousness. Then the brain stem takes control, and normal breathing resumes. This mechanism allows humans to swim underwater for short periods and to sleep without concern for breathing.

SECTION 3 REVIEW

1. How does internal respiration differ from external respiration?
2. Outline the path of oxygen from the atmosphere to the bloodstream.
3. Explain the process of gas exchange in the lungs.
4. Differentiate between oxygen transport and carbon dioxide transport in the bloodstream.
5. Sequence the skeletal and muscular changes that take place when a person inhales.
6. What factors regulate the rate of breathing?

CRITICAL THINKING

7. **Relating Concepts** Predict the effect that increasing altitude would have on blood-oxygen saturation.
8. **Applying Information** Why does a single-celled organism not need a respiratory system?
9. **Predicting Results** Normally, arterial blood is about 98 percent saturated with oxygen. What are two conditions that could result in lower oxygen saturation?

CHAPTER HIGHLIGHTS

SECTION 1

The Circulatory System

- The human circulatory system is made up of the cardiovascular system and the lymphatic system.
- The heart is a muscular organ that pumps blood through an intricate network of blood vessels.
- Blood flows from the body into the heart, which then pumps blood to the lungs. After oxygenation, blood returns to the heart, which pumps blood to the rest of the body.
- Arteries carry blood away from the heart. Materials are exchanged at the capillaries. Veins contain valves and carry blood back to the heart.
- In pulmonary circulation, blood travels between the heart and lungs. In systemic circulation, blood travels between the heart and all other body tissues.
- The lymphatic system returns lymph, fluid that has collected in the tissues, to the bloodstream.

Vocabulary

cardiovascular system (p. 933)

lymphatic system (p. 933)

atrium (p. 933)

ventricle (p. 933)

valve (p. 934)

aorta (p. 935)

sinoatrial node (p. 935)

atrioventricular node (p. 935)

pulse (p. 935)

artery (p. 936)

blood pressure (p. 936)

hypertension (p. 936)

capillary (p. 936)

vein (p. 937)

pulmonary circulation (p. 937)

systemic circulation (p. 937)

atherosclerosis (p. 938)

lymph (p. 939)

SECTION 2

Blood

- Blood is composed of plasma (water, metabolites, wastes, salts, and proteins), red blood cells, white blood cells, and platelets.
- Red blood cells transport oxygen. White blood cells help defend the body against disease. Platelets are essential to the formation of a blood clot.
- Blood clotting occurs when platelets release a clotting protein, which causes a clotting reaction to occur. A fibrin net forms, trapping blood cells and platelets.
- Human blood can be grouped into four types—A, B, AB, and O—based on proteins on the surface of red blood cells. Another antigen called *Rh factor*, is sometimes present on red blood cells.

Vocabulary

plasma (p. 940)

red blood cell (erythrocyte)
(p. 940)

hemoglobin (p. 940)

white blood cell (leukocyte)
(p. 941)

phagocyte (p. 941)

antibody (p. 941)

platelet (p. 942)

fibrin (p. 942)

blood type (p. 943)

antigen (p. 943)

Rh factor (p. 943)

SECTION 3

The Respiratory System

- External respiration is the exchange of gases between the atmosphere and the blood. Internal respiration is the exchange of gases between the blood and the cells of the body.
- The lungs are the site of gas exchange between the atmosphere and the blood.
- Air enters through the mouth or nose, passes through the pharynx, larynx, trachea, bronchi and bronchioles and into alveoli. A network of capillaries surrounds each alveolus. All exchange of gases in the lungs occurs at the alveoli.
- Most oxygen is carried attached to hemoglobin. Some carbon dioxide is carried bound to hemoglobin. A small amount is dissolved in plasma. Most carbon dioxide is carried as bicarbonate ions.
- During inspiration, the diaphragm and rib muscles contract, the thoracic cavity expands, and air is pulled into the lungs. During expiration, the diaphragm and rib muscles relax, the thoracic cavity contracts, and air is forced out of the lungs.
- The rate of breathing is controlled by nerve centers in the brain that monitor the level of carbon dioxide in the blood.

Vocabulary

respiratory system (p. 946)

external respiration (p. 946)

internal respiration (p. 946)

lung (p. 946)

pharynx (p. 947)

epiglottis (p. 947)

trachea (p. 947)

larynx (p. 947)

bronchus (p. 947)

bronchiole (p. 947)

alveolus (p. 947)

inspiration (p. 949)

diaphragm (p. 949)

expiration (p. 950)

CHAPTER REVIEW

USING VOCABULARY

1. Distinguish between *systolic pressure* and *diastolic pressure*.
2. Choose the term that does not belong in the following group: *erythrocyte*, *hemoglobin*, *leukocyte*, and *platelet*. Explain why it does not belong.
3. For each pair of terms, explain the relationship between the terms.
 - a. *atrioventricular valve* and *semilunar valve*
 - b. *artery* and *vein*
 - c. *expiration* and *inspiration*
4. **Word Roots and Origins** The word *phagocyte* is derived from the Greek word *phagein*, which means “to eat.” The suffix *cyte* means “cell.” Using this information, explain why the term *phagocyte* is a good name for the biological process that the term describes.

UNDERSTANDING KEY CONCEPTS

5. **Identify** the parts of the human heart, and describe the function of each part.
6. **Outline** the route that blood takes through the heart, lungs, and body.
7. **Relate** the structure of arteries, veins, and capillaries to the function of each.
8. **Compare** the pulmonary arteries and the aorta.
9. **Compare** the pulmonary veins and the inferior vena cava.
10. **Summarize** the roles of the lymphatic system.
11. **Discuss** the function of each of the components of blood.
12. **Identify** the structure that red blood cells lack that limits their life span.
13. **Describe** three differences between white blood cells and red blood cells.
14. **Sequence** the process of blood-clot formation that occurs after a vessel is injured.
15. **Explain** the A-B-O blood-typing system.
16. **Identify** the role of the Rh factor in determining blood compatibility for transfusion.
17. **Compare** external respiration with internal respiration.
18. **Sequence** the path oxygen travels from the environment into the blood.
19. **Compare** the transport and exchange of oxygen and carbon dioxide.
20. **Describe** the movement of the diaphragm and the rib muscles during inspiration and expiration.

21. **Name** the factor that stimulates the brain stem to increase the breathing rate.

22.  **CONCEPT MAPPING** Use the following terms to create a concept map that shows the relationship between the cardiovascular, lymphatic, and respiratory systems: *artery*, *capillary*, *vein*, *lymphatic system*, *pulmonary circulation*, *systemic circulation*, *atrium*, *ventricle*, *aorta*, and *vena cava*.

CRITICAL THINKING

23. **Inferring Relationships** A person with anemia can have too few red blood cells or low hemoglobin levels. The most common symptom is a lack of energy. Why would anemia cause this symptom?
24. **Applying Information** Explain how the lymphatic system moves lymph through the body without the aid of a pumping organ like that of the cardiovascular system.
25. **Analyzing Concepts** One function of the cardiovascular system is to help maintain a uniform body temperature. Explain how the constant circulation of blood throughout the body can accomplish this task.
26. **Interpreting Graphics** Copy the blood-type table below on a sheet of paper. Fill in the missing information for each type.

TABLE 1 Blood Type

Blood type	Antigen on the red blood cell	Antibodies in plasma	Can receive blood from	Can donate blood to
A		B	O, A	A, AB
B	B		O, B	B, AB
AB	A, B	Neither A nor B	O, A, B, AB	
O	Neither A nor B	A, B		O, A, B, AB

27. **Calculating Data** Calculate the number of times a person's heart will beat if the person lives 75 years. Assume that the average heart beats 70 times per minute.
28. **Recognizing Relationships** Assuming that the heart of an overweight person beats an additional 10 times per minute, explain why being overweight can put additional strain on the heart.

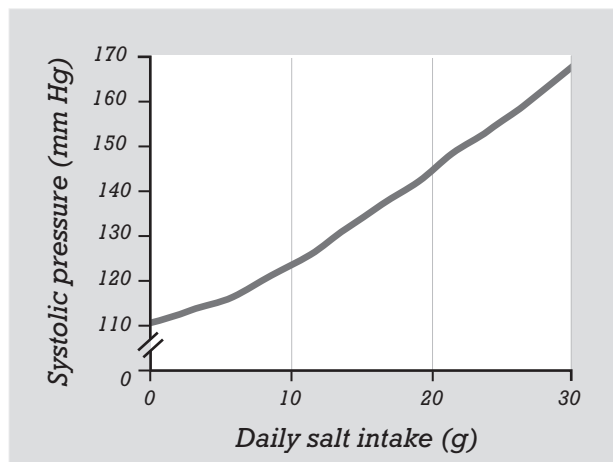


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- In what direction does blood move during ventricular systole?
 - from the atria to the veins
 - from the ventricles to the atria
 - from the atria to the ventricles
 - from the ventricles to the arteries
- What is the function of the lymphatic system?
 - It opens two-way vessels.
 - It helps the body fight infections.
 - It interacts with the respiratory system.
 - It transports intercellular fluid away from the heart.
- Fibrin is a protein that does which of the following?
 - transports oxygen
 - helps form a blood clot
 - destroys invading microorganisms
 - stimulates the production of antibodies

INTERPRETING GRAPHICS: The graph below shows how systolic pressure is affected by salt intake. Use the graph to answer the question that follows.

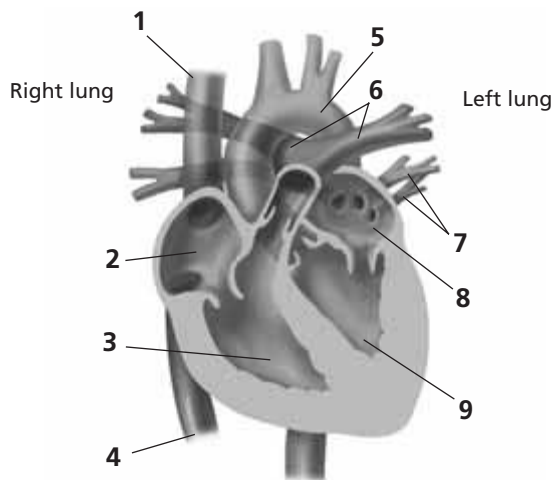


- What is the relationship between salt intake and blood pressure?
 - As salt intake increases, blood pressure increases.
 - As salt intake increases, blood pressure decreases.
 - Salt intake of 20 g per day results in stable blood pressure.
 - Salt intake of 30 g per day results in stable blood pressure.

DIRECTIONS: Complete the following analogy.

- superior vena cava : deoxygenated blood :: pulmonary veins :
 - type A blood
 - type B blood
 - oxygenated blood
 - deoxygenated blood

INTERPRETING GRAPHICS: The model below shows a cross section of the heart. Use the model to answer the question that follows.



- Which numbers point to vessels that bring blood into the heart?
 - 1, 4, and 7
 - 1, 5, and 6
 - 4, 5, and 6
 - 5 and 6 only

SHORT RESPONSE

Even a small increase or decrease in blood volume has an effect on blood pressure. When an accident victim suffers significant blood loss, the person is transfused with plasma rather than whole blood.

Why is plasma effective in meeting the immediate threat to life?

EXTENDED RESPONSE

Polio is a disease that paralyzes muscles by affecting the nerves that make the muscles move.

Part A List muscles involved in breathing.

Part B Explain how polio might affect breathing.

Test TIP

Slow, deep breathing helps a person relax. If you suffer from test anxiety, focus on your breathing in order to calm down.

Measuring Lung Volumes and CO₂ Production

OBJECTIVES

- Use indirect measurement to determine lung capacity.
- Determine the effect of exercise on breathing rate and CO₂ production.

PROCESS SKILLS

- measuring
- analyzing data
- hypothesizing
- experimenting
- collecting data

MATERIALS

- safety goggles
- lab apron
- disposable gloves
- 1 L bromothymol indicator solution
- drinking straws
- 100 mL Erlenmeyer flasks, 2 per group
- 100 mL graduated cylinders
- marker
- plastic wrap
- spirometer
- stopwatch or clock with second hand

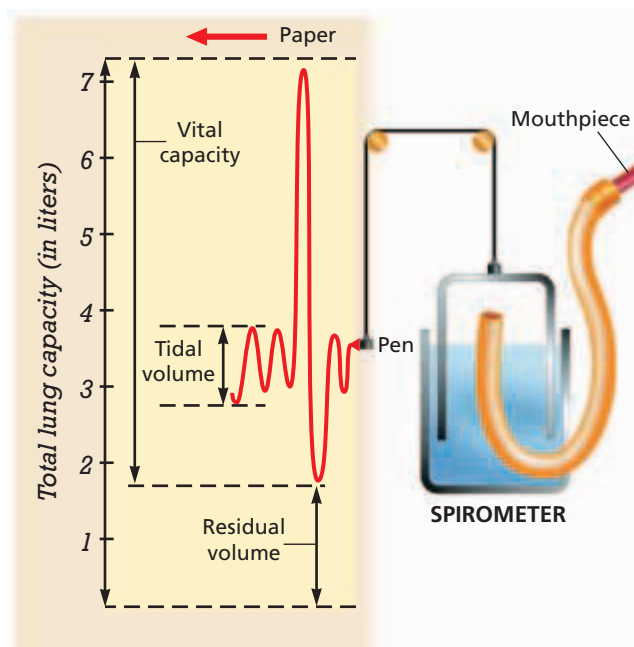
Background

1. A spirometer is an instrument used to measure the volume of air a person can breathe.
2. Compare the diagram of a spirometer on the right with the spirometer you will be using to complete this investigation. The marking pen creates a line that can be compared with the scale on the left side to measure liters of air.
3. Tidal volume is the volume of air inhaled or exhaled during a normal breath.
4. Lung capacity is the total volume of air that the lungs can hold. Total lung capacity is 5 to 6 L. What factors might increase or reduce lung capacity?
5. Expiratory reserve volume is the amount of air that can be forcefully exhaled after a normal exhalation.
6. Vital capacity is the maximum amount of air that can be inhaled or exhaled.

7. Carbon dioxide is soluble in water. You can determine the relative amount of CO₂ in your breath by using an indicator to react with the CO₂. Higher CO₂ levels will react with the indicator solution faster.

PART A Tidal Volume, Expiratory Volume, and Vital Capacity

1. Make a data table in your notebook like the one shown on the next page (Part A Lung Volumes).
2.  Place a clean mouthpiece in the end of the spirometer. **CAUTION** Many diseases are spread by body fluids, such as saliva. **Do NOT share a mouthpiece with anyone.** Inhale a normal breath. Hold your nose, then exhale a normal breath into the spirometer. Record your data in the table.
3. Measure your expiratory reserve volume by first breathing a normal breath and exhaling normally. Then put the spirometer tube to your mouth as you forcefully exhale whatever air is left in your lungs. Be sure to force out as much air as possible. Record your data in the table.



PART A Lung Volumes

	Average for young adult males	Average for young adult females	Average for athletes	Your readings
Tidal volume	500 mL			
Expiratory reserve volume	100 mL			
Vital capacity	4,600 mL			

- The table includes values for young adult males. The average volumes for young adult females are 20–25 percent lower than those of males. Calculate the average volumes for young adult females. Athletes can have volumes that are 30–40 percent greater than the average for their gender. Calculate the average volumes for athletes.
-  Dispose of your mouthpiece in the designated waste container.
- In your data table, record the amount of time in seconds that it took to get the same yellow color in flask 2 as you got in flask 1.
- Calculate the difference in the amount of time it took to see a color change in the two flasks. What can you infer about the amount of CO₂ you exhaled before and after exercise?

-   Clean up your materials. Pour the solutions down the sink, and rinse the sink thoroughly with water. Wash your hands before leaving the lab.

PART B Breathing Rate and CO₂ Production

- Discuss with your partners the use of bromothymol blue as an indicator of CO₂. Develop a hypothesis that describes a relationship between air volume exhaled during rest or exercise and the volume of CO₂ exhaled.
- Make a data table in your notebook like the one on this page, titled "Part B CO₂ Production".
- Label two flasks as 1 and 2.
-   **CAUTION** Wear safety goggles at all times during this procedure. If you get the indicator solution on your skin or clothing, wash it off at the sink while calling to your teacher. If you get the indicator solution in your eyes, immediately flush it out at the eyewash station while calling to your teacher.
- Add 100 mL of indicator solution to each flask. Cover the mouth of each flask with plastic wrap.
- Remove the plastic wrap from flask 1. Begin the stopwatch. Blow gently through one straw into flask 1 until the solution turns a yellowish color, exhaling slowly so that the solution does not bubble up. **CAUTION** Be careful not to inhale the solution or get it in your mouth. Stop the stopwatch.
- Record in your data table the time in seconds that it took to see a color change in flask 1.
- Exercise by jogging in place or doing jumping jacks for 2 min. Begin the stopwatch immediately. Blow gently through a new straw into flask 2 until the solution becomes the same yellowish color as the solution in flask 1. Stop the stopwatch.

Analysis and Conclusions

- How did your tidal volume compare with that of your classmates?
- What are the independent and dependent variables in Part B? How did you vary the independent variable and measure changes in the dependent variable?
- Why were the flasks covered with plastic wrap?
- Do your data support your hypothesis from Part B? Explain your answers.
- How do you know whether you produced more carbon dioxide before or after you exercised?
- What were some of the possible sources of error in your experiment?

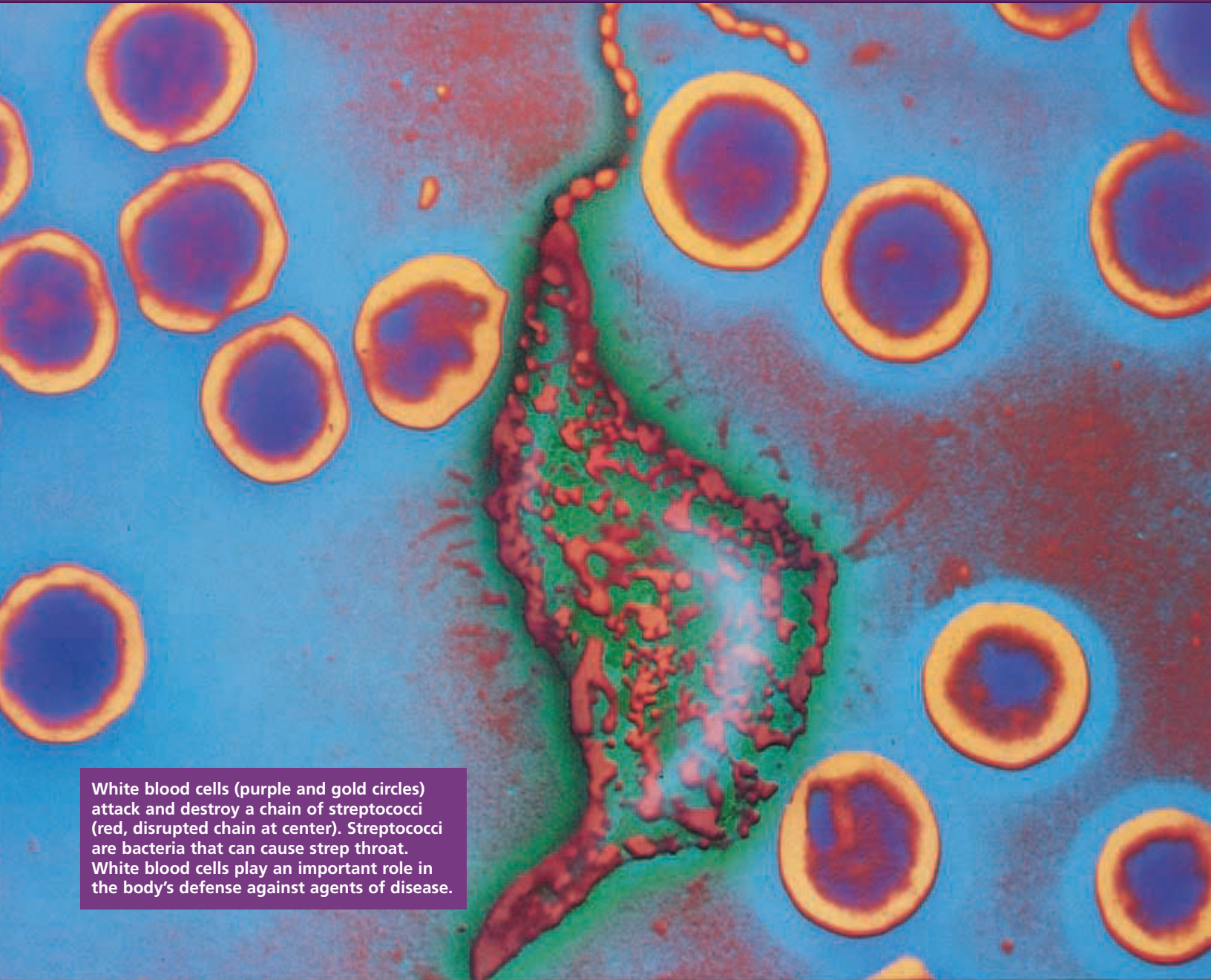
PART B CO₂ Production

Time for color change in flask 1	
Time for color change in flask 2	
Difference in time between flask 1 and flask 2	

Further Inquiry

Design an experiment to determine whether exercise affects heart rate in the same way it affects breathing rate and tidal volume.

THE BODY'S DEFENSE SYSTEMS



White blood cells (purple and gold circles) attack and destroy a chain of streptococci (red, disrupted chain at center). Streptococci are bacteria that can cause strep throat. White blood cells play an important role in the body's defense against agents of disease.

SECTION 1 *Nonspecific Defenses*

SECTION 2 *Specific Defenses: The Immune System*

SECTION 3 *HIV and AIDS*

NONSPECIFIC DEFENSES

When a type of virus called a rhinovirus enters the human body, it can cause the common cold. Diseases, such as colds, that are caused by agents that have invaded the body are called **infectious diseases**. This section explains how the human body identifies the agents that cause infectious diseases and defends itself against these agents.

IDENTIFYING PATHOGENS

A **pathogen** is any agent that causes disease. Robert Koch (KAWHK) (1843–1910), a German doctor, was the first person to establish a step-by-step procedure for identifying the particular pathogen that causes an infectious disease. In the 1870s, Koch studied anthrax, a disease of cattle that can spread to people. Koch observed that cattle with the illness had swarms of bacteria in their blood. He hypothesized that these bacteria caused anthrax.

To test his hypothesis, Koch isolated rod-shaped bacteria from a cow with anthrax and grew colonies of the bacteria to be sure he had isolated a single species. Then, he injected healthy cows with these bacteria. The cows developed anthrax. Koch found that the blood of these cows contained the same rod-shaped bacteria as the first cow. Furthermore, healthy cows that he had not injected lacked this type of bacteria. Koch concluded that the isolated species of bacterium causes anthrax. Through these studies, he developed **Koch's postulates**, which are “rules” for determining the cause of a disease. Figure 47-1 illustrates these postulates.

OBJECTIVES

- **Summarize** Koch's postulates for identifying a disease-causing agent.
- **Describe** how the skin and mucous membranes protect the body against pathogens.
- **Describe** the steps of the inflammatory response.
- **Analyze** the roles of white blood cells in fighting pathogens.
- **Explain** the functions of fever and proteins in fighting pathogens.

VOCABULARY

infectious disease
pathogen
Koch's postulates
mucous membrane
inflammatory response
histamine
phagocyte
neutrophil
macrophage
natural killer cell
complement system
interferon

FIGURE 47-1

By applying the four principles of Koch's postulates, scientists can identify the pathogen that causes an infectious disease.

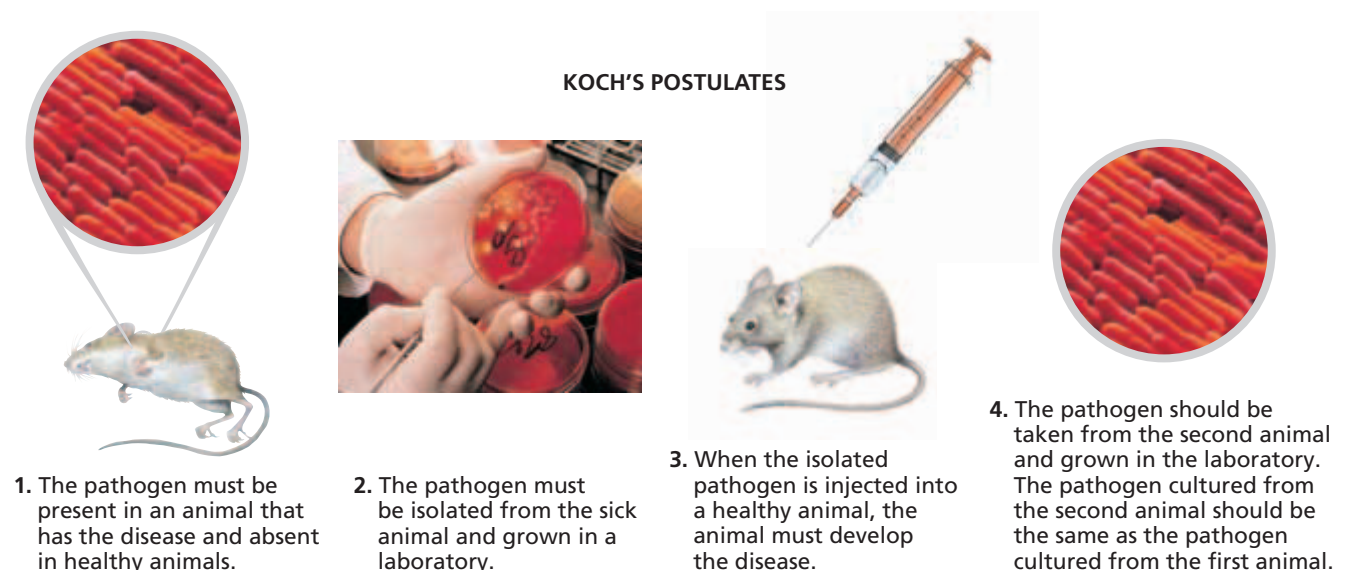


TABLE 47-1 Pathogens Responsible for Some Human Diseases

Disease	Pathogen	Method of transmission
Botulism	<i>Clostridium botulinum</i> (bacterium)	contaminated food
Lyme disease	<i>Borrelia burgdorferi</i> (bacterium)	tick bites
AIDS	HIV (human immunodeficiency virus)	sexual contact, contaminated needles, contact with contaminated fluids from a mother to a fetus or infant
Severe acute respiratory syndrome (SARS)	coronavirus (virus)	person-to-person contact, indirect contact through pathogens in air or on objects (from coughs or sneezes)
Amebic dysentery	<i>Entamoeba histolytica</i> (protist)	contaminated food and water
Athlete's foot	<i>Tinea</i> (fungus)	contact with contaminated surfaces, person-to-person contact
Head lice	Lice (invertebrate parasite)	person-to-person contact, sharing personal items

Scientists have used Koch's postulates to identify thousands of pathogens. Many human diseases are caused by bacteria, viruses, protists, fungi, and invertebrates. Pathogens can spread to humans in five main ways—through air, food, water, person-to-person contact, and the bites of animals. Table 47-1 lists examples of pathogens that cause different human diseases and the means by which each is commonly transmitted.

FIGURE 47-2

The passages of the respiratory system are lined with cells that are covered with beating cilia (purplish strands). Pathogens (bluish circles) that become trapped in mucus secreted by these cells are swept upward, away from the lungs. (5,325 $\times$)



FIRST LINE OF DEFENSE: BARRIERS

The body's nonspecific defenses help protect the body against any pathogen, regardless of the pathogen's identity. Nonspecific defenses include the skin and mucous membranes. **Mucous membranes** are epithelial tissues that protect the interior surfaces of the body that may be exposed to pathogens.

Most pathogens must enter the body to cause disease. The skin serves as a physical barrier to pathogens. Any break in the skin may allow pathogens to enter the body. In addition, the skin also releases sweat, oils, and waxes. These substances contain chemicals that are toxic to many pathogens. For example, sweat contains *lysozyme*, an enzyme that destroys some bacteria.

Mucous membranes serve as a barrier and secrete *mucus*, a sticky fluid that traps pathogens. Mucous membranes line the respiratory and digestive systems, the urethra, and the vagina. The passages of the respiratory tract are lined with cells that are covered with beating cilia, as shown in Figure 47-2. These cilia sweep mucus and pathogens up to the pharynx, where they are swallowed. Most swallowed pathogens are destroyed in the stomach by acids.

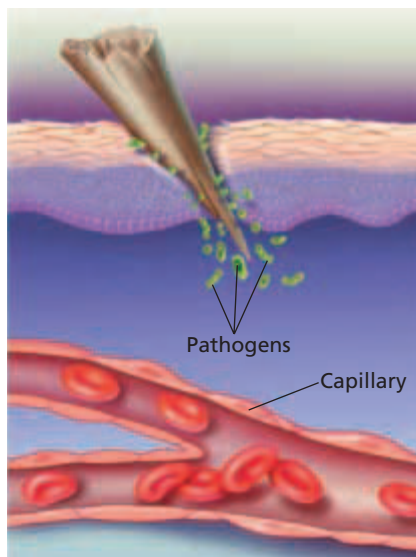
SECOND LINE OF DEFENSE: NONSPECIFIC IMMUNITY

If a pathogen gets past the skin and the mucous membranes, there is a second line of nonspecific defense inside the body—nonspecific immunity. Nonspecific immunity includes the inflammatory response, the temperature response, and proteins. Like the barriers of the first line of defense, these second-line defenses are nonspecific—they work the same way against any pathogen.

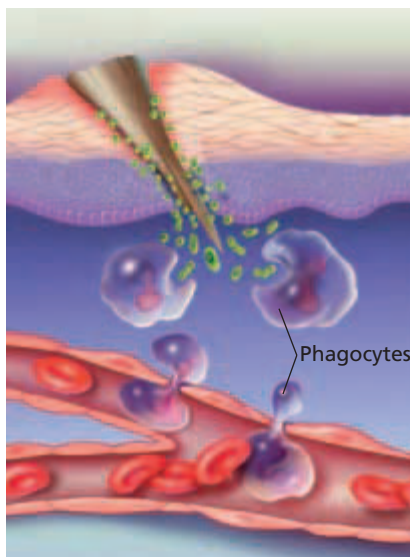
Inflammatory Response

Any pathogen that gets past the skin or mucous membranes will stimulate the **inflammatory response**, a series of events that suppress infection and speed recovery. An example is shown in Figure 47-3. When cells are damaged, whether through a cut on the skin or invasion by pathogens, some of the damaged cells release histamine (HIS-tuh-MEEN), as described in step ①. **Histamine** is a substance that increases blood flow to the injured area and increases the permeability of surrounding capillaries. The changes caused by histamine result in redness, swelling, warmth, and pain. If blood vessels have been damaged, platelets begin the blood-clotting process, sealing off surrounding tissues and stopping pathogens from entering the rest of the body.

White blood cells fight pathogens that have entered the body. In step ②, fluids and white blood cells called *phagocytes* pass through the capillary walls to the injured area. **Phagocytes** ingest and destroy pathogens and foreign matter, as shown in step ③. Phagocytes and some other types of white blood cells are attracted to the site of injury by histamine.



- ① An injury may allow pathogens to get past the barrier of the skin. Injured cells release chemical messengers, such as histamine.



- ② Nearby capillaries respond by swelling and leaking fluid. Phagocytes pass through capillary walls and attack the pathogens.

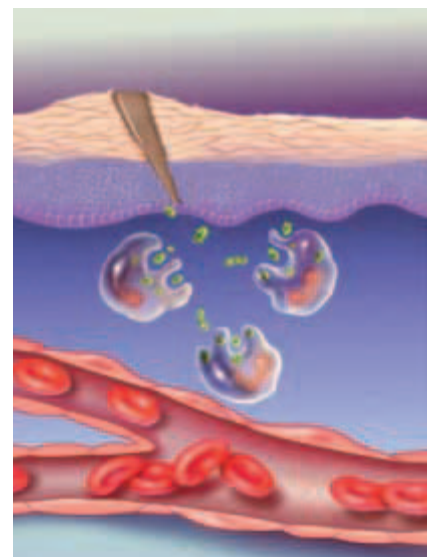
Eco Connection

Agriculture and Human Diseases

The beginning of farming and herding about 10,000 years ago changed the nature of human diseases. When humans began to keep herds of domesticated animals, such as cattle and sheep, humans were exposed to the pathogens that infect these animals. Some of these pathogens then began infecting humans. Measles, tuberculosis, smallpox, and flu are among the diseases that may have been transmitted to humans from domesticated animals.

FIGURE 47-3

Injury to cells triggers an inflammatory response.



- ③ Phagocytes destroy the pathogens, and the injury begins to heal.

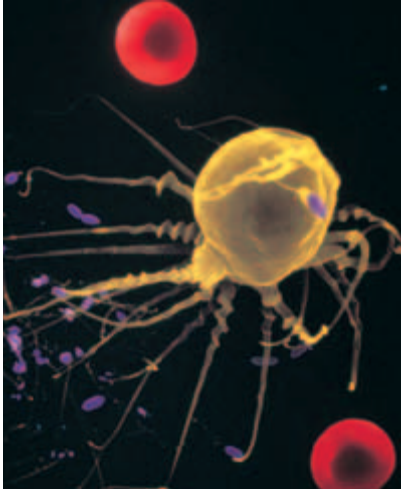


FIGURE 47-4

This macrophage (shown in yellow) is using cytoplasmic extensions to capture bacteria (shown in purple). (SEM 17,400 $\times$)

Word Roots and Origins

macrophage

from the Greek *makros*, meaning "large," and *phagein*, meaning "to eat"

The **neutrophil** (NOO-troh-fil) is the most abundant type of phagocyte in the body. Neutrophils circulate through blood vessels, and they can squeeze through capillary walls to reach the infection site. Once there, neutrophils ingest pathogens they encounter. Another type of phagocyte is the **macrophage** (MAK-roh-FAYJ), shown in Figure 47-4. Macrophages engulf pathogens and cellular debris. Some are stationed in body tissues, waiting for pathogens, while others seek out pathogens.

Natural killer cells are large white blood cells that attack pathogen-infected cells—not the pathogens themselves. Natural killer cells are effective at killing cancer cells and virus-infected cells. A natural killer cell pierces the cell membrane of its target cell, allowing water to rush in and causing the cell to burst.

Temperature Response

When the body begins to fight pathogens, body temperature may increase several degrees. A rise in body temperature above the normal 37°C (98.6°F) is called a *fever*. Fever is a symptom of illness that shows the body is responding to an infection. Some pathogens trigger fever, as do chemicals released by macrophages. A moderate fever may slow bacterial and viral growth and promote white blood cell activity. However, very high fever is dangerous because extreme heat can destroy important cellular proteins. Temperatures greater than 39°C (103°F) can be dangerous, and those greater than 41°C (105°F) can lead to death.

Proteins

Proteins also provide nonspecific defenses. About 20 different proteins make up the **complement system**. Complement proteins circulate in the blood and become active when they encounter certain pathogens. Some of these proteins form a ring-shaped structure that punctures the membranes of infected cells, causing the cells to die. Another nonspecific defense is **interferon**, a protein released by cells infected with viruses. Interferon causes nearby cells to make a protein that helps them resist viral infection.

SECTION 1 REVIEW

1. Explain how Koch tested his hypothesis about the cause of anthrax.
2. How does the body's first line of defense function?
3. What role does greater permeability of capillaries play in the inflammatory response?
4. How do natural killer cells differ from macrophages?
5. What is the role of interferon?

CRITICAL THINKING

6. **Analyzing Information** Scientists can't always apply all of Koch's postulates to determine the cause of a disease. Explain why.
7. **Forming Reasoned Opinions** Should a fever always be treated? Why or why not?
8. **Inferring Relationships** Explain how cold symptoms show that the body is using both lines of nonspecific defenses to fight pathogens.

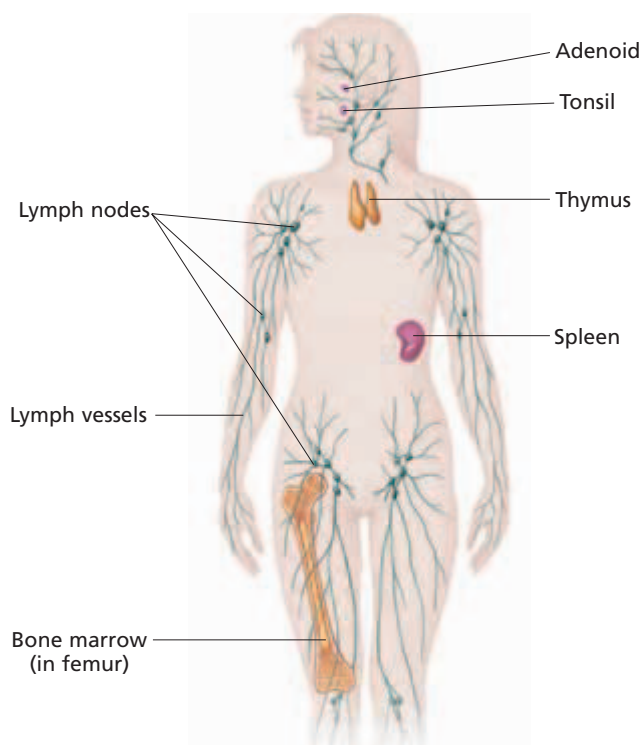
SPECIFIC DEFENSES: THE IMMUNE SYSTEM

Although the nonspecific defenses usually keep pathogens from harming the body, a pathogen sometimes breaks through. In response, the body begins its third line of defense—a response aimed specifically at the pathogen.

THE IMMUNE SYSTEM

The **immune system**, the cells and tissues that recognize and attack foreign substances in the body, provides the body's specific defenses. The immune system fights pathogens and helps to stop the growth and spread of cancers. The immune system is made up of several tissues and white blood cells. The components of the immune system, shown in Figure 47-5, are found throughout the body. The tissues include the bone marrow, thymus, lymph nodes, spleen, tonsils, and adenoids. The white blood cells of the immune system are called **lymphocytes** (LIM-foh-sietz).

Each part of the immune system plays a special role in defending the body against pathogens. *Bone marrow*, the soft material found inside long bones, such as the femur, makes the billions of new lymphocytes needed by the body every day. The **thymus**, a gland located above the heart, helps produce a special kind of lymphocyte.



OBJECTIVES

- **Identify** and describe the parts of the immune system.
- **Explain** how the immune system recognizes pathogens.
- **Compare** the actions of T cells and B cells in the immune response.
- **Relate** vaccination to immunity.
- **Distinguish** between allergy, asthma, and autoimmune disease.

VOCABULARY

immune system
lymphocyte
thymus
spleen
B cell
T cell
antigen
immune response
helper T cell
cell-mediated immune response
cytotoxic T cell
humoral immune response
plasma cell
antibody
memory cell
immunity
vaccination
allergy
asthma
autoimmune disease

FIGURE 47-5

The cells and tissues of the immune system recognize and attack foreign substances in the body.

Word Roots and Origins

antigen

from the Greek *anti*,
meaning "against,"
and *gen*, meaning "producing"

Lymph nodes, located throughout the body along the vessels of the lymphatic system, contain lymphocytes. (Recall that the lymphatic system gathers and filters the fluid, called *lymph*, that leaks from the circulatory system.) Lymph nodes collect pathogens from the lymph and expose them to lymphocytes. The **spleen**, the largest lymphatic organ in the body, stores healthy blood cells, breaks down aging red blood cells, and helps develop lymphocytes and other types of white blood cells. The spleen also collects pathogens from the blood, and the lymphocytes in the spleen attack these trapped pathogens. The *adenoids* and *tonsils* are masses of lymph tissue found in the nose and throat.

There are two types of lymphocytes: B cells and T cells. **B cells** are made in the bone marrow and complete their development there. **T cells** are also made in the bone marrow but complete their development only after traveling to the thymus.

RECOGNIZING PATHOGENS

Lymphocytes can provide specific defenses because they recognize foreign invaders. An **antigen** (AN-tuh-juhn) is any substance that the immune system can recognize and react with. Antigens, as shown in Figure 47-6a, cause lymphocytes to react. A wide variety of substances can be antigens, including pathogens or parts of pathogens, bacterial toxins, insect venom, and pollen. In addition, almost any molecule that is not a natural part of an individual's body, such as that from transplanted tissue or transfused blood of an incompatible type, can act as a foreign antigen. When lymphocytes recognize an antigen, they bind to the antigen to start a specific attack. The reaction of the body against an antigen is known as an **immune response**.

FIGURE 47-6

(a) Antigens are found on a pathogen's surface. (b) The receptor proteins on the surface of lymphocytes (such as B cells, shown here) have a complex, three-dimensional structure. (c) The receptors can bind to antigens that have a complementary shape.

